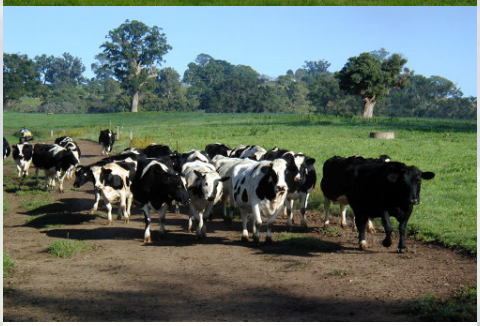













# Accounting for Nutrients on Australian Dairy Farms

**DAV 12307**

Final Report
June 2010

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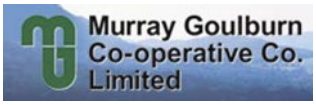












































This report was written and compiled by Cameron Gourley, Warwick Dougherty, Sharon Aarons, and Murray Hannah; with input from many other members of the Accounting for Nutrients project team.

For further information about 'Accounting for Nutrients on Dairy Farms' visit the project website at www.accounting4nutrients.com.au or contact Cameron Gourley on: 03 56 242226 (office).

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# 1. Executive Summary

Historically Australian farming relied on effectively recycling nutrients to sustain productivity. However modern agriculture increasingly depends on importing rather than reusing nutrients. This is particularly the case in the dairy industry where continuing intensification, reflected in increasing stocking rates and milk production per cow, has been largely driven by greater levels of feed and fertiliser inputs. The imbalance between the amount of nutrients imported onto the farm and that leaving the farm in product has consequently lead to significant on-farm nutrient accumulation and nutrient losses to water and the atmosphere.

There are increasing pressures on the Australian dairy industry to improve nutrient use on dairy farms, and in particular to reduce phosphorus (P) and nitrogen (N) losses. The costs of nutrients in fertiliser have increased substantially over the past decade, caused by an expanding demand in developing nations, diminishing high quality raw materials, and the strong link between fertiliser and energy costs. Additionally, dairy farming in Australia is largely consolidated in the cooler and wetter, or irrigated southern coastal areas which also support a rapidly growing urban population. Societal expectations for clean water and air, both domestically and internationally, along with competition for land and water resources are likely to increase. Despite these external pressures, dairy farming continues to intensify, and nutrient excesses are likely to further increase with even greater nutrient losses to the broader environment.

It is within this context that the national Accounting for Nutrients on Australian Dairy Farms project (A4N) was undertaken. The overall project objective was to provide the dairy and fertiliser industries and other stakeholders, with information and tools for improved nutrient use on dairy farms.

Specifically this project aimed to:

i. Develop consensus and commitment from the Australian dairy and fertiliser industries towards a nationally agreed framework for nutrient accounting.

ii. Improve our understanding of nutrient requirements, nutrient flows and potential losses at a paddock, farm and catchment scale.

iii. Develop, refine and validate a nutrient auditing framework for N, P, potassium (K), sulphur (S), calcium (Ca) and magnesium (Mg) at a range of complexities.

iv. Improve the knowledge and practices of the fertiliser industry, private and government agricultural consultants, dairy industry field officers, and farmers relating to nutrient management on dairy farms.

The project activities were structured within five modules, namely:

1) governance, partnerships, standardisation, and integration,
2) defining the key nutrient pools and transformations for Australian dairy farms,
3) addressing knowledge gaps within nutrient accounting frameworks,
4) constructing a nutrient accounting framework for Australian dairy farms, and
5) delivering information and tools.

A key part of the Accounting for Nutrients project was the measurement and monitoring of nutrient imports, exports and within-farm nutrient flows on dairy farming operations across Australia. Forty-four commercial dairy farms were involved in this study, representing a broad range of geographic locations, productivity, herd and farm size, reliance on irrigation, and soil types. Four of the selected 44 dairy farms were certified organic producers, and as such did not use conventional inorganic fertilisers or chemicals. On-farm data collection began in December 2007 and the final visit to report back to the participating farmers was in December 2009. The standardised production year for determining nutrient flows and whole-farm nutrient balances was between February 2008 and February 2009. Forty-one dairy farms provided comprehensive information which enabled whole-farm nutrient balances and nutrient use efficiencies to be determined.

**Engagement** Outside of the project farmers, more than 50 people were directly involved in project activities. These included scientists, technicians, field staff, fertiliser company agronomists, and industry and government policy players from across Australia. Sixteen regional teams were actively engaged in working with the participating dairy farmers and collecting farm information.

Early and continued engagement with more than 15 stakeholder organisations, including dairy and fertiliser companies, government policy, extension and research groups, and dairy farmers, provided important feedback and ensured the on-ground relevance of project activities and outcomes. Particular strengths of the project identified by these stakeholders included: (i) strong national leadership, (ii) sustained partnerships among scientists and key next-users, (iii) improved information regarding the diversity, complexity and impact of nutrients on dairy farms and the challenges facing the Australian dairy industry, and (iv) new research tools generated by the project.

This extensive national network provided large benefits at many levels, including skill development, enhanced knowledge, and standardisation of information and approaches, that benefited the project and project outcomes.

**Innovative on-farm research** The Accounting for Nutrients project demonstrated the benefits of a well structured on-farm research program. Preliminary planning and review resulted in an innovative project design that enabled the investigation of nutrient fluxes and transformations at the component, farm and industry scale. The broad range of dairy farms studied across Australia provided information that was highly relevant to current and future dairy systems.

Well-defined and thorough data gathering methods were developed and the time and effort assigned to developing and reviewing detailed protocols for all facets of data collection was essential. A number of these protocols, whilst designed specifically for the project, should serve as important templates for future on-farm research work. The collaborating farmers became active and enthusiastic partners in the research project and made an essential contribution to the collection of farm-based data, which was of overall high quality.

A custom built relational database was designed to manage the large amount of farm-based information, including information on farm bio-physical characteristics, productivity and nutrient management metrics. This database allowed for a systematic process of data quality assurance, collation and interpretation, and provides a valuable resource to other dairy projects; especially modelling based projects.

More than 2000 dairy feed, milk and manure samples from the participating dairy farms were collected. The subsequent collation and statistical interpretation of this data generated important new Australian knowledge about nutrient contents and sources of variation, and will serve as a valuable resource for other current and future projects.

The development of a nutrient balance empirical model allowed the research team to calculate nutrient balances and nutrient use efficiencies for the participating dairy farms. This model incorporates ‘best available’ data and algorithms and includes over 30 data input sheets that are designed to accept data generated in custom queries from the project database. While specifically designed as a research tool, the empirical model has the potential to form the basis of a nutrient budgeting tool for industry.

**Whole-farm nutrient balances and efficiencies** The efficiency of nutrient use in dairy production is limited by the biological potential of cows to transform feed nutrients into milk and of crops and pastures to convert applied nutrients into forage and other agronomic products. However, decisions made about fertiliser types, rates, timing of use, the types and nutrient concentration of feed provided to dairy cows, and the capture and reuse of excreted nutrients, are all management factors which can greatly influence nutrient use and nutrient losses.

The difference between nutrient imports and exports; termed nutrient surplus, is a measure of the amount of nutrients accumulating on the farm and/or lost to the broader environment. For comparison between farms, this is usually standardised to kg of nutrient per ha. The whole-farm nutrient use efficiency is a measure of the proportion of the imported nutrient on to a farm within a production year, which is exported in product.

The detailed quantification of imports and exports on dairy farms enabled an accurate assessment of the major sources of nutrients responsible for nutrient imbalances. An important potential source of error in calculating nutrient balances is the variability that can be associated with the mass and concentration of various sources of nutrients. By determining the uncertainty around the nutrient content of individual sources, particularly dairy feeds, this project provided guidance in the use of standardised 'book values' and a high degree of confidence in the balance and efficiency determinations.

Feed and fertiliser are generally the most significant contributors to nutrient imports and the amounts are principally determined by stocking rate and type of imported feed. However there may be a wide variety of other sources of nutrients on any individual dairy farm. For example, a significant source of N on farms with low N fertiliser use was from N fixation by legumes, while irrigation of bore water could be a significant source of calcium and magnesium.

The amounts of nutrients imported on to the dairy farms in this study varied greatly, as did the production benefit. While there was a strong and positive link between the amount of N imported in fertiliser and milk production overall, this was not the case for P, K, and S, despite the investment in fertiliser to deliver these nutrients. Ca and Mg imports were also poorly related to milk production, although they are rarely purchased directly in fertiliser.

The whole-farm nutrient use efficiencies were in general low and often less than 25%, although dairy farms that were not applying any fertilisers may have had P and S efficiencies of > 100%, when they were mining stored nutrients and exporting more than was imported.

Whole-farm nitrogen surplus for the 41 dairy farms ranged from 47 to 600 kg N/ha and was strongly related to the level of milk production. The whole-farm N use efficiency ranged from 14 to 50%, with a median of 26%. Based on these findings, an average of around 14 g of N is lost to the broader environment for each litre of Australian milk produced. While the potential N losses appear high, these Australian results are comparable to those measured in other industrialised dairy industries such as New Zealand, the EU and USA. It is interesting to note however that two thirds of the farms in this study had annual N surpluses above the current European Union compliance standard (180 kg N/ha/year) and this proportion is a likely reflection of the Australian dairy industry as a whole. While no such standards currently exist in Australia, international markets may assign similar standards in the future.

Whole-farm P surplus ranged from -7 to + 133 kg P/ha, with a median of 28 kg/ha. The phosphorus use efficiencies across the 41 dairy farms ranged from 6 to 158%, with a median of 35%. K surplus ranged between 13 and 452 kg K/ha, with a median of 78 kg K/ha. The K use efficiencies ranged from 9 to 48%, with a median value of 20%. Whole-farm S surplus ranged from -1 to 184 kg S/ha, with a median value of 27 kg S/ha. The overall sulphur use efficiency ranged from 6 to 110%, with a median value of 21%.

The poor relationship between P, K and S fertiliser inputs and milk production from home-grown pasture and crops is not surprising given the high soil P, K and S levels measured on these farms and the relative availability and nutrient supply capacity of these dairy farm soils.

**Nutrient intakes and nutrient loading in specific locations** The efficiency with which dairy cows utilise feed nutrients to make milk (i.e. the ratio of milk nutrients exported to total feed nutrients consumed) significantly influenced the amounts of nutrients excreted. On these dairy farms, supplementation of pasture-based diets to increase milk production substantially increased dietary crude protein and phosphorus, potassium and sulphur intake. As nutrient intakes increased, so did excretion of these nutrients, resulting in animal nutrient use efficiencies generally < 20%. On average across all the herds studied, almost half a kilogram of N and K was excreted daily by each cow, about 7 times the amount of P and S excreted.

The calculated nutrient excretion rates (g of nutrient/cow/day) for the lactating herds on each farm at each survey visit were combined with farm-specific spatial and temporal cow movement data to determine nutrient deposition loads (kg/day) to locations around dairy farms. Overall, cows spent the greatest proportion of their time in pasture paddocks, although this varied markedly between farms and in different seasons within the same

farm. As the reliance on imported feed increased, the proportion of time spent in other places such as feed pads increases. Paddocks closest to the dairy had a much greater proportion of 'cow time' than those further away and subsequently had much greater nutrient loading rates. While the overall proportion of nutrients excreted in holding yards, dairyshed, feed pads and laneways was generally less than that on paddocks, deposition rates in these small areas constituted very high loading rates on a 'per ha' basis and a significant source of nutrients losses from gaseous emissions, runoff and leaching processes. This improved understanding of the spatial and temporal patterns of animal movement and nutrient deposition have important implications for targeted improvements in nutrient capture and recycling, nutrient accumulation and potential losses.

**Soil nutrient status on Australian dairy farms** The wide-ranging sampling and analysis of soils from more than 2000 paddocks on the participating dairy farms provided important insight into current soil nutrient status and fertiliser requirements. The statistical analysis of relationships between soil fertility and various management factors allowed for a comprehensive exploration of the factors influencing nutrient accumulation on dairy farms.

More than three quarters of the paddocks sampled had available phosphorus, potassium and sulphur levels above agronomic requirements. At these levels most pastures and crops are unlikely to produce additional dry matter from fertiliser inputs. An exception to this was paddocks sampled on organic or biodynamic dairy farms. These farms generally have significantly lower soil phosphorus and sulphur fertility, with pastures and crops potentially responsive to the application of soluble fertiliser inputs.

Within farm nutrient heterogeneity was substantial irrespective of the characteristics of the dairy farm studied. High nutrient loading from the deposition of animal excreta is clearly a key driver of elevated soil nutrient levels and the diversity of soil fertility levels. Paddocks with high densities of animals per ha can have very high nutrient accumulation while those infrequently visited or with low stock densities will generally have lower soil nutrient levels.

These results demonstrate that there are substantial opportunities to reduce fertiliser inputs on most dairy farms without a loss of production. In many cases, high levels of soil P, K and S may supply necessary plant nutrients for a number of years before maintenance nutrient applications may be required. Where fertiliser applications are warranted for increasing pasture and crop productivity, a more targeted approach should be undertaken. The relatively small costs associated with undertaking a strategic and on-going soil sampling program are likely to be returned many times through the potential savings in unnecessary fertiliser expenditure. The clear link between spatial distribution of nutrient levels and animal management provides informed guidance on where nutrients are likely to be required and approaches that farmers should consider to minimise nutrient accumulation.

**Communication and delivery** A range of communication activities and supporting tasks have been delivered during the 4 years of the project. While the principal project focus has been on science methodology, data collection and interpretation of dairy farm nutrient flows and transformations, around 170 communication activities have been undertaken nationally. The majority have been written news articles published in specific rural newspapers or magazines, and presentations to stakeholder groups, including farmer discussion groups and field days, industry training forums, specialist industry stakeholder groups, and national industry conferences. Additionally the project team was involved in a number of complimentarily activities including developing on-farm rapid assessment techniques, soil nutrient mapping standards, and national dairy nutrient management planning guidelines.

Another key product from the project is the 'Dairy nutrient accounting technical manual'. This manual outlines and justifies the requirements for determining a nutrient balance on dairy farms, how to accurately calculate balances and efficiencies based on the best available data, and how to interpret the information.

Seven scientific publications have already been written, and peer reviewed for publication in scientific journals and science forums. The publishing of information from the project is an important step in consolidating the scientific basis of the project information and industry implications. It is anticipated that another 4 scientific papers will be produced over the forthcoming 12 months.

A number of next-users have actively sought and are already using project research outputs, sometimes with direct input from the A4N project team. Other key next user groups, such as some fertiliser companies, are preparing to update their current nutrient recommendation systems with the research outputs from the project. Many of the farmers directly involved in the project have made substantial changes to the way they make nutrient decisions. The overall relevance of the research work and the use of regional case-studies have provided an important vehicle for delivering information to other farmers.

**Recommendations** Improved nutrient management of Australian dairy farms requires solutions to complex farm system problems, such as unnecessary use of fertiliser, excess levels of nutrients in cow diets, poor manure collection and land-spreading techniques, heterogeneous nutrient distribution through animal management, and in some cases, farmers' unwillingness and/or inability to adopt improved knowledge and practices.

Despite increases in fertiliser prices in recent years, market signals do not appear to be resulting in improved nutrient management practices on dairy farms. In general, fertiliser decisions continue to be broad in scale and the economic return from fertiliser inputs appears to be of lesser concern than ensuring that nutrient levels are not restricting productivity.

Nutrient accounting provides a more holistic approach to dairy farm nutrient management, encompassing the needs of farmers to meet production goals, identifying management opportunities for improving nutrient use efficiencies, and reducing the risk of off-farm nutrient loss. In the USA and Europe, and more recently in New Zealand, nutrient accounting tools, used in association with traditional soil testing, have become a central policy instrument. The Australian dairy industry would be well served with a nationally standardised nutrient accounting decision support tool, which should be freely available for farmers, advisors and other stakeholders.

While there are clearly biological limits, there are some substantial opportunities to improve nutrient use efficiency on dairy farms and therefore enhance overall productivity and decrease the loss of nutrients to the broader environment. This has been well demonstrated by the quantification of various nutrient pools, fluxes and transformations that occur across a large number of Australian dairy farms. Decreasing the surplus and increasing the efficiency of P, K and S is likely to result from a reduction in fertiliser imports and provide important cost savings. Despite the strong link between N use and milk production, the range in N use efficiency measured across these dairy farms demonstrate the potential for gains in pasture and crop productivity, and a decrease in the amount of purchased feed required.

Key opportunities to improve nutrient cycling and nutrient use efficiency identified within this study have included (i) a reduction in unnecessary dietary nutrient concentrations, (ii) improved spatial and temporal movement of animals within dairy farms to reduce heterogeneous nutrient deposition, accumulation and losses, (iii) increasing the capture, storage and redistribution of excreted nutrients, and (iv) more strategic fertiliser and effluent applications.

From a research perspective, there is a need to more thoroughly define the on-farm management decisions which may result in greater nutrient use efficiencies and to quantify the actual productivity gains from capturing more nutrients for productive use within dairy farms. This information will be critical in developing and applying recommendations that have a high probability of being implemented on dairy farms.

Specifically we recommend that:

i. All dairy industry stakeholders, including farmers, government and commercial players are aware of excess nutrients on dairy farms, the major causes, and opportunities for mitigation and enhanced production, particularly as the dairy industry continues to strive for increased productivity.

ii. Standardised approaches are used to determine nutrient budgets across Australia. This can initially be improved through the endorsement and promotion of the Nutrient Accounting Technical Manual produced by this project. In the future, there would be significant benefit in the availability of a comprehensive web-based tool, such as is freely available in New Zealand.

iii. The uneven distribution of nutrients within dairy farm is highlighted to dairy farmers and service providers. Of particular note are nutrient hotspots which may contribute to significant nutrient exports to the environment. There is an ongoing need to promote strategic soil testing and more spatially and temporally specific fertiliser and effluent applications.

iv. Specific farm systems with likely high nutrient surpluses and losses in environmentally sensitive areas are identified and provided with targeted assistance to increase nutrient use efficiency and decrease nutrient losses.

v. Specific management strategies that have the potential to improve nutrient use efficiency on dairy farms should be investigated on commercial dairy farms to determine the economic, environmental and social costs and benefits. Using industry models such as Udder and DairyMod in the first instance would offer a cost effective research strategy prior to any field studies.

vi. New and existing research projects should take the opportunities to link production and environmental outcomes and the implications at the farm level through the use of the nutrient accounting empirical model.

vii. Trials should be established that demonstrate that there are financial benefits and no reduction in productivity when fertiliser inputs are reduced on soils with high soil fertility levels.

viii. The potential productivity, profitability and N efficiency of pasture legumes as a nitrogen source in contrast to nitrogen fertiliser should be investigated.

ix. A review be undertaken of innovative N fertiliser technologies being developed and trialled internationally, and where appropriate evaluate their applicability in Australian dairy systems.

x. The comprehensive information collected from the participating dairy farms be further utilised to evaluate the corresponding energy, carbon, greenhouse and water footprints of these systems, to ensure that industry recommendations on nutrient management don't conflict with goals relating to other environmental objectives.

# 2. Scope of this report

This final report describes the activities and outcomes arising from the national 'Accounting for Nutrients on Australian dairy farms' project for the period July 2006 to June 2010 inclusive. It is not possible within this report to present all data, due to the large amounts collected and analysed, particularly from the on-farm research work. Therefore selected examples and comprehensive summaries are used to provide the project based evidence to justify the key outcomes and recommendations.

The format of this report has been structured to reflect the 'modules' in the original project proposal, the relevant project activities and how they relate to the project objectives and outputs. The final sections aim to provide a detailed 'discussion' of the project findings, implications for the industry, recommendations for improvements in current nutrient management practices, and future research needs for the Australian dairy industry.

The key topic areas addressed within this final project report are as follows:

- The structure and function of the national A4N project
- The development of a nutrient accounting framework for Australian dairy farms
- The approach used to define nutrient flows, pools and transformations on dairy farms
- Quantifying nutrient balances, nutrient use efficiencies, and distribution patterns at multiple scales
- The role played by A4N in national nutrient management programs
- Direct industry engagement and delivery of project information
- Opportunities and challenges for improved nutrient management on dairy farms

# 3. Project background

Along with most other dairy producing countries, the Australian dairy industry continues to undergo significant change. The number of dairy farms has declined over the last 30 years, from over 22,000 in 1980 to around 8,000 in 2010. Over the same period, average farm herd size has increased from 77 cows in 1980 to 258 in 2010 whilst average annual milk production per cow has increased from 2,750 to 5,500 L.

A key driver of increased per cow productivity has been the increase in supplementary feeding and increasing forage yields and quality due to fertiliser use, particularly N. In 1980 most dairy farms were almost totally reliant on 'home grown' pasture and conserved forage. In recent times > 90% of all dairy farms used imported concentrates or grain, with the average dairy farm supplementation greater than 1.5 tonne per cow per year. The other major supplement brought on to dairy farms in Australia is hay, usually fed in equivalent amounts to grain. As expected, there is considerable variation in the amount and type of diet supplementation of lactating dairy cows, with grain inputs varying from 0 to 2.5 tonne DM per cow per year and forage inputs varying from 0 to 1.5 tonne DM per cow per year.

While nitrogen (N) and phosphorus (P) inputs are required for most dairy operations, when used in excess, they can significantly degrade air (N) and water quality (P & N). Phosphorus losses from dairy farms occur mainly through surface water transport and to a lesser extent leaching to ground water, leading to eutrophication of water storages, lakes and rivers. Nitrogen losses include the volatilization of ammonia ($NH_3$), resulting in particulate formation (haze) in the atmosphere and subsequent re-deposition, acidification and eutrophication of surface waters, the emission of the potent greenhouse gases NO and $NO_2$, and nitrate leaching into surface and ground water.

The risk of nutrient pollution from a dairy farm increases when nutrient inputs exceed the amount of nutrients leaving the farm in products. Total N and P inputs onto dairy farms, mainly in the forms of feed, fertiliser and N fixation by legumes, are usually much greater than the outputs of P and N in milk, animals, and crops. These surpluses tend to increase as farms intensify and stocking rates increase. Excess P on dairy farms can result in increasing soil P levels beyond agronomic requirements, which may also increase the concentration of dissolved P in surface runoff and leachate. Unlike P, N is not significantly buffered by soils, and where N is applied in high concentrations such as in dung, urine or fertiliser, losses though volatilization and leaching can be high. Excess potassium (K) can accumulate in soil and feed, and can cause severe metabolic disorders in ruminants.

Over the past 20 years, a range of environmental policies have been developed and implemented in Europe and the US, and more recently in New Zealand, with the aim of reducing nutrient losses from dairy farms to the environment. Central too many of these policy approaches has been the development and on-farm implementation of nutrient budgeting tools.

The changing nature of Australian dairy operations, the increasing societal pressure on the farming community to reduce nutrient losses to water and air, and the need to provide evidence that farm practises are meeting environmental standards, justifies the need for the development and implementation of a nutrient budgeting approach to improve nutrient management practices on Australian dairy farms. Nutrient budgeting will assist farmers to meet production goals and identify opportunities for improvements in nutrient use efficiencies, decrease nutrient surplus and accumulation on dairy farms and reduce the risk of off-farm nutrient losses. A standardised and well adopted nutrient budgeting approach will enable the Australian dairy industry to take a pro active approach in addressing nutrient management issues using strong science outcomes rather than waiting to be driven by environmental policy.

# 4. Project objectives and activities

To provide the dairy and fertiliser industries and NRM stakeholders, with improved information and tools for multi-nutrient management enabling greater nutrient use efficiencies on farm that will contribute to a reduction in off-site nutrient losses.

Specifically this project will:

- Develop consensus and commitment from the Australian dairy industry and the Australian fertiliser industry towards a nationally agreed framework for nutrient accounting.
- Improve our understanding of nutrient requirements, nutrient flows and potential losses at a paddock, farm and catchment scale.
- Develop, refine and validate a nutrient auditing framework for N, P, K, S, Ca and Mg at a range of complexities.
- Improve the knowledge and practices of the fertiliser industry, private and government agricultural consultants, dairy industry field officers, and farmers relating to nutrient management within dairy farming systems.

To address these objectives, the Accounting for Nutrients project has undertaken the following activities:

I. Foster strong and meaningful partnerships across key stakeholder groups, including research, extension and policy-based agencies, milk processing and fertiliser companies and dairy farmer organisations.

II. Review the components, structures, strengths and weaknesses of existing multi-nutrient accounting approaches used locally and in the EU, USA and New Zealand.

III. Describe nutrient inputs/outputs, comparative nutrient efficiencies, components of nutrient flows, and nutrient distribution for N, P, K, S, Ca, and Mg for a cross section of 44 Australian dairy farms.

IV. Identify key knowledge gaps associated with nutrient pools, transformations, retention and losses in dairy farming systems, at the paddock, farm and catchment scale.

V. Undertake specific desktop reviews and research on key knowledge gaps to refine the nutrient accounting framework.

VI. Develop and refining a multi-nutrient auditing framework for N, P, K, S, Ca and Mg, allowing nutrient auditing to be conducted at a number of depths of complexity, ranging from a simple whole farm assessment to a detailed spatial and temporal audit of the farm at an individual paddock scale.

VII. Integrate the nutrient accounting framework, with soil fertility targets, nutrient distribution, and the Farm Nutrient Loss Index to inform on-farm nutrient management decisions, and 'road test' the framework through the national network on selected farms, at the paddock and farm scale.

VIII. Assist front-line users, such as fertiliser companies and dairy processors, in the development and capture of tools and approaches, including opportunities with Web-based information and tools.

IX. Prepare and disseminate targeted articles, publications, and newsletters, as well as scientific publications to peer-reviewed journals and conferences, based on the outcomes of these research activities.

X. Ensure that project outputs are used to inform policy options for driving improved nutrient management practices on Australian dairy farms.

XI. Deliver information and tools so that they are well integrated with current dairy farmer training programs that are offered by government extension staff, private consultants, fertiliser companies, CMA's and dairy processors.

These activities are structured in 5 project modules, namely

(i) governance, partnerships, standardisation, and integration,

(ii) defining the key nutrient pools and transformations for Australian dairy farms,

(iii) addressing knowledge gaps within nutrient accounting frameworks,

(iv) constructing a nutrient accounting framework for Australian dairy farms, and

(v) delivering information and tools.

# 5. Project teams

The project is lead by the Project Science Management team, which has been responsible for attracting funding, project planning, managing resources, method development and implementation, data collation, data interpretation and project delivery. Additionally, the on-farm project team (OFT) has played a crucial technical advisory and delivery role, while other players from associated collaborating organisations, notably represented by the National Network team (NN), and the Science Advisory group, have provided technical advice. The Project Advisory Committee (PAC) has played a key role in strategic advice, review of project progress and deliverables, and endorsement through their industry and government networks.

## 5.1 Project Science Team

| | |
|---|---|
| Dr Cameron Gourley | Senior Research Scientist and Project Leader |
| Dr Warwick Dougherty | Research Scientist and Deputy Project Leader |
| Ivor Awty | Senior Technical Officer |
| Jenny Collins | Technical Officer |
| Paul Durling | Senior Technical Officer |
| Donna Gibson | Technical Officer |
| Dr Sharon Aarons | Senior Research Scientist |
| Lianne Dorling | Technical Officer |
| Sally Sceney | Project Officer |
| Andrew Smith | PhD student |
| Murray Hannah | Biometrician |

**Dr Cameron Gourley**

**Project Leader**



**Dr Warwick Dougherty**

**Deputy Project Leader**



## 5.2 Project Advisory Committee

The role of this committee has been to provide strategic advice and endorsement of activities relating to the project. Specifically it has had responsibility for reviewing progress at different stages of the project according to agreed timelines and milestone reports, communicating with various stakeholders to capitalise on the strong awareness associated with the project, build on project products and technical and extension information, and identifying further opportunities for information transfer and access to various audiences.

Membership includes representatives from the principal funding bodies, the fertiliser industry, and various State and Commonwealth agencies across Australia.

*Project Advisory Committee members and their affiliation*

| | |
|---|---|
| Nick Drew (Chairperson) | Fertiliser Industry Federation of Australia |
| Nigel Bodinnar / Charlie Walker | Incitec/Pivot |
| Ken Peverill | Dairy Australia |
| Cathy Phelps | Dairy Australia |
| David Marland | DPI Victoria |
| Peter Orchard | DPI NSW |
| David Windsor | Ag WA |
| Catherine Viljoen (resigned) | Land and Water Australia |
| Ian Grant | Megafert, Australia |
| Karen Romano / Annette Zurrer | GippsDairy |
| Malcolm Sellen | Dairy farmer, Gippsland |
| Joe Chittick | Dairy farmer NSW and DIDCO board member |
| Beverly Wallace | Australian Dairy Farmers Federation |
| Joanne Caminiti | West Gippsland Catchment Management Authority |
| Andrew Gallagher | Murray Goulburn Co-op |
| Ken Garner | Bega Cheese |
| Marlos DaSouza | Victorian EPA |



**Accounting for Nutrients National Project Advisory Committee team members**

## 5.3 National Network

A key mechanism to ensure that the project delivered relevant outcomes at a national level was the establishment of a National Network (NN) team during the project development phase of the project. The NN consisted of representatives from research, extension and policy staff from state agencies across Australia, national dairy industry policy groups, private dairy consultants, dairy farmers and the fertiliser industry.

The NN team contributed to the project through the provision of technical input, information and data generation, interpretation and the development of a nutrient accounting framework. The NN team required that regionally specific issues were addressed, and contributed to the direction of communication activities.

*National Network members and affiliation*

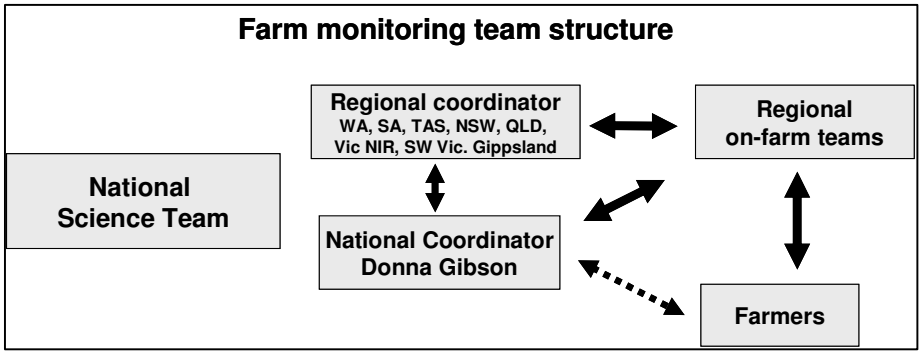
| | |
|---|---|
| Ashley Senn | DPI NSW |
| Nigel Fleming | SARDI |
| David Weaver, Martin Staines, Rob Summers, Nardia Keipert | WA Dept of Agriculture & Food |
| Rob Chataway, Kevin Lowe | DPI Qld |
| Anne Crawford, Richard Eckard, Andrew Smith | The University of Melbourne |
| Lucy Burkitt, Jessica Coad | TIAR |
| Jeanette Howie, Gary Maslen, Chris Bagot | Dairy farmers |
| Neil Lane | Interlac |
| Wayne Pluskie | Back Paddock Pty Ltd |
| Alan Fletcher | Incitec Pivot Ltd |
| Andrew Gallagher | Murray Goulburn Co-op |
| Ken Garner | Bega Cheese |
| Scott McDonald, Graeme Ward, Kerry Greenwood | DPI Victoria |
| Janine Price, Wayne Bath | EPA Victoria |



**Accounting for Nutrients National network team members at a project workshop in May 2007**

## 5.4 On-farm monitoring team

A key part of the Accounting for Nutrients project was the measuring and monitoring of nutrient imports, exports and within-farm nutrient flows on a broad range of dairy farming operations across Australia. This involved quarterly farm visits to 44 participating dairy farms by the Accounting for Nutrients 'on-farm' teams across Australia. The on-farm monitoring teams were responsible for liaising with farmers, organising visits, ensuring that they are familiar with the sampling requirements, and collecting information and samples during the designated quarterly and other occasional visits.




The national on-farm monitoring team included:

**Victoria:** Donna Gibson, Paul Durling, Ivor Awty, Jenny Collins, Lianne Dorling, Cameron Gourley, Sharon Aaron, Leah de Vries – Department of Primary Industries Victoria; Andrew Smith - University of Melbourne.

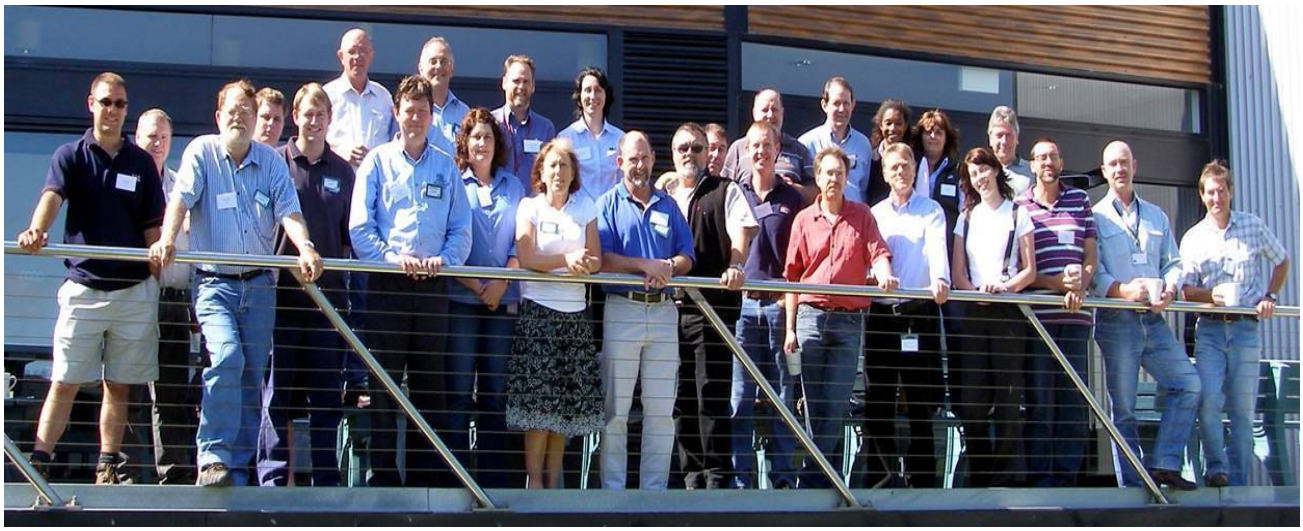
**New South Wales:** Warwick Dougherty, Ashley Senn, Rick Jennings, Craig Muir, Michael Davy, Scott Richards, Ken Giddings, Katrina Sinclair - NSW Department of Primary Industries.

**South Australia:** Nigel Fleming, Phil Lewis -SARDI

**Tasmania:** Jessica Coade, Lucy Burkitt -TIAR

**Western Australia:** David Weaver, Martin Clarke, John Grant -Department of Agriculture and Food, WA.

**Queensland:** Kevin Lowe, John Lindsay, Rob Chataway; Department of Primary Industries QLD.



**Accounting for Nutrients 'On-farm' monitoring team members, Ellinbank workshop Nov 2007.**

## 5.5. Science Reference Group

An important part of the overall A4N project, and in particular the development of the Nutrient Accounting Framework, has been the thorough peer review process. This ensured that the best scientific approaches were utilised, data interpretation was the most rigorous possible, and that all necessary assumptions were reviewed and based on the best available scientific information. This peer review involved two key activities: (i) referral of issues to leading scientists and industry experts, and (ii) publishing of project information in peer reviewed journals.

This peer review process was initiated during the early project development phase, was an important part of developing a national set of protocols for the on-farm research, and has continued throughout operational activities, data collation and interpretation stage within the A4N project.

The Science Reference Group consisted of key scientists and industry experts drawn from various state agencies, university institutions and commercial dairy industry players and provided critical scientific oversight of the development of the Nutrient Accounting Framework. Where necessary expertise from outside the Science Reference Group was sought. This group continues to provide advice on peer review strategy and where appropriate contribute to the publication of technical and scientific articles relating to the Nutrient Accounting Framework.



**Accounting for Nutrients 'Science Reference Group' Ellinbank meeting July 2008.**

The Science reference group included:

Dr Cameron Gourley, Dr Sharon Aarons, Murray Hannah– Department of Primary Industries Victoria
Dr Richard Eckard, Andrew Smith - University of Melbourne.
Dr Warwick Dougherty - NSW Department of Primary Industries.
Nigel Fleming -SARDI
Dr Lucy Burkitt, Jess Coad –Tasmanian Institute Agricultural Research
Dr David Weaver - Department of Agriculture and Food, WA.
Kevin Lowe, Rob Chataway - Department of Primary Industries QLD.
Dr Ian Fillery – CSIRO, WA.
Professor Mark Powell - USDA Dairy Forage Research Centre, Wisconsin, USA.

# 6. Project Activities and Outputs

## 6.1 Governance, partnerships, standardisation, and integration.

Project activities with this module aimed to develop partnerships, and ensure standardisation and integration at a regional, state and national level for the long term implementation of improved nutrient management across the Australian dairy industry. The proposed national players, relationships and responsibilities are provided in Appendix 2.

**Key activities and achievements**

**Module 1. Governance, partnerships, standardisation, and integration.**

- ✓ Funding and other contributor's collaboration agreements finalised with $3 m in cash investment and $4.7 m in-kind over the 4 years of the project
- ✓ Establishment and expansion of National Network, Project Advisory Committee and On-farm project teams
- ✓ Establishment of Science Advisory Group
- ✓ Presentations and engagement with high level industry and government stakeholders
- ✓ Development of detailed 'next-user' strategy
- ✓ Development of 'Nutrient Accounting Package – product specifications' document.
- ✓ Project Advisory Committee 6 monthly meetings (9 meetings overall)
- ✓ Formal independent review (Nov 08), review report and project team response
- ✓ Development of national nutrient management planning guidelines
- ✓ Agreement with principal investors (DPI Victoria and DA) regarding A4N Consolidation and Delivery year (2010/11)

### 6.1.1 Meeting Next-User needs

Accounting for Nutrients (A4N) is a large national research project with the aim of generating new knowledge and specialist research outputs to improve nutrient management on dairy farms. The project team and major stakeholders recognised that the project's research outputs are principally delivered to next-users such as fertiliser and dairy companies, other scientists, government extension agencies and policy analysts. In particular, next- user groups such as agribusiness firms and staff (fertiliser industry, milk companies) will be the major delivery mechanisms for the project's outputs to dairy farmers. It will be these next-users who will generally directly interface with the end-users (i.e. dairy farmers) and ultimately influence the management of nutrients on dairy farms. Consequently, representatives of these key next-user groups predominantly make up the PAC committee.

Early in the project, a next-user utilisation plan was developed and actively implemented. This plan is based on section 7 in the Communication, Evaluation, Reporting & Next-User Utilisation - Draft Strategy & Plans (November 2006), downloadable from the project website www.accounting4nutrients.com.au.

The PAC agreed that achievable success in the life of the Accounting for Nutrients project would be that targeted next-users are aware of and understand the knowledge and tools generated by the Accounting for Nutrients research outputs to get on-farm practice change.

As part of these activities, the project team initiated a technical meeting with representatives from key next-user groups (September 2008) to define the project outputs and products (Nutrient Accounting Package, Appendix 3).

In summary, the 'Nutrient Accounting Package', includes four potential supporting outputs:

- Human capital development
- Scientific and technical papers
- Guidelines and recommendations
- Technical conceptual and empirical models

While wide-scale practice change by dairy farmers (as expected from an extension program) was beyond the scope of the current Accounting for Nutrients project, the project team has also continued to provide project information and knowledge about improved nutrient management to dairy farmers directly throughout the life of the project through general media articles and informal and formal industry forums.

### 6.1.2 Informing policy

A key objective within this module is to actively inform industry and government policy groups regarding the role of nutrient accounting systems as a low cost and politically acceptable tool to deliver improved nutrient resource management outcomes for the Australian Dairy Industry. To this end the following activities (Table 1) were undertaken to provide key industry and government players with an understanding of the issues and approaches being addressed by the Accounting for Nutrients project.

**Table 1. Key A4N presentations to key industry and government groups**

| Audience | Type of interaction |
|---|---|
| DPI Victoria – Animal Industries policy group | Presentation and discussions |
| EPA Victoria | Presentation and discussions |
| Land and Water Australia - Healthy Soils program | Invited presentation at national workshop |
| GippsDairy board | Presentation and discussions |
| National Dairy Industry Council – NRM committee | Presentation and discussions |
| Fertiliser Industry Federation | Presentation and discussions |
| DIDCO Members Council | Presentation |
| Formal independent project review | 2 day review process |
| Victorian Dairy Extension centre<br>- annual meeting | Presentation and discussions |
| Dairy Australia board | Presentation and discussions |
| National Dairy Industry Council – NRM committee | Presentation and discussions |
| All national dairy boards | Presentation and discussions |
| DIDCO board | Presentation |

### 6.1.3 Evaluating partnerships, standardisation, and integration

The approach to evaluation utilised a theory of action (sometimes called 'project logic') that describes how the A4N project's resources and activities were directed towards producing a series of outputs that will contribute to outcomes (Figure 1).

The focus of project evaluation within the life of the A4N project has been to concentrate linkages between 'resources, research activities, research outputs and next users'. It is assumed that by meeting the needs of the project's targeted next-users then the achievement of the project's higher level outcomes (through end users) will be maximized.

It should be noted that the major activities of the A4N project have involved mobilising the project's resources (the network of national scientific and technical staff and farmers), refining the project's methodologies and undertaking the detailed on-farm monitoring, data collection and interpretation. The previous and current project milestones reports have been the key tool to monitor project progress in these areas.

A key component of the 'governance, partnerships, standardisation, and integration' associated with the A4N project has been through the establishment and subsequent role of the Project Advisory Committee (PAC). This committee included representatives from industry and government such as Dairy Australia, regional dairy boards, national dairy industry policy, State Departments of Primary Industries, Federal Agencies, national fertiliser companies and next-users groups such as dairy factories (Murray Goulburn, Bega Cheese) and government regulatory agencies (Victorian EPA) and policy groups. As such, the PAC is a key 'Next-User' group. These players brought strong industry and government linkages which will continue to strengthen the delivery and acceptance of project information and tools. Additionally, two farmers provided representation for dairy farmers at a national level.

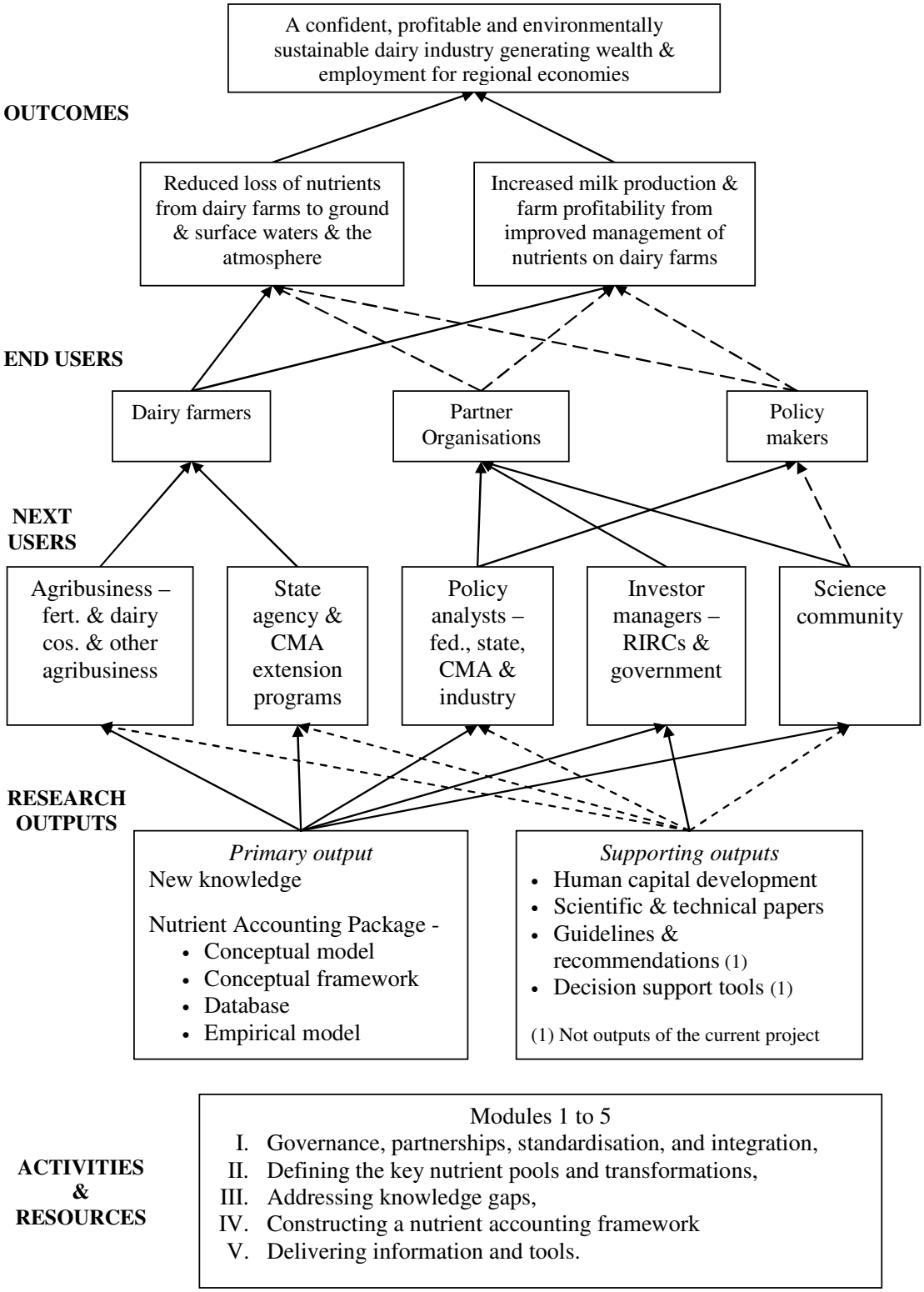
The engagement and contributions by members of the PAC to the project has been significant. The PAC reviewed the project activities, commented on the project milestone reports and provided important feedback to the project team, as well as revisions to the format and role of the new A4N website. Members have promoted the project throughout their networks and increased awareness of the project issues, objectives and activities. This resulted in a number of invitations to the project leadership team to make presentations to regional and national industry group meetings, national programs and invitations to contribute to other project activities.

The November 2008 PAC meeting was planned to coincide with the 2-day independent review. This provided the independent review team with the opportunity to hold discussions with the project PAC, while also participating in a regular PAC meeting.

The National Network (NN) of technical experts, including scientists, advisors and farmers, and Science reference group are other key 'next-user' groups. The NN has provided important input into the technical details of the Accounting for Nutrients project, notably reviewing and revising the on-farm protocols (i.e. sampling guide, farm survey forms and farmer diaries), training of on-farm project teams, data collection and quality assurance but also have been strongly influenced by the project development, activities and findings.

The A4N Science Reference Group, comprising 12 key scientific experts, has also played an important peer-review role to ensure the scientific integrity of the project. This included input into data collection, data management, collation and interpretation, and the steps in developing the model and approach to calculating nutrient accounts and efficiencies (the Nutrient Accounting framework). Throughout the project, a number of workshops and formal and informal discussions provided this important input. The benefits associated with the development of such a closely linked science network, with the overall aim of improving nutrient management for the national dairy industry are discussed in section 6.1.4.

**Figure 1. Accounting for Nutrients Project logic**



**Project Advisory Committee Evaluation.**

An opinion survey was used to measure and monitor the attitude of the Project Advisory Committee (PAC) towards the Accounting for Nutrients project. The questionnaire was administered at each meeting of the PAC.

The opinion survey asked respondents to rate 19 statements in a scale that goes from strongly disagree (scale 1) to strongly agree (scale 6) with the option of "don't know" for all questions. These 19 statements were then used to assess five domains of interest –

1. Quality of project management (draws on 4 statements)
2. Value of the project to me and my organization (draws on 4 statements)
3. The projects contribution to improved nutrient management on dairy farms (draws on 4 statements)
4. General satisfaction (draws on 1 statement)
5. Project progress and impact (draws on 6 statements)

Figure 2 provides results for all survey periods (May 2007, May 2008, November 2008, June 2009 and December 2009).

Overall, the scores and associated written comments recognised the high performance of the project team and the potential impacts flowing from the project. There was little change in the assessments throughout the life of the project with the PAC consistently scoring the 'Quality of the project', 'Value of the project to your organisation', 'Potential contribution to improved nutrient management', and 'General satisfaction' around 5 out of 6. Similarly, the assessment of the 'Potential project impacts' was scored around 5 out of 6 overall, with the exception resulting from a belief than dairy farmers themselves would not be primary users of project outputs.

These scores were supported with some written comments, which also recognised some of the complexity and challenges the project team faced:

*"Hard to fault the current approach"*

*"An important project for efficient dairy systems in the future"*

*"This project has the opportunity to make a big impact but the team has to transition to a new phase to max this out."*

*"I think timelines are going to be tight with research activities planned up. It's bigger than initially planned (my understanding) yet still trying to fit into the same end date with little additional resources."*

*"Need to ensure access to biometrician throughout the project"*

*"Potentially there will be value but highly dependent on simplicity of outcomes to enable efficient low cost delivery to the target market – dairy farmers."*

*"Progress good, impact still up to others who may yet to be involved"*

A significant number of comments also reinforced the importance and urgency for the project to develop products that meet the needs of next-users and ultimately farmers:

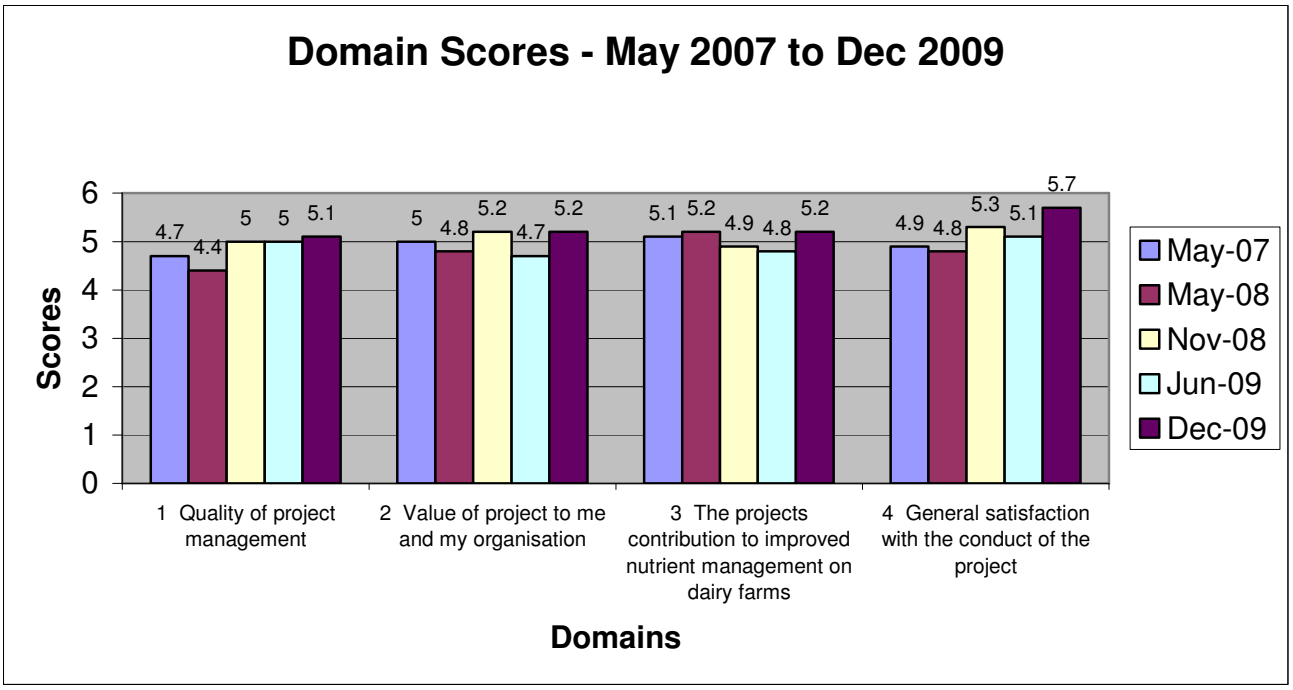
*"Perhaps some very specific focus on outputs that can be integrated into FertCare".*

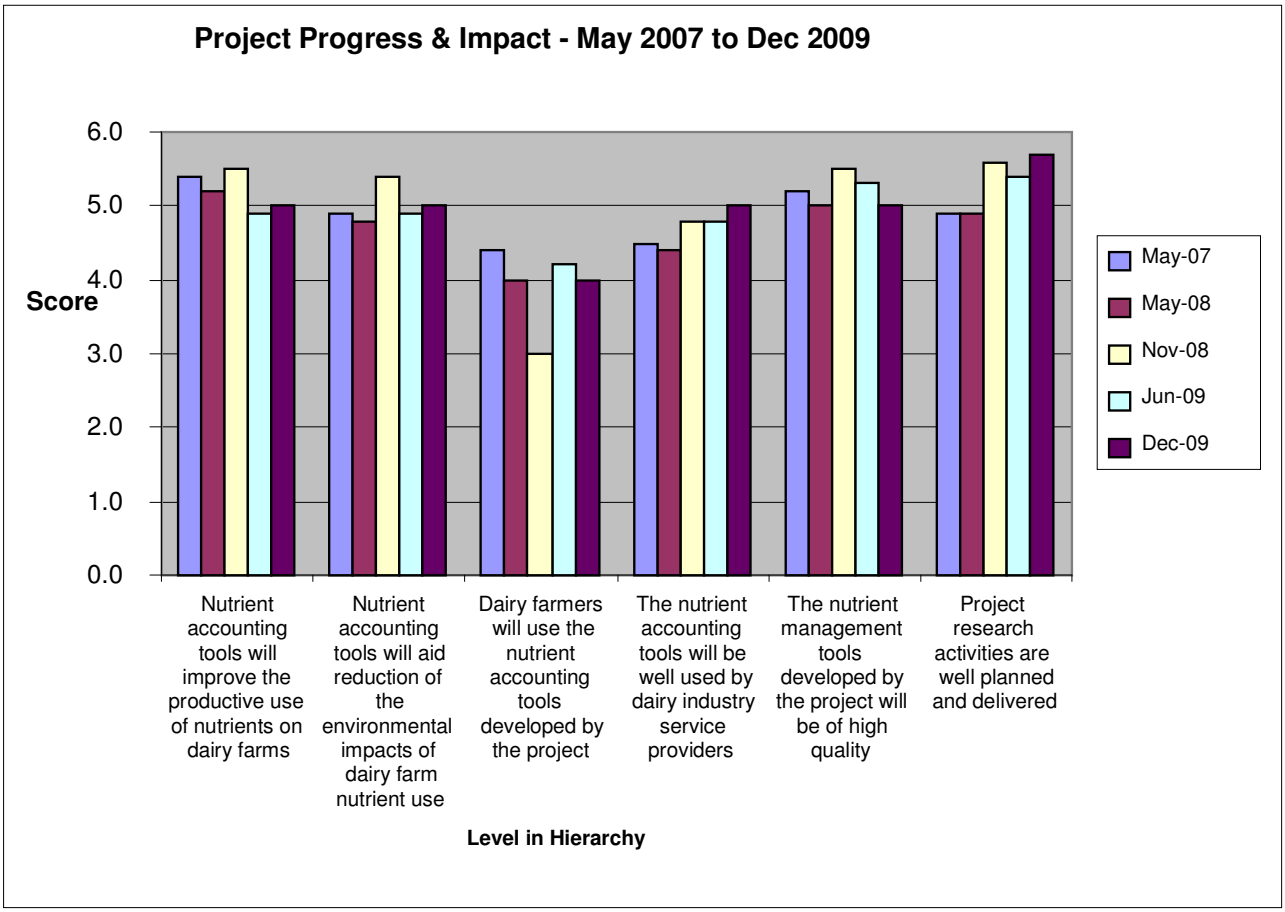
*"Getting engagement and commitment from advisors: commercial, public and fertilizer industry will be critical"*

*"This project has great potential and ownership from the dairy industry which I think is important. It will be interesting to see the level of uptake/project impact measured"*

*"It is of value - but will need to go the next step to be of interest to stakeholders. On its own the empirical model is of limited value."*

**Figure 2. Domain scores from the PAC opinion surveys.**



**Independent project review and recommendations.**

A formal independent project review and associated A4N Project Advisory Committee meeting in November 2008 discussed in detail the current project deliverables, achievements against projected timelines and future needs from industry stakeholders.

The 3 independent reviewers included Professor David Chapman, Chair (University of Melbourne), Dr Bruce Thorold (DairyNZ) and Mr John Gallienne (Dairy Industry consultant). The review was undertaken over 2 days with day-2 arranged to coincide with the 6 monthly Project Advisory Committee (PAC) meeting, with 17 members in attendance.

The project team made a number of presentations and participated in a series of question and answer sessions over these 2 days. The November 2008 Milestone report was also a key input into the review process.

While the review team was highly supportive of the need for the work and complimentary of the project planning and approaches undertaken, they also raised concerns about the ambitious timelines and work load necessary to achieve successful project outputs. Reflecting the views expressed by the PAC in particular, the reviewers proposed a greater focus on the next phase of work beyond June 2010, including the development of a nationally uniform nutrient accounting decision support tool to address industry and government needs.

A summary of the key recommendations from this review process included:

1. Continued future support for the A4N project from project stakeholders in line with the need to substantially improve nutrient management on Australian dairy farms.
2. Redirecting the proposed on-farm monitoring efforts beyond February 09 to data analysis, interpretation and provision of results back to participating farmers, so that the current project meets existing timelines.
3. Identify individuals or groups with expertise in biophysical research modelling with a view toward forming a collaboration that enables the project team begin the development of the empirical model which will ultimately develop into an industry DSS.
4. Jointly develop a strategic plan for an A4N Phase 2 beyond June 2010, which explicitly aims to produce a nutrient accounting decision support tool that aligns with next-user needs.

Consequently, in late November 2008, the PAC requested that the A4N Science Management Team prepare a discussion paper to provide a number of options pertaining to the current A4N project and deliverables, and the scope and direction of a future A4N project (A4N phase 2).

The extraordinary PAC meeting held in March 2009 recommended that the project team finalise the existing project as originally proposed by June 2010, curtail the current on-farm monitoring in February 09 and direct greater effort into data interpretation and model development. The PAC also requested that the project leadership team initiate the development of a full project submission for A4N Phase 2, which incorporates the development of a national and industry supported Nutrient Accounting Decision Support Tool.

However, in February 2010, Dairy Australia indicated that (at this stage) it does not accept the primary recommendation from the mid-term review, i.e. that a second phase of A4N should be developed with the explicit goal of producing a decision support tool. Instead it proposed a technical transfer year in 2010/2011 where the major effort is directed towards drawing out the lessons from the A4N project and embedding them into improved nutrient management decisions within the dairy industry. This recommendation recognised that the current A4N project, budget and timeline were insufficient to deliver these activities and outcomes.

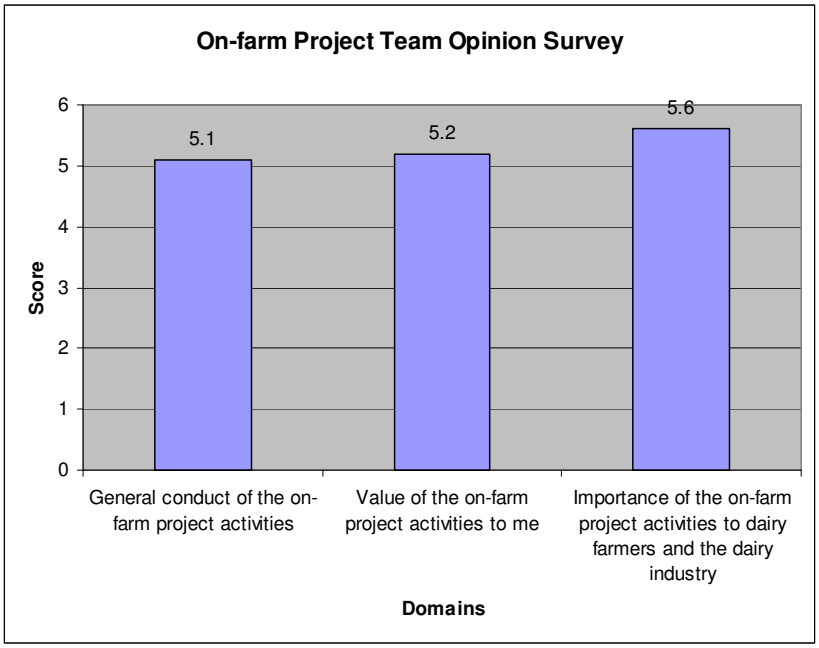
**On-farm team evaluation.**
The on-farm project team met at Ellinbank in June 2009 to review and plan the final phase of their involvement in the A4N project. Included in the workshop program was a session on the evaluation of the on-farm data collection phase.

Each team member completed a questionnaire considering three domains or areas of interest –

- General conduct of the on-farm project activities,
- Value of the project activities to me, and
- Importance of the on-farm project activities to dairy farmers and the dairy industry.

Eighteen people participated in the evaluation. The summary score for each domain is shown in Figure 3.

**Figure 3. Domain scores from on-farm project team survey**




Overall, the on-farm project team members viewed the project and their role in a very positive light with comments such as –

*"I now have a lot better understanding of management of animal movements & therefore nutrient management"*

*"Model will provide farmers & industry with a useful tool for NRM"*

*"The equipment & materials required for each farm visit was always well supplied. It made doing the visits simple"*

*"Felt part of the team ….."*

Some weaknesses were also identified –

*"Feedback/results to dairy farmers sometimes had long time delays"*

*"The extent of records or data collected as part of the project would be too extensive for farmers to do on their own - may limit the application of any tools"*

**Next-User Evaluation**

Feedback has been recently sought from a cross section of project next users and nutrient management scientists who have had involvement with the A4N project. This included senior research scientists who have been a part of the A4N project team (primarily outside Victoria), senior extension staff (industry, agency and catchment), agribusiness companies, members of the Project Advisory Committee and some participating dairy farmers.

Fourteen people provided written responses to the following questions –

- Is the A4N project team regarded as being relevant and credible in nutrient management at a national level?
- What role has the A4N project had in highlighting the need for improved nutrient management on dairy farms (profitability and environmental sustainability)?
- Have nutrient management specialists increased capability and confidence in dealing with contemporary dairy farm nutrient management issues?
- Are next users planning to use and adapt Accounting for Nutrients research outputs and publications for their specific needs?
- What has been the response from dairy farmers who have participated in your programs?

The full evaluation report is available as a separate document. The key findings and some examples of supporting statements are provided below.

(1) The A4N research project is providing strong national leadership and through its approach and structure has formed powerful and sustained partnerships around the network of nutrient research scientists and key next user deliverers such as the fertiliser industry, fertiliser companies, dairy companies and extension agents (industry, catchment and government).

> *"The A4N team, and its predecessor, BFD, is unique in that it gathers together nutrient management specialists from around the country and provides a focus for a specific topic over a period of time. ….. This approach is extremely powerful, not only for the scientific outcomes, but also for those involved, whether they be part of the team, or other stakeholders involved in the project." Research Scientist*
>
> *"… the national cooperation and the organisation of data collection could not have been accomplished without the established credibility of the project management team." Fertilizer Industry Executive*

(2) There is a high level of awareness of the A4N research project amongst nutrient management scientists and key next user groups. The project has highlighted the diversity, complexity and impact of nutrient movement on dairy farms and the challenges facing the Australian dairy industry.

> *A4N has clearly demonstrated the complexity of nutrient movement which adds a new dimension to managing nutrients on the farm for both profitability and environmental risk management." Fertilizer Industry Executive*
>
> *The project has provided the best "map" of the nutrient flows on Australian dairy farms, and shown where the opportunities to improve efficiencies (profitability) and reduce environmental impacts exist." Fertiliser Company Agronomist*

(3) A4N research has already generated significant new knowledge and understanding of nutrient levels, balances and movements in dairy farming systems across Australia with implications for both productivity and environmental impacts. That new knowledge is regarded as highly relevant and credible by dairy industry nutrient management specialists. The research tools used by the A4N project have identified opportunities and options to improve nutrient management on dairy farms.

*"A4N has pulled majority of the factors of nutrient management neatly together, where previously what was available was scattered and almost impossible to compare. A4N has produced a very strong data set that can lend itself to many uses. Real farm figures and real farm situations are what farmers can relate to, particularly the participating farmers." State Dairy Extension Officer*

*"This is a fundamental piece of work that should be seen as a benchmark and starting point for improved nutrient management on dairy farms around Australia." Fertiliser Company Agronomist*

(4) The level of capability and confidence of nutrient management specialists in dealing with contemporary dairy farm nutrient management issues appears to be directly related to the degree of involvement in the A4N project and/or the active utilisation of research outputs from the A4N project for extension programs. Those nutrient management specialists who either have been directly involved in the research project or have actively drawn on the emerging knowledge being generated by the A4N (and earlier projects) have improved their capability and confidence in dealing with contemporary dairy farm nutrient management issues.

*"…having 2 of our staff directly involved with the on-farm data collection has been a great asset in their further development, with both staff more confident when dealing with farmers following their involvement." NRM Project Manager- Dairy*

*"The specialists that have been exposed to this project now have a much greater understanding of the complexity and magnitude of nutrient cycling on dairy farms. This will contribute to an increased capability and confidence in the decisions and advice that they provide." Agriculture Extension Services Manager*

(5) A number of next users have actively sought and are already using the A4N research outputs as they progressively emerge from the project, sometimes with direct input from the A4N project team to assist their use with specific next users. Other key next user groups are preparing to use the research outputs from the A4N project.

*"The nutrient element (in our package) is now complete and relies heavily on a nutrient calculator which we have developed, at our expense, from the output of the A4N project. In developing this tool, we have learnt from being involved personally in the A4N project and have been able to use the A4N team extensively to make sure the science and calculations in the tool are correct" Dairy Company Field Services Manager*

*"One of the key investment criteria considered by the Australian Government Reef Rescue program is each industries' "solvability" toolkit. The dairy industry has been well positioned in solving nutrient management and water quality issues by having access to the A4N research and decision support tools." Dairying for Tomorrow Coordinator*

*"The outputs are already in use in WA in a number of areas. This includes the use of the generic framework developed in A4N and its application to the understanding of Phosphorus use efficiency in Australian agriculture, and how this efficiency may be improved. ….. Additionally, the whole farm nutrient mapping, and farmgate nutrient budget approaches are being applied as part of WA's Fertiliser Action Plan (FAP). Notably the FAP also picks up on previous work undertaken in a similar national project, Better Fertiliser Decisions, and adopts the nationally agreed critical soil test levels as the standards by which assessments of soil Phosphorus status are made." Research Scientist*

*"Fertcare is using A4N outputs as part of the Accredited Advisor training program." Fertilizer Industry Executive*

(6) The influence of the A4N project goes across nutrient management programs in the industry, government and agribusiness sectors. These next users see the value of the A4N generated knowledge and will be keen to make further use of the research outputs in the future.

> *"… a review of the Fertcare training material is about to commence and Dr Gourley has again (kindly) agreed to be a member of the technical committee responsible for updating the material. This review will enable all of the relevant results from the A4N project to be incorporated into the program, which will be delivered to fertilizer and plant nutrition specialists across the country." Environment Manager, Fertilizer Industry*
>
> *"The QLD Dairy NRM Program will continue to direct service providers to use the A4N research outputs and tools included in the SNMP service provider kit." Dairying for Tomorrow Coordinator*
>
> *"The key elements of (our company) environmental management module are nutrients, water and greenhouse gas management. The nutrient element is now complete and relies heavily on a nutrient calculator which we have developed, at our expense, from the output of the A4N project. The tool is currently being piloted with over 100 (of our milk) suppliers ……." Dairy Company Field Services Manager*
>
> *"The dairy extension centre regards the A4N project as very valuable with significant importance to our future direction in nutrient extension. We have been watching the space with great interest and are keen not only to upgrade our existing projects and tools but continue to collaborate to design new projects and decision support tools for the future needs of the industry. Our strategy group recently allocated $45k to link one of our staff with the A4N project to scope a new direction in extension services" Statewide NRM Project Manager- Dairy*

(7) Next users and nutrient management specialists who have either been actively involved in the A4N project or actively drawn on the emerging knowledge being generated by the A4N (and other projects) have improved their awareness, understanding, capability and confidence in dealing with contemporary dairy farm nutrient management issues.

> *"The ….. Dairy NRM program, with assistance from Dr Cameron Gourley…. has developed a SNMP service provider kit. This kit ensures service providers have access to standard decision support tools and are using a consistent planning logic in providing fertiliser recommendations to farmers. ….. The Program has now engaged 4 service providers who are familiar with and have developed the capacity to use the decision support tools developed through the A4N project." Dairying for Tomorrow Coordinator*
>
> *"Yes, the team in the A4N project certainly have passed on to a broader level of users such as consultants and agronomists so that they can then influence what's happening on ground." Dairy Company Milk Supply and Farm Services Manager*
>
> *"…having 2 of our staff directly involved with the on-farm data collection has been a great asset in their further development, with both staff more confident when dealing with farmers following their involvement." Statewide NRM Project Manager- Dairy*

(8) There appears to be another group of next users who are either taking a more passive approach (such as individual fertiliser companies) or are expecting this knowledge to be packaged and communicated and specific nutrient management tool(s) to be developed by the A4N project.

> *"… understanding of the complete nutrient picture has clearly improved, however tools to assist advisors and farm managers are required before consistent systematic consideration can be given to the complex interactions" Fertilizer Industry Executive*

*"The projects deliverables must be supported with extension activities to ensure best practice adoption at the farm gate" Fertiliser Company Agronomist*

*"The tool for on ground application has yet to be developed and it will not be till that occurs that we will see real improvement. At this stage we have only seen initial awareness raising." Dairy Company Milk Supply and Farm Services Manager*

(9) All farmers involved in the A4N project were very positive about their participation in the project being impressed by the conduct of the project and the reports, maps, feedback etc. they received. A number of farmers have already made good use of information supplied for their farm by the A4N project team. For some farmers it appears to have raised their awareness and interest but they will need further support to make effective use of the feedback they have received.

*"A very positive response, with immediate financial results through a streamlined, targeted fertiliser program in some cases. The provision of farm nutrient maps was extremely well received, … Generally, the increased knowledge of their farms nutrient status was greatly appreciated, and universally it was anticipated that that knowledge will be utilised in forward planning." Research Scientist*

*"…. the participating dairy farmers have shown real appreciation for the depth of the project. ….. Feedback from farmers was that info presented at the field days was fantastic, and alerts all to where we are at in regards to nutrient budgeting." State Dairy Extension Officer*

## 6.2. Nutrient accounting for Australian dairy farms.

This section describes the development and refinement of the multi-nutrient auditing framework for N, P, K, S, Ca and Mg, which enables nutrient accounting to be conducted on (conventional and organic) dairy farms at a range of complexities, including whole farm assessment and the more detailed farm-system scale.

**Key activities and achievements**

**Module 4. Constructing a nutrient accounting framework for Australian dairy farms**

- ✓ Fine tuning of individual components of a nutrient budget
- ✓ Refinement of data capture methods, sampling and record keeping by farmers
- ✓ Data management procedures developed and refined
- ✓ Development of conceptual framework for nutrient accounting at specific spatial scales
- ✓ Development of empirical model for whole farm nutrient accounting
- ✓ Development of algorithms for nutrient flows, transformations and losses

In 2007, the National Network defined the requirements of the nutrient accounting framework as:

i. Determine a whole farm nutrient balance and associated output/input efficiencies,
ii. Identify, quantify and prioritise key nutrient imports and exports at the farm scale,
iii. Determine inputs and outputs at the paddock scale,
iv. Determine nutrient stores held on the farm (i.e. feed, manure, etc),
v. Identify and predict potential nutrient losses and their associated pathways,
vi. Estimate nutrients applied to unproductive areas, and highlight inefficiencies,
vii. Define the uncertainty around nutrient calculations and budget predictions,
viii. Establish appropriate benchmarks for surpluses and potential nutrient efficiencies at the whole-farm and component level,
ix. Integrate nutrient balances at the field level with Australian agronomic soil test levels and use this information to assist with fertiliser recommendations
x. Provide recommendations on management practices which will improve nutrient budgets and efficiencies.

Over the course of the project, there have been several key areas of development of this framework. These include:

- On-going process of review and revision of the nutrient accounting framework specifically directed towards the individual components – this development is described in Section 6.3. This revision has been occurring in tandem with the refinement of protocols, farmer interviews, sample and data collection and examination of survey data.
- Refinement of data capture processes which relates both to refinement of questionnaires and diaries to ensure that data gathered is useable in a model, i.e. it either has to be quantitative or if qualitative that can be converted simply to quantitative.
- The management of collected data through a specially designed relational database.

### 6.2.1 Review of existing nutrient accounting approaches

During the early part of the Accounting for Nutrients project, a detailed review of Australian and international nutrient management approaches and tools was undertaken by members of the project science team. This review process culminated in the production of three peer reviewed scientific publications provided below (i, ii, and iii), which provide a sound scientific justification for the approaches and methodologies developed for the Accounting for nutrients project. Subsequent to this, another refereed journal paper (iv) was produced examining the application of nutrient use efficiency estimates for dairy farms. A number of other science papers have also been produced that have presented information and interpretation from the A4N project.

#### Scientific publications resulting from A4N project to date

i. Gourley CJP, Powell JM, Dougherty WJ, Weaver DM (2007) Nutrient budgeting: an approach to improving nutrient management on Australian dairy farms. Australian J Experimental Agriculture 47: 1064-1074.

ii. Gourley CJP and Powell JM (2007). Nutrient management approaches and tools for dairy farms in Australia and the USA. The Babcock Institute for International Dairy Research and Development, University of Wisconsin-Madison, USA. http://babcock.cals.wisc.edu/publications/environment.lasso

iii. Gourley CJP, Aarons SR, Powell JM, Dougherty WJ and Weaver DM (2007). Revising the paradigm for improved nutrient management on Australian dairy farms. Australian and New Zealand Dairy Science Conference, Melbourne, Victoria, Sept 2007.

iv. Powell JM, Gourley CJP, Rotz CA and Weaver DM (2010). Nitrogen Use Efficiency: A Measurable Performance Indicator for Dairy Farms. Environmental Science and Policy 13: 217- 228.

v. Gourley CJP, Aarons SR, Hannah MD, Dougherty WJ (2010). Soil nutrient concentrations and variations on dairy farms in Australia. 19th World Congress of Soil Science, Soil Solutions for a Changing World 1 – 6 August 2010, Brisbane, Australia.

vi. Aarons SR, Gourley CJP (2010) Precision dairy farming: cows in space and time. 19th World Congress of Soil Science, Soil Solutions for a Changing World 1 – 6 August 2010, Brisbane, Australia.

vii. Andrew Smith. Ph. D. Thesis title: "Predicting nitrogen losses from dairy farming catchments at multiple spatial scales." University of Melbourne.

#### Technical Exchange trip to New Zealand

The Project Leader - Cameron Gourley, Deputy Project Leader - Warwick Dougherty and visiting scientist Professor Mark Powell, travelled to New Zealand for a week in June 2008 to gain a better understanding of approaches to improving nutrient management on dairy farms. The visit focused on the gaining a better understanding of the development and implementation of the widely used nutrient budgeting model OVERSEER®, recent policy developments in addressing excess nutrients at the farm and catchment scale, and current research projects examining mitigation strategies.

The A4N team met with key AgResearch researchers, OVERSEER® developers, industry and policy staff engaged in the field of nutrient management and accounting. The A4N scientists presented seminars to groups at AgResearch (Hamilton), Massey University (Palmerston North) and Lincoln University/Dairy NZ (Christchurch). These seminars provided an overview of the 'Accounting for Nutrients on Australian dairy farms' project and the proposed research methods for calculating nutrients fluxes on the 44 dairy farms involved in the study.

The NZ groups were impressed with and supportive of the national and inclusive approach adopted in the A4N project, and the detailed on-farm monitoring work currently underway. Most of the approaches currently proposed for calculating nutrient fluxes in the Accounting for Nutrients project were widely endorsed by technical experts in NZ. Additionally, a number of other key technical issues relating to on-farm nutrient

monitoring and information sources were resolved and a clearer pathway for particular project components has been developed.

The information gathered from this NZ visit had an important impact on the future activities and delivery of the Accounting for Nutrients project. There is much to be gained by sharing knowledge and insights from the researchers and modellers who have had a decade or more of experience in developing the OVERSEER® model. This exchange of knowledge and dialogue should be continued to ensure efficiencies and the best possible scientific approach and practical relevance of the A4N.

There is a strong regulatory environment for managing nutrient surpluses and losses in NZ. While drivers for similar regulations currently do not appear as strong or imminent in Australia, the need for a similar approach to improve nutrient management for the Australian dairy industry and the broader community's environmental standards is clear. Understanding this policy environment is particularly important as the current Accounting for Nutrients project progresses from the development of a nutrient accounting 'framework' to a 'ready for market' nutrient accounting tool. Moreover, there is a need for the Project Team and the Project Advisory Committee to more carefully consider the delivery of a national nutrient accounting tool so as to maximise acceptance, utilisation and potential impact across the Australian Dairy Industry and its service providers.

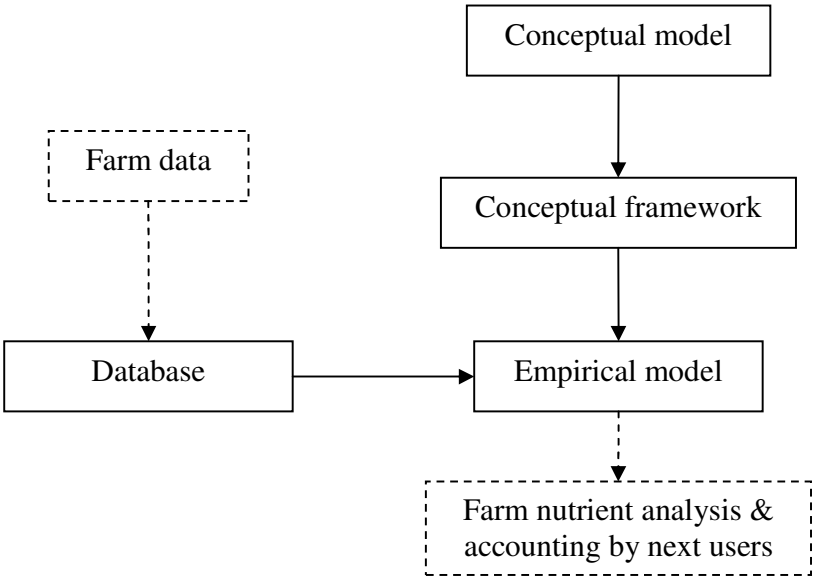
A full report of this study tour is available on the A4N website www.accounting4nutrients.com.au\ project reports.

Further to this visit, key project staff (Cameron Gourley, Warwick Dougherty and Andrew Smith) again visited New Zealand in early 2010 to further discuss the nutrient budgeting approach adopted in the A4N project and the data arising from this research. New Zealand research, extension and commercial experts again endorsed the approach taken in the Accounting for Nutrients project. They were particularly impressed with the amount of data gathered, its comprehensiveness and the approach taken to interpreting these data. Important discussions were also held regarding strategies to improve nitrogen use efficiency on dairy farms, based on the outcomes of the A4N project. These discussions influenced the strategies for improving efficiency discussed elsewhere in this report.

### 6.2.2 Development of the conceptual framework

The various steps in the development of a nutrient accounting approach were identified in the early stages of the A4N project, in conjunction with key stakeholders. Figure 4 shows the various components in a hierarchical layout starting from a relatively simplistic conceptual model to an empirical model. The operational model was used by the A4N team to calculate nutrient balances and efficiencies on the participating dairy farms in the project. The components of this layout in dashed boxes are outside the scope of the current project (see Accounting for Nutrients – Products' Appendix 3).

**Figure 4. The various A4N project products and progression of modelling approach.**




The '**conceptual model'** (Figure 5) is a generalised description of the flows of nutrients imported, exported and within farms. This model was used as the fundamental basis for determining necessary data collection requirements for the 44 monitoring farms as well as serving as the basis for the development of the more detailed conceptual framework. The conceptual model was reported in the published review (Gourley et al 2007).

The **'conceptual framework'** (Figure 6) is a central component of this project which has contributed to the calculation of nutrient flows on the 44 monitoring farms. It will also serve as a key resource for development of nutrient budgeting tools in the future. The conceptual framework includes a detailed description of the flows of nutrients in to, out of and within farms and a detailed technical manual that describes all of the algorithms, their use, source and justification including all scientific supporting documentation. The Nutrient Accounting Technical Manual including the schematic conceptual framework is provided in a separate document.

The **'empirical model'** is a purpose built MS Excel computer program, based on the nutrient accounting framework that utilises farm-based information. It comprises of > 30 separate data entry pages where data are imported via a number of queries in the MS Access database database. It has defined parameters and algorithms which calculate nutrient flows, transformations and balances on a dairy farm. In its current form it should be viewed as a research tool and requires a high level of user understanding to successfully input and manage data. The parameters and algorithms are described in the conceptual framework technical manual and have been derived from prior knowledge (in the literature and from consultation with appropriate experts) and relationships established as a part of the Accounting for Nutrients research project. A typical nutrient balance output from the Empirical Model is presented and described in Figure 7.

**Figure 5. The conceptual model of the flows of nutrients in to, out of and within farms (adapted from Gourley *et al.* 2007).**

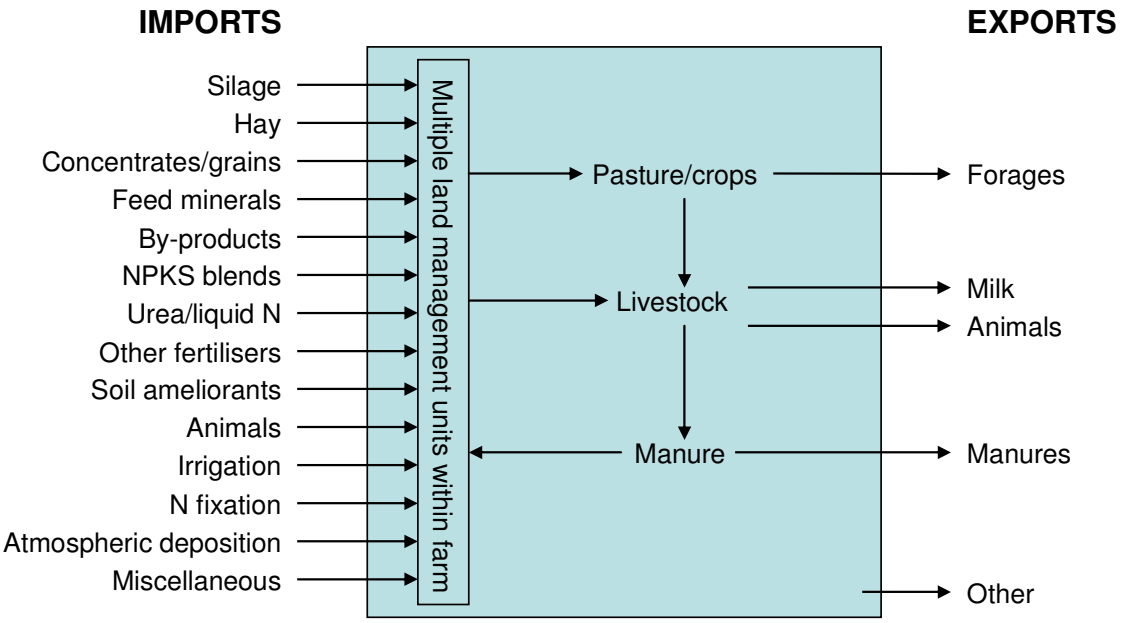
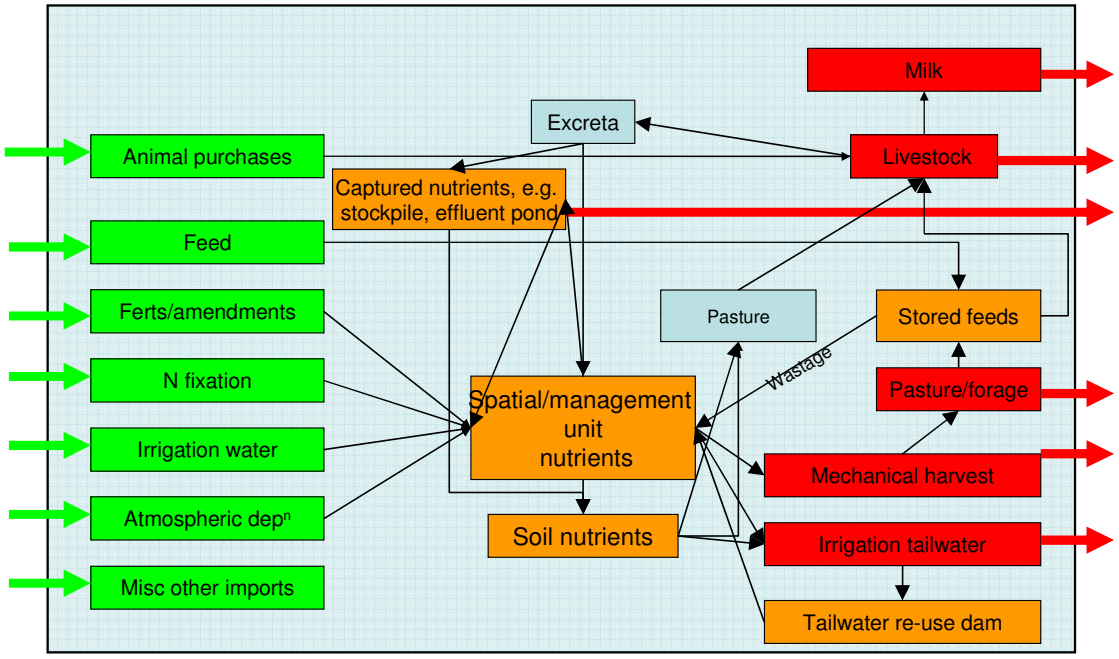
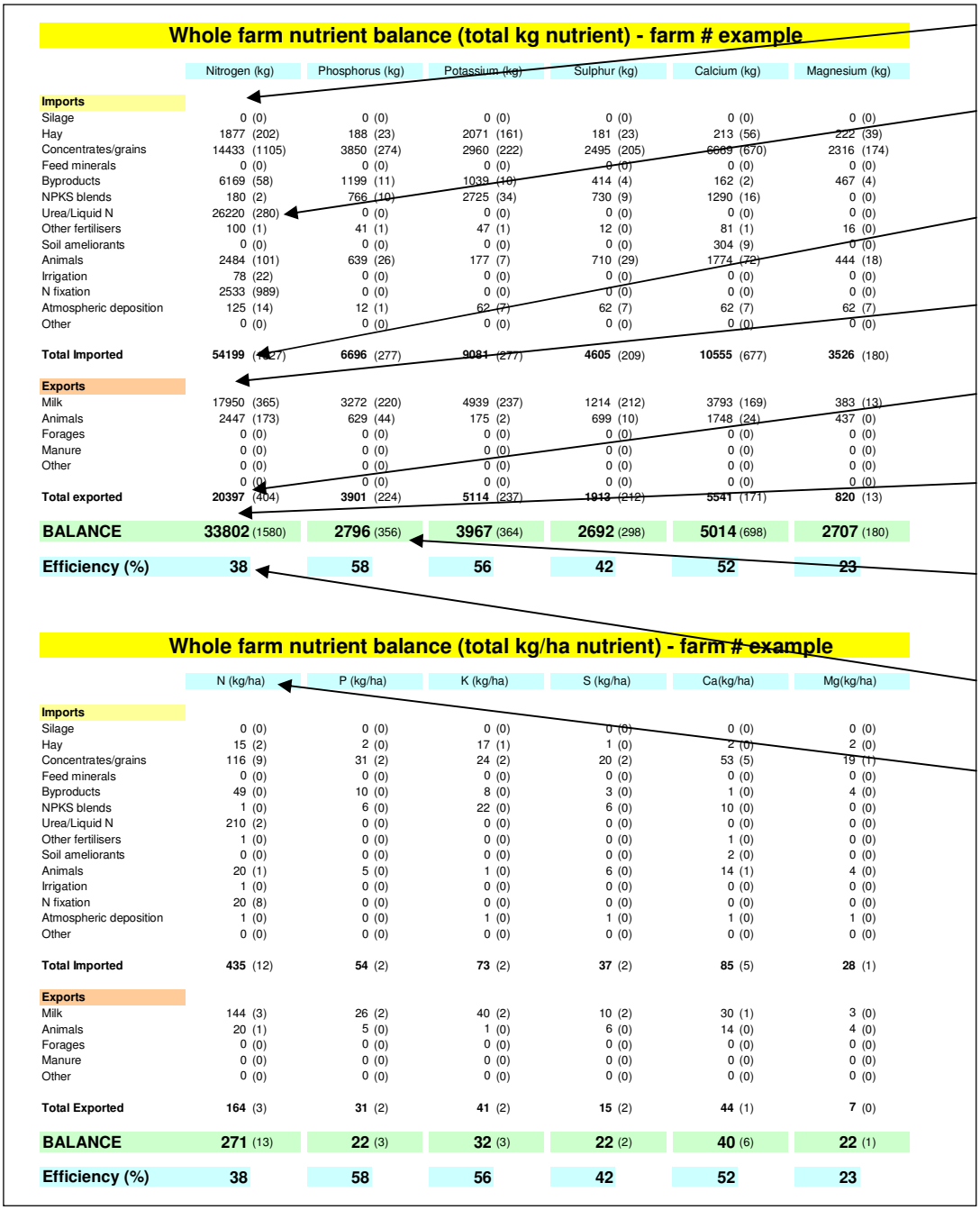



**Figure 6. Revised conceptual framework of flows of nutrient imports, exports and within dairy farms.**



## Figure 7. Typical output from the developed A4N empirical model



**Whole farm nutrient balance (total kg nutrient) - farm # example**

| | Nitrogen (kg) | Phosphorus (kg) | Potassium (kg) | Sulphur (kg) | Calcium (kg) | Magnesium (kg) |
|---|---|---|---|---|---|---|
| **Imports** | | | | | | |
| Silage | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Hay | 1877 (202) | 188 (23) | 2071 (161) | 181 (23) | 213 (56) | 222 (39) |
| Concentrates/grains | 14433 (1105) | 3850 (274) | 2960 (222) | 2495 (205) | 6669 (670) | 2316 (174) |
| Feed minerals | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Byproducts | 6169 (58) | 1199 (11) | 1039 (10) | 414 (4) | 162 (2) | 467 (4) |
| NPKS blends | 180 (2) | 766 (10) | 2725 (34) | 730 (9) | 1290 (16) | 0 (0) |
| Urea/Liquid N | 26220 (280) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Other fertilisers | 100 (1) | 41 (1) | 47 (1) | 12 (0) | 81 (1) | 16 (0) |
| Soil ameliorants | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 304 (9) | 0 (0) |
| Animals | 2484 (101) | 639 (26) | 177 (7) | 710 (29) | 1774 (72) | 444 (18) |
| Irrigation | 78 (22) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| N fixation | 2533 (989) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Atmospheric deposition | 125 (14) | 12 (1) | 62 (7) | 62 (7) | 62 (7) | 62 (7) |
| Other | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| **Total Imported** | **54199** (1427) | **6696** (277) | **9081** (277) | **4605** (209) | **10555** (677) | **3526** (180) |
| **Exports** | | | | | | |
| Milk | 17950 (365) | 3272 (220) | 4939 (237) | 1214 (212) | 3793 (169) | 383 (13) |
| Animals | 2447 (173) | 629 (44) | 175 (2) | 699 (10) | 1748 (24) | 437 (0) |
| Forages | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Manure | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Other | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| **Total exported** | **20397** (404) | **3901** (224) | **5114** (237) | **1913** (212) | **5541** (171) | **820** (13) |
| **BALANCE** | **33802** (1580) | **2796** (356) | **3967** (364) | **2692** (298) | **5014** (698) | **2707** (180) |
| **Efficiency (%)** | **38** | **58** | **56** | **42** | **52** | **23** |

**Whole farm nutrient balance (total kg/ha nutrient) - farm # example**

| | N (kg/ha) | P (kg/ha) | K (kg/ha) | S (kg/ha) | Ca(kg/ha) | Mg(kg/ha) |
|---|---|---|---|---|---|---|
| **Imports** | | | | | | |
| Silage | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Hay | 15 (2) | 2 (0) | 17 (1) | 1 (0) | 2 (0) | 2 (0) |
| Concentrates/grains | 116 (9) | 31 (2) | 24 (2) | 20 (2) | 53 (5) | 19 (1) |
| Feed minerals | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Byproducts | 49 (0) | 10 (0) | 8 (0) | 3 (0) | 1 (0) | 4 (0) |
| NPKS blends | 1 (0) | 6 (0) | 22 (0) | 6 (0) | 10 (0) | 0 (0) |
| Urea/Liquid N | 210 (2) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Other fertilisers | 1 (0) | 0 (0) | 0 (0) | 0 (0) | 1 (0) | 0 (0) |
| Soil ameliorants | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 2 (0) | 0 (0) |
| Animals | 20 (1) | 5 (0) | 1 (0) | 6 (0) | 14 (1) | 4 (0) |
| Irrigation | 1 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| N fixation | 20 (8) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Atmospheric deposition | 1 (0) | 0 (0) | 1 (0) | 1 (0) | 1 (0) | 1 (0) |
| Other | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| **Total Imported** | **435** (12) | **54** (2) | **73** (2) | **37** (2) | **85** (5) | **28** (1) |
| **Exports** | | | | | | |
| Milk | 144 (3) | 26 (2) | 40 (2) | 10 (2) | 30 (1) | 3 (0) |
| Animals | 20 (1) | 5 (0) | 1 (0) | 6 (0) | 14 (0) | 4 (0) |
| Forages | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Manure | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Other | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| **Total Exported** | **164** (3) | **31** (2) | **41** (2) | **15** (2) | **44** (1) | **7** (0) |
| **BALANCE** | **271** (13) | **22** (3) | **32** (3) | **22** (2) | **40** (6) | **22** (1) |
| **Efficiency (%)** | **38** | **58** | **56** | **42** | **52** | **23** |

Total nutrient imported e.g. in concentrates or grain

Uncertainty (Std Dev) around the total nutrient estimate for each source

Sum of all nutrients imported from all sources

Total nutrient exported e.g. milk shipped off farm

Sum of all nutrients exported from all sources

Difference between total nutrients imports and exports

Integrated uncertainty (Std Dev) for the total nutrient balance

Ratio of exported nutrient and imported nutrient *100

All nutrient imported presented on a per ha basis

### 6.2.3 Overcoming data collection challenges

**Key activities and achievements**

**Module 3.**

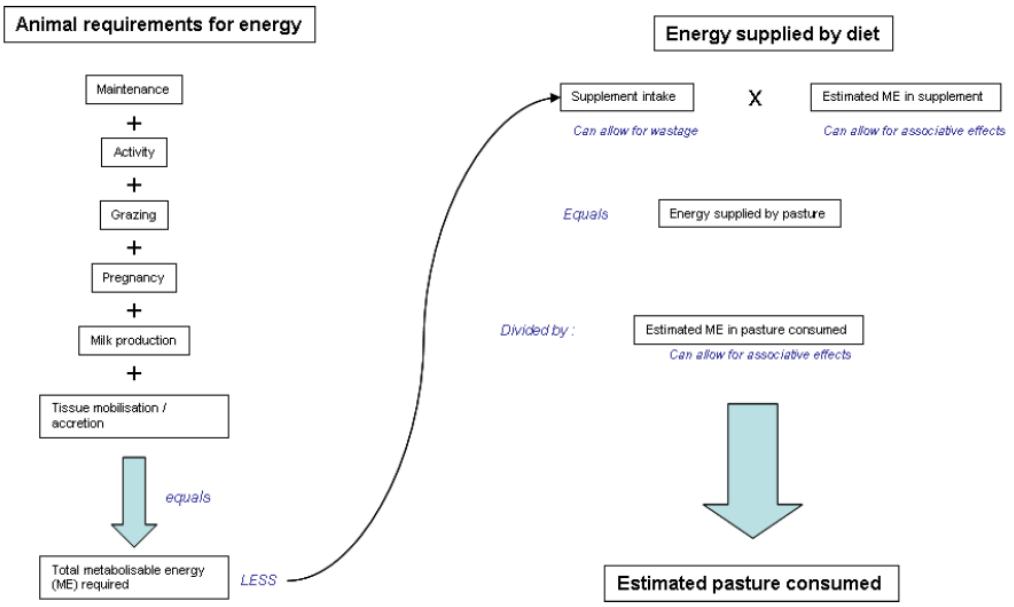
- ✓ Key information gaps identified
- ✓ Strategy for resolving these developed
- ✓ Science exchange visit to NZ in July08
- ✓ Literature reviews on specific topic areas completed.
- ✓ Methodologies incorporated into Nutrient Accounting Technical Manual

During the development of the nutrient accounting framework and farm monitoring protocols, a number of data collection challenges were identified. While the importance of these nutrient inputs varied from major to minor, a thorough and scientifically defendable estimation of all identified nutrient fluxes was required. Consequently different strategies were used to deal with these, i.e. literature review, model development, and/or consultation with appropriate experts. The key challenges identified and a brief summary of the strategy for resolving them are listed below. Further details are provided in the Nutrient Accounting Technical Manual.

**Paddock pasture productivity.** This issue relates to the calculation of total pasture consumed/forage harvested. This input is required to calculate individual paddock dry matter production, a necessary input for determining nitrogen fixation. In the case of farm system budgets, total pasture consumed/forage harvested is also required to determine individual paddock productivity and removal.

The selected approach, known as a 'back calculation' relies on the difference in metabolisable energy (ME) requirements for cows (and other stock) of known live-weight and milk production, and the ME consumed from supplementary feed consumed by stock. The method assumes that the difference in required ME, after specific adjustments for loss in body weight, walking, grazing, etc, came from consumed pasture (Figure 8). A modified pasture back-calculation tool developed by DPI Victoria can be accessed from http://www.dpi.vic.gov.au/pastureconsumptioncalculator.

**Figure 8. Conceptual diagram of pasture back calculation (Heard and Wales 2009).**



While the approach appears simple, the calculation of pasture consumption from particular paddocks/zones is complex. In order to achieve a balance between data needs, confidence in the results and practicalities, it is recommended that the concept of zoning is employed. This means grouping together areas of farms that have similar characteristics, such as soil type, history, land use, productivity, irrigation, etc. The experience of the Accounting for Nutrients project and other projects is that for grazing and cropping parts of the farm, no more than 8 zones are required. These zones then should be ranked using a rating system to calculate a productivity index. The index for each zone is a score from 0-10, 0 being the least visited or productive and 10 being the most visited or productive. Zones can have equal ratings assigned to them. The productivity index should be on a per hectare basis, as distinct for a whole of zone productivity index.

Once these indices have been estimated, an area weighted (AW) index should be calculated by multiplying each index by the area of the paddock or zone to which it relates. The AW index is then used to calculate pasture and subsequently nutrient removal. Because of differences in productivity between particular zones, different amounts of pasture will be consumed and hence different quantities of nutrients consumed by grazing livestock.

$$\text{DM consumed Pdk}_i\text{ (t)} = \left( \frac{\text{AW Productivity index Pdk}_i}{\sum \text{AW Productivity indices}} \right) \times \text{total pasture DM consumed (t)}$$

Where AW Productivity index $Pdk_i$ is area weighted productivity index for paddock 'i' as described above. Nutrient removal can then be calculated by multiplying pasture removal by nutrient content for whole of farm or individual zones. This approach was reviewed in discussions with AgResearch scientists, and DairyNZ staff, as well as at our Science Reference group meeting in July 2008 and after modifications, subsequently endorsed.

**N fixation by legumes.** There are two distinct situations where this pathway can be a substantial N input. The first is in mixed pastures - via clover, and the second in legume crops such as lucerne. An estimate of the legume % of a pasture or crop is then required to calculate the biomass of clover or legume produced on an annual basis. Note that this production figure should be for the feed on offer component, i.e. that legume able to be grazed as distinct from the whole above ground legume. The legume production can be generated on a paddock by paddock basis if such data is available. Alternatively a farm pasture or crop average can be used. The N fixation is then calculated based upon the method of Ledgard et al. (2001) as follows:

Total fixed N = clover production $\times$ (0.061 – (6.1 $\times$ $10^{-5}$ $\times$ N fertilizer rate))

Where total fixed N (kg N/ha/yr) includes the below grazing height component which is 70% greater than the feed on offer biomass component, clover production (kg/ha/yr), and N fertilizer rate is kg N/ha/yr.

**Atmospheric deposition.** Atmospheric deposits are generally a very small contributor to nutrient inputs, although farms located closer to the coast may have higher deposition rates, depending on prevailing winds. Moreover, some farms in close proximity to heavy industry e.g. downwind of coal fired power stations, may have relatively high loads of specific nutrients like sulphur. Based on a number of Australian literature sources (Blackburn and McLeod 1983; Hingston and Gailitis 1976; Hutton and Leslie 1958; Probert 1976), estimated atmospheric deposition rates for N, P, K, S, ca and Mg are shown in Table 2.

**Table 2. Atmospheric deposition rates (kg/ha/yr)**

| | N | P | K | S | Ca | Mg |
|---|---|---|---|---|---|---|
| >30 km | 1 | 0.1 | 1 | 1 | 2 | 2 |
| <30 km | 2 | 0.4 | 4 | 4 | 8 | 8 |

## 6.3. Defining and quantifying nutrient pools, fluxes and transformations

This project module aimed to identify the key nutrient pools and transformations on Australian dairy farms. This was achieved through a detailed study of 44 dairy farms representing the range and diversity of dairy operations across Australia. This process included determining the appropriate methods for quantification, detailed farm descriptions and monitoring over a 2 year period, the development of a custom built relational database, and appropriate statistical analysis.

**Key activities and achievements**

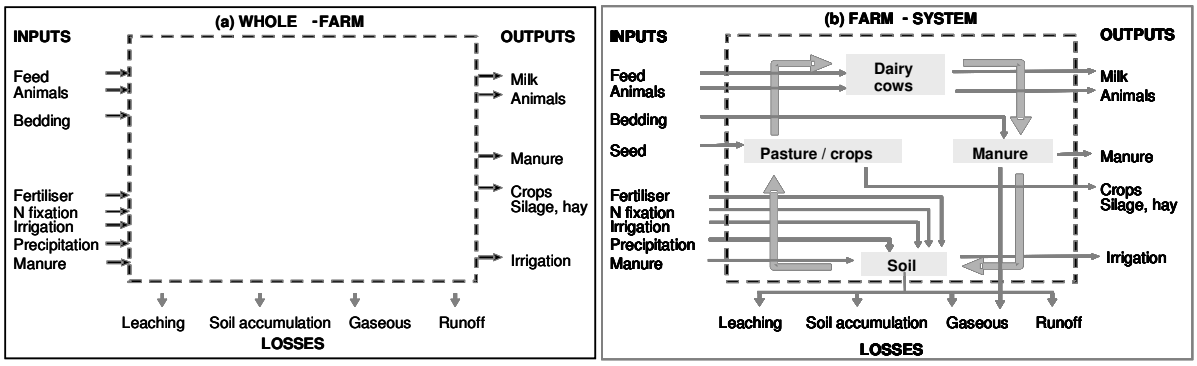
**Module 2. Defining key nutrient pools and transformations on Australian dairy farms**

- ✓ On-farm protocols developed and reviewed
- ✓ Establishment of national on-farm teams
- ✓ On-farm questionnaires for each visit created and revised
- ✓ Technical training workshops to ensure standardisation of methods and data collection.
- ✓ Custom designed relational database for data management
- ✓ Initial planning and consultation (Dec 2007)
- ✓ On-farm monitoring and sampling (Feb 2008, May 08, Aug 2008, Nov 2008, Feb 2009)
- ✓ Verification visit and feedback visit (June 2009, Nov 2009)
- ✓ Spatial analysis of land-use on farms and farm layout maps
- ✓ Survey information entered into relational database
- ✓ Greater than 2200 samples submitted to laboratory
- ✓ Greater than 10,000 analytical results entered into database
- ✓ Final on-farm team workshop (June 2009)
- ✓ Statistical analysis of all milk, dung, feed, effluent, irrigation water, soil results completed
- ✓ Paddock soil nutrient levels mapped for each farm
- ✓ Assessment of paddock legume content, sampling and mapping for each farm
- ✓ Determination of whole-farm nutrient use efficiencies
- ✓ Assessment of uncertainty around predicted whole-farm balances
- ✓ Variability associated with various inputs and importance on sampling
- ✓ Importance of defining the land base
- ✓ Determination of within-farm nutrient use efficiencies
- ✓ Determination of herd nutrient excreta for each farm
- ✓ Cows in Space and Time – quarterly assessments
- ✓ Nutrient loads in management zones – quarterly
- ✓ Cows in Space and Time – daily assessments
- ✓ Nutrient loads in management zones – quarterly
- ✓ Farm Nutrient Loss Index assessment of individual paddocks for participating dairy farms
- ✓ Bench marking of individual farms
- ✓ Documented nutrient accounting process in technical manual
- ✓ Determination of predictive relationships between dairy farm characteristics, nutrient balances and surpluses

### 6.3.1 On-farm research approach

1. A detailed description of nutrient pools and flows was undertaken for a cross-section of 44 Australian dairy farms, inclusive of all states and regions where possible.
2. There were two tiers of farm monitoring. The first involved the collection of data required for the assessment of a 'whole-farm' (often referred to as 'farm-gate') nutrient budget (Figure 9a). The second level of monitoring, referred to as 'farm-system' monitoring, involved more detailed monitoring and measurements to assess internal nutrient cycling and nutrient use efficiencies, as well as a 'whole-farm' nutrient budget (Figure 91b).
3. This two-tiered approach allowed for a more comprehensive characterisation of nutrient budgets and cycling across the spectrum of farming operations within the Australian dairy industry. Robust comparisons of the two approaches and their relative merits will assist in defining the necessary characteristics of a national dairy industry nutrient accounting framework. Additionally, the two-tiered approach enabled more direct linkage with existing regional projects which are currently undertaking nutrient budgeting activities and allow an opportunity for reciprocal co-investment.
4. Detailed data were collected from a range of sources including farm records (i.e. stock numbers, animal age and live weight, milk production, fertiliser inputs, imported forage), direct measurements (i.e. pasture, feed, manure, effluent, milk, soil, irrigation water, and where necessary model estimates (biological N fixation, atmospheric deposition).
5. For each of the input and output factors, components were sampled and laboratory analysis undertaken where possible. The variability around the nutrient concentration and amount of each component was determined and results compared against estimated 'book' values and expected variability, derived from published information. Where this was not possible, these coefficients were obtained from literature or expert assessment.
6. The various input and output factors on each individual farm were then prioritised from most to least important, and their impact on the overall nutrient budget determined.
7. Importantly, priority was given to the user-friendliness of information gathered by and for farmers.
8. Specific gaps in knowledge were also identified. These issues may include quantifying nutrient losses through leaching and drainage, nutrient concentrations in feed inputs and animal products, and retention of nutrients in soil.

**Figure 9. Nutrient inputs, outputs, losses and within-farm flow pathways for a typical dairy farm as required for (a) whole-farm, and (b) farm-system nutrient accounting.**



## Development of experimental protocols and monitoring tools

A detailed 'on-farm' experimental protocol document was developed to provide a comprehensive set of instructions for collecting data from the participating 44 dairy farms. For each different nutrient pool, there is a description of the issues associated with collecting data, the methods for collecting data and a summary table of the data requirements and responsibilities.

In addition to the protocol documents and surveys, customised diaries were provided to all farmers at the commencement of monitoring to allow them to record farm activities. Standard questionnaires are also developed and will be used by the A4N 'on-farm' team members during the quarterly farm visits and interviews. Detail about sample handling, storage and shipping, are all outlined in the project documentation. All of these documents are available online from the A4N website www.accounting4nutrients.com.au.

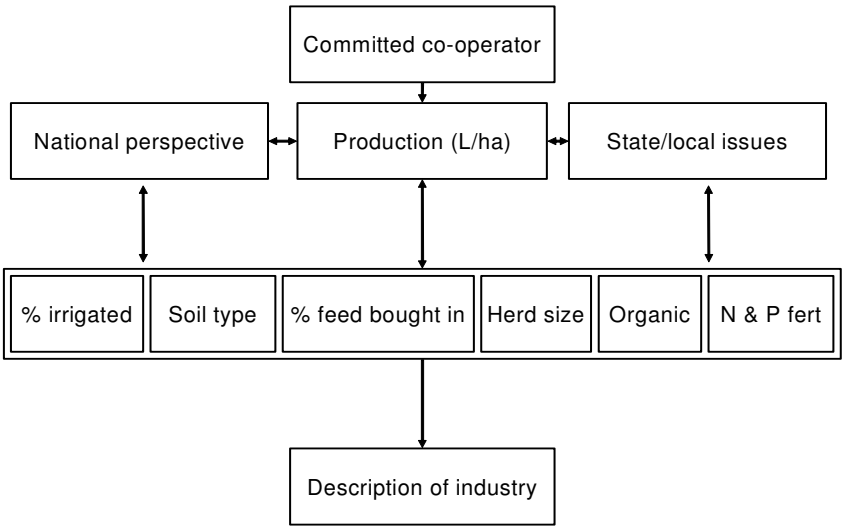
## Selection of monitoring farms

The selection of monitoring farms for the Accounting for Nutrients project was undertaken in conjunction with all project collaborators and stakeholders. The process was based upon the farm selection protocol contained in Milestone report #2. The broad objective of the farm selection process was to select a suite of 44 farms that would be monitored on a quarterly basis over 18 months to determine the range and variability of nutrient pools and fluxes that occur on dairy farms across Australia.

Nominations for participating farms were sought from all National Network and Project Advisory Committee members, and other stakeholders such as funding bodies, farmers, regional development programs, consultants, milk factories etc. Approximately twice as many farms as needed were nominated from the 8 dairy regions to allow the project team to then select a final set of farms that met the objective of achieving a balanced distribution of farm characteristics within the suite of study farms.

At the National Network meeting in May 2007 the use of a hierarchical selection process was endorsed (Figure 10). This process is outlined in the figure below. The process of selection used a statistical approach designed by the project biometrician in conjunction with the Project Science Team. Five key criteria were considered in the farm selection process. These were: (i) geographic location (region), (ii) litre milk/grazed hectare, (iii) grazed hectares, (iv) % of grazed area of farm irrigated, and (v) soil P sorption capacity (predicted from texture classes). The geographic location of farms was determined by agreement between the project team, key stakeholders and investors. An iterative optimisation routine was used within the regional constraints, resulted in a statistically 'best-possible' selection of farms with as wide as possible range of combination of the desired key characteristics.

**Figure 10. Hierarchical farm selection process**

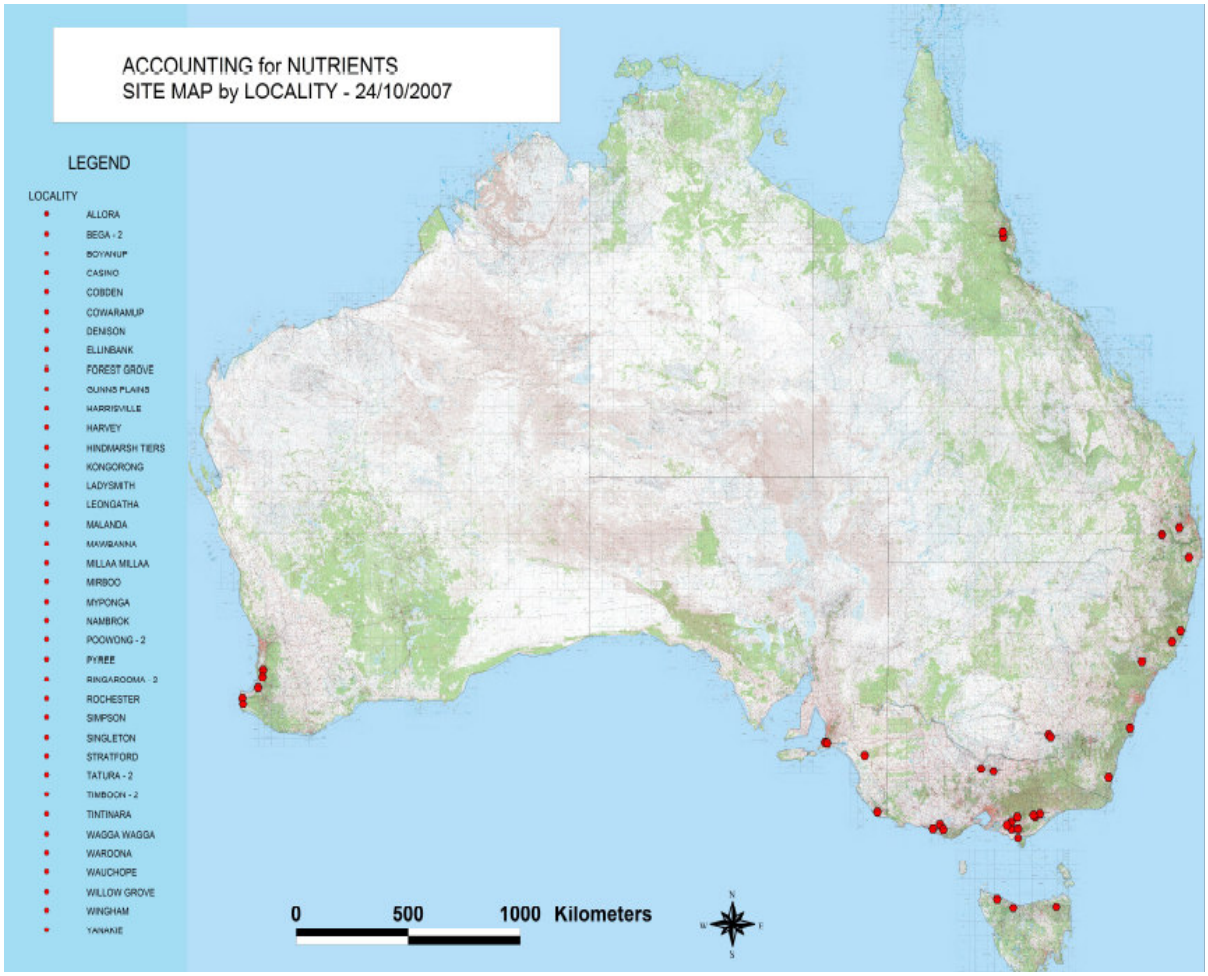


Once the proposed 44 farms had been identified by this process the regional coordinators were consulted to ensure that the selections were appropriate (e.g. distribution of farms was representative of the local industry). The individual farmers selected were then contacted to advise them that they had been selected in the initial round and seeking confirmation of their willingness to still be involved. When this process was completed the farmers were contacted by phone and mail to confirm their involvement and advise them of the next steps and timetable of activities. The farmers that were not selected were also contacted to thank them for their nominations and advising them that they had not been chosen at this stage.

Farms selected were subject to two different types of monitoring, the first being 'Farm-system' nutrient monitoring (includes internal cycling), and the second being 'Whole-farm' nutrient budgeting (akin to current farm-gate nutrient budgeting).

The locations and distribution of the 44 farms across Australia is shown in Figure 11. This spatial distribution of farms in conjunction with the wide range of production characteristics of farms (illustrated in Figures 12 - 15) demonstrates the diversity of the industry that will be examined in the study. Capturing this diversity ensures that the Nutrient Accounting Framework is relevant to all farming systems, that we understand the diversity of nutrient accounting on Australian dairy farms, and that the whole of the industry is potentially exposed to this project and its findings.

**Figure 11. Location of the 44 selected dairy farms participating in the A4N study.**



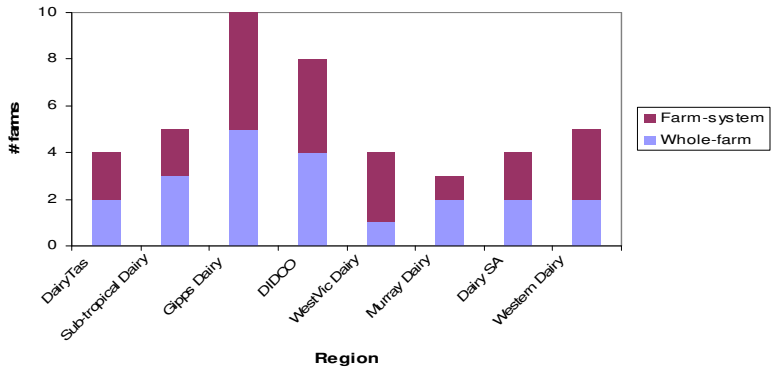



**Figure 12. Distribution of farm system and whole farm monitoring sites within each region.**

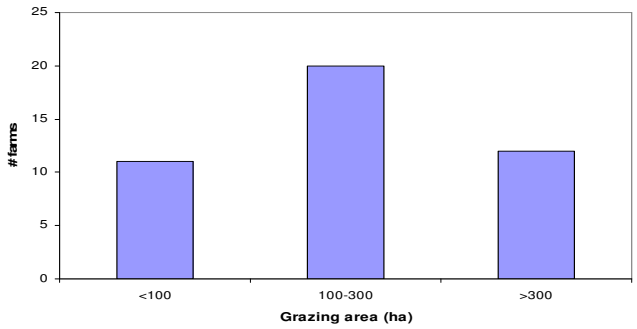



**Figure 13. Distribution of farms by grazing area.**




**Figure 14. Distribution of farms by # grazed area irrigated**

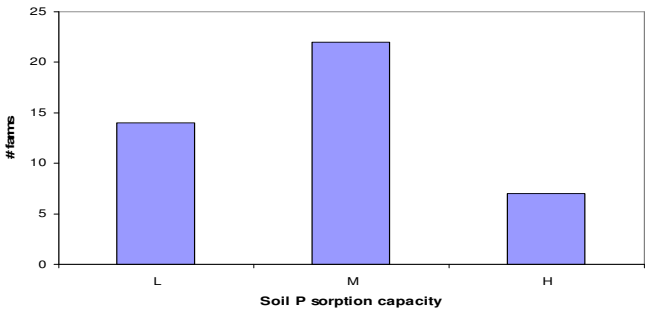



**Figure 15. Distribution of farms by soil P sorption capacity (based on texture class)**

## Farm visits and data collection

A key part of the Accounting for Nutrients project involved the measuring and monitoring of nutrient imports, exports and within-farm nutrient flows on a broad range of dairy farming operations across Australia. This involved farm visits to all 44 participating dairy farms by the Accounting for Nutrients 'on-farm' teams across Australia quarterly. Detailed protocols, revised questionnaires and sampling equipment were provided to all on-farm monitoring teams prior to each round of farm visits to ensure consistency of sample and data collection. Feedback on this process has been regularly provided back to the project leadership team from various farm monitoring team members.

The aims of the quarterly farm visits were to:

- Review farm descriptive information, including herd numbers and structure, pasture-cropping programs.
- Review and update farm paddock and infrastructure layouts.
- Collect quarterly information specifically relating to feeding practices and milk production.
- Identify new purchases and stores of feed and fertiliser
- Collect samples of current feed components, milk and manure samples.
- Review the previous quarter of farmer diary information and distribute the new set of farmer diary sheets.

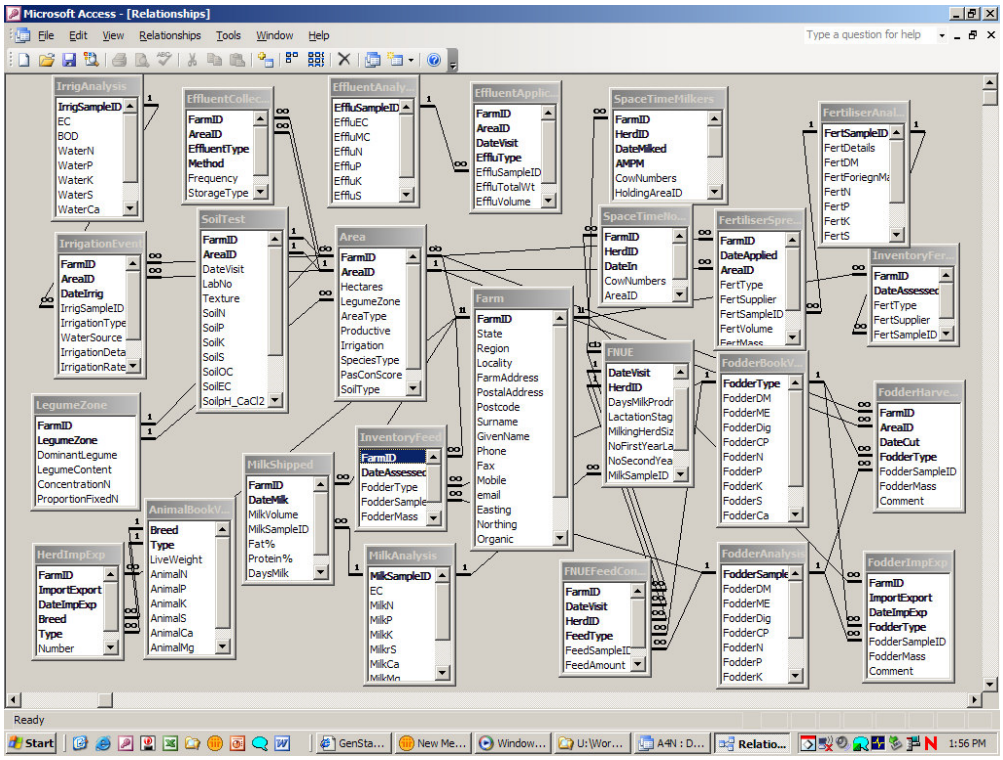
The reduced on-farm data collection period with a focus on a 12 month assessment between February 2008 and February 2009, resulted in a revised sampling schedule and the completion of the quarterly farm visits, sample collection and farmer record-keeping in February/March 09 (T4, visit #6). Farm visits began with a preliminary visit in December 2007 (T-1) to introduce the on-farm teams to the farmers, get an overview of each farm, review the provided farm maps, and importantly, answer any questions the farmers may have about the project. February 2008 marked the beginning of the sample collection and monitoring phase. Farm visits and sample collections were completed for February (Time zero - T0), May (T1), August (T2), November (T3) and February (T4), providing a full 12 months of data. The scheduled July 09 (T5) farm visit was used to collect any identified information gaps and verify the compiled data for each farm. The final visit, undertaken in December 09 (T6), was used to provide a comprehensive one-on-one feedback session, including the provision of collated farm-based information, farm balances, nutrient distribution and legume maps, component nutrient efficiencies, nutrient loading rates, as well as interpretation guidelines. Additionally, all of the participating 44 dairy farms have been comprehensively sampled for soil and legumes, and information has been processed and farm maps provided back to the farmers.

While there have been a number of challenges impacting on the generation of analytical data from the farm collected samples, the revisions to the on-farm data collection has provided the necessary time and effort to address most of these issues. The slow turnaround time for analysis from the feed testing laboratory as well as problems with the electronic formatting of analytical outputs, did cause delays in the provision of analytical data back to the on-farm teams and farmers. There were also occasional delays in forwarding on completed questionnaires and collected samples from some of our on-farm team members. However, all samples were ultimately provided, laboratory analysis was completed, and data was compiled, formatted and provided back to the on-farm teams and farmers by the December 2009 due date.

## Collation and quality assurance of data sets.

All of the data collected in this project were centrally managed and stored on a custom designed and built national database. The data collected came in many forms, having differing spatial scales and time steps, but also having common elements identifying the units (farms, areas, animals, samples, dates, etc.) to which measurements belong. Such data were best managed in a relational database (Figure 16) in order to impose identifier consistency, to ensure data integrity and protection, and to facilitate linking between data sourced from land, feed, animal, and other sources. The database was constructed using Microsoft Access and data entry was centrally managed by staff from DPI Victoria.

**Figure 16. Conceptual map of the relational database and its elements.**




Completed questionnaires from each of the farm visits were forwarded to staff at DPI Ellinbank where they were reviewed, and matched with the corresponding samples. Where inconsistencies were present, follow-up information from the on-farm team members was sought. Sample information was collected and each sample given a unique sample ID. All this information as well as initial processing information such as dry matter determination was entered onto specially designed spreadsheets. These were then used to track each individual sample through collection, laboratory processing and analysis, data entry, and presentation of results back to the farmers. The farmer diaries of occasional records such as fertiliser and feed deliveries, animal sales, irrigation schedules, etc. were also loaded into the relational database, as they were collected from each quarterly visit.

All information from the completed questionnaires, as well as the farmer completed occasional records, went through a quality assurance process and then entered manually into the database. As the related analytical data was returned from the testing laboratory, it was matched to collected sample information and also loaded into the data base. While data entry has been simplified with the development of the relational database the interpretation and process of entering data from hand written questionnaire posed some not unexpected challenges. These included difficulty in interpreting information, inconsistencies in provided information and incomplete answers. Improvements in the questionnaires and sampling were provided to the on-farm teams at each quarter, as well as guidance and one-on-one discussions on improving information and sample collection.

**Uncertainty & sensitivity analysis of farm nutrient budgets**

Conceptually, the farm gate nutrient budget model is simple. The farm nutrient balance is the difference between nutrient loads coming onto a farm and nutrient loads leaving the farm. Because of the impracticalities of measuring certain nutrient loads (nutrients leaving the farm in particular, such as nitrogen volatilisation or phosphorus lost to run-off into waterways) in practice the balance is defined as the difference between identifiable nutrient loads coming onto a farm and identifiable nutrient loads leaving the farm.

It is usually possible to measure or predict all of the nutrient loads coming onto a farm, even such quantities as atmospheric nitrogen fixation and deposition. Hence, even if it is not possible to quantify environmental losses, a positive nutrient balance would imply a build-up and eventual environmental loss of nutrients and would be of interest to the industry and regulators.

The A4N project aimed firstly to measure, as well as possible, the nutrient balance on a range of Australian dairy farms and, secondly, to investigate the measurement process. Given a measured whole-farm nutrient balance, it is pertinent to ask how accurate it is likely to be. We also need to understand the causes, or potential causes, of nutrient balance imprecision/inaccuracy (Oenema et al., 2003). That is, how much is it necessary to sample and measure inputs and outputs, and which ones? These issues become the subject of "uncertainty" and "sensitivity" analyses.

**Analysis of the anatomy of variation of feed imports** Feed and fertiliser are usually the largest sources of nutrient imports onto a dairy farm. As fertiliser is a highly regulated product with standard nutrient concentrations, variations in the nutrient concentrations of feed provides important information about the importance of measuring feed nutrient concentrations. Depending on the quantity and origin of data for each measurement type we propose to estimate components of variance for:

1. Sampling/measurement error (between "replicate samples" same farm, same time period, same paddock)
2. Between loads/paddocks/etc. and times within farms
3. Between farms and time within regions
4. Between regions

Such information will show to what extent and in what circumstances data vary. This will be useful in understanding if and when sampling and measurement of feed inputs and outputs are required for the construction of reliable farm nutrient budgets. Variances are also needed as input to uncertainty analyses.

**Comparison between book and measured values of input and output concentrations** Measured concentration values can be compared directly with published "book values" by comparing their means and variances, superimposing book values on distributions of measured values observed in the A4N project. Differences between book and measured values can be interpreted in the light of sampling and measurement precision where adequate data are available. Similarly, the loads (i.e., concentrations multiplied by mass/volume), using measured and book values, can similarly be compared.

We substituted book values for measured values in the nutrient balance model on selected dairy farm nutrient balances to observe their impact on the resulting balance calculations. This was done for major inputs/outputs and a selection of minor ones, according to availability of data and to keep the task manageable.

**Uncertainty analysis** Here we are interested in what the uncertainty of the individual input/output data implies collectively for the uncertainty of the overall nutrient budget. The replicate variances (or measurement error variances) can be used to imply a "best-case" uncertainty distribution for the balance that allows for measurement/sampling error alone. Other configurations, using within or between farm, time or region variances could be used to predict the uncertainty associated with lower sampling intensities, or of using published, or A4N derived, "book" values. The rules below were used to calculate the individual uncertainty for each import or export, the sum of the imports and exports, and overall whole-farm balance. Actual CV values for feed nutrient concentrations estimated from 1 above were used. Additionally, other CV values were assigned to the nutrient concentration or load weights of other imports or exports based on previous research measurements or expert opinion.

## Simple rules for uncertainty calculations.

1. Coefficient of variation $CV = \frac{\sigma}{\mu}$ i.e. ratio of standard deviation to mean
   (Often expressed as percentage, e.g. $CV = 0.10 = 10\%$. So, in essence, $CV$ is the standard deviation expressed as a percentage of the mean.)
2. If two quantities, X and Y (e.g. a mass and a concentration) are independent then the $CV$ of their **product** (i.e. load) is $CV(XY) = \sqrt{CV^2(X) + CV^2(Y)}$
3. The same applies for a **ratio** (e.g. if $X$ is input N, and $Y$ is output N, and the ratio $Y/X$ is "N efficiency") then $CV(Y/X) = \sqrt{CV^2(X) + CV^2(Y)}$
4. If two quantities, X and Y (e.g. two loads) are independent then the *standard deviation* of their **sum** is $\sigma(X+Y) = \sqrt{\sigma^2(X) + \sigma^2(Y)}$
5. The same applies for a **difference** (e.g. if $X$ is inputs and $Y$ is outputs, the balance is $X$-$Y$) then $\sigma(X-Y) = \sqrt{\sigma^2(X) + \sigma^2(Y)}$
6. **Summing loads**: Suppose we have $n$ independent masses $M_i$, $i=1\ldots n$, with $CV_M$, and corresponding independent concentrations, $C_i$, $i=1\ldots n$, with $CV_C$, (e.g. 10 truck loads of grain) then the Total nutrient load is $\sum_{i=1}^{n} M_i C_i$. If the individual nutrient loads, $M_i C_i$, are all equal then,
   $CV(Total\ Load) = \frac{1}{\sqrt{n}}\sqrt{CV_M^2 + CV_C^2}$ (which is the $CV$ of an individual load on root $n$.)
7. However, if we have $n$ independent masses $M_i$, $i=1\ldots n$, with $CV_M$, but just one concentration, $C$ with $CV_C$, then Total nutrient load is $\left[\sum_{i=1}^{n} M_i\right] \times C$ and if the $M_i$ are all equal then $CV(Total\ Load) = \sqrt{\frac{CV_M^2}{n} + CV_C^2}$.
   If the $M_i$ are not all equal then $CV(Total\ Load) = \sqrt{\frac{CV_M^2 \sum M_i^2}{\left(\sum M_i\right)^2} + CV_C^2}$.

The detailed sampling of major components within the individual farms involved in the study, and the determination of variances (for feed see appendix 5) were used to imply a "best-case" uncertainty distribution that allows for measurement/sampling error alone. Although more precise variance configurations (within or between farm, time and region) were determined for specific feed types where there were large numbers of samples collected, for example pasture silage, pasture hay, ryegrass pasture, and milk, this was not possible with most feed types or other nutrient sources due to smaller overall data sets.

Therefore a more conservative approach of variance estimates was adopted for all measured feed, milk, bedding, manure sources, where overall variance was determined from all pooled data generated within the project. Where no measured data was available, variance estimates from other published data were used. As no direct measurements of load weight were undertaken during the study, the variance of mass estimates was primarily based on previous research measurements or expert opinion.

### 6.3.2 Characteristics of participating dairy farms

General farm information about the 44 participating dairy farms is provided in Table 3. Herd size across the farms ranged from 51 to 1263 cows, with an average of 296.

The major breed of dairy cattle was Holstein-Friesian, with a smaller number of cross-bred herds, and one Jersey herd. Interestingly, less common breeds such as Illawara and Australian Reds were also represented. All dairy farms imported feed and have effluent management systems, 16 farms used feed pads, and 19 used sacrifice paddocks. All dairy farms used fertiliser, 8 used organic fertilisers, while 40 used inorganic fertilisers. All but 3 dairy farms milk twice a day, 2 milked once a day, the remaining farm milks 3 times per day. Fifteen of the dairy farms had different feeding strategies for different milking groups.

**Table 3. General characteristics of all 44 dairy farms involved in the on-farm component**

| Farm characteristic | No. of farms | Farm characteristic | No. of farms |
|---|---|---|---|
| Main breed | | Effluent systems | |
| ○ Holstien/fresian | 34 | ○ 1 pond | 14 |
| ○ Cross breed | 5 | ○ 2 pond | 21 |
| ○ Jersey | 1 | ○ 3 pond | 3 |
| ○ Australian red | 2 | ○ Sump and pump | 5 |
| ○ Illawarra | 2 | ○ compost | 1 |
| Feed pad | 16 | Different feeding groups | 15 |
| Fertiliser use | | Milking frequency | |
| ○ Inorganic* | 40 | ○ Once/day | 2 |
| ○ Organic | 8 | ○ Twice/day | 41 |
| Importing feed | 44 | ○ Thrice/day | 1 |
| Sacrifice areas | 19 | Using total mixed rations | 4 |

* Many also include organically certified inorganic fertiliser materials and blends.

Information from the 7 farm visits during the farm monitoring period (Dec 07 – Dec 09) has been used to gather 'real-time' information about the participating dairy farms. This formed the basis of the meta-data for each farm which has been used to facilitate the industry wide interpretation of nutrient balances and efficiencies.

Table 4 provides land-based information and stocking rates for each of the 44 participating dairy farms involved in the farm monitoring program. Land groups include the total land-base used for dairy production, the area within the boundary of the dairy 'home' farm, and the actual land-base used for pasture and crop production and cow management. Overall, the average total land, dairy-farm land and contact land areas were 336, 235, 194 ha, respectively, but varied widely, ranging from 67 - 1046, 47 - 612, and 40 - 460 ha, respectively (Table 5). The average cow number throughout the study period across all the 44 dairy farms (including lactating cows, dry cows and Springer heifers) was 296 cows, but ranged from 51 - 1263 cows across the 41 dairy farms which completed the study period.

Only 12 of the 44 dairy farms did not use irrigation as a means of increasing pasture or crop production (Table 5). The proportional area irrigated on each dairy farm ranged from 0 to 95% of the contact area, with a mean value of 34%.

The reliance on imported feed is based on the whole-farm metabolisable energy (ME) imported (and feed) as a proportion of the total ME requirements for milk and livestock productivity within the production period. The reliance on imported feed values ranged from 3 to 66%, with an overall mean of 35% and a median of 32%. Not surprisingly, as stocking rate increased, so did the reliance on imported feed.

**Table 4. Specific characteristics for the 44 participating farms involved in the study**

| Farm ID | State/Region | WF (0) or FS (1) | Organic (OR) Conventional (CO) | Total[1] Land | Dairy[2] Farm land | Contact Land[3] | Cow #s[4] | Total[5] Land Stocking Rate | Dairy[6] Farm Land Stocking Rate | Contact Land[7] Stocking Rate | Irrigation Y/N | Irrigation[8] (ha) | Irrigation[9] % | % Feed[10] Imported |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Tas | 0 | CO | 381 | 381 | 306 | 622 | 1.6 | 1.6 | 2.0 | Y | 133 | 44 | 3.14 |
| 2 | Tas | 1 | CO | 185 | 169 | 125 | 330 | 1.8 | 2.0 | 2.6 | Y | 72 | 58 | 21.4 |
| 3 | Tas | 0 | CO | 211 | 211 | 192 | 386 | 1.8 | 1.8 | 2.0 | Y | 94 | 49 | 18.19 |
| 4 | Tas | 1 | CO | 625 | 395 | 365 | 731 | 1.2 | 1.9 | 2.0 | Y | 329 | 90 | 15.62 |
| 5 | South East QLD | 0 | CO | 599 | 496 | 227 | 117 | 0.2 | 0.2 | 0.5 | Y | 42 | 19 | 65.5 |
| 6 | North Queensland | 1 | CO | 307 | 307 | 256 | 234 | 0.8 | 0.8 | 0.9 | N | 0 | 0 | 52.49 |
| 7 | North Queensland | 0 | OR | 135 | 135 | 126 | 100 | 0.7 | 0.7 | 0.8 | Y | 15 | 12 | 9.58 |
| 8 | South East QLD | 1 | CO | 195 | 139 | 134 | 180 | 0.9 | 1.3 | 1.3 | Y | 83 | 62 | 41.94 |
| 9 | Far North Coast NSW | 0 | CO | 201 | 201 | 171 | 238 | 1.2 | 1.2 | 1.4 | Y | 93 | 54 | 14.93 |
| 10 | Gippsland | 1 | CO | 331 | 143 | 127 | 421 | 1.3 | 2.9 | 3.3 | N | 0 | 0 | 42.43 |
| 11 | Gippsland | 0 | CO | 488 | 388 | 364 | 603 | 1.2 | 1.6 | 1.7 | N | 0 | 0 | 21.84 |
| 12 | Gippsland | 0 | CO | 148 | 124 | 115 | 119 | 0.8 | 1.0 | 1.0 | N | 0 | 0 | 14.42 |
| 13 | Gippsland | 0 | CO | 187 | 174 | 168 | 201 | 1.1 | 1.2 | 1.2 | Y | 6 | 4 | 40.23 |
| 14 | Gippsland | 0 | OR | 168 | 135 | 127 | 131 | 0.8 | 1.0 | 1.0 | N | 0 | 0 | 26.45 |
| 15 | Gippsland | 1 | CO | 137 | 98 | 93 | 198 | 1.4 | 2.0 | 2.1 | Y | 22 | 24 | 20.11 |
| 16 | Gippsland | 0 | CO | 115 | 76 | 65 | 172 | 1.5 | 2.3 | 2.6 | Y | 34 | 52 | 32.72 |
| 17 | Gippsland | 0 | CO | 206 | 123 | 112 | 318 | 1.5 | 2.6 | 2.6 | Y | 107 | 95 | 33.57 |
| 18 | Gippsland | 1 | CO | 141 | 101 | 72 | 214 | 1.5 | 2.1 | 3.0 | N | 0 | 0 | 40.05 |
| 19 | Gippsland | 1 | CO | 67 | 67 | 58 | 167 | 2.5 | 2.5 | 2.9 | Y | 55 | 95 | 28.68 |
| 20 | Far South Coast NSW | 1 | CO | 99 | 71 | 66 | 127 | 1.3 | 1.8 | 1.9 | Y | 55 | 84 | 42.55 |
| 21 | NSW | 0 | CO | 705 | 381 | 361 | 254 | 0.4 | 0.7 | 0.7 | Y | 80 | 22 | 61.32 |
| 22 | Far South Coast NSW | 1 | CO | 257 | 257 | 221 | 251 | 1.0 | 1.0 | 1.1 | Y | 63 | 28 | 50.31 |
| 23 | NSW | 1 | CO | 619 | 212 | 186 | 440 | 0.7 | 2.1 | 2.4 | Y | 38 | 21 | 28.98 |
| 24 | NSW | 1 | CO | 1046 | 366 | 315 | 155 | 0.1 | 0.4 | 0.5 | Y | 53 | 17 | 51.2 |
| 25 | Hunter Valley | 0 | CO | 141 | 56 | 53 | 51 | 0.4 | 0.9 | 1.0 | Y | 37 | 70 | 31.67 |
| 26 | NSW | 1 | CO | 96 | 47 | 40 | 101 | 1.1 | 2.2 | 2.5 | Y | 5 | 12 | MD |
| 27 | NSW | 0 | OR | 268 | 192 | 162 | 180 | 0.7 | 0.9 | 1.1 | Y | 63 | 39 | 29.01 |
| 28 | South West Vic | 1 | CO | 487 | 487 | 460 | 540 | 1.1 | 1.1 | 1.2 | N | 0 | 0 | 41.72 |
| 29 | South West Vic | 1 | CO | 522 | 464 | 430 | 550 | 1.1 | 1.2 | 1.3 | N | 0 | 0 | |
| 30 | South West Vic | 1 | CO | 210 | 213 | 143 | 139 | 0.7 | 0.7 | 1.0 | N | 0 | 0 | 25.89 |
| 31 | South West Vic | 1 | CO | 116 | 115 | 100 | 133 | 1.1 | 1.2 | 1.3 | N | 0 | 0 | 31.85 |
| 32 | Northern Irrigation District Vic | 1 | CO | 948 | 356 | 236 | 571 | 0.6 | 1.6 | 2.4 | Y | 87 | 37 | 60.83 |
| 33 | Northern Irrigation District Vic | 0 | CO | 108 | 108 | 102 | 162 | 1.5 | 1.5 | 1.6 | Y | 88 | 86 | 45.15 |
| 34 | Northern Irrigation District Vic | 1 | CO | 152 | 88 | 72 | 228 | 1.5 | 2.6 | 3.2 | Y | 57 | 78 | 51.62 |
| 35 | South Australia | 0 | CO | MD | MD | MD | MD | MD | MD | MD | Y | 180 | MD | MD |
| 36 | South Australia | 1 | CO | 612 | 612 | 307 | 1263 | 2.1 | 2.1 | 3.7 | Y | 280 | 91 | 61.14 |
| 37 | South Australia | 1 | CO | 526 | 349 | 307 | 163 | 0.3 | 0.5 | 0.5 | Y | 154 | 50 | 45.86 |
| 38 | South Australia | 0 | CO | MD | 363 | 338 | 315 | | 0.9 | 0.9 | Y | 80 | 24 | 42.31 |
| 39 | Western Australia | 1 | CO | 703 | 389 | 371 | 531 | 0.8 | 1.4 | 1.4 | Y | 122 | 33 | 27.21 |
| 40 | Western Australia | 1 | CO | 296 | 196 | 186 | 231 | 0.8 | 1.2 | 1.2 | Y | 106 | 57 | 27.54 |
| 41 | Western Australia | 1 | OR | 344 | 330 | 209 | 106 | 0.3 | 0.3 | 0.4 | N | 0 | 0 | 31.63 |
| 42 | Western Australia | 0 | CO | 168 | 98 | 89 | 116 | 0.7 | 1.2 | 1.3 | Y | 32 | 36 | 31 |
| 43 | Western Australia | 0 | CO | 669 | 372 | 275 | 441 | 0.7 | 1.2 | 1.6 | Y | 111 | 40 | 26.59 |
| 44 | Far North Coast NSW | 0 | CO | 197 | 104 | 98 | 211 | 1.1 | 2.0 | 2.2 | N | 0 | 0 | 47.84 |

* Data from farms 26, 29 and 35 was not sufficient to complete the whole-farm nutrient balances and efficiency assessment

**Table 5. Summary dairy farm information for the overall 41 participating farms, the 37 conventional farms and 4 organic farms.**

| | | Total[1] Land | Dairy Farm[2] land | Contact Land[3] | Cow #s [4] | Total Land[5] Stocking Rate | Dairy Farm Land[6] Stocking Rate | Contact Land [7] Stocking Rate | Irrigation [8] (ha) | Irrigation [9] % | % Feed Imported |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **All participating dairy farms n = 41** | **Min** | 67 | 47 | 40 | 51 | 0.15 | 0.24 | 0.40 | 0 | 0 | 3 |
| | **Max** | 1046 | 612 | 460 | 1263 | 2.48 | 2.95 | 3.70 | 329 | 95 | 66 |
| | **Average** | 336 | 235 | 194 | 296 | 1.04 | 1.42 | 1.66 | 63 | 34 | 35 |
| **All conventional dairy farms n = 37** | **Min** | 67 | 47 | 40 | 51 | 0.15 | 0.24 | 0.49 | 0 | 0 | 3 |
| | **Max** | 1046 | 612 | 460 | 1263 | 2.48 | 2.95 | 3.70 | 329 | 95 | 66 |
| | **Average** | 347 | 238 | 198 | 313 | 1.08 | 1.49 | 1.75 | 67 | 37 | 36 |
| **All organic dairy farms n = 4** | **Min** | 135 | 135 | 126 | 100 | 0.31 | 0.32 | 0.40 | 0 | 0 | 10 |
| | **Max** | 344 | 330 | 209 | 180 | 0.78 | 0.97 | 1.11 | 63 | 39 | 32 |
| | **Average** | 229 | 198 | 156 | 129 | 0.63 | 0.74 | 0.83 | 20 | 13 | 24 |

**Definitions**

1 - All Land area under management which contributes to dairy operation. Includes lease blocks, dry blocks and out blocks

2 - Overall land within the "Home" farm boundary. Includes some non productive areas such as wetlands and vegetation

3 - Total land area used for fodder production or feeding of lactating cows. Includes grazed paddocks, cropping, feeding areas such as feedpads and sacrifice paddocks

4 - Total number of lactating, dry and springing cows averaged from data collected at 5 quarterly on farm visits

5 - Cow numbers divided by Total Land hectares (1)

6 - Cow numbers divided by Dairy Farm Land hectares (2)

7 - Cow numbers divided by Production and Contact Land hectares (3)

8 - Hectares of Production and Contact Land irrigated

9 - Irrigation hectares divided by Production and Contact Land hectares (3)

## Comparisons between conventional and organic dairy farms

An important part of the on-farm monitoring was the 'between-farm' comparisons between organic and conventional dairy operations. This was done by comparing the results determined from the 4 organic farms with 4 conventional farms in similar geographical and climatic zones.

A key component of this comparison was the soil analyses from selected areas within each farm. Additionally, many other quantitative analyses within the farms were investigation, such as feed nutrient concentrations, redistribution and recycling via dung, nutrient transfers through forage harvesting, feed nutrient imports, animal and milk nutrient exports (for more details see the project protocol documents).

Overall the organic farms studies were smaller and were more self reliant on home grown feed. The total land used for dairy production on the 4 organic dairy farms ranges from 135 to 344 ha, while the home farm land areas ranged from 135 to 330 ha (Table 5). In contrast, the total land used for dairy production on the 40 conventional dairy farms ranges from 67 to 1046 ha, while the home farm land areas ranged from 47 to 612ha. Out blocks or leased land contributing to around 15% of land used to contribute to dairy production on the organic farms and around 32% on the conventional farms.

Herd size across the 4 organic dairy farms was generally smaller than the conventional herds, and ranged from 100 to 180 cows with an average of 129 cows, compared to 51 -1263 cows with an average of 313 cows (Table 5). Stocking rates, determined as dairy farm land stocking rates, averaged 0.74 cows/ha on the organic farms and 1.49 on the conventional farms.

Three out of the 4 organic dairy farms use organic fertiliser products and approved inorganic fertiliser such as rock phosphate, or specialty blends. One of the organic farms uses effluent and compost as its only means of fertiliser (Table 6). All 4 organic dairy farms rely on imported feed to varying degrees with two of the dairy farms importing more than double the feed than that of the other 2 farms.

Two of the organic dairy farms also manage additional animal enterprises (poultry, pigs, beef) while the other two were solely dairy producers. Two of the organic farms had 2-pond effluent systems, one had a sump/pump system and the remainder piled and compost scraped manure. Three organic farms had Holstein-Friesian as the major breed of dairy cattle while one had Australian Reds. All the organic dairy farms relied to some extent on imported feed, while only one used sacrifice paddocks.

### Table 6. Selected characteristics and examples of feed and fertiliser from the organic dairy farms.

| Farm # | Herd size | Farm size | Imported feed types | Fertiliser types |
|---|---|---|---|---|
| 7 | 100 | 135 | Molasses, Pasture hay, pellets, Superior minerals, sorghum grain | Compost |
| 14 | 131 | 135 | Lucerne hay and silage, Barley, Safflower, Kelp, Sea salt, Sugar, Straw | Boron, Lime, sugar, Worm juice, fish |
| 27 | 180 | 192 | Barley, Wheat, Soy meal, , molasses, | Lime, Rock phosphate, compost, gypsum, Natramin, Zoelite, Balance & grow |
| 41 | 106 | 330 | Barley/oat mix | Lime, Pasture power, Potassium sulphate |

## Spatial analysis of land use on participating dairy farms

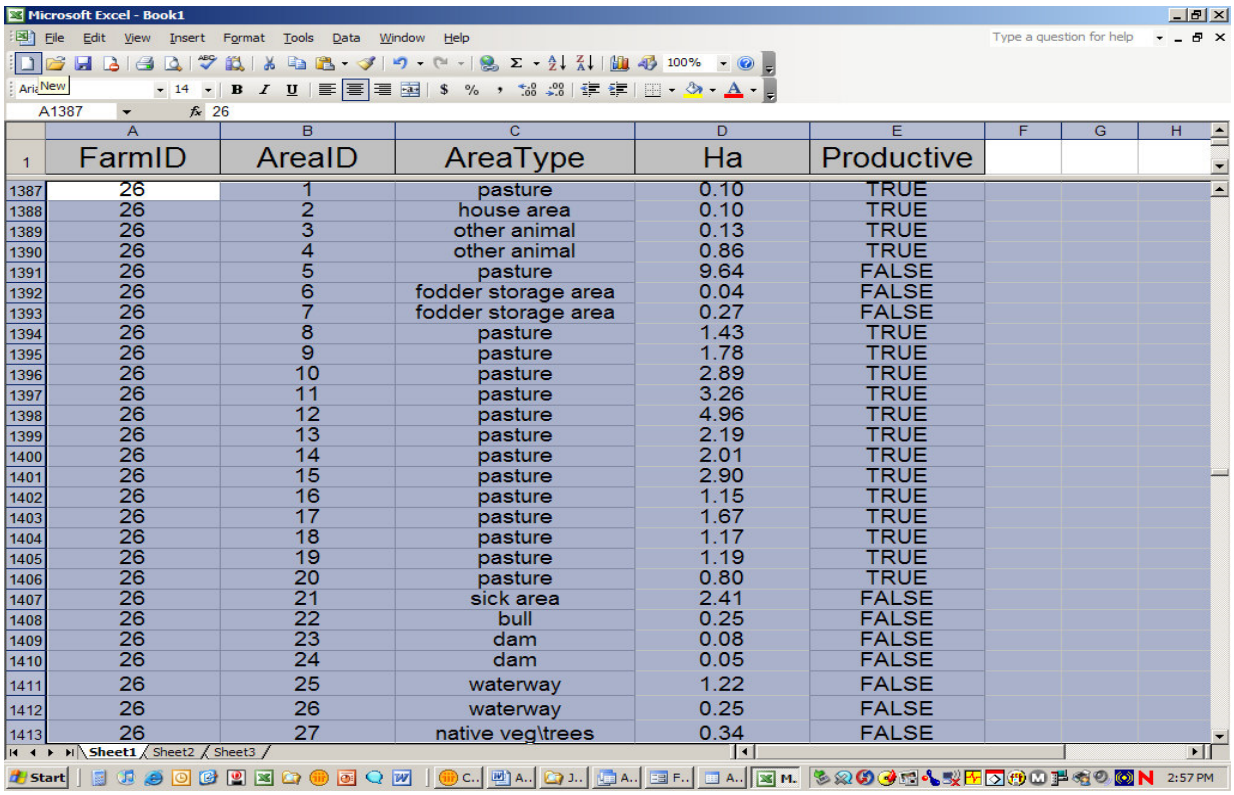
Digital mapping of all 44 dairy farms participating in the project used aerial photographs and schematics of current farm layouts to generate farm maps for collecting information about land management and sample collection. These spatial layers included paddock boundaries, fence lines, watering points, housing and infrastructure, irrigation systems, utilised and non-utilised areas (i.e. bush, riparian, etc), and other landscape features. Copies of these maps were provided to the on-farm teams and farmers along with the data recording diaries.

More detailed spatial data interpretation involved determining the total milking area on each farm, the area of individual paddocks, areas of non-productive land within individual paddocks (such as shelter belts, riparian and wetland areas), laneways and total lengths, and distances walked to each paddock from the dairy. These data assist in calculating nutrient loading rates from fertiliser, effluent, and animal excreta, as well as estimates of the time spent walking by lactating cows each day.

Figure 17 below provides an example of the spatial data sets finalised for each of the 44 dairy farms involved in the study. This data defines the specific areas and whether the land use is used for productive purposes (pasture/crop land). This data is an integral part of the farm system nutrient accounting process.

Table 7 provides examples of the total land-use areas and relative proportions for 3 participating dairy farms. The proportion of the farm used for pasture/crop production is surprisingly similar amongst the 3 farms total paddock areas (81-88%), as was the areas used for laneways (2.5-2.8%). In contrast, the irrigation farm (farm 19) had 11% of land use allocated to irrigation channels, while a rain-fed farm (farm 10) had 3.6% of its land occupied by fenced riparian areas. All farms had small areas such as feed pads, calving or sacrifice paddocks, where animal concentrations are likely to be high throughout the year.

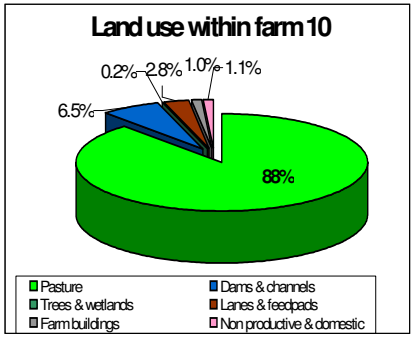
**Figure 17. An example of the spatial information determined for each individual dairy farm involved in the study, available from the relational data base.**

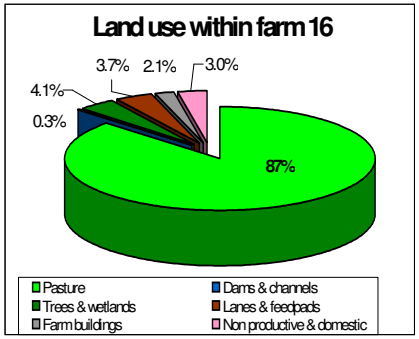


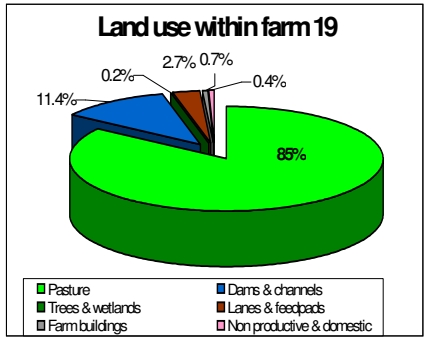

| | A | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|---|
| 1 | FarmID | AreaID | AreaType | Ha | Productive | | | |
| 1387 | 26 | 1 | pasture | 0.10 | TRUE | | | |
| 1388 | 26 | 2 | house area | 0.10 | TRUE | | | |
| 1389 | 26 | 3 | other animal | 0.13 | TRUE | | | |
| 1390 | 26 | 4 | other animal | 0.86 | TRUE | | | |
| 1391 | 26 | 5 | pasture | 9.64 | FALSE | | | |
| 1392 | 26 | 6 | fodder storage area | 0.04 | FALSE | | | |
| 1393 | 26 | 7 | fodder storage area | 0.27 | FALSE | | | |
| 1394 | 26 | 8 | pasture | 1.43 | TRUE | | | |
| 1395 | 26 | 9 | pasture | 1.78 | TRUE | | | |
| 1396 | 26 | 10 | pasture | 2.89 | TRUE | | | |
| 1397 | 26 | 11 | pasture | 3.26 | TRUE | | | |
| 1398 | 26 | 12 | pasture | 4.96 | TRUE | | | |
| 1399 | 26 | 13 | pasture | 2.19 | TRUE | | | |
| 1400 | 26 | 14 | pasture | 2.01 | TRUE | | | |
| 1401 | 26 | 15 | pasture | 2.90 | TRUE | | | |
| 1402 | 26 | 16 | pasture | 1.15 | TRUE | | | |
| 1403 | 26 | 17 | pasture | 1.67 | TRUE | | | |
| 1404 | 26 | 18 | pasture | 1.17 | TRUE | | | |
| 1405 | 26 | 19 | pasture | 1.19 | TRUE | | | |
| 1406 | 26 | 20 | pasture | 0.80 | TRUE | | | |
| 1407 | 26 | 21 | sick area | 2.41 | FALSE | | | |
| 1408 | 26 | 22 | bull | 0.25 | FALSE | | | |
| 1409 | 26 | 23 | dam | 0.08 | FALSE | | | |
| 1410 | 26 | 24 | dam | 0.05 | FALSE | | | |
| 1411 | 26 | 25 | waterway | 1.22 | FALSE | | | |
| 1412 | 26 | 26 | waterway | 0.25 | FALSE | | | |
| 1413 | 26 | 27 | native veg\trees | 0.34 | FALSE | | | |

**Table 7. Examples of spatial data analysis from three dairy farms participating in the 'on-farm' monitoring component.**

| GIS layer | Farm 10 | | Farm 16 | | Farm 19 | |
|---|---|---|---|---|---|---|
| **Description** | **ha** | **% total** | **ha** | **% total** | **ha** | **% total** |
| Grazed Paddocks (inc crop) | 126.4 | 88.0% | 64.6 | 84.7% | 55.4 | 81.2% |
| Calf paddocks | 0.5 | 0.3% | 1.9 | 2.5% | 0.7 | 1.1% |
| Sick paddocks | | | | | 1.6 | 2.4% |
| Channel | | | | | 7.7 | 11.3% |
| Dam | 3.7 | 2.6% | | | | |
| Effluent dam | 0.4 | 0.3% | 0.2 | 0.3% | 0.1 | 0.1% |
| Waterways (fenced) | 5.2 | 3.6% | | | | |
| Feedpad | | | 0.9 | 1.2% | | |
| Laneway | 4.0 | 2.8% | 1.9 | 2.5% | 1.8 | 2.7% |
| Dairy infrastructure | 1.2 | 0.8% | 0.3 | 0.4% | 0.3 | 0.4% |
| Fodderstore | | | | 0.0% | 0.1 | 0.2% |
| Building | 0.3 | 0.2% | 1.3 | 1.7% | 0.1 | 0.1% |
| Trees (fenced) | 0.2 | 0.2% | 0.2 | 0.3% | 0.1 | 0.2% |
| Wetlands (fenced) | | | 2.9 | 3.8% | | |
| Roadway | 0.2 | 0.2% | 0.1 | 0.1% | 0.1 | 0.1% |
| Other animal use | | | 1.4 | 1.8% | | |
| House | 1.4 | 1.0% | 0.6 | 0.8% | 0.1 | 0.1% |
| Ungrazed other | | | 0.2 | 0.3% | 0.1 | 0.2% |
| **Total area** | 143.6 | 100% | 76.5 | 100% | 68.2 | 100% |



### 6.3.3 Quantifying nutrient imports and exports

**Feed**

Dairy farms often import large amounts of grain, grain-based concentrates, and dietary supplements on a regular basis. Similarly, bulk forage, usually varieties of hay, and a range of different by-products may also be purchased and fed to lactating cows, dry cows and other stock on the farm.

During the quarterly visits the on-farm team members in association with the farmers identified different feed types imported onto the farm, as well as stores of conserved hay, silage, and cereals, and collected representative samples of these feeds. Collected samples were initially dried to determine the %DM of the samples, and then forwarded on to DPI Ellinbank for further processing, prior to submitting to the commercial laboratory for analysis.

Over the life of the project more than 1500 separate feed samples were collected and analysed. The data provides an important insight into the types and diversity of feeds fed, grown and purchased by participating dairy farms. These feed samples were analysed for feed quality characteristics such as dry matter content, metabolisable energy, acid detergent fibre, neutral detergent fibre, % crude protein, % crude fat, % lignin, % starch, % water soluble carbohydrates, as well as mineral contents: % calcium, % phosphorus, % magnesium, % potassium, % sulphur.

The feed used or purchased on all 44 dairy farms varies in terms of types and amounts fed, with a broad variety of grain, hay, silage and by-products imported (Table 8). Grain types included wheat, barley, corn, lupins, triticale and sorghum. Silage types included lucerne, sorghum, millet, and a variety of pasture based silages. By-products included black cake, brewers grain, canola meal, citrus pulp, cotton seed meal, fresh tomatoes, mash, palm kernels, pea pollard and soybean meal. The feed quality and mineral characteristics of these feeds also varied substantially.

Raw data was initially reviewed by segmenting sample analysis into broad and specific feed types and then plotting distributions. Outliers were investigated through rechecking data entry and descriptions and where deemed appropriate, reallocated to different feed types. Robust Z scores were calculated according to the NATA standards (> 4 Standard deviations away from the mean) and therefore used to identify extreme values. Where these data points were unexplainable, they were excluded from the data set. This applied to only 74 of the 9480 sample analysis (0.8% of sample analysis).

A detailed summary of the mineral content of the feed types identified and sampled during the 5 quarterly farm visits between February 2008 and February 2009 is provided in Appendix 4. This includes the minimum, maximum, mean, standard deviations and coefficient of variation values (%) for CP, N, P, K, S, Ca, Mg and ME, which was derived for each of these feed groups. This important data set provides a comprehensive database of dairy feed characteristics while the variability of mineral contents in various feeds has informed the uncertainty estimates of dairy feed mineral analysis.

While it was clearly not possible to sample every load of feed imported, or vat of milk produced, the samples collected provided a representative description of the imported feeds. The overall proportion of all feed imports which were directly sampled by the on-farm teams was 33%. On a 'feed type' basis this included 43% of all bedding, 17% of by-products, 26% of pellets, 31% of grain, 31% of feed minerals, 53% of hay and 73% of silage imports.

When determining nutrient imports and exports for particular sources and individual loads on each farm, the following rules were therefore applied and implemented in order:

(i) actual nutrient concentrations when a load had been directly sampled
(ii) nutrient concentrations from a similar source from that farm, with the closest matching data
(iii) average nutrient concentrations from a similar source from all A4N study farms
(iv) average nutrient concentrations from a similar source from published Australian data.

**Table 8. Feed classes and types sampled from the 44 dairy farms during quarterly farm visits.**

| SampleClass | SampleSubtype | SampleClass | SampleSubtype |
|---|---|---|---|
| Byproduct | Almond shells | | |
| Byproduct | Bread | Grain & TMR | Barley grain |
| Byproduct | Brewers grain | Grain & TMR | Corn grain |
| Byproduct | Canola meal | Grain & TMR | Lupin grain |
| Byproduct | Citrus pulp | Grain & TMR | Safflower grain |
| Byproduct | Cotton seed meal | Grain & TMR | Sorghum grain |
| Byproduct | Ex sugarcane | Grain & TMR | Triticale grain |
| Byproduct | Fruit | Grain & TMR | Wheat grain |
| Byproduct | Linseed meal | Grain & TMR | Mixed grain |
| Byproduct | Palm kernels | Grain & TMR | TMR forage only |
| Byproduct | Pea pollard | Grain & TMR | TMR with minerals |
| Byproduct | Soybean meal | | |
| | | Hay | Canola hay |
| Concentrate & Mineral | Grain + Min | Hay | Cereal hay |
| Concentrate & Mineral | Mix Beef | Hay | Cereal straw |
| Concentrate & Mineral | Mix Calf | Hay | Clover hay |
| Concentrate & Mineral | Pellets Calf | Hay | Forage blend |
| Concentrate & Mineral | Pellets Dairy | Hay | Lucerne hay |
| Concentrate & Mineral | Pellets Springer | Hay | Oat Hay |
| Concentrate & Mineral | Pellets Weaner | Hay | Pasture hay |
| Concentrate & Mineral | Mineral additive | Hay | Sorghum hay |
| Concentrate & Mineral | Mineral mined | Hay | Sorghum/millet hay |
| Concentrate & Mineral | Salt | | |
| | | Silage | Canola silage |
| Crop & Pasture | Brassica crop | Silage | Cereal silage |
| Crop & Pasture | Millett crop | Silage | Kikuyu silage |
| Crop & Pasture | Sorghum crop | Silage | Lucerne silage |
| Crop & Pasture | Turnip crop | Silage | Maize silage |
| Crop & Pasture | Kikuyu pasture | Silage | Oats & peas silage |
| Crop & Pasture | Legume pasture | Silage | Paspalum silage |
| Crop & Pasture | Lucerne pasture | Silage | Pasture silage |
| Crop & Pasture | Lucerne&sorghum past | Silage | Prairie grass silage |
| Crop & Pasture | Oats pasture | Silage | Seteria silage |
| Crop & Pasture | Ryegrass pasture | Silage | Sorghum/millet silage |
| **Sample numbers** | | | **1499** |

**Differences in mineral content of different feed types.** The large number of samples and analysis of mineral content, along with knowledge about each farm system (organic, conventional, irrigated, dryland), sampling time and location, has allowed for a detailed investigation of the influence of these factors on nutrient concentrations in various feeds.

In particular this investigation addresses the question "should we differentiate feed samples, based on individual farms, whether the farms are organic or irrigated farms, regional differences or sampling time, to determine the most accurate feed nutrient content?"

**Within-farm and between-farm.** A comparison of the two basic components of variation, namely within-farm and between-farms, was undertaken for sample type and measurements that have sufficient data (Table 9). The within-farm variance is the pooled variance between all samples of that type within a farm. The between-farm variance estimates the additional variance between farms, unaccounted for by the variance within farms. In essence, this is the difference between the total variance of the data and the pooled within-farm variance. The ratio of within-farm to total variance indicates the magnitude of the within-farm variance. A value greater than 60% has been chosen to identify sample types and mineral contents where the within-farm variance is large.

In almost all feeds and nutrients measured (79 of the 125 variance estimates), the within-farm variance was large (>60% of the total variance). However, grains, mixed grains and minerals, cow or calf pellets, mineral additives and mixed grain blends, often had within-farm variation as a smaller component of the total variance. These feed types are likely to be more regulated with specific blends provided to particular farms.

These results suggest that to most accurately determine nutrient concentrations; some feed types should be sampled directly from the farm being assessed. However, since few dairy feed types are sampled and analysed for mineral (or any) analysis once they arrive on the farm, this is likely to be an impractical approach. Therefore, the overall high within-farm variation in feed mineral content justifies standard nutrient contents and use of appropriate uncertainties, where individual feed analysis is not available.

**Table 9. Within farm and between farm components of variance**

| Sample type | Variable | Between farm variance | Pooled within farm vaiance | Within Farm / Total variance (%) | >60% |
|---|---|---|---|---|---|
| Barley grain | CP% | 3.605 | 2.926 | 44.80 | FALSE |
| | N% | 0.092 | 0.075 | 44.80 | FALSE |
| | P% | 0.000 | 0.007 | 100.00 | TRUE |
| | K% | 0.002 | 0.003 | 63.04 | TRUE |
| | S% | 0.000 | 0.000 | 50.24 | FALSE |
| | Ca% | 0.000 | 0.000 | 38.84 | FALSE |
| | Mg% | 0.000 | 0.000 | 66.21 | TRUE |
| Brewers grain | CP% | 0.960 | 5.813 | 85.82 | TRUE |
| | N% | 0.025 | 0.149 | 85.82 | TRUE |
| | P% | 0.016 | 0.013 | 44.19 | FALSE |
| | S% | 0.000 | 0.003 | 100.00 | TRUE |
| | Ca% | 0.003 | 0.000 | 6.88 | FALSE |
| | Mg% | 0.007 | 0.003 | 33.08 | FALSE |
| Canola meal | CP% | 1.837 | 5.582 | 75.24 | TRUE |
| | N% | 0.047 | 0.143 | 75.24 | TRUE |
| | P% | 0.001 | 0.011 | 91.25 | TRUE |
| | K% | 0.002 | 0.007 | 82.88 | TRUE |
| | S% | 0.000 | 0.006 | 100.00 | TRUE |
| | Ca% | 0.000 | 0.004 | 96.07 | TRUE |
| | Mg% | 0.003 | 0.002 | 38.22 | FALSE |
| Cereal hay | CP% | 1.455 | 8.682 | 85.65 | TRUE |
| | N% | 0.037 | 0.222 | 85.65 | TRUE |
| | P% | 0.000 | 0.004 | 95.15 | TRUE |
| | K% | 0.097 | 0.240 | 71.19 | TRUE |
| | S% | 0.001 | 0.001 | 69.78 | TRUE |
| | Ca% | 0.006 | 0.003 | 35.50 | FALSE |
| | Mg% | 0.000 | 0.001 | 61.20 | TRUE |
| Cereal silage | P% | 0.005 | 0.003 | 38.05 | FALSE |
| | K% | 0.570 | 0.046 | 7.53 | FALSE |
| | S% | 0.000 | 0.002 | 92.70 | TRUE |
| Cereal straw | CP% | 0.000 | 2.400 | 100.00 | TRUE |
| | N% | 0.000 | 0.061 | 100.00 | TRUE |
| | P% | 0.001 | 0.003 | 78.31 | TRUE |
| | K% | 0.000 | 0.078 | 100.00 | TRUE |
| | S% | 0.000 | 0.001 | 100.00 | TRUE |
| | Ca% | 0.000 | 0.003 | 98.66 | TRUE |
| | Mg% | 0.000 | 0.001 | 88.73 | TRUE |
| Grain + Min | CP% | 14.094 | 11.956 | 45.90 | FALSE |
| | N% | 0.384 | 0.288 | 42.89 | FALSE |
| | P% | 0.018 | 0.020 | 52.47 | FALSE |
| | K% | 0.003 | 0.012 | 82.76 | TRUE |
| | S% | 0.001 | 0.003 | 67.95 | TRUE |
| | Ca% | 0.121 | 0.162 | 57.35 | FALSE |
| | Mg% | 0.017 | 0.019 | 53.95 | FALSE |
| Lucerne hay | CP% | 0.497 | 10.095 | 95.30 | TRUE |
| | N% | 0.013 | 0.258 | 95.30 | TRUE |
| | P% | 0.001 | 0.013 | 94.93 | TRUE |
| | K% | 0.106 | 0.212 | 66.60 | TRUE |
| | S% | 0.001 | 0.003 | 79.31 | TRUE |
| | Ca% | 0.034 | 0.087 | 72.06 | TRUE |
| | Mg% | 0.007 | 0.013 | 63.25 | TRUE |
| Mineral additive | CP% | 187.941 | 8.510 | 4.33 | FALSE |
| | N% | 4.811 | 0.218 | 4.33 | FALSE |
| | P% | 0.514 | 0.964 | 65.21 | TRUE |
| | K% | 0.127 | 0.018 | 12.17 | FALSE |
| | S% | 0.594 | 0.295 | 33.20 | FALSE |
| | Ca% | 22.828 | 9.776 | 29.98 | FALSE |
| | Mg% | 4.035 | 2.786 | 40.84 | FALSE |
| Mixed grain | CP% | 27.824 | 15.007 | 35.04 | FALSE |
| | N% | 0.712 | 0.384 | 35.04 | FALSE |
| | P% | 0.011 | 0.007 | 38.27 | FALSE |
| | K% | 0.004 | 0.008 | 66.62 | TRUE |
| | S% | 0.001 | 0.001 | 45.16 | FALSE |
| | Ca% | 0.013 | 0.025 | 65.19 | TRUE |
| | Mg% | 0.005 | 0.009 | 63.40 | TRUE |
| Oat Hay | CP% | 1.743 | 4.038 | 69.84 | TRUE |
| | N% | 0.045 | 0.103 | 69.84 | TRUE |
| | P% | 0.002 | 0.005 | 72.85 | TRUE |
| | K% | 0.137 | 0.278 | 66.92 | TRUE |
| | S% | 0.000 | 0.002 | 90.69 | TRUE |
| | Ca% | 0.002 | 0.005 | 68.15 | TRUE |
| | Mg% | 0.001 | 0.001 | 47.95 | FALSE |
| Pasture hay | CP% | 1.550 | 8.492 | 84.56 | TRUE |
| | N% | 0.040 | 0.217 | 84.56 | TRUE |
| | P% | 0.001 | 0.006 | 80.53 | TRUE |
| | K% | 0.065 | 0.298 | 82.02 | TRUE |
| | S% | 0.001 | 0.003 | 77.83 | TRUE |
| | Ca% | 0.008 | 0.026 | 75.97 | TRUE |
| | Mg% | 0.001 | 0.002 | 68.17 | TRUE |
| Pasture silage | CP% | 5.748 | 11.248 | 66.18 | TRUE |
| | N% | 0.147 | 0.288 | 66.18 | TRUE |
| | P% | 0.003 | 0.008 | 70.91 | TRUE |
| | K% | 0.156 | 0.517 | 76.85 | TRUE |
| | S% | 0.002 | 0.003 | 64.35 | TRUE |
| | Ca% | 0.027 | 0.023 | 46.11 | FALSE |
| | Mg% | 0.001 | 0.001 | 66.69 | TRUE |
| Pellets Calf | CP% | 0.000 | 9.373 | 100.00 | TRUE |
| | N% | 0.000 | 0.240 | 100.00 | TRUE |
| | P% | 0.011 | 0.030 | 74.08 | TRUE |
| | K% | 0.000 | 0.010 | 99.68 | TRUE |
| | S% | 0.003 | 0.002 | 37.76 | FALSE |
| | Ca% | 0.191 | 0.197 | 50.77 | FALSE |
| | Mg% | 0.001 | 0.002 | 65.97 | TRUE |
| Pellets Dairy | CP% | 5.052 | 3.563 | 41.36 | FALSE |
| | N% | 0.129 | 0.091 | 41.36 | FALSE |
| | P% | 0.008 | 0.028 | 78.10 | TRUE |
| | K% | 0.008 | 0.005 | 39.02 | FALSE |
| | S% | 0.001 | 0.001 | 51.37 | FALSE |
| | Ca% | 0.423 | 0.075 | 15.10 | FALSE |
| | Mg% | 0.039 | 0.022 | 36.06 | FALSE |
| Pellets Springer | CP% | 10.833 | 2.489 | 18.69 | FALSE |
| | N% | 0.277 | 0.064 | 18.69 | FALSE |
| | P% | 0.019 | 0.010 | 34.78 | FALSE |
| | K% | 0.030 | 0.003 | 8.42 | FALSE |
| | S% | 0.057 | 0.039 | 40.45 | FALSE |
| | Ca% | 0.142 | 0.093 | 39.58 | FALSE |
| | Mg% | 0.041 | 0.004 | 9.34 | FALSE |
| Ryegrass pasture | CP% | 1.770 | 32.104 | 94.77 | TRUE |
| | N% | 0.028 | 0.869 | 96.93 | TRUE |
| | P% | 0.003 | 0.014 | 83.54 | TRUE |
| | K% | 0.105 | 0.575 | 84.52 | TRUE |
| | S% | 0.002 | 0.006 | 79.35 | TRUE |
| | Ca% | 0.012 | 0.025 | 68.32 | TRUE |
| | Mg% | 0.001 | 0.005 | 76.96 | TRUE |
| Triticale grain | CP% | 0.000 | 1.491 | 100.00 | TRUE |
| | N% | 0.000 | 0.038 | 100.00 | TRUE |
| | P% | 0.000 | 0.004 | 100.00 | TRUE |
| | K% | 0.000 | 0.001 | 100.00 | TRUE |
| | S% | 0.000 | 0.001 | 100.00 | TRUE |
| | Ca% | 0.000 | 0.000 | 78.64 | TRUE |
| | Mg% | 0.000 | 0.000 | 100.00 | TRUE |
| Wheat grain | CP% | 3.649 | 3.195 | 46.68 | FALSE |
| | N% | 0.093 | 0.082 | 46.68 | FALSE |
| | P% | 0.001 | 0.002 | 62.66 | TRUE |
| | K% | 0.000 | 0.003 | 100.00 | TRUE |
| | S% | 0.000 | 0.001 | 86.03 | TRUE |

Potential differences in feed mineral contents between organic v conventional farms, irrigated v dryland farms, dairy farming region and sample time, was investigated using a Residual Estimate of Maximum Likelihood (REML) variance components analysis which tested for fixed effects of organic, irrigation, sample time, region and an interaction term sample time and region. This analysis could only be undertaken on sample types with >100 data points, and therefore was applicable to only pasture hay, pasture silage and ryegrass pasture feed types. Pasture silage and ryegrass pasture are generally home-grown feed sources and would therefore not be associated with a whole-farm nutrient balance as it is neither imported or exported but utilised within the farm. Pasture hay may be either imported from another farm or out block, or home grown and fed within the farm.

**Nutrient concentrations for organic and conventional, irrigated and dryland dairy farms.** In general, the N, P, and K concentrations of pasture hay, pasture silage and ryegrass pasture were lower on organic dairy farms (in most cases significantly lower), while Ca concentrations were higher (Table 10). The S and Mg contents of pasture hay, pasture silage and ryegrass pasture were similar. There results reflect the lower P and K soil fertility found within the organic dairy farms, a higher Ca contribution to the overall cation exchange capacity, and also a potentially higher use of lime.

Interestingly, samples of pasture hay and ryegrass pasture from irrigated farms, in general had significantly higher N, P, K and S concentrations. There was little difference between irrigated and dryland farms in the pasture silage nutrient content.

When the most accurate nutrient concentrations for forage are required, these results suggest that a common set of forage nutrient concentrations should not be used for both organic and conventional farms, nor for irrigated and dryland farms. Rather, specific standards of nutrient concentrations should be derived for these specific forages within the different farm systems and feed types.

**Nutrient concentrations for dairy farms in different regions and sampling times**. Overall, the N, P, K, S, Ca and Mg concentrations of pasture hay, pasture silage and ryegrass pasture were not statistically different across all the dairy regions. Exceptions included K, S and Ca in pasture hay, Ca and Mg in pasture silage, and S in ryegrass pasture (Table 11). Specific soil fertility differences could in part at least be responsible for some of these differences, i.e. the elevated S levels may reflect the higher soil sulphur levels found in Gippsland soils. Pasture composition and fertiliser practices are a potential cause of regional differences as well as the timing of conservation practices.

In contrast, the N, P, K, S, Ca and Mg concentrations of pasture hay, pasture silage and ryegrass pasture did, in general, appear to be influenced by the season when it was sampled (Table 12). Nutrient concentrations were usually highest in pasture silage and ryegrass pasture during the cooler seasons and lowest during the warmer seasons, perhaps reflecting seasonal growth patterns. However, for pasture hay, the converse applied (Table 12). Albeit that many of these nutrient concentrations were significantly different at different sampling times, the absolute differences in nutrient concentrations were generally small and fluctuated around the mean value (Table 12).

These results suggest that a standard set of these specific forage nutrient concentrations (though specific for organic or irrigated farms) can be used across the different dairy farming regions and applied irrespective of season.

**Table 10. Nutrient concentration and sample number (N) for pasture hay, pasture silage and ryegrass pasture sampled from conventional or organic, and irrigated or dryland dairy farms.**

| | | N% | | P% | | K% | | S% | | Ca% | | Mg% | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Mean | N | Mean | N | Mean | N | Mean | N | Mean | N | Mean | N |
| Pasture hay | Conventional | 1.73 | 176 | 0.265 | 180 | 1.78 | 182 | 0.217 | 182 | 0.482 | 176 | 0.207 | 175 |
| | Organic | 1.60 | 25 | **0.205** | 24 | 1.64 | 24 | 0.243 | 22 | **0.542** | 21 | **0.180** | 25 |
| | Dry | 1.55 | 56 | 0.217 | 55 | 1.61 | 56 | 0.191 | 55 | 0.482 | 52 | 0.189 | 55 |
| | Irrigated | **1.77*** | 145 | **0.273** | 149 | **1.82** | 150 | **0.230** | 149 | 0.491 | 145 | 0.209 | 145 |
| Pasture silage | Conventional | 2.67 | 217 | 0.394 | 212 | 2.79 | 211 | 0.285 | 217 | 0.569 | 217 | 0.242 | 214 |
| | Organic | 2.30 | 23 | **0.292** | 21 | **1.94** | 22 | 0.220 | 22 | **0.887** | 18 | 0.250 | 22 |
| | Dry | 2.59 | 84 | 0.393 | 80 | 2.77 | 80 | 0.273 | 84 | 0.566 | 80 | 0.230 | 84 |
| | Irrigated | 2.66 | 156 | 0.381 | 153 | 2.67 | 153 | 0.282 | 155 | **0.608** | 155 | **0.250** | 152 |
| Ryegrass pasture | Conventional | 3.74 | 199 | 0.452 | 198 | 2.81 | 200 | 0.349 | 203 | 0.534 | 194 | 0.295 | 196 |
| | Organic | **2.99** | 23 | **0.418** | 23 | 2.47 | 23 | **0.293** | 23 | **0.685** | 23 | 0.288 | 23 |
| | Dry | 3.30 | 50 | 0.399 | 51 | 2.62 | 51 | 0.311 | 51 | 0.505 | 51 | 0.252 | 50 |
| | Irrigated | **3.77** | 172 | **0.464** | 170 | 2.82 | 172 | **0.352** | 175 | 0.564 | 166 | **0.307** | 169 |

*** values in bold are significantly different (P < 0.05) from corresponding pair.**

**Table 11. Nutrient concentration and sample number (N) for pasture hay, pasture silage and ryegrass pasture sampled from different dairy farming regions.**

| | | N% | | P% | | K% | | S% | | Ca% | | Mg% | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Mean | N | Mean | N | Mean | N | Mean | N | Mean | N | Mean | N |
| **Pasture hay** | **Far North Coast NSW** | 1.26 | 2 | 0.289 | 2 | 1.24 | 2 | 0.202 | 2 | 0.484 | 1 | 0.187 | 2 |
| | **Far South Coast NSW** | 1.89 | 5 | 0.283 | 4 | 2.13 | 5 | 0.278 | 5 | 0.347 | 5 | 0.214 | 4 |
| | **Gippsland** | 1.65 | 50 | 0.247 | 49 | 1.72 | 48 | 0.204 | 49 | 0.413 | 50 | 0.214 | 49 |
| | **Hunter Valley** | 1.73 | 5 | 0.373 | 6 | 2.08 | 7 | 0.245 | 6 | 0.509 | 6 | 0.236 | 4 |
| | **North Queensland** | 1.33 | 12 | 0.192 | 12 | 1.61 | 11 | 0.316 | 9 | 0.435 | 11 | 0.137 | 12 |
| | **Northern Irrigation District Vic** | 1.81 | 4 | 0.310 | 4 | 2.15 | 4 | 0.203 | 4 | 0.473 | 4 | 0.200 | 4 |
| | **NSW** | 2.05 | 9 | 0.267 | 15 | 2.36 | 14 | 0.250 | 15 | 0.438 | 11 | 0.229 | 13 |
| | **South Australia** | 1.89 | 13 | 0.279 | 13 | 1.99 | 13 | 0.208 | 13 | 0.472 | 13 | 0.228 | 13 |
| | **South East QLD** | 2.93 | 1 | | 0 | 2.88 | 3 | 0.280 | 3 | 0.752 | 2 | 0.338 | 1 |
| | **South West Vic** | 1.41 | 23 | 0.229 | 23 | 1.67 | 23 | 0.191 | 22 | 0.438 | 23 | 0.164 | 22 |
| | **Tas** | 1.69 | 21 | 0.255 | 21 | 1.88 | 21 | 0.199 | 21 | 0.434 | 21 | 0.197 | 21 |
| | **Western Australia** | 1.86 | 56 | 0.270 | 55 | 1.48 | 55 | 0.222 | 55 | 0.640 | 50 | 0.211 | 55 |
| | Significant differences? | **no** | | **no** | | **yes** | | **yes** | | **yes** | | yes | |
| **Pasture silage** | **Far North Coast NSW** | 2.39 | 3 | 0.459 | 2 | 2.15 | 3 | 0.286 | 3 | 0.591 | 3 | 0.283 | 3 |
| | **Far South Coast NSW** | 2.94 | 7 | 0.454 | 7 | 2.78 | 7 | 0.294 | 7 | 0.590 | 7 | 0.303 | 5 |
| | **Gippsland** | 2.92 | 68 | 0.418 | 65 | 2.95 | 65 | 0.306 | 68 | 0.482 | 68 | 0.250 | 68 |
| | **Hunter Valley** | | 0 | | 0 | | 0 | | 0 | | 0 | | 0 |
| | **North Queensland** | 1.28 | 1 | 0.130 | 1 | 1.30 | 1 | | 0 | | 0 | | 0 |
| | **Northern Irrigation District Vic** | 2.83 | 5 | 0.367 | 5 | 3.22 | 4 | 0.261 | 5 | 0.360 | 5 | 0.256 | 5 |
| | **NSW** | 2.65 | 26 | 0.390 | 25 | 2.96 | 24 | 0.267 | 26 | 0.547 | 26 | 0.261 | 25 |
| | **South Australia** | 2.91 | 32 | 0.418 | 32 | 2.69 | 32 | 0.278 | 32 | 0.784 | 32 | 0.271 | 32 |
| | **South East QLD** | 2.56 | 1 | 0.497 | 1 | 1.76 | 1 | 0.230 | 1 | 0.912 | 1 | 0.366 | 1 |
| | **South West Vic** | 2.30 | 34 | 0.398 | 34 | 2.92 | 34 | 0.272 | 34 | 0.512 | 34 | 0.202 | 34 |
| | **Tas** | 2.57 | 25 | 0.332 | 25 | 2.67 | 25 | 0.288 | 25 | 0.511 | 25 | 0.221 | 25 |
| | **Western Australia** | 2.22 | 38 | 0.307 | 36 | 2.01 | 37 | 0.240 | 38 | 0.841 | 34 | 0.229 | 38 |
| | Significant differences? | **no** | | **no** | | **no** | | **no** | | **yes** | | **yes** | |
| **Ryegrass pasture** | **Far North Coast NSW** | 3.85 | 8 | 0.508 | 8 | 2.66 | 8 | 0.314 | 8 | 0.497 | 8 | 0.287 | 8 |
| | **Far South Coast NSW** | 3.65 | 10 | 0.491 | 10 | 2.60 | 10 | 0.322 | 10 | 0.518 | 10 | 0.351 | 9 |
| | **Gippsland** | 3.68 | 48 | 0.437 | 46 | 2.68 | 47 | 0.370 | 47 | 0.506 | 46 | 0.278 | 47 |
| | **Hunter Valley** | 3.41 | 8 | 0.435 | 8 | 2.59 | 8 | 0.306 | 8 | 0.599 | 6 | 0.394 | 8 |
| | **North Queensland** | 3.08 | 10 | 0.413 | 10 | 3.07 | 10 | 0.233 | 10 | 0.518 | 10 | 0.247 | 10 |
| | **Northern Irrigation District Vic** | 4.32 | 9 | 0.518 | 9 | 3.35 | 8 | 0.384 | 9 | 0.435 | 9 | 0.298 | 9 |
| | **NSW** | 3.67 | 35 | 0.507 | 35 | 2.97 | 34 | 0.334 | 36 | 0.539 | 35 | 0.310 | 35 |
| | **South Australia** | 3.83 | 24 | 0.507 | 24 | 2.67 | 25 | 0.334 | 25 | 0.699 | 23 | 0.314 | 24 |
| | **South East QLD** | 3.48 | 8 | 0.524 | 8 | 2.33 | 9 | 0.331 | 9 | 0.484 | 7 | 0.356 | 6 |
| | **South West Vic** | 3.14 | 16 | 0.376 | 18 | 2.47 | 18 | 0.326 | 18 | 0.509 | 18 | 0.238 | 17 |
| | **Tas** | 3.87 | 31 | 0.380 | 30 | 2.97 | 31 | 0.389 | 31 | 0.567 | 31 | 0.273 | 31 |
| | **Western Australia** | 3.62 | 15 | 0.372 | 15 | 2.79 | 15 | 0.325 | 15 | 0.655 | 14 | 0.308 | 15 |
| | Significant differences? | **no** | | **no** | | **no** | | **yes** | | **no** | | **no** | |

**Table 12. Nutrient concentration and sample number (n) for pasture hay, pasture silage and ryegrass pasture sampled at different time periods.**

| | | Date of sampling | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | Feb-08 | May-09 | Aug-09 | Nov-09 | Feb-10 | Mean | Significance? |
| **Pasture hay** | **N%** | 1.79 | 1.57 | 1.40 | 1.86 | 1.60 | 1.71 | Yes |
| | **n** | 82 | 18 | 14 | 39 | 48 | | |
| | **P%** | 0.26 | 0.32 | 0.24 | 0.27 | 0.23 | 0.26 | Yes |
| | **n** | 85 | 18 | 13 | 39 | 49 | | |
| | **K%** | 1.82 | 1.82 | 1.52 | 1.79 | 1.69 | 1.76 | No |
| | **n** | 85 | 20 | 14 | 39 | 48 | | |
| | **S%** | 0.22 | 0.20 | 0.19 | 0.24 | 0.21 | 0.22 | Yes |
| | **n** | 84 | 17 | 13 | 41 | 49 | | |
| | **Ca%** | 0.50 | 0.47 | 0.39 | 0.56 | 0.44 | 0.49 | Yes |
| | **n** | 79 | 18 | 12 | 39 | 49 | | |
| | **Mg%** | 0.21 | 0.19 | 0.19 | 0.22 | 0.19 | 0.20 | Yes |
| | **n** | 80 | 17 | 14 | 40 | 49 | | |
| **Pasture silage** | **N%** | 2.58 | 2.67 | 2.50 | 2.76 | 2.63 | 2.64 | Yes |
| | **n** | 101 | 22 | 9 | 51 | 57 | | |
| | **P%** | 0.38 | 0.46 | 0.46 | 0.39 | 0.36 | 0.38 | Yes |
| | **n** | 99 | 20 | 8 | 50 | 56 | | |
| | **K%** | 2.69 | 2.64 | 3.06 | 2.69 | 2.74 | 2.71 | No |
| | **n** | 98 | 22 | 8 | 50 | 55 | | |
| | **S%** | 0.28 | 0.28 | 0.28 | 0.28 | 0.28 | 0.28 | No |
| | **n** | 101 | 22 | 9 | 50 | 57 | | |
| | **Ca%** | 0.60 | 0.61 | 0.63 | 0.62 | 0.56 | 0.59 | Yes |
| | **n** | 100 | 22 | 9 | 47 | 57 | | |
| | **Mg%** | 0.24 | 0.27 | 0.26 | 0.25 | 0.23 | 0.24 | Yes |
| | **n** | 100 | 21 | 8 | 50 | 57 | | |
| **Ryegrass pasture** | **N%** | 3.05 | 4.13 | 4.37 | 3.43 | 3.36 | 3.67 | Yes |
| | **n** | 42 | 44 | 43 | 49 | 44 | | |
| | **P%** | 0.37 | 0.52 | 0.54 | 0.45 | 0.35 | 0.45 | Yes |
| | **n** | 41 | 44 | 42 | 49 | 45 | | |
| | **K%** | 2.50 | 2.99 | 3.28 | 2.57 | 2.55 | 2.77 | Yes |
| | **n** | 42 | 44 | 43 | 50 | 44 | | |
| | **S%** | 0.33 | 0.36 | 0.37 | 0.33 | 0.32 | 0.34 | Yes |
| | **n** | 42 | 45 | 44 | 50 | 45 | | |
| | **Ca%** | 0.54 | 0.57 | 0.50 | 0.59 | 0.54 | 0.55 | No |
| | **n** | 42 | 41 | 43 | 50 | 41 | | |
| | **Mg%** | 0.32 | 0.32 | 0.25 | 0.28 | 0.31 | 0.29 | Yes |
| | **n** | 41 | 43 | 43 | 50 | 42 | | |

**Milk, Animals, Fertiliser, N fixation and Irrigation**

Milk nutrients exported were recorded through access to monthly milk statements, as well as the collection of milk samples for analysis during each quarterly farm visit for all 44 farms. The annual milk statement was also used to back-check the collected information from the quarterly visits.

The nutrient concentrations of milk were determined from quarterly samples collected during the farm visits between February 2008 and February 2009. Raw nutrient concentration data for milk samples are presented in Appendix 5. Overall, the N, P, K, S, Ca and Mg concentrations of milk were similar across all the participating dairy farms and between sampling times (Table 13). Apart from the milk sulphur levels, variations in nutrient concentrations were small (coefficient of variation values >15%). These results demonstrate the general consistency of milk nutrient levels and suggest that a standard set of milk nutrient concentrations can be used across different farms, regions and applied irrespective of season.

**Table 13. Average nutrient concentrations for milk sampled during quarterly visits from the participating dairy farms.**

| | Protien | N% | P% | K% | S% | Ca% | Mg% |
|---|---|---|---|---|---|---|---|
| Average value | 3.305 | 0.522 | 0.091 | 0.142 | 0.038 | 0.113 | 0.011 |
| Standard Dev | 0.233 | 0.037 | 0.014 | 0.015 | 0.015 | 0.011 | 0.001 |
| Coeff. of var. (%) | 7.0 | 7.1 | 14.9 | 10.5 | 39.0 | 9.7 | 7.4 |
| Sample no. | 199 | 198 | 146 | 219 | 218 | 218 | 219 |

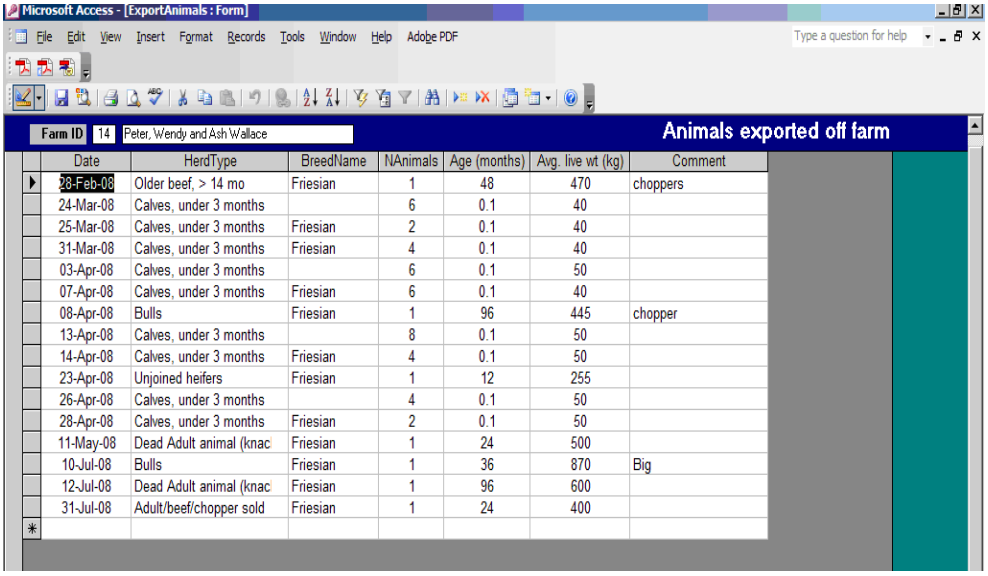
Stock purchases, sales and mortality were recorded in the farmer occasional diaries, and reviewed at each quarterly visit. Figure 18 provides an example of information of stock sales, purchases and deaths being collected on the participating dairy farms. As no animal nutrient concentrations were determined during the project, previously published standards of nutrient contents were used in the balance calculations.

Similarly, inorganic fertiliser imported (such as urea, NPKS blends, superphosphate, etc) were not analysed during the project as these products have highly regulated nutrient concentrations and must meet government standards. Manure and organic fertiliser products were sampled and analysed where possible and actual nutrient concentrations from these samples were applied to the nutrient balance calculations.

The legume content of pastures (or legume crops) was directly determined on all participating dairy farms at a sampling time corresponding with the likely maximum legume present (usually spring or summer). Figure 19 provides the varying legume % (dry weight basis) for each of five legume zones on each farm, nominated by the participating farmers. While many paddocks had no measurable legumes, the highest legume content of a block of paddocks (legume zone) was 68%. Overall, legume contents in pastures were very low with a median legume content of only 2%. The mean value of 11%, reflected a small number of very high (> 50%) legume contents measured on specific farms. These results highlight the general lack of legumes in dairy pastures, potentially due to high nitrogen inputs on most dairy farms (i.e. nitrogen fertiliser) but also dry seasonal conditions in many dairy regions. However, those farms and areas within farms with high legume contents contribute to a relatively high N input from N fixation and therefore legume assessment should not be overlooked when assessing N balances on Australian dairy farms.

Nutrients imported in irrigation water were also assessed through the recording of irrigation applications and the sampling and laboratory analysis for nutrient concentrations where possible. Overall there were 57 independent irrigation water analyses of samples collected from irrigation channels and bores. Recycled water used for irrigation was not included in this summary as this was not providing an external input. Not surprisingly, nutrient contents varied considerably (Table 14). Of particular note are the large differences in Ca and Mg, reflecting the sources of water (bore water) and the regions (high soil and rock carbonates) in some regions.

**Figure 18. An example of data on stock sales, purchases and deaths being collected on the participating dairy farms.**




| Date | HerdType | BreedName | NAnimals | Age (months) | Avg. live wt (kg) | Comment |
|---|---|---|---|---|---|---|
| 28-Feb-08 | Older beef, > 14 mo | Friesian | 1 | 48 | 470 | choppers |
| 24-Mar-08 | Calves, under 3 months | | 6 | 0.1 | 40 | |
| 25-Mar-08 | Calves, under 3 months | Friesian | 2 | 0.1 | 40 | |
| 31-Mar-08 | Calves, under 3 months | Friesian | 4 | 0.1 | 40 | |
| 03-Apr-08 | Calves, under 3 months | | 6 | 0.1 | 50 | |
| 07-Apr-08 | Calves, under 3 months | Friesian | 6 | 0.1 | 40 | |
| 08-Apr-08 | Bulls | Friesian | 1 | 96 | 445 | chopper |
| 13-Apr-08 | Calves, under 3 months | | 8 | 0.1 | 50 | |
| 14-Apr-08 | Calves, under 3 months | Friesian | 4 | 0.1 | 50 | |
| 23-Apr-08 | Unjoined heifers | Friesian | 1 | 12 | 255 | |
| 26-Apr-08 | Calves, under 3 months | | 4 | 0.1 | 50 | |
| 28-Apr-08 | Calves, under 3 months | Friesian | 2 | 0.1 | 50 | |
| 11-May-08 | Dead Adult animal (knac | Friesian | 1 | 24 | 500 | |
| 10-Jul-08 | Bulls | Friesian | 1 | 36 | 870 | Big |
| 12-Jul-08 | Dead Adult animal (knac | Friesian | 1 | 96 | 600 | |
| 31-Jul-08 | Adult/beef/chopper sold | Friesian | 1 | 24 | 400 | |

**Figure 19. Pasture or crop legume content for five productivity zones on each dairy farm**

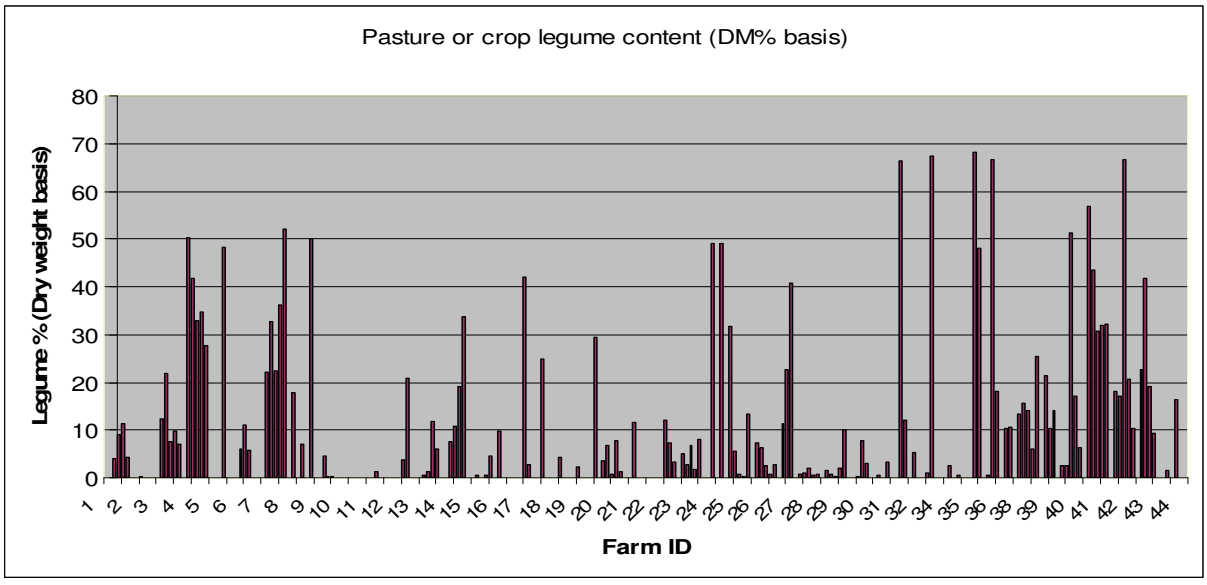



**Table 14. Average, minimum and maximum nutrient concentrations of irrigation water.**

| | Nitrogen mg/l | Phosphorus mg/l | Potassium mg/l | Sulphur mg/l | Calcium mg/l | Magnesium mg/l |
|---|---|---|---|---|---|---|
| Mean | 0.93 | 0.04 | 2.00 | 2.89 | 13.14 | 7.62 |
| | | | | | | |
| Max | 4.8 | 0.22 | 10 | 32 | 120 | 52 |
| min | 0.2 | 0.01 | 0.6 | 0 | 0.85 | 0.76 |

### 6.3.4 Whole-farm nutrient balances

Following the completion of on-farm data collection between February 2008 and February 2009, the large amount of information was collated and compiled within the relational database. Missing or questionable data was identified and a follow up farm visit was undertaken in July 09 with the participating farmers to review, fill data gaps and endorse the final data. At the end of this process, there was insufficient data from 3 out of the 44 participating farms (farm 26, 29 and 35) to determine whole-farm balances and associated output/input efficiencies. The periodic assessments of cow nutrient use efficiency, cow locations, nutrient loading rates and nutrient distribution patterns were achieved for 43 of the 44 participating farms. The detailed statistical analysis of feed and other nutrient sources was used to determine the uncertainty around nutrient calculations and budget predictions.

The on-farm project teams prepared a detailed portfolio of nutrient management information for each participating dairy farm and presented and discussed the results with farmers in December 2009. An example of some of the information presented to each farm is presented in Appendix 6.

It is not practical within this report to present whole-farm balances for all 41 dairy farms. Examples have therefore been used to illustrate key messages and highlight differences in nutrient flows and transformations for a range of different dairy systems. Combined and integrated data from all farms is presented in the 'Industry interpretation and implications' chapter of this report (section 7).

#### Nutrient sources, nutrient balances and nutrient use efficiencies

Table 15 provides an example of a nutrient balance output for N, P, K, S, Ca and Mg, generated from the compilation of all the collected farm-based information from an example farm, presented on a 'total nutrient' basis. Table 16 presents the same data which has been standardised on a 'total nutrient per ha of contact dairy farm land' basis. This standardisation allows for all individual farms to be compared.

This example farm imported around 30% of feed requirements, and had a stocking rate of 2.4 cows/ha. The imported feed corresponded with a large amount of nitrogen also imported in the grain and hay (Table 15). While this also resulted in the import of substantial amounts of phosphorus, potassium, sulphur and calcium; inorganic fertiliser was the largest contributor. Other sources of nutrients included N fixation from legumes and atmospheric deposition, although these contributions were generally small and insignificant on a per ha basis (Table 16). On this particular farm, there were no animals imported during this 12 month period. Not surprisingly, the single largest source of nutrient export was in milk for all nutrients presented. The removal of animals as calves or culls was also important but only contributed about 10% of the total exports. There were no other products exported during the monitoring period.

The total nutrient inputs (the sum of the various inputs sources) minus the total nutrient exports (the sum of the various exports) equate to the overall whole-farm nutrient balance. In the case of nitrogen this amounted to a net import of 16.3 tonne and a net export in milk and animals of 6.4 tonne, with a net farm surplus of 9.9 tonne (Table 15). On a 'per-ha' basis, this was equivalent to 163 kg N/ha imported, 64 kg N/ha and a net farm surplus of 99 kg N/ha (Table 16). The corresponding whole-farm balances for P, K, S, Ca and Mg were 11, 87, 20, 13 and 3 kg/ha respectively, ass in net surplus.

The whole-farm nutrient use efficiency is determined by the proportion of nutrient exported, divided by that imported. In the case of nitrogen the whole-farm N use efficiency was 39%. The corresponding whole-farm efficiencies for P, K, S, Ca and Mg were 51, 15, 25, 57 and 41% respectively (Table 15).

The uncertainties around the various inputs were in general relatively small and usually less than 10% of the estimated nutrients imports or exported, for each source. An exception to this is N fixation, where estimates are recognised as having a high degree of uncertainty and reflected in its high predicted variance. However, as the total uncertainty of the balance estimates are determined from integrating the various sources of uncertainty and the relative weighting from individual contributions, the overall estimate of uncertainty was less that 10% of the balance values, with the exception of magnesium.

**Table 15. An example of a farm nutrient balance sheet for all determined nutrients and sources, presented on a kg of nutrient basis.**

| Whole farm nutrient balance (total kg nutrient) - farm # | | | | | | |
|---|---|---|---|---|---|---|
| | Nitrogen (kg) | Phosphorus (kg) | Potassium (kg) | Sulphur (kg) | Calcium (kg) | Magnesium (kg) |
| **Imports** | | | | | | |
| Silage | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Hay | 1138 (98) | 183 (18) | 1077 (67) | 114 (11) | 341 (72) | 110 (16) |
| Concentrates/grains | 8868 (862) | 902 (81) | 1501 (143) | 675 (71) | 190 (24) | 412 (39) |
| Feed minerals | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Byproducts | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| NPKS blends | 0 (0) | 1216 (34) | 7600 (215) | 1824 (52) | 2432 (69) | 0 (0) |
| Urea/Liquid N | 6028 (98) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Other fertilisers | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Soil ameliorants | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Animals | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Irrigation | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| N fixation | 138 (54) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Atmospheric deposition | 100 (11) | 10 (1) | 50 (6) | 50 (6) | 50 (6) | 50 (6) |
| Other | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| **Total Imported** | **16273** (875) | **2310** (90) | **10228** (267) | **2663** (88) | **3013** (102) | **572** (43) |
| **Exports** | | | | | | |
| Milk | 5717 (116) | 998 (67) | 1514 (73) | 456 (80) | 1230 (55) | 110 (4) |
| Animals | 695 (16) | 179 (4) | 50 (1) | 199 (3) | 497 (7) | 124 (0) |
| Forages | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Manure | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Other | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| **Total exported** | **6413** (117) | **1177** (67) | **1564** (73) | **654** (80) | **1727** (55) | **234** (4) |
| **BALANCE** | **9860** (883) | **1134** (112) | **8664** (276) | **2009** (119) | **1287** (116) | **337** (43) |
| **Efficiency (%)** | **39** | **51** | **15** | **25** | **57** | **41** |

**Table 16. An example of a farm nutrient balance sheet for all determined nutrients and sources, on a kg of nutrient per ha basis.**

| Whole farm nutrient balance (kg/ha nutrient) - farm # | | | | | | |
|---|---|---|---|---|---|---|
| | N (kg/ha) | P (kg/ha) | K (kg/ha) | S (kg/ha) | Ca(kg/ha) | Mg(kg/ha) |
| **Imports** | | | | | | |
| Silage | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Hay | 11 (1) | 2 (0) | 11 (1) | 1 (0) | 3 (1) | 1 (0) |
| Concentrates/grains | 89 (9) | 9 (1) | 15 (1) | 7 (1) | 2 (0) | 4 (0) |
| Feed minerals | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Byproducts | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| NPKS blends | 0 (0) | 12 (0) | 76 (2) | 18 (1) | 24 (1) | 0 (0) |
| Urea/Liquid N | 60 (1) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Other fertilisers | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Soil ameliorants | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Animals | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Irrigation | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| N fixation | 1 (1) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Atmospheric deposition | 1 (0) | 0 (0) | 1 (0) | 1 (0) | 1 (0) | 1 (0) |
| Other | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| **Total Imported** | **163** (9) | **23** (1) | **102** (3) | **27** (1) | **30** (1) | **6** (0) |
| **Exports** | | | | | | |
| Milk | 57 (1) | 10 (1) | 15 (1) | 5 (1) | 12 (1) | 1 (0) |
| Animals | 7 (0) | 2 (0) | 0 (0) | 2 (0) | 5 (0) | 1 (0) |
| Forages | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Manure | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Other | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| **Total Exported** | **64** (1) | **12** (1) | **16** (1) | **7** (1) | **17** (1) | **2** (0) |
| **BALANCE** | **99** (9) | **11** (1) | **87** (3) | **20** (1) | **13** (1) | **3** (0) |
| **Efficiency (%)** | **39** | **51** | **15** | **25** | **57** | **41** |

## Major sources of nutrients

The various sources of nutrients and the minimum, maximum and median values for N, P, K and S (kg/ha, contact area), determined from the 41 participating farms, have been summarised in Table 17. The median value has been used rather than the mean, as high input farms involved in the study have skewed specific input values, particularly for fertiliser and feeds.

While N, P, K, S, Ca and Mg are all contained in various products imported and exported, the contribution of any particular source between farms varies widely (often from a minimum of 0, when a source is not used). It is important to note that almost all of the identified inputs and outputs contribute significantly in at least some dairy operations.

There is not a limited number of nutrient sources but a wide variety of sources that contribute to nutrient inputs and ultimately to nutrient use efficiencies and surpluses (or deficits). These different nutrient sources demonstrates the diversity of farming systems and the range of necessary nutrient inputs required to accurately determine nutrient balances in Australian dairy production systems.

However, these sources of nutrients and the contribution to nutrient imports or exports are likely to be correlated; both negatively and positively. An example of a positive correlation would be the importation of hay and grain; as one increases it would be expected that so would the other. In contrast, a negative correlation may occur between nitrogen fertiliser and the amount of supplementary feed purchased; as fertiliser input increases, the reliance on imported feed may decline. Similarly, a negative correlation may occur with nitrogen fertiliser and N fixation; as fertiliser input increases, the legume content of pastures generally declines and so too the amount of N fixed by legumes from the atmosphere.

These factors may not be strongly linked however. A farm may grow surplus home-grown hay and balances the roughage with grain. Seasonal factors may necessitate the purchase of feed even though high rates of N fertiliser may have been applied.

***Nitrogen.*** Nitrogen is clearly an essential nutrient input, and this data indicated that Australian dairy systems can import N from a variety of sources (Table 17). In order of relative importance, the single largest source of N imported on to any one farm was inorganic fertiliser, either as a NPKS blends or commonly as urea. Interestingly, the next largest single input identified in the study was from N fixation by legumes. Imported feed, either as hay, grain, or as by-products all contributed very similar N imports on one farm or another. The lower, but still important, contribution from imported silage probably reflects the lower relative amount contributing to feed imports. Other fertilisers, such as poultry manure, were also found to provide an important source of N. When large numbers of animals are moved off the farm and then brought back again, as happens in 'off-wintering' systems, imports and exports of N in the form of animals can also be equally important. While an order of magnitude smaller, irrigation water can also contribute a significant amount of nitrogen, depending on the water source. For example, sourcing irrigation water from town sewage treatment works (as in this case) or dairy factory waste water may contribute significant nutrient loads. Finally, feed minerals, atmospheric deposition and soil ameliorants (lime, gypsum) contribute only a relatively small amount of N into the dairy systems studied.

The export of N is however much simpler. Milk shipped off farm was the single largest source of exported N, although animal movements (as mentioned above) can also significantly export N. In this study we found that there was limited home-grown feed shipped off the farm, with most harvested material stored and fed back within the farm boundary.

***Phosphorus and sulphur.*** Phosphorus and sulphur occur in similar concentrations in a variety of feeds and fertiliser and therefore the amounts of P and S from various sources are often similar (Table 17). Interestingly, S from irrigation water can be another important source. However, the most common source of imported P and S was inorganic fertiliser, accounted for within a NPKS blend, probably as superphosphate. 'Other fertiliser' products, which refer to non-traditional fertilisers such as rock phosphate and poultry manure, also provided an important source of P and S on some farms.

Imported feed was also an important contributor to total P and S imports, especially by-products due to the higher P contents in some products, followed by grain, hay and silage. Like N, P and S imports and exports in animals can appear large when large numbers of animals are moved off and back on to the farm. Feed minerals, atmospheric deposition and soil ameliorants only contributed a relatively small amount of P and S into the dairy systems studied.

While export of P and S is largely within milk shipped off farm, the lower concentration of P and S than N means that lower exports occur overall. There was little P and S shipped off the farm in exported feed.

***Potassium.*** Potassium concentrations are similar to nitrogen in forages, but lower in milk and animals (Table 17). In order of relative importance, the single largest source of K imported on to any one farm was in feed by-products, followed by hay, then inorganic fertiliser, and silage. Irrigation water was also identified as an important source, as well as grain and other fertilisers. Finally, animal movements, feed minerals, atmospheric deposition and soil ameliorants contributed only a relatively small amount of K into the dairy systems studied. Similarly to P and S, K is largely exported within milk shipped off farm, but there may also be significant export of K in removed forages.

***Calcium and Magnesium.*** Calcium and magnesium are also imported from a variety of sources (Table 17), but some very large inputs result from specific use of soil ameliorants such as lime, or regional characteristics of irrigation water with high cation concentrations. Apart from these specific inputs (in order of relative importance), Ca and Mg was also imported onto farms in important amounts in inorganic fertiliser, grain, animals, hay, by-products and silage. Similarly to P and S, Ca and Mg, is largely exported within milk shipped off-farm, but there may also be significant export of Ca in animals and forage.

## Differences in nutrient sources between conventional and organic dairy farms

While there were only 4 organic farms in comparison to 37 conventional dairy farms involved in the study, it is interesting to note differences in nutrient flows into and out of these contrasting systems. Nutrient imports of N, K, S, Ca and Mg for organic farms were generally at much lower levels than on conventional farms, reflecting the smaller farm sizes, limited use of inorganic manufactured fertiliser, lower cow numbers and stocking rates, and higher reliance on home-grown feed.

The organic farms did not use inorganic manufactured N or P fertilisers, though one used a potassium sulphate blend as a source of K and S. In contrast however, the application rates of rock phosphate on three of the 4 dairy farms studied was a major source of both P and Ca import on the organic farms and resulted in P imports being amongst the highest of any of the farms studied.

Most N was imported on to the organic farms through the process of N fixation, followed by imported grain, hay, by-products and silage. All other sources delivered very little N imports. Additional but small sources of P, K and S imports also came from imported grain, hay, by-products and silage. All other sources delivered very little P, K and S imports on these organic dairy farms, with the exception of some custom made compost blends and bedding. The sometimes high amount of Ca and Mg imported onto the organic farms also reflected the use of soil ameliorants such as lime.

**Table 17. Sources of nutrient imports and exports, ranges and median values per hectare of contact land**

| Conventional Farms | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Nitrogen (kg/ha) | | | Phosphorus (kg/ha) | | | Potassium (kg/ha) | | | Sulphur (kg/ha) | | | Clacium (kg/ha) | | | Magnesium (kg/ha) | | |
| **Imports** | **Min** | **Max** | **Median** | **Min** | **Max** | **Median** | **Min** | **Max** | **Median** | **Min** | **Max** | **Median** | **Min** | **Max** | **Median** | **Min** | **Max** | **Median** |
| Silage | 0.0 | 103.0 | 7.6 | 0.0 | 16.1 | 1.2 | 0.0 | 143.6 | 5.9 | 0.0 | 9.6 | 0.6 | 0.0 | 11.3 | 1.2 | 0.0 | 11.8 | 0.4 |
| Hay | 0.0 | 222.9 | 11.2 | 0.0 | 25.6 | 1.7 | 0.0 | 219.7 | 11.8 | 0.0 | 24.1 | 1.2 | 0.0 | 39.8 | 3.4 | -2.5 | 22.0 | 1.2 |
| Concentrates/ grain | 2.0 | 222.0 | 52.5 | 0.2 | 35.8 | 9.2 | 0.5 | 38.7 | 13.4 | 0.1 | 14.8 | 4.3 | 0.3 | 86.8 | 13.3 | 0.2 | 30.5 | 7.0 |
| Feed minerals | 0.0 | 6.0 | 0.0 | 0.0 | 1.6 | 0.0 | 0.0 | 2.3 | 0.0 | 0.0 | 1.5 | 0.0 | 0.0 | 6.3 | 0.0 | 0.0 | 5.0 | 0.0 |
| Byproducts | 0.0 | 214.2 | 0.0 | 0.0 | 43.3 | 0.0 | 0.0 | 224.0 | 0.0 | 0.0 | 48.3 | 0.0 | 0.0 | 29.7 | 0.0 | 0.0 | 14.9 | 0.0 |
| NPKS Blend | 0.0 | 321.6 | 10.0 | 0.0 | 81.8 | 13.0 | 0.0 | 176.9 | 31.5 | 0.0 | 84.7 | 14.8 | 0.0 | 307.6 | 15.7 | 0.0 | 2.0 | 0.0 |
| Urea/ Liquid N | 0.0 | 302.7 | 80.7 | 0.0 | 0.0 | 0.0 | 0.0 | 5.2 | 0.0 | 0.0 | 1.9 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| Other fertiliser | 0.0 | 65.4 | 0.0 | 0.0 | 23.6 | 0.0 | 0.0 | 24.0 | 0.0 | 0.0 | 6.2 | 0.0 | 0.0 | 32.1 | 0.0 | 0.0 | 6.6 | 0.0 |
| Soil ameliorant | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.6 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 1311.0 | 0.0 | 0.0 | 178.0 | 0.0 |
| Animal | 0.0 | 61.9 | 4.6 | 0.0 | 15.9 | 1.2 | 0.0 | 4.4 | 0.3 | 0.0 | 17.7 | 1.3 | 0.0 | 44.2 | 3.3 | 0.0 | 11.0 | 0.8 |
| Irrigation | 0.0 | 27.3 | 0.4 | 0.0 | 6.4 | 0.0 | 0.0 | 72.7 | 3.0 | 0.0 | 109.1 | 1.5 | 0.0 | 711.0 | 9.0 | 0.0 | 180.0 | 6.5 |
| N Fixation | 0.0 | 284.8 | 16.6 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| Atm deposition | 1.0 | 4.0 | 1.0 | 0.1 | 0.4 | 0.1 | 1.0 | 4.0 | 1.0 | 1.0 | 4.0 | 1.0 | 2.0 | 8.0 | 2.0 | 2.0 | 8.0 | 2.0 |
| Other | 0.0 | 1.8 | 0.0 | 0.0 | 0.2 | 0.0 | 0.0 | 3.1 | 0.0 | 0.0 | 0.2 | 0.0 | 0.0 | 0.7 | 0.0 | 0.0 | 0.3 | 0.0 |
| **Exports** | | | | | | | | | | | | | | | | | | |
| Milk | 17.3 | 190.4 | 58.3 | 2.6 | 35.4 | 10.2 | 3.8 | 50.7 | 16.6 | 1.1 | 12.4 | 4.4 | 3.6 | 38.9 | 12.3 | 0.3 | 3.8 | 1.2 |
| Animal | 0.2 | 78.1 | 11.8 | 0.1 | 20.1 | 3.0 | 0.0 | 5.6 | 0.8 | 0.1 | 22.3 | 3.4 | 0.2 | 55.8 | 8.5 | 0.0 | 13.9 | 2.1 |
| Forages | 0.0 | 11.0 | 0.0 | 0.0 | 1.7 | 0.0 | 0.0 | 11.6 | 0.0 | 0.0 | 1.1 | 0.0 | 0.0 | 3.0 | 0.0 | 0.0 | 1.3 | 0.0 |
| Manure | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |

| Organic Farms | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Nitrogen (kg/ha) | | | Phosphorus (kg/ha) | | | Potassium (kg/ha) | | | Sulphur (kg/ha) | | | Calcium (kg/ha) | | | Magnesium (kg/ha) | | |
| **Imports** | **Min** | **Max** | **Median** | **Min** | **Max** | **Median** | **Min** | **Max** | **Median** | **Min** | **Max** | **Median** | **Min** | **Max** | **Median** | **Min** | **Max** | **Median** |
| Silage | 0.0 | 9.8 | 3.1 | 0.0 | 1.3 | 0.6 | 0.0 | 9.2 | 2.2 | 0.0 | 0.8 | 0.3 | 0.0 | 2.9 | 0.6 | 0.0 | 1.3 | 0.4 |
| Hay | 1.0 | 27.9 | 23.0 | 0.1 | 3.7 | 2.4 | 1.1 | 35.7 | 13.6 | 0.2 | 2.6 | 1.9 | 0.4 | 11.1 | 4.5 | 0.1 | 3.1 | 2.8 |
| Concentrates/ grain | 6.6 | 26.0 | 14.2 | 1.4 | 4.6 | 2.8 | 1.6 | 6.5 | 4.6 | 0.5 | 1.5 | 1.1 | 0.4 | 2.5 | 0.9 | 0.6 | 1.8 | 1.1 |
| Feed minerals | 0.0 | 0.0 | 0.0 | 0.0 | 0.5 | 0.1 | 0.0 | 0.1 | 0.0 | 0.0 | 0.8 | 0.2 | 0.0 | 2.6 | 0.5 | 0.0 | 0.3 | 0.1 |
| Byproducts | 0.0 | 12.9 | 0.9 | 0.0 | 3.2 | 0.3 | 0.0 | 14.0 | 3.5 | 0.0 | 1.5 | 0.5 | 0.0 | 3.2 | 0.3 | 0.0 | 1.1 | 0.5 |
| NPKS Blend | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 23.5 | 0.0 | 0.0 | 9.8 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| Urea/ Liquid N | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| Other fertiliser | 0.0 | 2.1 | 0.0 | 1.0 | 66.1 | 53.3 | 0.0 | 7.7 | 0.9 | 0.0 | 9.3 | 2.9 | 0.0 | 323.7 | 64.0 | 0.0 | 0.0 | 0.0 |
| Soil ameliorant | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 374.5 | 147.0 | 0.0 | 36.7 | 0.0 |
| Animal | 0.0 | 6.0 | 1.8 | 0.0 | 1.5 | 0.5 | 0.0 | 0.4 | 0.1 | 0.0 | 1.7 | 0.5 | 0.0 | 4.3 | 1.3 | 0.0 | 1.1 | 0.3 |
| Irrigation | 0.0 | 3.5 | 0.1 | 0.0 | 0.1 | 0.0 | 0.0 | 5.3 | 0.1 | 0.0 | 4.5 | 0.0 | 0.0 | 7.5 | 2.5 | 0.0 | 10.6 | 3.6 |
| N Fixation | 40.3 | 113.6 | 61.2 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| Atm deposition | 1.0 | 5.6 | 1.0 | 0.1 | 0.6 | 0.1 | 1.0 | 5.6 | 1.0 | 1.0 | 5.6 | 1.0 | 2.0 | 11.1 | 2.0 | 2.0 | 11.1 | 2.0 |
| Other | 0.0 | 49.4 | 0.1 | 0.0 | 26.5 | 0.0 | 0.0 | 21.1 | 0.0 | 0.0 | 7.6 | 0.0 | 0.0 | 61.4 | 0.1 | 0.0 | 7.1 | 0.0 |
| **Exports** | | | | | | | | | | | | | | | | | | |
| Milk | 15.7 | 34.9 | 24.9 | 2.5 | 6.5 | 4.5 | 4.4 | 10.4 | 7.2 | 1.1 | 2.5 | 1.8 | 3.0 | 7.8 | 5.3 | 0.3 | 0.7 | 0.5 |
| Animal | 3.6 | 8.2 | 4.4 | 0.9 | 2.1 | 1.1 | 0.3 | 0.6 | 0.3 | 1.0 | 2.4 | 1.3 | 2.5 | 5.9 | 3.2 | 0.6 | 1.5 | 0.8 |
| Forages | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| Manure | 0.0 | 0.2 | 0.0 | 0.0 | 0.3 | 0.0 | 0.0 | 0.6 | 0.0 | 0.0 | 0.1 | 0.0 | 0.0 | 1.0 | 0.0 | 0.0 | 0.7 | 0.0 |

**Nutrient surpluses and use-efficiencies determined amongst the 41 dairy farms**

The comparison of the farm nutrient balances across the 41 dairy farms in this study allows for the determination of minimum, maximum, and median nutrient surpluses and efficiencies for the nutrients assessed in this study. The nutrient surpluses or deficits and nutrient use efficiencies for each nutrient are presented in Appendix 6. While whole-farm nitrogen use efficiency is commonly used as an environmental indicator, the environmental and productivity implications of the whole-farm efficiencies for P, K, S, Ca and Mg use should also be considered in light of the overall productivity of the dairy farm, existing nutrient stores available (i.e. soil fertility levels) and potential rate of decline.

**Nitrogen** Whole-farm nitrogen surplus across the 41 independent dairy farms ranged from 47 to 600 kg N/ha and followed a normal distribution (Appendix 6). Half the farms had nitrogen surplus values between 150 and 300 kg N/ha. Fourteen farms had nitrogen surplus values less than 150 kg N/ha, while five farms had nitrogen surplus values above 300 kg N/ha. The median N surplus was 193 kg N/ha. The overall nitrogen use efficiency across the 41 dairy farms ranged from 14 to 50%, with a median value of 25%.

**Phosphorus** Whole-farm phosphorus surplus across the 41 independent dairy farms ranged from -7 to 133 kg P/ha (Appendix 6). The median P surplus was 26 kg P/ha. Five dairy farms were in net deficit, with P balances < 0 kg P/ha, while in contrast 8 farms had P surplus estimates > 50 kg P/ha. The overall phosphorus use efficiency across the 41 dairy farms ranged from 6 to 158%, with a median value of 32%. The negative P balances and efficiency measures greater than 100%, indicate greater removal of P than importation (in any form), and subsequently a 'mining' of P from the system, presumably the soil. In contrast, P surpluses are likely to be increasing soil P content, though this will depend on the spatial distribution of deposited P and loading rates.

**Potassium** All the 41 dairy farms assessed had a potassium surplus, ranging from 13 to 452 kg K/ha (Appendix 6). Eleven of the farms had potassium surplus values of less than 40 kg K/ha and 14 farms had potassium surplus values greater than 100 kg K/ha. The median K surplus for the 41 farms was 74 kg K/ha. The potassium use efficiency across the 41 dairy farms ranged from 9 to 48%, with a median value of 20%.

**Sulphur** Whole-farm S surplus across the 41 independent dairy farms ranged from -1 to 184 kg S/ha. The median S surplus was 27 kg N/ha (Appendix 6). One farm had a small S deficits, while 16 farms had S surpluses less that 20 kg S/ha. While most farms had S surpluses in the range 20 – 50 kg S/ha, 7 farms had surplus values above 50 kg S/ha. The overall sulphur use efficiency across the 41 dairy farms ranged from 6 to 110%, with a median value of 21%.

**Calcium** Whole-farm Ca surplus across the 41 independent dairy farms ranged from 1 to 1327 kg Ca/ha with a median Ca surplus of 84 kg Ca/ha (Appendix 6). Ten farms had Ca surpluses greater than 200 kg Ca/ha, reflecting specific Ca sources applied to these farms rather than the normal feed and fertiliser. The overall Ca use efficiency across the 41 dairy farms ranged from 1 to 97%, with a median value of 18%.

**Magnesium** Whole-farm Mg surplus across the 41 independent dairy farms ranged from 6 to 222 kg Mg/ha, with a median value of 32 kg Mg/ha (Appendix 6). Five farms had Mg surplus values less than 10 kg Mg/ha, while 4 farms had Mg surplus values above 70 kg Mg/ha. Half of the farms had Mg surpluses in the range 20 – 50 kg Mg/ha. The overall Mg use efficiency across the 41 dairy farms ranged from 2 to 52%, with a median value of 10%.

## Nutrient sources, balances and efficiencies amongst contrasting farms

In order to illustrate differences in (i) sources of nutrients imported and exported, (ii) whole-farm balances and (iii) nutrient use efficiencies, amongst the 41 dairy farms studies, 5 contrasting farms have been selected. These 5 farms include 4 conventional farms and 1 organic farm, from different dairy regions. The conventional farms have been attributed 'low or high feed' and 'low, moderate or high fertiliser' descriptors, based on the measured inputs (Table 18). Summary tables for N, P, K, S, Ca and Mg for each of these farms are presented below.

**Nitrogen.** The major sources of N inputs were different between the 5 selected farms. For example, the main N input for the 'low feed and fertiliser' farm and the organic farm was from N fixation, with inputs of 285 and 65 kg N/ha respectively. In contrast, the 'low feed, high fertiliser' farm imported most of the N from blended fertiliser or urea, with N fixation providing only a small N input of 2 kg N/ha. Most of the N within the 2 'high feed' systems was imported within a combination of grain, by-products, hay and silage. While still important, both fertiliser N (as inorganic blends or urea) and N fixation provided a relatively smaller part of the overall N inputs. Total N inputs ranged between 103 kg N/ha (organic) and 798 kg N/ha (high feed, high fertiliser).

For all of these contrasting systems, shipped milk was the single largest source of total N exported, though culled animals and calf sales also contributed to N removal (Table 18). The total N exports were substantially different between farms and ranged from 20 to 198 kg N/ha, largely reflecting the overall intensity of the different systems.

The overall farm N balances for these selected examples of different dairy systems ranged from a net surplus of 83 kg N/ha for the organic system to 600 kg N/ha for the 'high fertiliser, high feed' system (Table 18). This net surplus reflects the amount of N not captured in production. Because of the transient nature of N in agricultural systems, this N is likely to be lost to surface and ground water and the atmosphere.

The proportion of imported N that ends up in exported products (N use efficiency) also varied between these farms and was not related to the net surplus (Table 18). For example the organic farm had the lowest N surplus of 83 kg N/ha, but the N use efficiency was only 20%, resulting largely from the low milk production per ha and therefore a lower amount of N exported. Similarly, the 'low feed, high fertiliser' system had a net surplus of 413 kg N/ha and a low N use efficiency of 16%, reflecting high N inputs in fertiliser but proportionately low N exports. In contrast, the 'high feed, high fertiliser' system had a net surplus of 600 kg N/ha, and N use efficiency of 25%. The 'high feed, medium fertiliser' system had a net surplus of 351 kg N/ha and the highest N use efficiency of 28%, reflecting a higher proportion of N exported, despite the high N inputs.

**Phosphorus and Sulphur.** The concentrations of P and S are similar in many fertiliser products, feeds, and animals, but S is lower in milk. The major sources of both P and S inputs were therefore reflected in the types of systems examined (Table 19). For example, the main P and S input were from inorganic fertiliser for the 'low feed, low fertiliser' and 'low feed, high fertiliser' conventional farms, and from alternative fertilisers (rock phosphate) on the organic farm. In contrast, the 'high feed, high fertiliser' and 'high feed, medium fertiliser' farms imported a greater proportion of their P and S from a combination of grain, by-products, hay and silage. Total P inputs ranged between 44 kg P/ha (low feed, high fertiliser) and 171 kg P/ha (high feed and fertiliser). Total S inputs ranged between 25 kg S/ha (organic) and 199 kg S/ha (high feed and fertiliser).

Shipped milk was the single largest source of total P and S exported, though culled animals and calf sales also contributed to P and S removal (Table 19). However, relative to total N exports, total P and S exports were substantially less and ranged from only 4 to 37 kg P/ha, and 2 to 18 kg S/ha.

The overall farm P balances for these selected dairy systems ranged from a net surplus of 30 kg P/ha for the 'low feed, high fertiliser' system to 133 kg P/ha for the 'high fertiliser, high feed' system (Table 19). Interestingly, the organic farm had a relatively high P surplus of 53 kg P/ha, reflecting the high input of P in the form of rock phosphate. The P use efficiency also varied between these farms. The highest P use efficiency amongst these selected farms was 32% on the 'low feed, high fertiliser farm' while the lowest was 6% on the organic farm, reflecting the high levels of P input but relatively low P exports.

The overall farm S balances for these dairy systems ranged from a net surplus of 22 kg S/ha for the organic system, to 184 kg S/ha for the 'high fertiliser, high feed' system (Table 20). The S use efficiency also varied between these farms. The highest S use efficiency amongst these selected farms was 26% on the 'high feed, medium fertiliser' system while the lowest was 7% on the 'high fertiliser, high feed' system.

Unlike N, these net surpluses of P and S do not reflect an estimate of environmental losses directly. Phosphorus and to a lesser extent S are generally well held and stored in soils, with only a small proportion usually lost to surface and ground water and the atmosphere. However, ongoing surpluses, an uneven redistribution of P and S within the landscape, and soils with a limited capacity to retain P and S, may lead to significant environmental losses; particularly relevant for P contamination of waterways.

**Potassium.** Potassium fertiliser application rates on these farms were high in relation to P and S, and in a number of cases comparable to N. While forage K concentrations were also generally similar to N (> 3%), they are lower in grains, milk and animals. The largest contribution to imported K was from fertiliser for the 'low feed, low fertiliser', 'low feed, high fertiliser' and organic farm systems (Table 21). In contrast, the 'high feed, high fertiliser' and 'high feed, medium fertiliser' system both had a greater proportion of imported K coming from a combination of grain, by-products, hay or silage. On one farm ('high feed, high fertiliser') irrigation water also contributed a significant amount of K.

As with other nutrients, shipped milk was the single largest source of K exported, with only minor removal of K in animal removals (Table 21). The total K exports were substantially different and ranged from 5 (organic farm) to 51 kg K/ha (high feed, high fertiliser), largely reflecting the overall intensity of the different systems.

The overall farm K balances for these selected dairy systems ranged from a net surplus of 38 kg K/ha for the organic system to 451 kg K/ha for the 'high fertiliser, high feed' system (Table 21). The proportion of imported K that ends up in exported products (K use efficiency) also varied between these farms. The organic farm system had a K use efficiency of 11%, similar to the efficiency of 10% determined from the 'high feed, high fertiliser' farm system, despite their contrasting K surpluses (Table 21). Most of the selected farm systems shown in Table 21 had K use efficiencies of between 9 and 12%, with the exception of the 'low feed, low fertiliser' farm system, which had a K use efficiency of 21%.

**Calcium and Magnesium.** The major sources of Ca and Mg determined on these contrasting farms highlight the substantial contribution from some specific sources, as well as the common imports that result from feed and fertiliser (Table 22 and 23). For example, large amounts of Ca resulted from soil ameliorant imports (lime) and other fertilisers (rock phosphate) on the organic farm, while irrigation water was a very large source of Ca and Mg imports on the 'high feed, high fertiliser' farm.

Additionally, NPKS fertiliser blends also was responsible for significant imports of Ca, while to a lesser extent, Ca and Mg was also imported onto farms in grain, animals, hay, by-products and silage. Total Ca inputs ranged between 54 kg Ca/ha (high feed, medium fertiliser) and 980 kg Ca/ha (high feed and fertiliser). Total Mg inputs ranged between 11 kg Mg/ha (low feed, low fertiliser) and 227 kg Mg/ha (high feed, high fertiliser).

Calcium is largely exported in milk shipped off farm, but there may also be significant export of Ca in removed animals and forage. The Mg concentration in both milk and animals is small and subsequently net exports of Mg are generally low. Total Ca exports ranged between 7 kg Ca/ha (organic) and 44 kg Ca/ha (high feed, high fertiliser). Total Mg exports ranged between 1 kg Mg/ha (organic) and 7 kg Mg/ha (high feed, medium fertiliser) (Table 23).

The overall farm Ca balances for these selected dairy systems ranged from a net surplus of 12 kg Ca/ha for the 'high feed, medium fertiliser' system to 936 kg Ca/ha for the 'high fertiliser, high feed' system (Table 22). The Mg farm balances ranged from a net surplus of 7 kg Mg/ha for the 'low feed, low fertiliser' system to 222 kg Mg/ha for the 'high fertiliser, high feed' system (Table 23).

The large amounts of imported Ca and Mg specific to some farms, as a result of soil ameliorants or high cation concentrations in irrigation water, resulted in a wide range of Ca and Mg use efficiencies. For example the organic system had a Ca surplus of 510 kg Ca/ha and a Ca use efficiency of only 1%; largely resulting from the high Ca imported and small Ca exported (Table 22).

In contrast, the 'high feed, medium fertiliser' farm has the smallest total Ca import (54 kg Ca/ha), and relatively high Ca exported (42 kg Ca/ha), resulting in a Ca use efficiency of 78%. The Mg use efficiencies were more consistent, ranging from 2% for the 'high feed, high fertiliser' system (large amounts of Mg in irrigation water) to 38% for the 'low feed, low fertiliser' system, with low Mg inputs and exports (Table 23).

## Table 18. Comparison of whole-farm nitrogen balances for 5 contrasting farms

| Farm and system type | Farm #4 | Farm #11 | Farm #32 | Farm #36 | Farm #41 |
|---|---|---|---|---|---|
| | Conventional | Conventional | Conventional | Conventional | Organic |
| | LFeed/LFert | LFeed/HFert | HFeed/MFert | HFeed/HFert | LFeed/LFert |
| Imported feed (%) | 16 | 22 | 61 | 61 | 32 |
| | | | **Nitrogen** | | |
| **Imports** | (kg/ha) | (kg/ha) | (kg/ha) | (kg/ha) | (kg/ha) |
| Silage | 14 | 8 | 103 | 17 | 0 |
| Hay | 11 | 21 | 56 | 89 | 21 |
| Concentrates/grains | 41 | 31 | 84 | 222 | 10 |
| Byproducts | 3 | 0 | 125 | 214 | 0 |
| Feed minerals | 0 | 0 | 0 | 2 | 0 |
| NPKS blends | 8 | 322 | 28 | 2 | 0 |
| Urea/Liquid N | 0 | 102 | 39 | 147 | 0 |
| Other fertilisers | 0 | 0 | 0 | 0 | 0 |
| Soil ameliorants | 0 | 0 | 0 | 0 | 0 |
| Animals | 5 | 5 | 0 | 5 | 1 |
| Irrigation | 1 | 0 | 1 | 5 | 0 |
| N fixation | 285 | 2 | 49 | 92 | 65 |
| Atmospheric deposition | 1 | 1 | 1 | 4 | 6 |
| Other | 0 | 0 | 1 | 0 | 0 |
| **Total Imported** | **369** | **492** | **487** | **798** | **103** |
| | | | | | |
| **Exports** | | | | | |
| Milk | 77 | 72 | 108 | 190 | 16 |
| Animals | 16 | 7 | 28 | 7 | 4 |
| Forages | 0 | 0 | 0 | 0 | 0 |
| Manure | 0 | 0 | 0 | 0 | 0 |
| **Total Exported** | **93** | **78** | **136** | **198** | **20** |
| | | | | | |
| **BALANCE (kg/ha)** | **276** | **413** | **351** | **600** | **83** |
| | | | | | |
| **Efficiency (%)** | **25** | **16** | **28** | **25** | **20** |

HFert = High Fertiliser Input
MFert = Medium Fertiliser Input
LFert = Low Fertiliser Input
HFeed = High Feed Input
LFeed = Low Feed Input

## Table 19. Comparison of whole-farm phosphorus balances for 5 contrasting farms

| Farm and system type | Farm #4 | Farm #11 | Farm #32 | Farm #36 | Farm #41 |
|---|---|---|---|---|---|
| | Conventional | Conventional | Conventional | Conventional | Organic |
| | LFeed/LFert | LFeed/HFert | HFeed/MFert | HFeed/HFert | LFeed/LFert |
| Imported feed (%) | 16 | 22 | 61 | 61 | 32 |
| | | | **Phosphorus** | | |
| **Imports** | (kg/ha) | (kg/ha) | (kg/ha) | (kg/ha) | (kg/ha) |
| Silage | 3 | 1 | 16 | 3 | 0 |
| Hay | 2 | 2 | 8 | 11 | 2 |
| Concentrates/grains | 3 | 8 | 16 | 27 | 2 |
| Byproducts | 0 | 0 | 16 | 43 | 0 |
| Feed minerals | 0 | 0 | 1 | 2 | 0 |
| NPKS blends | 57 | 32 | 31 | 82 | 0 |
| Urea/Liquid N | 0 | 0 | 0 | 0 | 0 |
| Other fertilisers | 0 | 0 | 0 | 0 | 52 |
| Soil ameliorants | 0 | 0 | 0 | 0 | 0 |
| Animals | 1 | 1 | 0 | 1 | 0 |
| Irrigation | 0 | 0 | 0 | 1 | 0 |
| N fixation | 0 | 0 | 0 | 0 | 0 |
| Atmospheric deposition | 0 | 0 | 0 | 0 | 1 |
| Other | 0 | 0 | 0 | 0 | 0 |
| **Total Imported** | **66** | **44** | **89** | **171** | **57** |
| | | | | | |
| **Exports** | | | | | |
| Milk | 14 | 12 | 19 | 35 | 2 |
| Animals | 4 | 2 | 7 | 2 | 1 |
| Forages | 0 | 0 | 0 | 0 | 0 |
| Manure | 0 | 0 | 0 | 0 | 0 |
| Total Exported | **18** | **14** | **26** | **37** | **4** |
| | | | | | |
| **BALANCE** | **48** | **30** | **63** | **133** | **53** |
| | | | | | |
| Efficiency (%) | **27** | **32** | **29** | **22** | **6** |

HFert = High Fertiliser Input
MFert = Medium Fertiliser Input
LFert = Low Fertiliser Input
HFeed = High Feed Input
LFeed = Low Feed Input

## Table 20. Comparison of whole-farm sulphur balances for 5 contrasting farms

| **Farm and system type** | Farm #4 | Farm #11 | Farm #32 | Farm #36 | Farm #41 |
|---|---|---|---|---|---|
| | Conventional | Conventional | Conventional | Conventional | Organic |
| | LFeed/LFert | LFeed/HFert | HFeed/MFert | HFeed/HFert | LFeed/LFert |
| Imported feed (%) | 16 | 22 | 61 | 61 | 32 |
| | | | **Sulphur** | | |
| **Imports** | (kg/ha) | (kg/ha) | (kg/ha) | (kg/ha) | (kg/ha) |
| Silage | 2 | 1 | 10 | 1 | 0 |
| Hay | 1 | 2 | 5 | 9 | 3 |
| Concentrates/grains | 4 | 3 | 6 | 13 | 1 |
| Byproducts | 0 | 0 | 15 | 48 | 0 |
| Feed minerals | 0 | 0 | 0 | 2 | 0 |
| NPKS blends | 62 | 39 | 3 | 85 | 10 |
| Urea/Liquid N | 0 | 0 | 0 | 0 | 0 |
| Other fertilisers | 0 | 0 | 0 | 0 | 6 |
| Soil ameliorants | 0 | 0 | 0 | 0 | 0 |
| Animals | 1 | 1 | 0 | 1 | 0 |
| Irrigation | 2 | 0 | 30 | 36 | 0 |
| N fixation | 0 | 0 | 0 | 0 | 0 |
| Atmospheric deposition | 1 | 1 | 1 | 4 | 6 |
| Other | 0 | 0 | 0 | 0 | 0 |
| **Total Imported** | **73** | **47** | **69** | **199** | **25** |
| | | | | | |
| **Exports** | | | | | |
| Milk | 7 | 5 | 10 | 12 | 1 |
| Animals | 5 | 2 | 8 | 2 | 1 |
| Forages | 0 | 0 | 0 | 0 | 0 |
| Manure | 0 | 0 | 0 | 0 | 0 |
| Total Exported | **12** | **7** | **18** | **15** | **2** |
| **BALANCE** | **61** | **40** | **51** | **184** | **22** |
| **Efficiency (%)** | **17** | **15** | **26** | **7** | **10** |

HFert = High Fertiliser Input
MFert = Medium Fertiliser Input
LFert = Low Fertiliser Input
HFeed = High Feed Input
LFeed = Low Feed Input

## Table 21. Comparison of whole-farm potassium balances for 5 contrasting farms

| **Farm and system type** | Farm #4 | Farm #11 | Farm #32 | Farm #36 | Farm #41 |
|---|---|---|---|---|---|
| | Conventional | Conventional | Conventional | Conventional | Organic |
| | LFeed/LFert | LFeed/HFert | HFeed/MFert | HFeed/HFert | LFeed/LFert |
| Imported feed (%) | 16 | 22 | 61 | 61 | 32 |
| | | | **Potassium** | | |
| **Imports** | (kg/ha) | (kg/ha) | (kg/ha) | (kg/ha) | (kg/ha) |
| Silage | 15 | 6 | 144 | 16 | 0 |
| Hay | 12 | 18 | 77 | 59 | 11 |
| Concentrates/grains | 8 | 9 | 22 | 39 | 3 |
| Byproducts | 0 | 0 | 17 | 224 | 0 |
| Feed minerals | 0 | 0 | 0 | 0 | 0 |
| NPKS blends | 73 | 177 | 0 | 141 | 24 |
| Urea/Liquid N | 0 | 0 | 0 | 0 | 0 |
| Other fertilisers | 0 | 0 | 0 | 0 | 0 |
| Soil ameliorants | 0 | 0 | 0 | 0 | 0 |
| Animals | 0 | 0 | 0 | 0 | 0 |
| Irrigation | 3 | 0 | 6 | 20 | 0 |
| N fixation | 0 | 0 | 0 | 0 | 0 |
| Atmospheric deposition | 1 | 1 | 1 | 4 | 6 |
| Other | 0 | 0 | 0 | 0 | 0 |
| **Total Imported** | **113** | **211** | **267** | **502** | **43** |
| | | | | | |
| **Exports** | | | | | |
| Milk | 21 | 19 | 29 | 51 | 4 |
| Animals | 1 | 0 | 2 | 1 | 0 |
| Forages | 1 | 0 | 0 | 0 | 0 |
| Manure | 0 | 0 | 0 | 0 | 0 |
| Total Exported | **23** | **20** | **31** | **51** | **5** |
| **BALANCE** | **89** | **191** | **236** | **451** | **38** |
| **Efficiency (%)** | **21** | **9** | **12** | **10** | **11** |

HFert = High Fertiliser Input
MFert = Medium Fertiliser Input
LFert = Low Fertiliser Input
HFeed = High Feed Input
LFeed = Low Feed Input

## Table 22. Comparison of whole-farm calcium balances for 5 contrasting farms

| **Farm and system type** | Farm #4 | Farm #11 | Farm #32 | Farm #36 | Farm #41 |
|---|---|---|---|---|---|
| | Conventional | Conventional | Conventional | Conventional | Organic |
| | LFeed/LFert | LFeed/HFert | HFeed/MFert | HFeed/HFert | LFeed/LFert |
| Imported feed (%) | 16 | 22 | 61 | 61 | 32 |
| **Imports** | (kg/ha) | (kg/ha) | **Calcium** (kg/ha) | (kg/ha) | (kg/ha) |
| Silage | 3 | 1 | 11 | 7 | 0 |
| Hay | 3 | 4 | 7 | 32 | 2 |
| Concentrates/grains | 10 | 16 | 2 | 13 | 0 |
| Byproducts | 0 | 0 | 8 | 30 | 0 |
| Feed minerals | 0 | 0 | 6 | 6 | 0 |
| NPKS blends | 141 | 72 | 1 | 172 | 0 |
| Urea/Liquid N | 0 | 0 | 0 | 0 | 0 |
| Other fertilisers | 0 | 0 | 0 | 0 | 128 |
| Soil ameliorants | 0 | 0 | 0 | 0 | 375 |
| Animals | 4 | 3 | 0 | 3 | 1 |
| Irrigation | 5 | 0 | 17 | 710 | 0 |
| N fixation | 0 | 0 | 0 | 0 | 0 |
| Atmospheric deposition | 2 | 2 | 2 | 8 | 11 |
| Other | 0 | 0 | 1 | 0 | 0 |
| **Total Imported** | **168** | **99** | **54** | **980** | **517** |
| | | | | | |
| **Exports** | | | | | |
| Milk | 16 | 16 | 22 | 39 | 3 |
| Animals | 11 | 5 | 20 | 5 | 3 |
| Forages | 0 | 0 | 0 | 0 | 0 |
| Manure | 0 | 0 | 0 | 0 | 0 |
| Total Exported | **27** | **20** | **42** | **44** | **7** |
| | | | | | |
| **BALANCE** | **141** | **78** | **12** | **936** | **510** |
| | | | | | |
| **Efficiency (%)** | **16** | **21** | **78** | **5** | **1** |

HFert = High Fertiliser Input
MFert = Medium Fertiliser Input
LFert = Low Fertiliser Input
HFeed = High Feed Input
LFeed = Low Feed Input

## Table 23. Comparison of whole-farm magnesium balances for 5 contrasting farms

| **Farm and system type** | Farm #4 | Farm #11 | Farm #32 | Farm #36 | Farm #41 |
|---|---|---|---|---|---|
| | Conventional | Conventional | Conventional | Conventional | Organic |
| | LFeed/LFert | LFeed/HFert | HFeed/MFert | HFeed/HFert | LFeed/LFert |
| Imported feed (%) | 16 | 22 | 61 | 61 | 32 |
| **Imports** | (kg/ha) | (kg/ha) | **Magnesium** (kg/ha) | (kg/ha) | (kg/ha) |
| Silage | 2 | 1 | 12 | 2 | 0 |
| Hay | 1 | 3 | 5 | 8 | 3 |
| Concentrates/grains | 3 | 6 | 5 | 11 | 1 |
| Byproducts | 0 | 0 | 7 | 15 | 0 |
| Feed minerals | 0 | 0 | 3 | 2 | 0 |
| NPKS blends | 0 | 0 | 1 | 0 | 0 |
| Urea/Liquid N | 0 | 0 | 0 | 0 | 0 |
| Other fertilisers | 0 | 0 | 0 | 0 | 0 |
| Soil ameliorants | 0 | 0 | 0 | 0 | 0 |
| Animals | 1 | 1 | 0 | 1 | 0 |
| Irrigation | 3 | 0 | 26 | 180 | 0 |
| N fixation | 0 | 0 | 0 | 0 | 0 |
| Atmospheric deposition | 2 | 2 | 2 | 8 | 11 |
| Other | 0 | 0 | 0 | 0 | 0 |
| **Total Imported** | **11** | **12** | **60** | **227** | **15** |
| | | | | | |
| **Exports** | | | | | |
| Milk | 1 | 1 | 2 | 4 | 0 |
| Animals | 3 | 1 | 5 | 1 | 1 |
| Forages | 0 | 0 | 0 | 0 | 0 |
| Manure | 0 | 0 | 0 | 0 | 0 |
| Total Exported | **4** | **3** | **7** | **5** | **1** |
| | | | | | |
| **BALANCE** | **7** | **10** | **53** | **222** | **14** |
| | | | | | |
| **Efficiency (%)** | **38** | **22** | **12** | **2** | **7** |

HFert = High Fertiliser Input
MFert = Medium Fertiliser Input
LFert = Low Fertiliser Input
HFeed = High Feed Input
LFeed = Low Feed Input

## Importance of defining the dairy land-base when determining nutrient balances

The whole-farm balances previously presented demonstrate the usefulness of identifying key sources of nutrients, farm-scale nutrient flows and surpluses/deficits from readily available data that are likely to be fairly accurate. However the diversity of dairy systems across Australia, relating to the intensity of inputs and productivity, and the size of each operation, necessitates the standardization of nutrient balance information if comparisons between farms are required. This is usually done by assessing the nutrient flows and balances on a 'per ha' basis; i.e. dividing the nutrient quantities (usually kilograms) by the number of ha contributing to the production system under study.

However, defining 'what land area contributes to the production system?' is not a straight forward task. Australian dairy production is increasingly reliant on separate land units which may directly contribute to milk production or alternatively may serve a specialist purpose. These areas may include runoff blocks, dryland blocks, and separate cropping land, and may be owned, leased or rented. Non lactating animals may also be agisted or 'parked' on other farms.

Nutrient transfers between these other land areas and the 'core' milking area may include:

- Silage and hay harvested and removed from other blocks and fed on the core milking area,
- Dry cows, heifers and calves moving on and off the core milking area
- Fertiliser delivered to the core milking area but then applied to other areas
- Feed delivered to the core milking area but fed to animals in other areas
- Manure collected (i.e. scrapings from feed pads, dairy effluent) and transferred to other areas

While a farmer is likely to know the area of land owned or leased, it is more difficult to accurately determine the land-base that directly contributes to milk production. There are likely to be a number of land areas within the 'home-farm' that may not directly contribute to milk production. These areas include fenced riparian areas, wetlands, bull paddocks, feed storage areas, houses, farm buildings and dams, and other animal grazing areas (horses, beef cows, etc).

Moreover, the intensity of use of land also creates challenges when defining the land-base used to determine nutrient balances. For example, a large but rarely used back paddock which only gets grazed once is still contributing to overall milk production; while in contrast, paddocks close to the dairy provide the major sources of pasture throughout the year.

The detailed spatial analysis undertaken to accurately quantify the various land-bases contributing to dairy production, and the understanding of the function of these other areas, was an important part of standardising and quantifying nutrient balance information in this study. Nutrient transfers such as those described above were identified through the questionnaires and the records of imports and exports of various nutrient sources.

The following land-bases were therefore defined and determined for each participating dairy farm:

- **Total Land** - All land area under management which contributes to dairy operation. Includes areas described as lease blocks, dry blocks and out blocks.
- **Dairy Farm Land** - Overall land within the 'Home Farm' boundary. Includes some non-productive areas such as wet lands, riparian areas, bushland, feed storage, shedding and housing.
- **Contact Land** - Total land area contacted by lactating and dry cows within the home farm. Includes grazed paddocks, laneways, feeding areas such as feedpads and sacrifice paddocks, calving areas, sick areas.

The importance of a well defined land-base is illustrated in Table 24, where nitrogen sources, balances and efficiencies are presented from three dairy farms and 3 different land areas, based on the previous definitions. Farm 15, is a relatively small farm with a minor difference between the contact land and

home farm land, but which also uses a large runoff block for dry cows. Farm 36 is a large farm, with only around half of the home farm area directly used for dairy production and there is no other land under management. Farm 43 has a large home farm area, of which about 75% is contact land, but also has a large outblock used for dry stock and cropping.

These farms vary in the amount of N imported and exported and the overall nitrogen balance, on a kg N basis. The whole-farm N use efficiencies for these farms, which ranged from 23 - 37%, are not influenced by land area (Table 24).

Not surprisingly, as the land area considered increases on each dairy farm, the net nitrogen surplus on a per ha basis decreases (Table 24). For example, the N surplus for farm 15 declined marginally from 181 to 172 kg/ha when the contact-land hectares or the home-farm hectares were used. However, the inclusion of the outblock area resulted in an N surplus of 125 kg N/ha. Similarly farms 36 and 43 have markedly different N surplus values, depending on which land-base was used in the denominator (Table 24).

The contact-land area definition above was selected to compare and contrast the 41 dairy farms within the current study. This land-base best reflected the intensity of the existing dairy system, and described the land area where the greatest deposition, removal, recycling and losses of nutrients was likely to occur. This assessment does however highlight the difficulties in comparing nutrient balance information from different studies (i.e. kg/ha), particularly when the land-base used is different or poorly defined.

## Using 'book values' rather than individual farm-defined nutrient concentrations

In the current study, detailed sample collection and subsequent laboratory analysis of feeds, milk and organic fertilisers from each of the 44 study farms, was undertaken to determine the most accurate estimates of nutrient flows and balances. However the overall integration of whole-farm nutrient flows and the assessment of nutrient balance may reduce the need for such measurement precision when the approach is applied more broadly.

To assess differences in whole-farm nutrient imports, exports, balances and efficiencies, comparisons were made on 5 contrasting dairy farms where the specific farm nutrient concentrations or generic (book value) nutrient concentrations were used (Table 25). The book values used were mean and standard deviation values calculated from previously existing Australian data, and pooled with information determined during the current study.

The impact of using book values rather the determined farm-based nutrient concentrations varied between farms and the sources identified (Table 25). For example, there was a 7 kg N/ha reduction in N imported grain on Farm 4, and a small increase in N imported in hay, resulting in a net change in N balance of only 5 kg N/ha overall. In contrast, there was a large reduction in N imported in silage on Farm 32, and an increase in N imported in hay and grain, which resulted in a net reduction of 36 kg N/ha in the N balance overall.

Not surprisingly, the magnitude of these differences was largely driven by the reliance of imported feed onto each farm, as other key nutrient sources such as fertiliser imports were determined from published nutrient concentrations on both sides of the equation. There were negligible changes in nutrient exports.

The difference in estimated N surplus, when using farm-based or the book value estimates of nutrient concentrations, were in most cases small, both in absolute and relative terms. For example the relative differences in whole-farm N surplus between the farm-derived nutrient concentrations and generic nutrient concentrations were -1.8%, -0.8%, +6.7% and – 3.6%, for farms 4, 11, 36 and 41 respectively. However, the relative difference in whole-farm N surplus on farm 32 was -10.3%, reflecting a significant decrease in N import principally from silage when a book value rather than the actual farm derived data was used. There were also only minor changes in the N use efficiency, with estimates increasing or decreasing by a maximum of 1%.

**Table 24. Impact of different land bases on whole farm N balance determination**

| | Farm #15 | | | | Farm #33 | | | | Farm #36 | | | | Farm #43 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Imports** | **Total** | **Contact Land** | **Home Farm** | **Total Land** | **Total** | **Contact Land** | **Home Farm** | **Total Land** | **Total** | **Contact Land** | **Home Farm** | **Total Land** | **Total** | **Contact Land** | **Home Farm** | **Total Land** |
| | | 93 ha | 98 ha | 137 ha | | 102 ha | 108 ha | 108 ha | | | | | | 275 ha | 372 ha | 669 ha |
| | | **Nitrogen (kg/ha)** | | | | **Nitrogen (kg/ha)** | | | | **Nitrogen (kg/ha)** | | | | **Nitrogen (kg/ha)** | | |
| Silage | 788 | 8 | 8 | 6 | 0 | 0 | 0 | 0 | 5201 | 17 | 8 | 8 | 0 | 0 | 0 | 0 |
| Hay | 540 | 6 | 6 | 4 | 9714 | 95 | 90 | 90 | 27238 | 89 | 45 | 45 | 3849 | 14 | 10 | 6 |
| Concentrates/grains | 4155 | 44 | 42 | 30 | 4652 | 46 | 43 | 43 | 68162 | 222 | 111 | 111 | 16251 | 59 | 44 | 24 |
| Feed minerals | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 677 | 2 | 1 | 1 | 0 | 0 | 0 | 0 |
| Byproducts | 0 | 0 | 0 | 0 | 1915 | 19 | 18 | 18 | 65765 | 214 | 107 | 107 | 0 | 0 | 0 | 0 |
| NPKS blends | 4050 | 43 | 41 | 30 | 0 | 0 | 0 | 0 | 476 | 2 | 1 | 1 | 12500 | 46 | 34 | 19 |
| Urea/Liquid N | 15088 | 162 | 154 | 110 | 2723 | 27 | 25 | 25 | 45156 | 147 | 74 | 74 | 50600 | 185 | 136 | 76 |
| Other fertilisers | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Soil ameliorants | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Animals | 938 | 10 | 10 | 7 | 407 | 4 | 4 | 4 | 1462 | 5 | 2 | 2 | 2800 | 10 | 8 | 4 |
| Irrigation | 31 | 0 | 0 | 0 | 123 | 1 | 1 | 1 | 1437 | 5 | 2 | 2 | 58 | 0 | 0 | 0 |
| N fixation | 1110 | 12 | 11 | 8 | 6414 | 63 | 59 | 59 | 28356 | 92 | 46 | 46 | 21525 | 79 | 58 | 32 |
| Atmospheric deposition | 93 | 1 | 1 | 1 | 102 | 1 | 1 | 1 | 1229 | 4 | 2 | 2 | 1095 | 4 | 3 | 2 |
| Other | 10 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | | | | | | | | | | | | | | | | |
| **Total Imported** | **26803** | **287** | **273** | **196** | **26049** | **256** | **241** | **241** | **245160** | **798** | **401** | **401** | **108678** | **397** | **292** | **162** |
| | | | | | | | | | | | | | | | | |
| **Exports** | | | | | | | | | | | 0 | 0 | | | | |
| Milk | 6096 | 65 | 62 | 44 | 5717 | 56 | 53 | 53 | 58446 | 190 | 95 | 95 | 18944 | 69 | 51 | 28 |
| Animals | 3309 | 35 | 34 | 24 | 350 | 3 | 3 | 3 | 2301 | 7 | 4 | 4 | 5695 | 21 | 15 | 9 |
| Forages | 503 | 5 | 5 | 4 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Manure | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | | | | | | | | | | | | | | | | |
| **Total exported** | **9908** | **106** | **101** | **72** | **6067** | **60** | **56** | **56** | **60747** | **198** | **99** | **99** | **24640** | **90** | **66** | **37** |
| | | | | | | | | | | | | | | | | |
| **BALANCE** | **16895** | **181** | **172** | **123** | **19982** | **196** | **185** | **185** | **184413** | **600** | **301** | **301** | **84039** | **307** | **226** | **126** |
| | | | | | | | | | | | | | | | | |
| **Efficiency (%)** | **37%** | **37%** | **37%** | **37%** | **23%** | **23%** | **23%** | **23%** | **25%** | **25%** | **25%** | **25%** | **23%** | **23%** | **23%** | **23%** |

**Table 25. Whole-farm nitrogen imports, exports, balances and efficiencies when individual farm or generic nitrogen concentrations are used.**

| Farm and system type | Farm #4 | Farm #4 | Farm #11 | Farm #11 | Farm #32 | Farm #32 | Farm #36 | Farm #36 | Farm #41 | Farm #41 |
|---|---|---|---|---|---|---|---|---|---|---|
| | Conventional | Conventional | Conventional | Conventional | Conventional | Conventional | Conventional | Conventional | Organic | Organic |
| | LFeed/LFert | LFeed/LFert | HFeed/HFert | HFeed/HFert | HFeed/MFert | HFeed/MFert | HFeed/HFert | HFeed/HFert | LFeed/LFert | LFeed/LFert |
| | Farm analysis | Grand means | Farm analysis | Grand means | Farm analysis | Grand means | Farm analysis | Grand means | Farm analysis | Grand means |
| Imported feed (%) | 16 | 16 | 22 | 22 | 61 | 61 | 61 | 61 | 32 | 32 |
| | | | | | **Nitrogen** | | | | | |
| **Imports** | (kg/ha) | (kg/ha) | (kg/ha) | (kg/ha) | (kg/ha) | (kg/ha) | (kg/ha) | (kg/ha) | (kg/ha) | (kg/ha) |
| Silage | 14 | 13 | 8 | 8 | 103 | 47 | 17 | 16 | 0 | 0 |
| Hay | 11 | 14 | 21 | 17 | 56 | 63 | 89 | 89 | 21 | 20 |
| Concentrates/grains | 41 | 34 | 31 | 32 | 84 | 96 | 222 | 246 | 10 | 10 |
| Byproducts | 3 | 2 | 0 | 0 | 125 | 120 | 214 | 209 | 0 | 0 |
| Feed minerals | 0 | 0 | 0 | 0 | 0 | 1 | 2 | 8 | 0 | 0 |
| NPKS blends | 8 | 8 | 322 | 322 | 28 | 28 | 2 | 2 | 0 | 0 |
| Urea/Liquid N | 0 | 0 | 102 | 102 | 39 | 39 | 147 | 147 | 0 | 0 |
| Other fertilisers | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Soil ameliorants | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Animals | 5 | 5 | 5 | 5 | 0 | 0 | 5 | 5 | 1 | 1 |
| Irrigation | 1 | 4 | 0 | 0 | 1 | 1 | 5 | 8 | 0 | 0 |
| N fixation | 285 | 284 | 2 | 2 | 49 | 50 | 92 | 105 | 65 | 65 |
| Atmospheric deposition | 1 | 1 | 1 | 1 | 1 | 1 | 4 | 4 | 6 | 6 |
| Other | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 |
| **Total Imported** | **369** | **363** | **492** | **488** | **487** | **446** | **798** | **838** | **103** | **101** |
| | | | | | | | | | | |
| **Exports** | | | | | | | | | | |
| Milk | 77 | 76 | 72 | 71 | 108 | 104 | 190 | 191 | 16 | 16 |
| Animals | 16 | 16 | 7 | 7 | 28 | 28 | 7 | 7 | 4 | 4 |
| Forages | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Manure | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| **Total Exported** | **93** | **92** | **78** | **78** | **136** | **131** | **198** | **199** | **20** | **21** |
| | | | | | | | | | | |
| **BALANCE (kg/ha)** | **276** | **271** | **413** | **410** | **351** | **315** | **600** | **640** | **83** | **80** |
| | | | | | | | | | | |
| **Efficiency (%)** | **25** | **25** | **15** | **16** | **28** | **29** | **25** | **24** | **20** | **21** |

### 6.3.5 Within-farm nutrient transformations

Internal farm nutrient transfers form an essential part of nutrient accounting frameworks by augmenting farm-gate (or whole-farm) budgets. The internal nutrient transfers define the processes whereby nutrients are re-distributed, accumulate on-farm or are susceptible to polluting the environment. By quantify within farm nutrient transformations opportunities to improve on-farm nutrient management decisions can be identified, including potential limitations to productivity and risks of nutrient losses to the environment.

Applications of nutrients in fertiliser and effluent, fodder harvesting and removal and grazing and feed management all contribute to internal nutrient transfers on dairy farms. The movement of cows and the locations where nutrients are excreted (directly or after storage and collection) are key components of within-farm nutrient transformations, because dairy cows are efficient recyclers of nutrients. Only a small proportion of ingested nutrients are exported in milk while in excess of 50% is excreted by the animal.

#### Cow movement on dairy farms – 'Cows in Space and Time'

The locations lactating dairy herds visited over a 24 hour period (paddocks, dairy shed, yards, laneway feed pad, and holding area) and the time the animals spent there were recorded on all farms on each of the 5 quarterly visits (Figure 20), with forty-three farmers providing adequate data for analysis. Cows spent most of their time in paddocks (73.7%) which was significantly greater than time spent elsewhere on dairy farms. Cows spent the least amount of time in dairy sheds with significantly increasing amounts of time spent in feed pads, holding areas, laneways and yards. The mean time in feed pads (3.9%) and holding areas (6.1%) was low when averaged over all farms, as these management units are not present on all farms. Of the 44 farms, 20 had feed pads, 11 identified holding areas, with 7 farms having both. When the data was analysed taking into account the farms where feed pads and / or holding areas were present, the mean proportion of time cows spent in these areas are 11.3 and 25.8% respectively (see for example Farms 22, 29, Figure 20). Cows only spent a mean of 1.7 and 8.6 % of their time in dairy sheds and yards, respectively, where excreted nutrients are routinely collected. Consequently the majority of excreted nutrients are available for pasture production or can be lost from the production system.

Preliminary regression analysis indicated that management factors influence the time farmers leave animals in locations around dairy farms. Percent time in paddocks decreased with increases in the percentage of feed imported. Increases in the time spent on feed pads and holding areas where the brought-in feed would be fed out to cows was also observed. It is likely that decreases in time in paddocks is related to availability of pasture, although no relationship appeared to exist between estimated pasture growth rate and percent time in particular locations. Pasture growth rate was estimated using data from the 'Pasture from Space' website (Pasture growth rate) for four weeks leading up to each farm visit.

Time on feed pads decreased whilst that in paddocks increased as the proportion of the land area under irrigation increased; suggesting that availability to water for pasture production increased the time cows spent in paddocks rather than in holding areas. The impacts of environmental factors (e.g. climate impacting on the use of feed pads) were less obvious and require further analysis.

Soil data collected in this project (see section 6.3.6) illustrated that paddocks and locations nearest the dairy shed had the highest nutrient concentrations. The use of 'night' paddocks, where cows are held overnight in paddocks usually close to the dairy shed, is thought to contribute to these nutrient accumulation zones; unlike 'day' paddocks.

The detailed spatial data collected for each farm was used to calculate the distance cows walked between the dairy shed and the locations they visited on each quarterly visit (Figure 21a). Night paddocks (where project farmers collected cows first in the morning before milking and placed cows last at night) were significantly closer to the dairy shed than the day paddocks (other paddocks where cows were held during the day). Importantly feed pads and holding areas were in general adjacent to the dairy shed (< 100 m). The times spent in these locations also indicate the potential for return of large amounts of nutrients (Figure 21b).

**Figure 20. Mean percent time spent by lactating dairy herds in 6 management units (paddocks, laneway, dairy shed, yards, feed pad, holding area) on 43 dairy farms.**

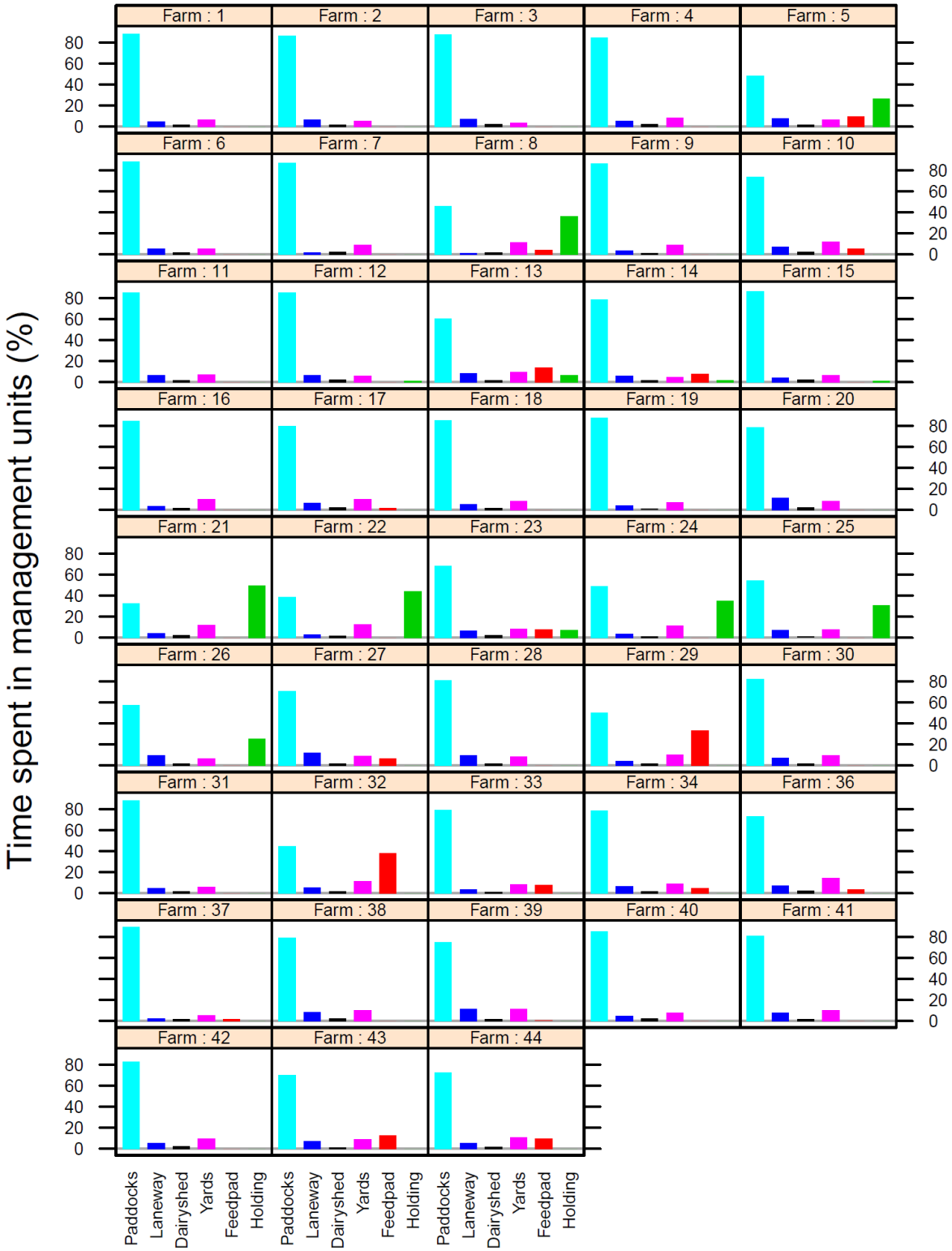
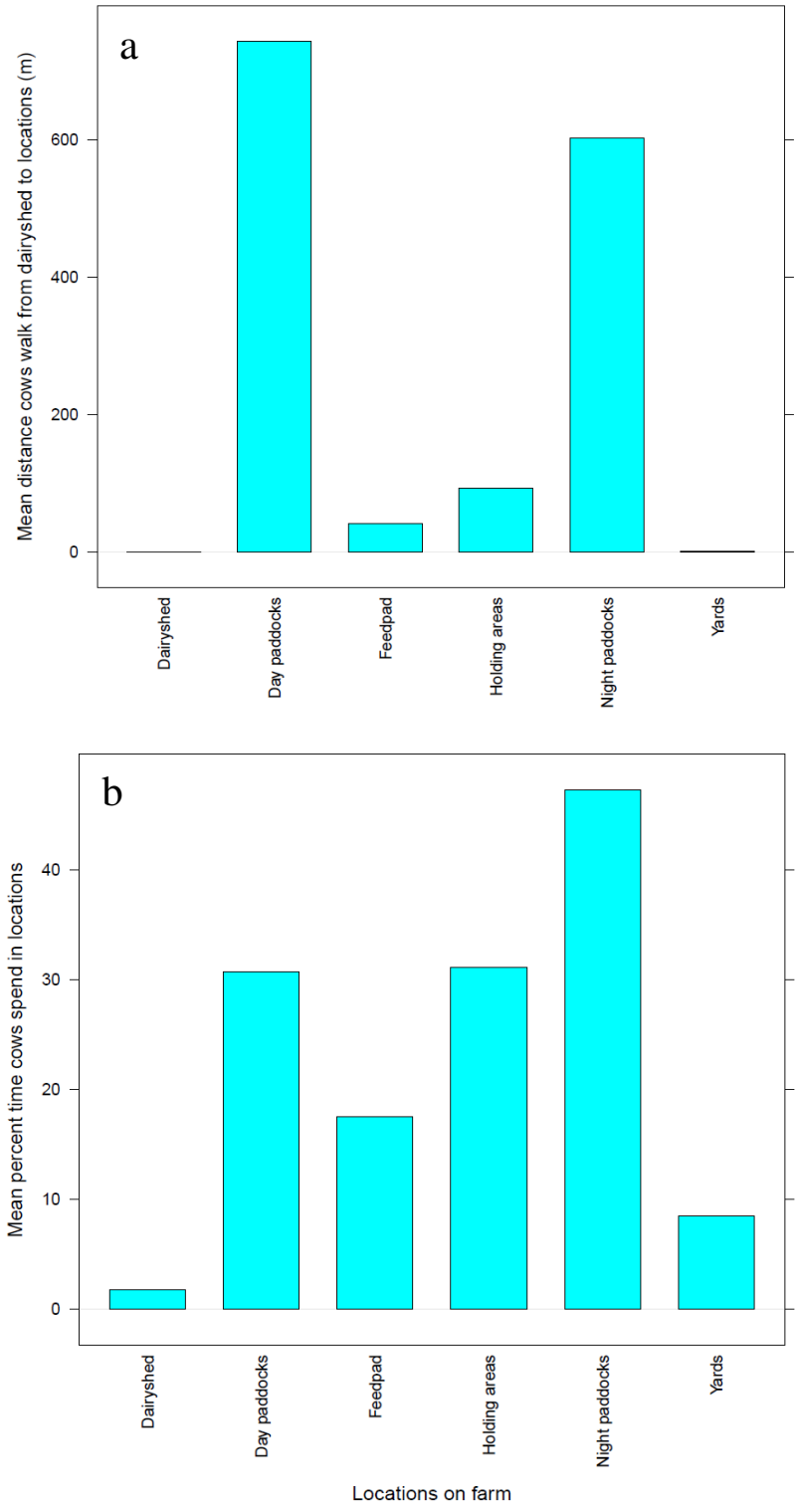


**Figure 21. Mean distance dairy cows walk to (a) and mean time cows spend in (b) locations on dairy farms each day.**

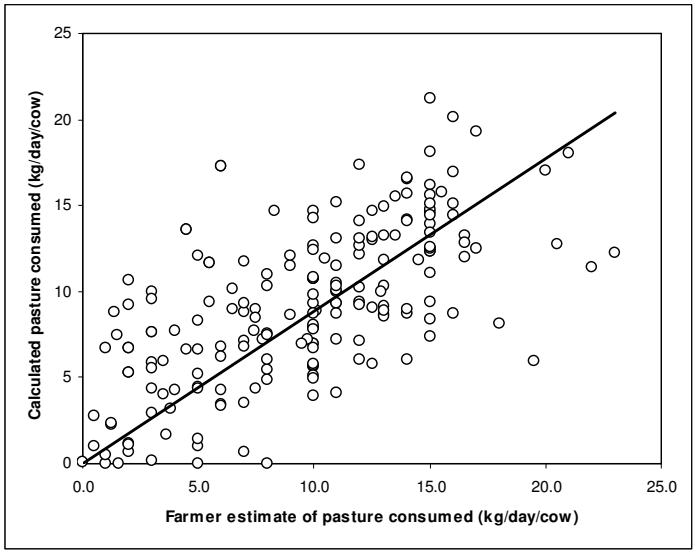


## Nutrient excretion by dairy cows

The efficiency with which dairy cows utilise feed nutrients to make milk (i.e. the ratio of milk nutrients exported to total feed nutrients consumed) will influence the amounts of nutrients excreted. Within confinement systems nutrients are managed by measuring feed intakes and subtracting nutrients exported in milk. The calculated nutrient returns can be used on-farm to minimise fertiliser use and environmental pollution.

Quantifying pasture dry matter consumed by cows is a challenge for grazing dairy systems, especially when data is collected on commercial dairy farms. The 'Pasture Consumption Calculator' (Pasture calculator) was modified, in conjunction with the developer (Janna Heard, 2008) to enable calculation of pasture consumed by the lactating herds on the day of each quarterly visit. The data required for the Calculator included the volume of milk produced, the cows' diet, the age structure, stage of lactation and pregnancy of the herd, cow live weight, distance walked and the terrain covered. Samples of all feeds and of the milk produced were collected for analysis. Farmers were also asked to give their estimate of pasture DM consumed by each cow. The calculated pasture consumed showed a strong 1:1 relationship ($C_b$ = 0.9939) with farmer-estimated pasture intake with of 65% of the data in agreement (Figure 22).

**Figure 22. Relationship between calculated pasture DM consumed and farmer estimate of pasture consumption.**




The calculated pasture intakes, in conjunction with supplement nutrients consumed were used to calculate the nutrient excretion rates (g nutrients excreted / cow / day) by dairy herds on all farms (Table 26). Almost half a kilogram of nitrogen was excreted daily by each cow, about 7 times the amount of phosphorus excreted. Therefore, for an average herd size of 250 cows over a 300-day lactation, 32.4 and 4.6 T of N and P are excreted around a dairy farm.

**Table 26. Daily nutrient excretion by lactating cows on 43 farms at 5 visits**

| | Excreted nutrients (g/cow.day) | | | | | |
|---|---|---|---|---|---|---|
| | N | P | K | S | Ca | Mg |
| Mean | 432.3 | 61.1 | 339.6 | 43.8 | 91.3 | 52.4 |
| Median | 430.9 | 59.4 | 329.4 | 42.1 | 88.2 | 49.9 |
| Minimum | 199.0 | 19.9 | 120.2 | 18.5 | 9.6 | 20.8 |
| Maximum | 792.0 | 131.6 | 670.8 | 101.7 | 210.3 | 274.2 |
| s.e.m.[§] | 7.39 | 1.39 | 6.90 | 0.88 | 2.69 | 1.39 |

[§] s.e.m. standard error of the mean

Supplementation of pasture-based diets by farmers aiming to increase milk production results in greater dietary crude protein and phosphorus intake. However, as N and P intakes increased, so did excretion of these nutrients (Figure 23), associated with reduced efficiency of use of N and P to produce milk (Figure 24).

**Figure 23. Changes in rates of excreted nitrogen (a) and phosphorus (b) as feed nutrient intake increases.**

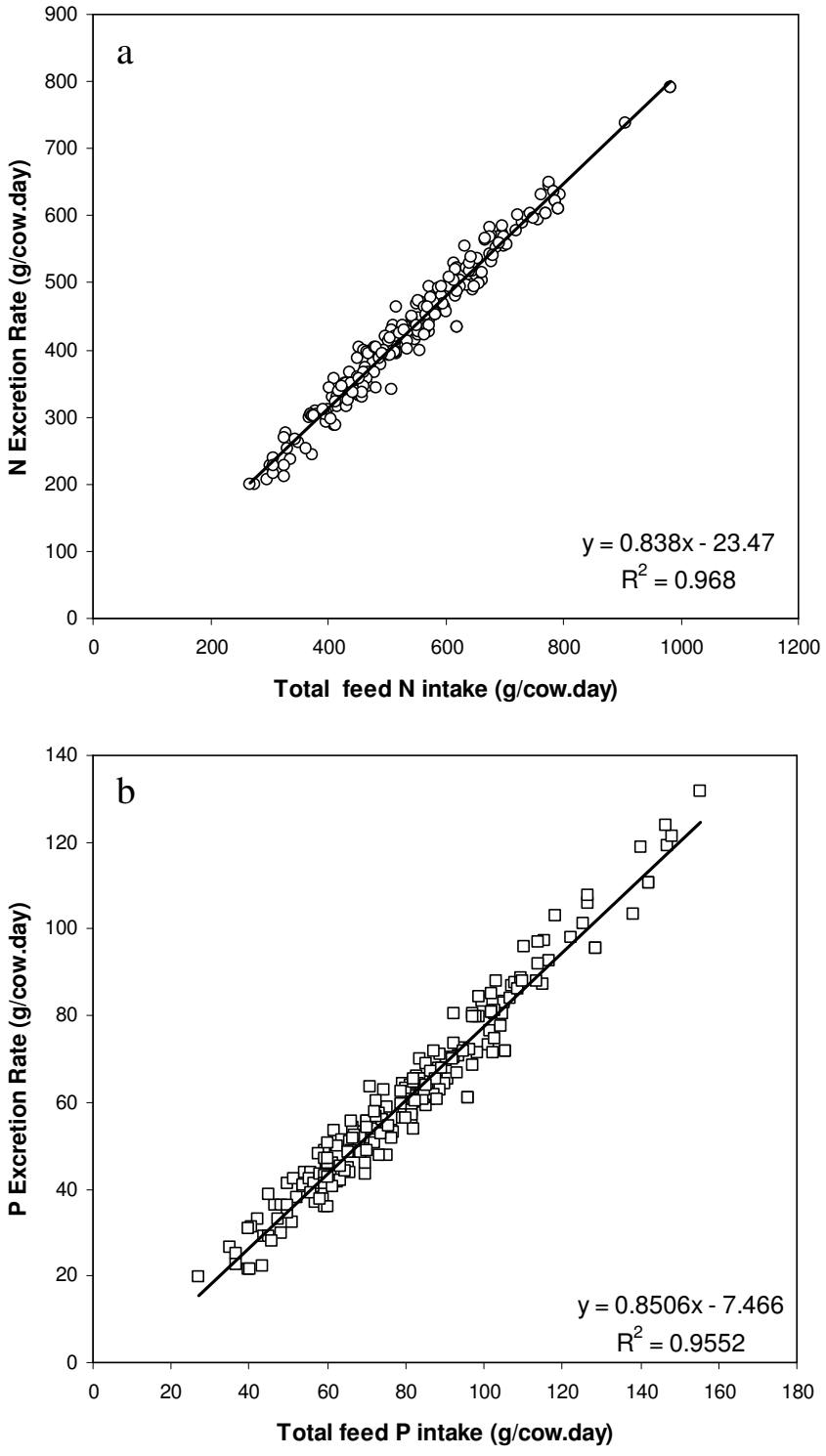
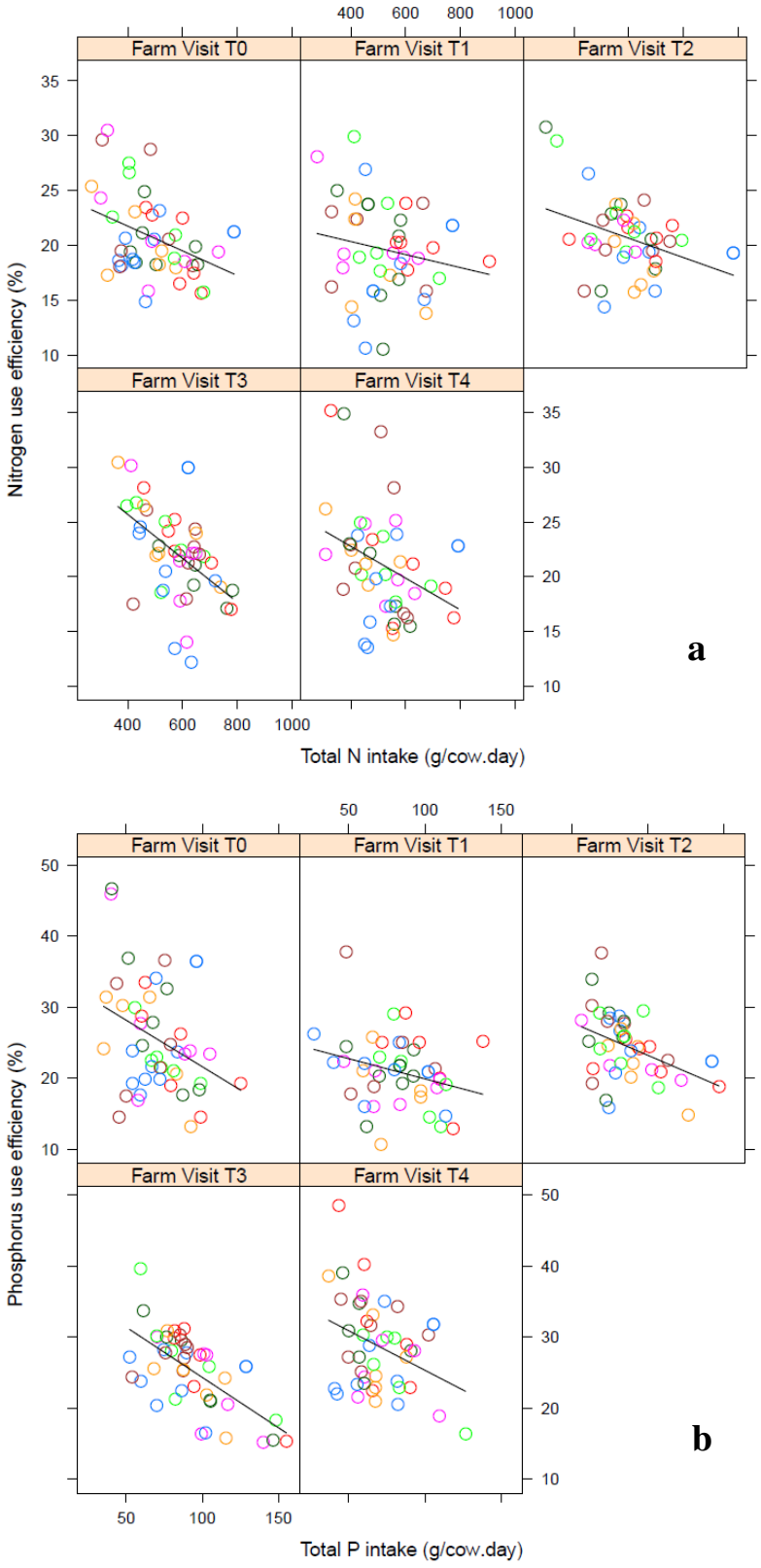


**Figure 24. Changes in the efficiency of use of feed nitrogen (a) and phosphorus (b) by dairy cows to produce milk as nutrient intake increases, calculated for all farms for each survey visit (T0-February 2008, T1-May 2008, T2-August 2008, T3-November 2008, T4-February 2009).**



On average close to 80% of intake N and 75% of P is excreted and if not carefully managed can potentially be wasted (Table 27). Despite this, on some farms as much as 35% of feed intake N was used to produce milk (ie Farms 21, 24 and 30), although these high N use efficiencies usually occurred only once for each farm.

The ranges (minimum to maximum) in utilisation of N and P by Australian dairy herds are comparable to that cited internationally. Feed N use efficiencies less than 20% is considered to be low, while 30 to 35% is considered above average (Chase 2003). Even so, there is scope to increase nutrient use efficiency on Australian dairy farms to maximise milk production, while minimising nutrient inputs.

**Table 27. Nutrient use efficiency calculated for herds on 43 farms at 5 visits**

| | Efficiency of nutrient utilisation (%) | | | | | |
|---|---|---|---|---|---|---|
| | N | P | K | S | Ca | Mg |
| Mean | 20.8 | 25.0 | 8.8 | 16.2 | 23.3 | 4.3 |
| Median | 20.5 | 24.3 | 8.3 | 14.9 | 21.0 | 4.1 |
| Minimum | 10.5 | 10.7 | 2.1 | 3.0 | 0.0 | 1.0 |
| Maximum | 35.1 | 48.5 | 19.7 | 47.9 | 75.5 | 8.7 |
| s.e.m.[§] | 0.28 | 0.43 | 0.22 | 0.48 | 0.73 | 0.08 |

[§] s.e.m. standard error of the mean

## Nutrient deposition and loading rates

The calculated nutrient excretion rates for the lactating herds on each farm at each survey visit were combined with farm-specific spatial and temporal cow movement data to determine nutrient deposition loads (kg/day) to locations around dairy farms. Not only were night paddocks on average closer to the dairy shed than day paddocks, but the amounts of nitrogen deposited there, for example, was 1.5 times greater than that deposited in day paddocks (Table 28), confirming the role of this management practice in contributing to nutrient re-distribution and accumulation on dairy farms.

When day and night paddocks were grouped for statistical analysis, paddocks received the greatest return of nitrogen in excreta (79 kg). The 79 kg returned daily to paddocks was generally returned to at least 2 different paddocks on most farms. However, where cows visit the same paddocks repeatedly the accumulation of nitrogen would need to be managed to ensure productive use of this resource. Although not presented here, the impact of repeated return of potassium on animal health (ie grass tetany) must also be considered where animals return to grazing paddocks frequently.

Nitrogen excreted in the yards was significantly greater than that returned to the dairy shed, feed pads and holding areas, and laneways. Consequently, nutrients deposited in yards need to be managed carefully as, based on project data, some farmers only wash their yards once daily despite collecting excreta deposited in the dairy shed at each milking. Nitrogen excreted in yards is particularly susceptible to volatilisation losses, and could constitute a large contribution to greenhouse gases based on the nutrient loads returned by dairy cows in this location. International research suggests the potential for large losses of gaseous nitrogen from concreted yards and other areas where animal are held for long periods (Kohn et. al., 1997).

Although smaller mean nitrogen loads were deposited in feed pads and holding areas, when only those farms with these management units were analysed, the mean depositions were 10.3 and 26.5 kg/day respectively. Like yards, gaseous nitrogen losses from concreted feed pads can be large. For feed pads that are not concreted, as well as holding areas, excreta are very infrequently collected, and nutrients such as nitrogen pose a potential for gaseous as well as run-off and leaching losses.

**Table 28. Mean nitrogen loads (kg/day) deposited by lactating cows in locations on 43 farms at 5 visits**

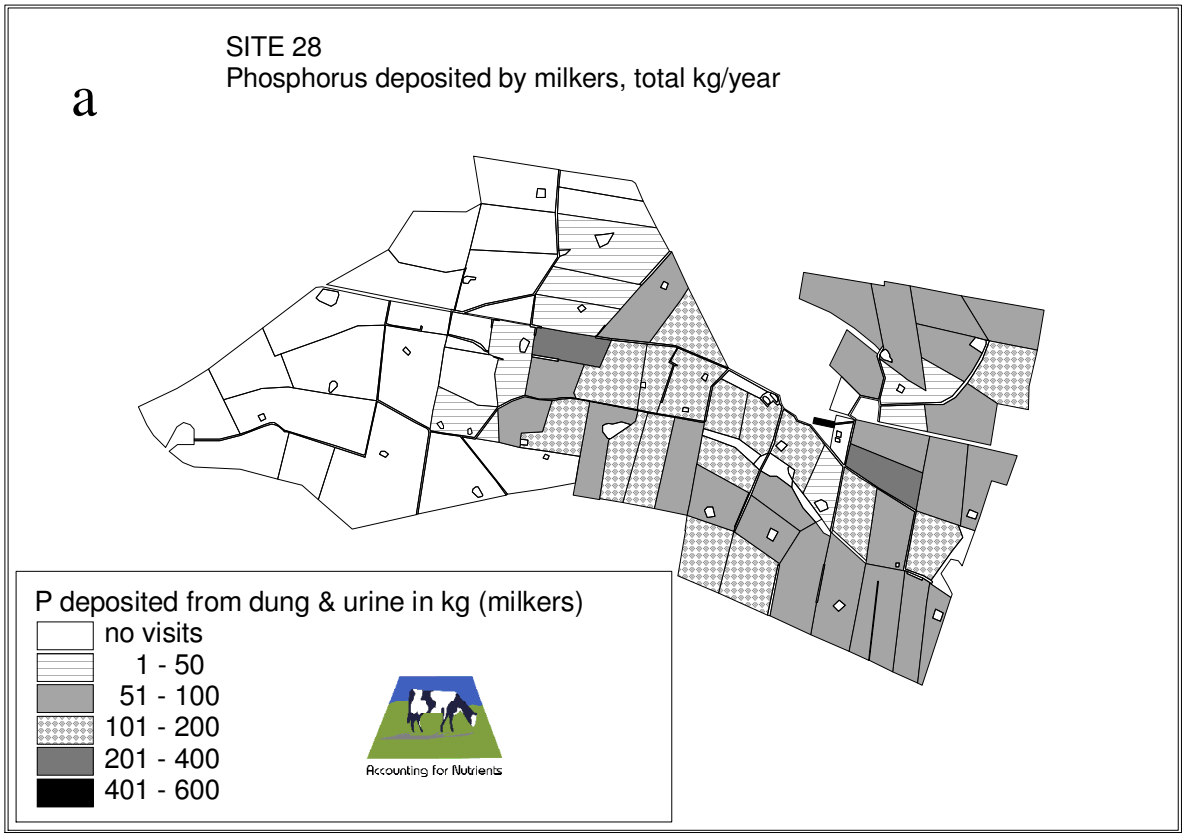
| Locations | Mean | Minimum | Maximum | Median | s.e.m.[§] |
|---|---|---|---|---|---|
| Day paddocks | 32.3 | 0.0 | 220.9 | 24.9 | 2.08 |
| Laneway | 7.1 | 0.0 | 168.3 | 4.2 | 0.83 |
| Dairyshed | 2.0 | 0.0 | 16.2 | 1.4 | 0.14 |
| Yards | 10.1 | 0.0 | 101.0 | 6.3 | 0.84 |
| Feedpad | 6.3 | 0.0 | 102.5 | 0.0 | 1.17 |
| Holding area | 4.1 | 0.0 | 79.6 | 0.0 | 0.82 |
| Night paddocks | 46.9 | 0.0 | 308.4 | 33.6 | 2.95 |

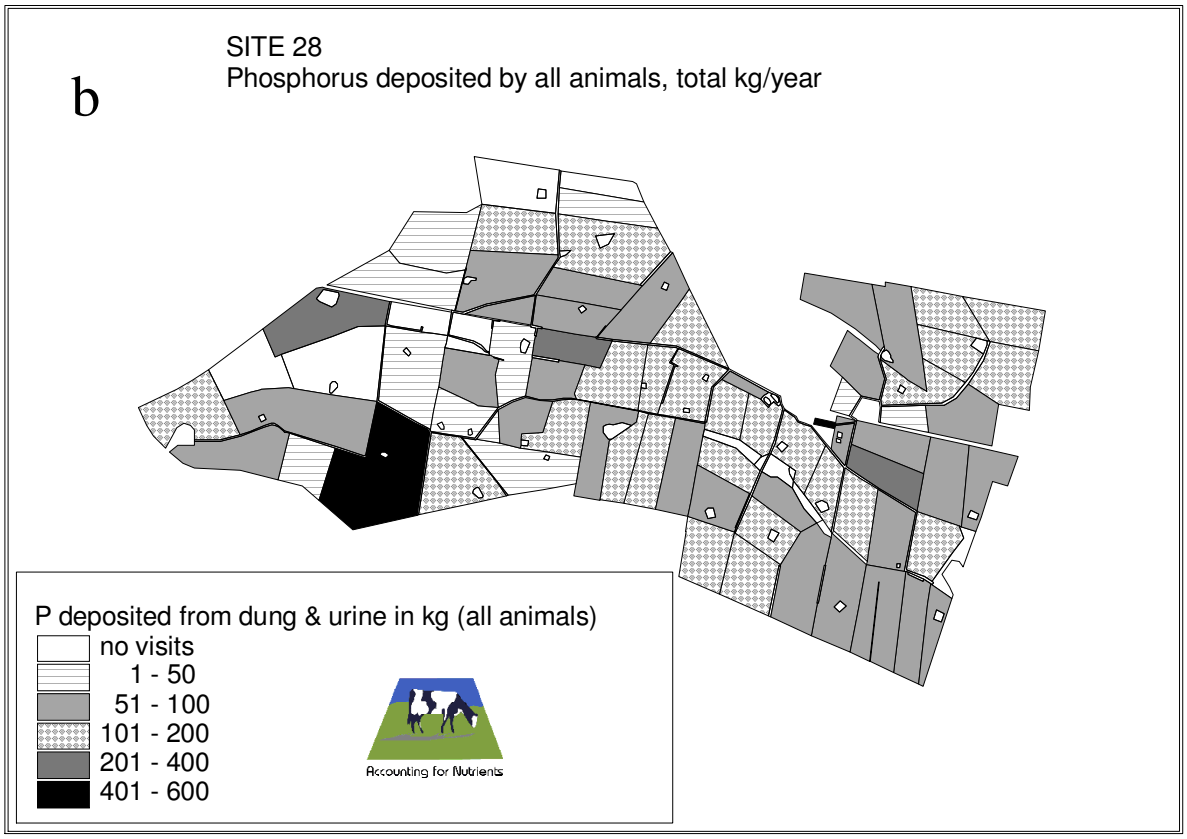
[§] s.e.m. standard error of the mean

The nutrients returned by dairy cows are seldom deposited uniformly across the paddocks on farms. From the annual records of the daily locations of dairy cows, nutrient deposition patterns across some of the farms were examined. On Farm 28, cows returned more frequently to certain paddocks, depositing nutrients unevenly across the farm (Figure 25a). The heterogeneous distribution of nutrients was exacerbated when the movement of other animals such as dry cows, springers and heifers were considered (Figure 25b).

Patterns of return such as these can explain the heterogeneous nature of soil test values on farms despite the long-standing tendency of farmers to blanket-apply fertilisers across farms. An awareness of the locations dairy cows visit, and calculated nutrient excretion rate for the lactating herd can assist farmers in reducing their reliance on fertilisers by accounting for the nutrients returned by grazing cows.

**Figure 25. Phosphorus (kg) deposited in dung and urine around Farm 28 over 421 days by a) lactating dairy cows and by b) all cows (lactating, dry, heifers, springers).**



### 6.3.6 Soil fertility and spatial nutrient distribution on dairy farms

All participating dairy farms were extensively soil sampled to determine the soil fertility levels of individual paddocks and other selected areas. This intensive soil sampling involved collecting samples from all paddocks used for pasture and crop production as well as areas where stock are confined (sacrifice paddocks, feeding areas, sick paddocks), but not laneways or areas where manure is collected. Prior to sampling, a soil sampling plan was developed between the on-farm team members and the farmer to determine which paddocks and areas were to be sampled and the appropriate sampling paths. The sampling path has now been added to the digitised farm maps and can be used by the farmers when future soil sampling is undertaken.

Collected soil was analysed for pH (in $CaCl_2$ and water), Olsen and Colwell P and K, phosphorus buffering index, $KCl_{40}$ S, organic carbon, and extractable cations (ammonium acetate). This information was used to create soil fertility maps of each dairy farm, using colour coding for each area and paddock that align with current agronomic soil fertility targets. These maps were provided with appropriate interpretation to all participating farms (an example is shown in Appendix 7).

Additionally, selected soil samples collected from 4 organic and 4 conventional dairy farms (matched by region and soil type), and representing a range in soil fertility levels (low to high Olsen P and Colwell K levels), were analysed for labile carbon. Labile C is thought to be representative of the pool of rapidly available C and may indicate greater potential for biological activity and soil health.

This comprehensive set of soil fertility data has enabled a detailed analysis and description of existing soil fertility levels on a broad range of Australian dairy farms and an exploration of the importance of farm system characteristics, management, and land-use. While mean values of individual paddocks are presented, additional analysis using median values or 'area weighted' means demonstrated similar results.

In general, the average soil nutrient levels, including P, K and S were found to be well above agronomic requirements on the dairy farms involved in this project (Table 29). For example, the average Olsen P level on pasture paddocks from the 37 conventional dairy farms was 36 mg/kg, well above the standard recommendation of 20 mg/kg. There were also substantial differences in soil fertility levels within and between dairy farms. While organic and conventional dairy farms had similar average pH and available K S levels in pasture soils, the organic dairy farms had substantially lower available P and S levels.

**Table 29. Mean soil pH, available P, K and S and PBI levels of different land uses from 40 conventional and 4 organic dairy farms.**

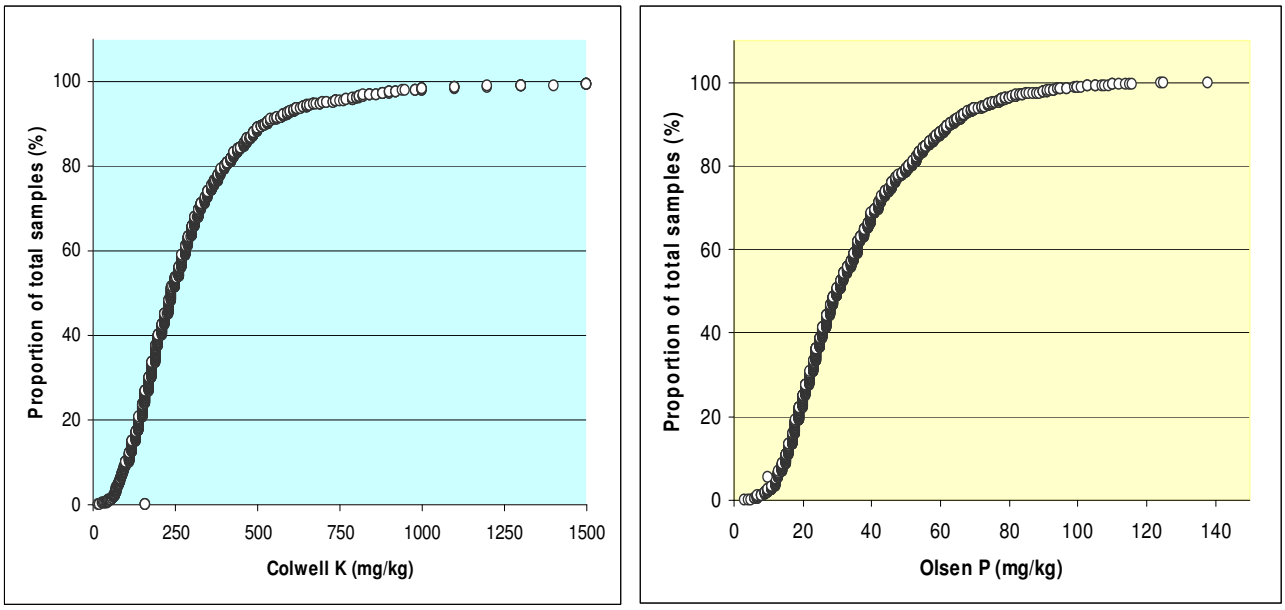
| Management/Use | Distance to dairy (m) | pH (CaCl2) | Olsen P (mg/kg) | Colwell P (mg/kg) | Colwell K (mg/kg) | KCl40 S (mg/kg) | PBI (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Organic** | | | | | | | |
| Pasture n=141* | 625.5 | 5.4 (0.6)^ | 16.7 (13.8) | 65 (62) | 271 (199) | 18.4 (20) | 302 (166) |
| **Conventional** | | | | | | | |
| Pasture n=1773 | 881.4 | 5.3 (0.7) | 35.6 (20) | 127 (76) | 296 (224) | 23.4 (33) | 263 (230) |
| Bull paddock n=6 | 444.0 | 5.3 (0.9) | 48.8 (26) | 169 (82) | 703 (602) | 45.3 (29) | 262 (24) |
| Feeding areas n=12 | 53.1 | 6.8 (1.2) | 319.9 (285) | 1151 (1286) | 4471 (3945) | 263.5 (263) | 321 (277) |
| Holding area n=13 | 400.4 | 5.8 (0.9) | 143.5 (171) | 510 (685) | 1515 (1271) | 73.8 (71) | 251 (252) |
| Sick paddock n=16 | 46.9 | 5.6 (0.9) | 71.4 (61) | 280 (282) | 771 (711) | 27.5 (19) | 178 (180) |
| Other animal n=104 | na | 5.1 (0.6) | 27.4 (15) | 100 (58) | 269 (180) | 14.6 (12) | 301 (281) |

* n= number of areas sampled ^ Standard Deviation in parenthesis

Land use had a substantial impact on the existing soil fertility levels. Areas with high animal densities, such as calving paddocks, feed pads, holding area and sick paddocks, had substantially elevated soil nutrient levels when compared to the overall pasture paddocks (Table 29). In contrast, low intensity areas such as 'other animal' and treed areas had much lower fertility levels.

The variation in soil P and K across all of the conventional dairy farms is shown in Figure 26. This data is only inclusive of grazed pasture paddocks and does not include other areas. While the Olsen P levels ranged between 3 and 189 mg/kg and the Colwell K levels ranged from 14 to 3400 mg/kg, the patterns of sample distribution for Olsen P and Colwell K were very similar. Only 20% of the paddocks sampled had soil P or K values below the recommended agronomic optimum (Olsen P of 20 mg/kg and Colwell K of 140 mg/kg), while 50% of the paddocks sampled were 1.5 times or more times the recommended agronomic optimum (Figure 26). At the high fertility end, 20% of paddocks sampled had Olsen P or Colwell K levels at least 3 times the agronomic requirements.

**Figure 26. The proportional distribution of soil P levels determined from dairy pastures (n= 1768) from 40 conventional dairy farms across Australia.**

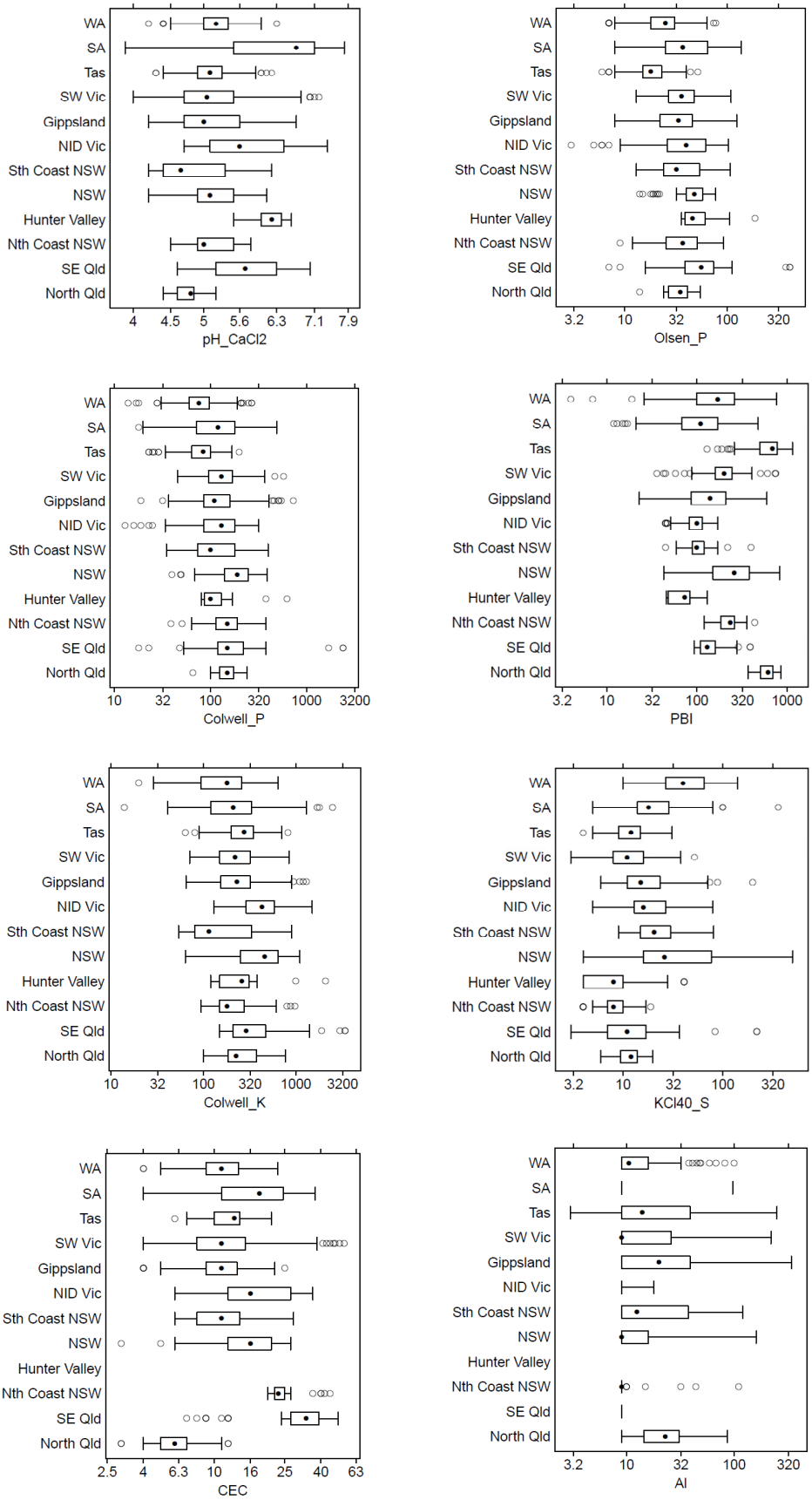



**Regional soil nutrient levels** The range of soil test values from paddocks sampled on conventional farms in specific dairy regions is presented as box and whisker plots in Figure 27. The dots represent the median value; box defines 50% of the data range and whiskers 100% of data, with the exception of the outlier paddocks shown. The varying number of dairy farms in each region should be noted.

Not surprisingly, there were specific regional differences in some soil chemical characteristics reflecting differences in the inherent soil types present. For example, soil pH was highest on the calcium rich soils of South Australia and lowest on soils in the south coast of New South Wales. Similar, PBI levels were generally higher in Northern Queensland and Tasmania, where kraznosem soils dominate, and lower on sandy loam soils in central New South Wales, South Australia and Western Australia.

There were only relatively small differences in the median soil P and K levels between regions, reflecting the influence of management rather than any inherent soil characteristic. An exception to this was the lower Olsen P level in Tasmania, probably reflecting the higher fixation capacity of these northern Tasmania soils. There were also regional differences in available sulphur, most likely as a result of different S retention of the soil types in each region.

**Figure 27. Box and whisker plots for various soil chemical characteristics of pasture soils collected from conventional dairy farms in different regions.**



**Comparison between conventional and organic dairy farms**

***Soil nutrient levels***. As a result of the greater number of paddocks from conventional dairy farms (n=40) compared with organic dairy farms (n=4), a smoothed frequency distribution curve was used to compare soil nutrient levels between these farming systems (Figure 28). Soil characteristics such as PBI and cation exchange capacity had similar distributions, suggesting a similar range of soil types included in both systems. Not surprisingly, conventional dairy farms overall had much larger Olsen P, Colwell P and available S distributions and median values, reflecting the use of inorganic P fertiliser and greater inputs of feed. In contrast, available K levels were similar between the different systems, while soil pH appeared lower for the conventional systems, with corresponding aluminium levels, higher (Figure 28).

***Labile carbon.*** Pair-wise comparisons of labile soil results from the conventional and organic dairy farms in each state (Table 30) indicate the variable nature of the labile carbon measurement. The measured labile carbon levels were high (usually >2000 mg/kg) and reflect the large amounts of easily degradable organic matter present in these pasture soils. In Victoria and NSW, the organic farms had higher labile carbon levels overall than conventional farms, while in Queensland and Western Australia, labile C levels were lower on the organic farms. When all data was pooled at a national level, the conventional farms had a higher average labile carbon level (Table 30).

The collected data was pooled to determine whether the intensity of land use, reflected in the existing soil fertility status was related to the measured labile carbon levels. This analysis indicated that there was no significant relationship between labile carbon and associated Olsen P (or Colwell K, data not presented) levels. Not surprisingly however, there was a significant positive linear relationship between total carbon and labile carbon (Figure 29).

The demonstrated differences in labile carbon measured between the different farming systems may warrant further investigation of this data set. However, the apparent variable nature of labile carbon values determined in this study, the strong relationship between total carbon and labile carbon on these dairy soils, and the high cost and labour intensiveness of labile carbon analysis, suggests that the current total organic carbon test provides similarly useful information.

**Figure 28. Smoothed frequency distribution diagrams for various soil chemical characteristics of pasture soils collected from conventional and organic dairy farms.**

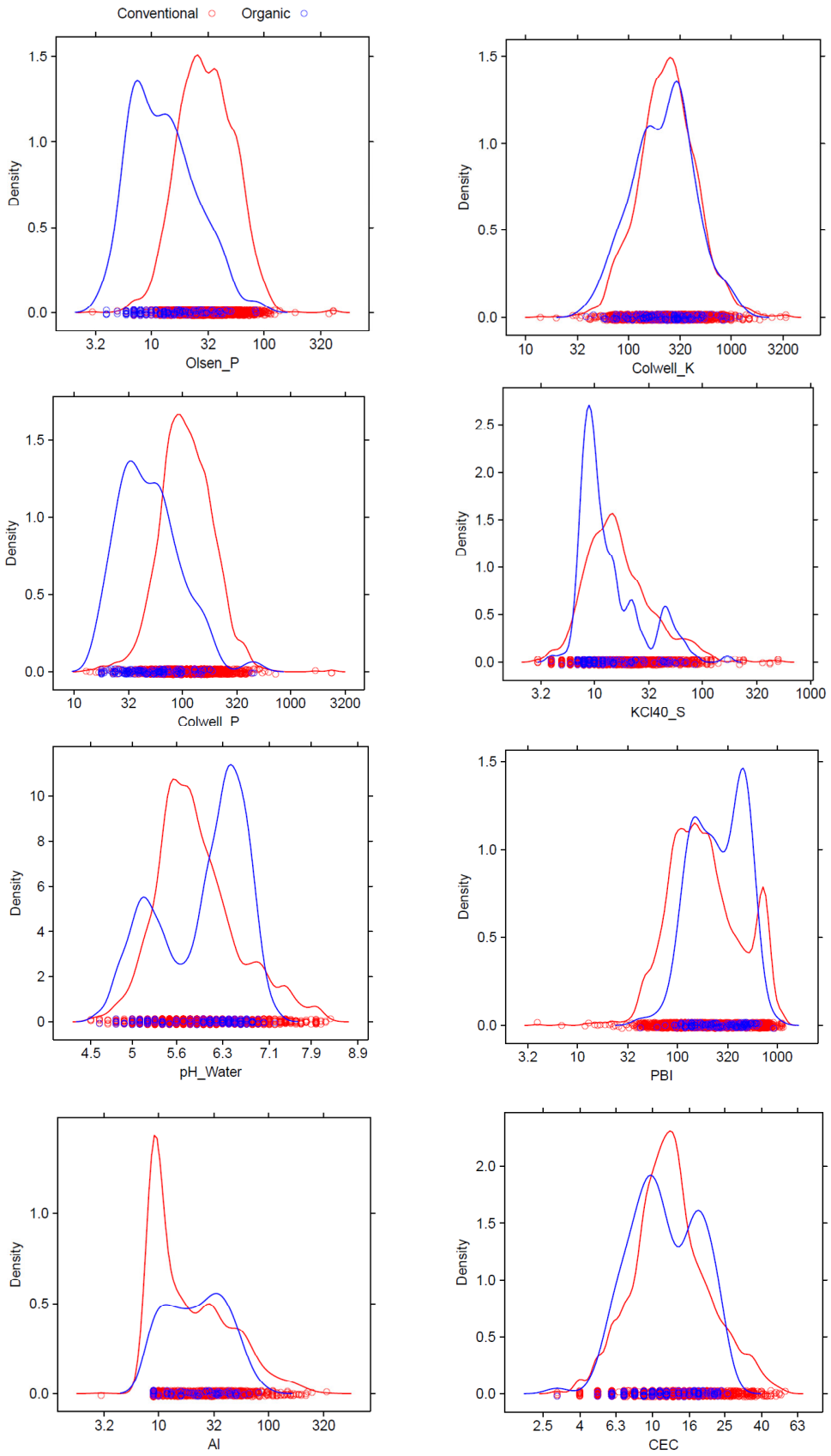
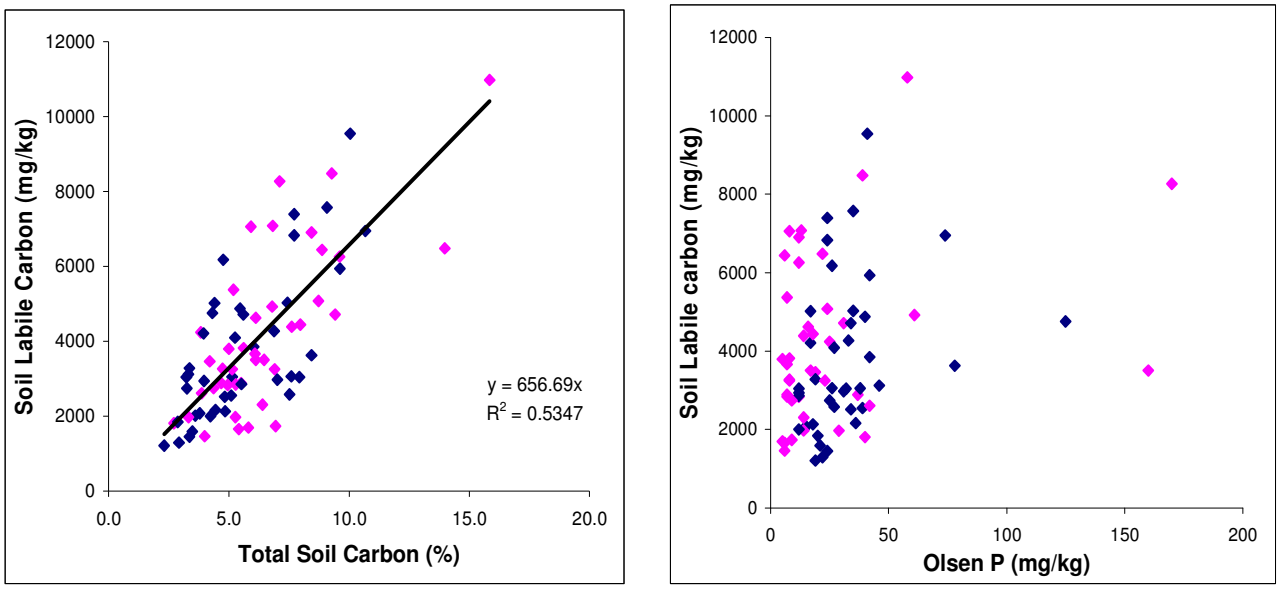


**Table 30. Average Colwell K, Olsen P and Labile carbon results from 10 selected paddocks from matched one organic and one conventional dairy farm in each state, and national averages.**

| | | | Colwell K (mg/kg) | Olsen P (mg/kg) | Total Carbon (%) | Labile Carbon (mg/kg) |
|---|---|---|---|---|---|---|
| Victoria | Conventional | Mean | 422 | 30.9 | 3.6 | 2963 |
| | | Std Dev. | 328 | 34.6 | 0.4 | 997 |
| | Organic | Mean | 395 | 12.4 | 6.7 | 4598 |
| | | Std Dev. | 218 | 5.9 | 2.0 | 2016 |
| | All Vic data | Mean | 409 | 21.7 | 5.2 | 3781 |
| | | Std Dev. | 272 | 25.9 | 2.1 | 1761 |
| WA | Conventional | Mean | 227 | 19.2 | 7.7 | 4395 |
| | | Std Dev. | 106 | 10.1 | 2.8 | 1913 |
| | Organic | Mean | 213 | 35.4 | 6.1 | 4331 |
| | | Std Dev. | 153 | 23.9 | 2.1 | 1621 |
| | All WA data | Mean | 228 | 27.9 | 7.1 | 4426 |
| | | Std Dev. | 129 | 19.3 | 2.6 | 1759 |
| NSW | Conventional | Mean | 232 | 31.1 | 4.3 | 2494 |
| | | Std Dev. | 166 | 9.4 | 1.3 | 1167 |
| | Organic | Mean | 431 | 59.9 | 6.2 | 4374 |
| | | Std Dev. | 490 | 58.3 | 3.7 | 2977 |
| | All NSW data | Mean | 332 | 45.5 | 5.2 | 3434 |
| | | Std Dev. | 370 | 43.2 | 2.9 | 2403 |
| QLD | Conventional | Mean | 344 | 31.7 | 8.2 | 5396 |
| | | Std Dev. | 221 | 6.6 | 1.0 | 2432 |
| | Organic | Mean | 252 | 11.3 | 5.4 | 3501 |
| | | Std Dev. | 149 | 9.8 | 0.8 | 1767 |
| | All QLD data | Mean | 298 | 21.5 | 6.8 | 4449 |
| | | Std Dev. | 190 | 13.2 | 1.7 | 2286 |
| Australia | Conventional | Mean | 326 | 25.7 | 6.5 | 4217 |
| | | Std Dev. | 287 | 35.3 | 2.6 | 2175 |
| | Organic | Mean | 299 | 31.7 | 5.5 | 3759 |
| | | Std Dev. | 236 | 21.0 | 2.2 | 1955 |
| | All Aust data | Mean | 313 | 28.7 | 6.0 | 3988 |
| | | Std Dev. | 263 | 29.0 | 2.6 | 2134 |

**Figure 29. Relationship between labile carbon, total soil carbon and Olsen P level on 4 organic (pink) and 4 conventional (black) dairy farms.**



**Predicting soil nutrient levels** The influence of farm management on specific soil nutrient levels was investigated using a random effects variance component analysis (REML). Characteristics associated with each paddock such as organic / conventional, existing soil PBI, described paddock use, intensity of grazing practises, whether effluent or irrigation water was applied, and distance from gate to dairy shed, was investigated.

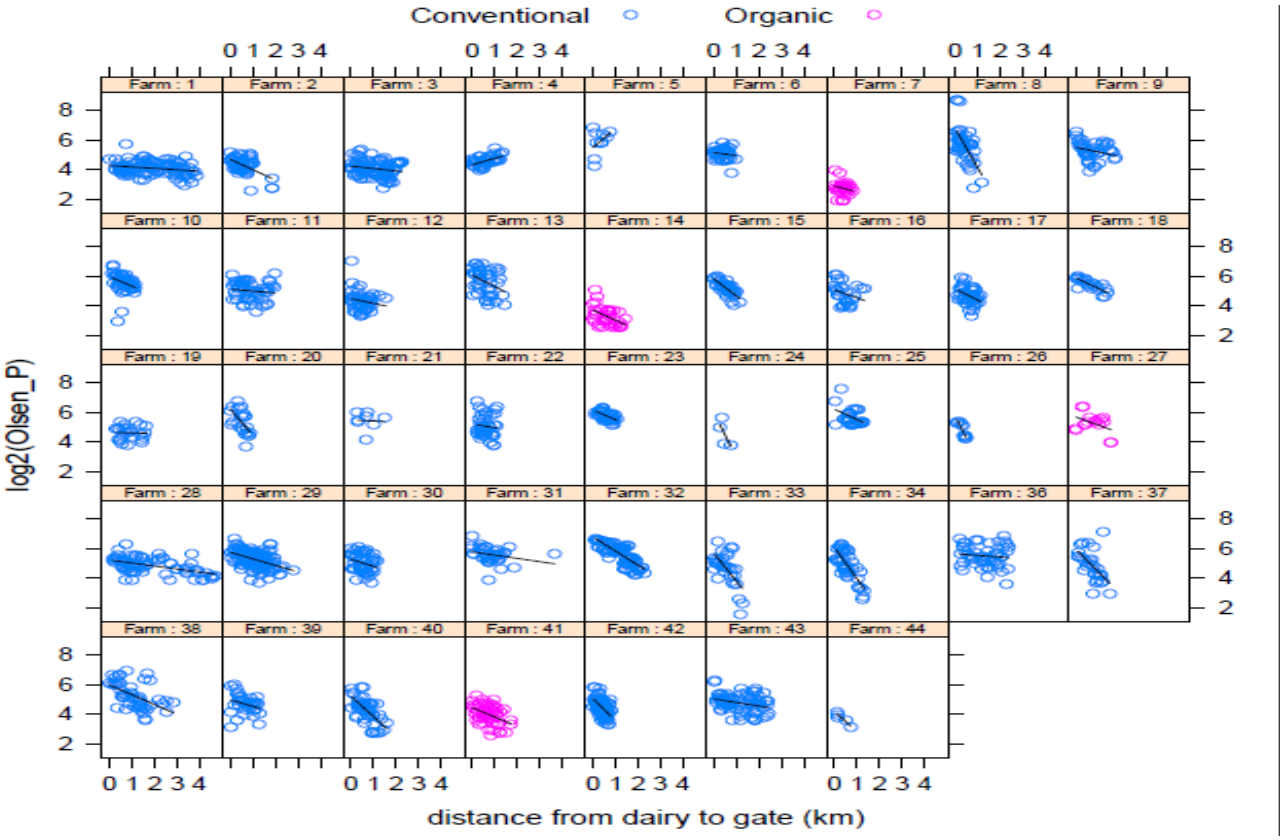
This analysis and subsequent regression relationships measured the association between these factors and the corresponding soil nutrient status. For example there was a highly significant negative relationship between Olsen P level and distance to dairy (Figure 30) on most dairy farms.

Overall the ***Olsen P*** and **Colwell P** level determined from pasture soils on dairy farms nationally was significantly ($P < 0.05$) related to the following factors: existing PBI level (-ve for Olsen P, +ve for Colwell P), distance of that paddock to the dairy (-ve), whether it was organic (-ve), whether the paddock was used for feeding (+ve), and the relative frequency of visits by grazing cows (+ve).

The ***Colwell K*** level determined from pasture soils on dairy farms nationally was significantly ($P< 0.05$) related to the following factors: existing PBI level (-ve), distance of that paddock to the dairy (-ve), whether it received dairy effluent (+ve), whether it was irrigated (-ve), whether the paddock was routinely harvested (-ve), and the relative frequency of visits by grazing cows (+ve).

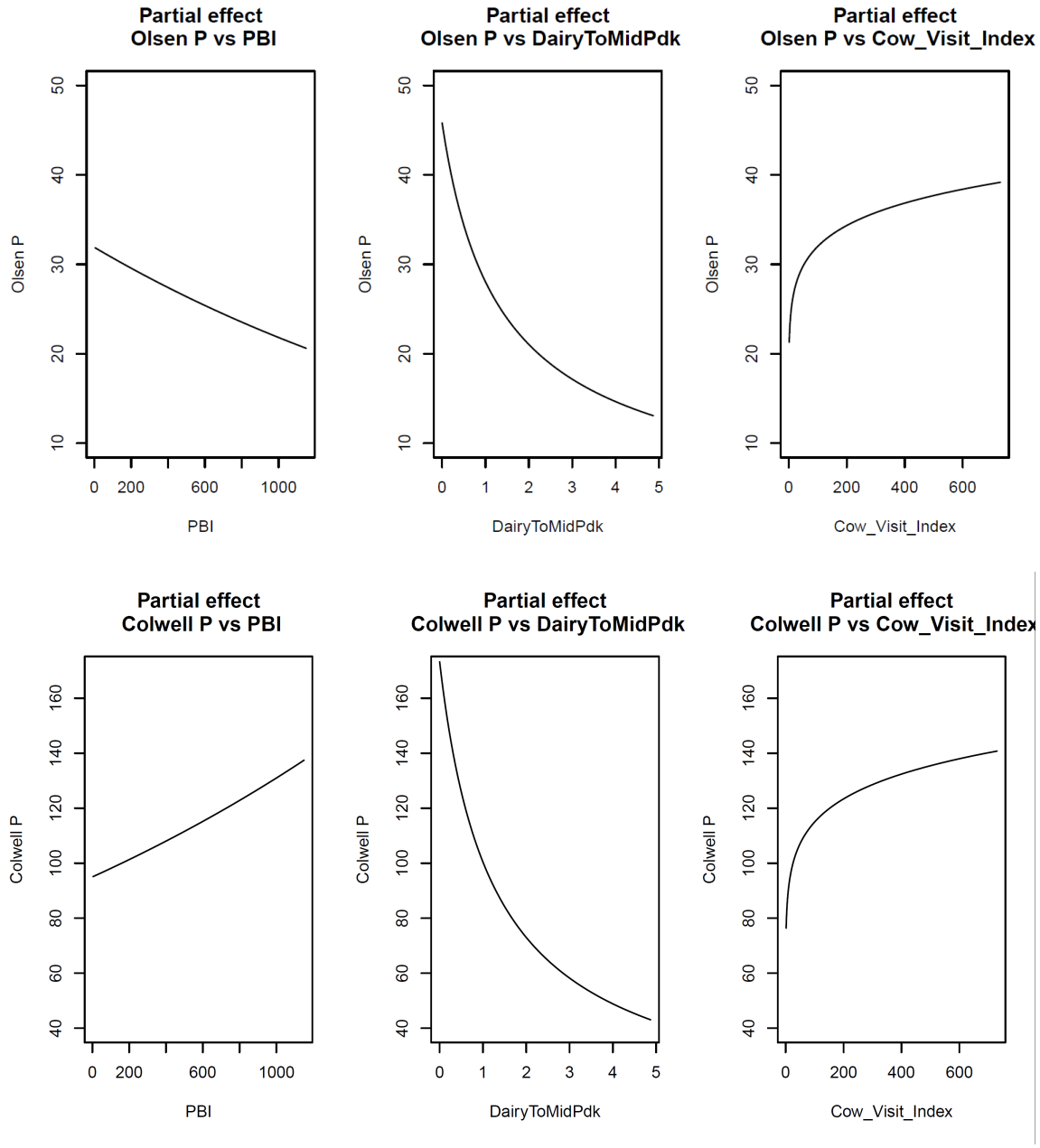
The ***Available S*** level determined from pasture soils on dairy farms nationally was significantly ($P< 0.05$) related to the following factors: existing PBI level (-ve), whether it was organic (-ve), distance of that paddock to the dairy (-ve), whether it received dairy effluent (+ve), whether it was irrigated (+ve), whether the paddock was routinely harvested (-ve), and the relative frequency of visits by grazing cows (+ve).

**Figure 30. Relationship between Olsen P soil test value and distance of the sampled paddock to the dairy (both log transformed) for pasture soils collected the 44 conventional and organic dairy farms.**



While individual levels varied markedly between farms, soil P, K and S levels were consistently and significantly related ($P < 0.001$) to distance-to-dairy and frequency of visits by grazing cows. This demonstrates that a key driver of within-farm nutrient heterogeneity appears to be stocking density of land and the resultant nutrient deposition from urine and dung. For example, there was an overall 3-fold difference in Olsen P, between paddocks close to, and those furthest from, the dairy (Figure 31).

**Figure 31. Partial effect of PBI, distance from dairy to gate, and cow visit index on Olsen P and Colwell P soil test levels.**



## 6.4. Delivering information and tools

**Key activities and outputs**

**Delivering information and tools - Module 5.**

- ✓ Development and review of Communication, Next user and Evaluation plan
- ✓ 16 targeted presentations to key policy and industry groups
- ✓ Establishment of Accounting 4 Nutrients website
- ✓ Project awareness and new information presented
  - 90 articles, radio interviews
  - 48 project presentations provided nationally
- ✓ Science based information presented nationally and internationally
  - 16 presentations and papers delivered at national and international conferences
- ✓ Road-testing of standardised nutrient budgeting and soil nutrient distribution tools on more than 250 dairy farms nation-wide
- ✓ Wide-spread collaboration with key dairy industry, extension, and research staff nationally.
- ✓ Direct involvement in 10 independent dairy nutrient management projects, nation-wide

Information and approaches from the A4N project have the potential to improve recommendations by key next-users such as fertiliser advisors, extension staff and dairy factory field officers, and also respond to broader community concerns relating to the environmental impact of nutrient use and nutrient losses.

The project communication plan developed in November 2006 identified a suite of tactical elements to establish the communication infrastructure and capability for the project. This is available on the A4N website. The approach used by the Accounting for Nutrients project team was based on establishing and delivering on a rolling six month task list to tackle priority actions associated with communication and evaluation.

The key principles for communication activities for the project were to:

- Mobilise support from project collaborators (e.g. industry groups, state agencies) and investors and their organisations communication capability.
- Use established media and communication channels and outlets where ever possible.
- Develop project advocates in established media and communication channels and outlets.
- Use communication tools appropriate to the needs and importance of the project stakeholder groups, organisations and individuals
- Use mass media (such as newspapers) for more general stakeholders and audiences where the goal is raising awareness and understanding
- Use focused media (such as newsletters) for stakeholder groups of direct relevance and importance to the project where the goal is awareness, understanding and support.
- Use person to person communication for organisations and individuals of direct importance to the project where the goal is active support for and commitment to the project.

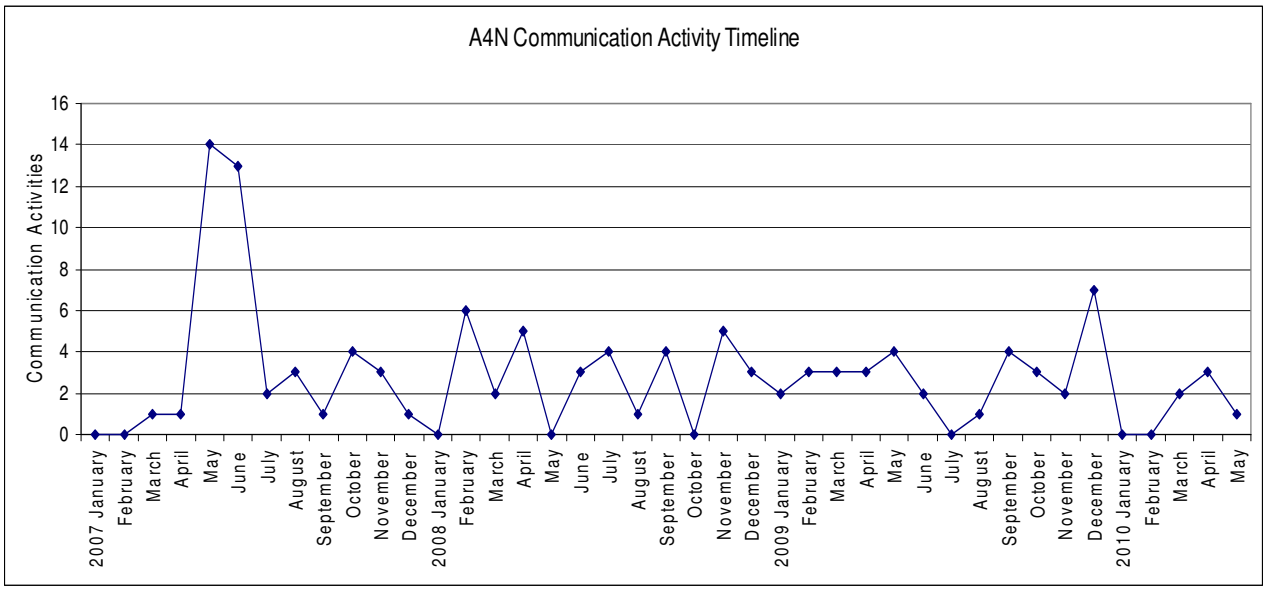
### 6.4.1 Communication activities and publications

A broad range of communication activities and supporting tasks have been undertaken during the 4 years of the Accounting for Nutrients on Australian Dairy Farms project. While the principal project focus has been on science methodology, data collection and interpretation, from the on-farm monitoring of nutrient flows and transformations, there have been around 170 communication activities undertaken nationally over the past 4 years.

These project activities have been an a number of forms but mostly involved written news articles published in specific rural newspapers or magazines, and presentations to key stakeholder groups. These groups included farmer discussion groups and field days, industry training forums, specialist industry stakeholder groups, and national industry conferences. The information delivered varied depending on the audience.

While there was an initial peak in media activities in the early stages as the project was 'launched', there has been a steady stream of information from the project team throughout the past 3 years (Figure 32). On average there have been around 2 media activities per month.

**Figure 32. Communication outputs throughout the life of the Accounting for Nutrient project.**




Newspaper articles and radio interviews have been the most easily accessible and effective way to deliver project information to a broader audience (Table 31). The project team has also published 7 journal and conference papers that have been targeted at the scientific community both nationally and internationally. Presentations, industry meetings and project reports have also been an important part of the project communication activities, with each aligned to target specific audiences (next-user, industry, government).

**Table 31. List of communication activities undertake during the A4N project.**

| Publications | Total = 90 | Targeted Audience | Total = 80 |
|---|---|---|---|
| Published Journal and Conference Papers | 7 | Industry Meetings | 14 |
| Radio interviews | 11 | Farmer presentations | 31 |
| Newspaper articles | 48 | Industry presentations | 11 |
| Project Newsletters | 8 | Industry Seminars | 4 |
| Website links | 4 | Next-user training | 4 |
| Industry Bulletins | 6 | Scientific conference | 16 |
| Magazine articles | 6 | | |

### Project website as an information source

A major communication output which was initiated in 2008 was the development and launching of the 'Accounting for Nutrients on Australian Dairy Farms' project website 'accounting4nutrients.com.au' (Figure 33). The project website aims to serve as an information source primarily for stakeholders and project participants.

The website includes an overview of the project; justification for work undertaken; project teams and members; technical details about on-farm research work, and serves as a repository for communication and technical documents and project reports. The website is not designed to be an extension tool for those interested in nutrient management, although useful project information and media outputs are also included.

**Figure 33. Accounting for Nutrients on Australian Dairy Farms project website**




## 6.4.2 Influencing nutrient management programs and road-testing

The project team recognised that an important part of achieving a high level of credibility and adoption of information generated from the project was to work collaboratively with other researchers, catchment managers, fertiliser agronomists, consultants, extension officers and industry. While many of these players were directly exposed to the A4N project through the extensive formal networks that have been established (see section 6.1), there were a number of independent projects focusing on the adoption of nutrient management tools and approaches on dairy farms, operating across Australian.

In many cases, this provided an excellent opportunity to directly collaborate with these projects, provide input into their planning processes, develop nationally standardised approaches, tools, and interpretation and importantly imbed A4N project information directly into these independent projects. This collaboration was also of significant benefit to the A4N project as it provided an important review of project outcomes in light of the specific needs of these independent projects and also enabled an early road-testing of nutrient accounting and how it fits within whole-farm nutrient management planning. This road-testing also ensured that the nutrient accounting approach was practical and can be adopted across the range of climates, landscapes and grazing practices on Australian dairy farms.

The initial project proposal had indicated that the project team would road-test this approach on at least 30 dairy farmers towards the end of the project. Through this extensive collaboration, more that 250 dairy farms nation-wide have had whole-farm nutrient budgets and soil nutrient distribution maps

generated and interpreted using the national standards provided through the A4N project. The project team has directly undertaken around 100 of these farm nutrient assessments, while the remaining 150 have been coordinated by the various project leaders across the country.

A brief description of key nutrient management programs, and the role the A4N project team played, is provided below.

## Improving the environmental credentials of milk companies

**Murray Goulburn** Murray Goulburn (MG) is Australia's largest dairy company, with one third of Australian dairy farms supplying milk to its processing and manufacturing plants. Over the past decade MG has lead the field in promoting and delivering best environmental practise to milk suppliers. This has been achieved through strong partnerships with Government, academic and industry organisations, and technical initiatives such as the development of company web-based tools and services.

During the past 4 years, the A4N project team have worked closely with the MG environment team and their programmers to design, develop and road-test a whole-farm nutrient accounting tool, which is currently available to MG suppliers through the MGf@rm website. This partnership has provided an important ground-truthing and a valuable route-to market for A4N project outputs.

The web-based IT initiatives developed by MG to house, update and distribute on-farm best practise tools and information has been successfully applied and evaluated in a recent joint project funded by DAFF, and supported by DPI Victoria, and the West Gippsland CMA, involving 25 dairy farmers. An expansion of this web-based approach is currently underway and is targeting 100 Murray Goulburn suppliers' nationally. This program will subsequently have a much greater impact and benefit across both MG suppliers and set national standards across the entire Australian dairy industry.

**Bega Cheese** The Accounting for Nutrients project team have worked closely with Bega Cheese Farm Services to assist in the development of 'standardised dairy farm nutrient management planning and interpretation' guidelines. Project team members also helped to deliver the program. The objectives of this program were to:

- help farmers understand where on their farms nutrients accumulate at high levels and may cause environmental contamination
- help farmers understand what is driving accumulation or reduction of nutrients in particular areas
- give farmers information and greater confidence to make informed decisions regarding fertiliser and effluent applications within the farm.

Nearly all Bega Cheese suppliers in Southern NSW (91 dairy farms) participated in the program between 2007 and 2009. The process involved soil testing of every paddock, assessing the risk of nutrient loss and in association with the farmers, developing whole-farm nutrient budgets for each farm.

Dissemination of the soil test information, soil nutrient maps, risk assessment and nutrient balance information was delivered back to the participating farmers during a series of farmer-group sessions in Bega. While involvement in the program was voluntary farmers were enthusiastic about the project and the information they received. A series of follow up sessions have also been undertaken with dairy consultants in the Bega area so that consistent information and interpretation are applied by next-users and that information pertaining to the project outcomes is well understood by local advisors.

## The role of A4N in national nutrient projects

A number of catchments across Australia with nutrient contamination problems have identified nutrient excess on dairy farms as a contributing factor. These areas include Far North Queensland, the Peel Harvey catchment in Western Australia, Gippsland Lakes, Shallow Inlet and Tarago catchments in Gippsland Victoria, Curdies River catchment in South-West Victoria, Sydney Basin catchment in New South Wales, and the Montagu catchment in North-West Tasmania.

Projects within these catchments are principally supported by state or federal grants, and managed by industry stakeholders, and often administered (and sometimes delivered) by the local CMA. Most of these projects aim to improve the delivery of appropriate information and tools to land managers and advisors, which subsequently improve on-farm nutrient management decisions and therefore reduce nutrient excess and losses to the broader environment.

The A4N project team has significantly contributed to these projects through a range of activities. This has included assisting in project scoping, developing standardised protocols, designing 'Soil Nutrient Management Plans', providing nutrient budgeting tools, soil test calibrations, and methodologies for mapping nutrient distributions. The project team has also provided training to state-based dairy extension staff, dairy company field officers and private dairy consultants. Additionally, the value of the farm-based data collected during the A4N project has been recognised by other national projects and the provision of A4N data and its interpretation has benefited all concerned.

Specific examples of projects which the A4N project team has collaborated with include:

- Queensland Soil Nutrient Management Plan, part of the Reef Rescue Package
- Western Australia - Fertiliser Action Plan
- South West Victoria - Improving soil health
- TIAR - Nutrient management program
- Sydney Water Catchment – Nutrient risk assessment
- National DGas project,
- Dairy feed database project,
- CSIRO National phosphorus review.

# 7. Industry-wide interpretation

The compilation of the individual data sets collected from the 41 dairy farms has enabled an overarching assessment of nutrient flows, transformations and efficiencies occurring within the Australian dairy industry. Importantly, the diversity of these dairy farms has facilitated the assessment of relevant relationships between nutrient loading rates, nutrient use efficiencies and surpluses (or deficits) and important management factors influencing dairy production. These include within-farm management factors such as dietary nutrient intake, and the location of dairy animals and resultant nutrient deposition in excreta, and whole-farm management factors such as stocking rates and reliance on imported feed.

This improved understanding of the factors influencing nutrient loading rates, nutrient use efficiencies and surpluses (or deficits) will enable more targeted mitigation strategies, improving both profitability and environmental outcomes on dairy farms. At an industry and government level, these relationships will assist in identifying current and future challenges facing the dairy industry, specifically relating to nutrient losses from dairy farms, and help shape strategic policy directions.

## 7.1 Key sources of nutrient exports and imports

Milk shipped off farm is clearly the major source of nutrients exported, making up more than 90% of the total annual nutrient exports. The nutrient exports of nutrients in animals is generally small, and although large-scale periodic animal exports do occur, the net exports in animal nutrients relates to removal of culled cows and calves, unless a significant change in the farm system is implemented. The export of nutrients in stockpiled manure or dairy effluent has the potential to remove significant amounts of nutrients but no participating farm shipped off any manure during the study period. Similarly, harvested and exported forages can also result in the net removal of significant amounts of nutrients, particularly N and K, though this occurred infrequently, with almost all harvested forage being consumed by animals within the same home-farm area.

Imported nutrients (N, P, K, S, Ca and Mg) are all contained in various products imported and exported and the relative contribution of any particular source varied widely between farms. Almost all of the identified inputs and outputs contribute significantly in at least some dairy operations.

**Nitrogen** In general, the single largest source of N imported on to dairy farms was inorganic fertiliser, either as a NPKS blends or urea. Imported grain or grain-based concentrates was also an important source of N, while hay, silage and by-products also all contributed very similar N imports on one farm or another. On some farms, particularly those with limited or no N fertiliser applications, N fixation by legumes was a major contribution. In particular cases, poultry manure, and municipal waste water also contribute a significant amount of nitrogen.

**Phosphorus** The most common source of imported P was inorganic fertilisers and alternative fertiliser products, such as rock phosphate and poultry manure. Imported feed was also an important contributor to total P imports, especially grains and grain-based concentrates and specific by-products with relatively high P concentrations.

**Sulphur** The most common source of imported S was inorganic fertiliser, though occasionally large amounts of S can also be imported in irrigation water or in soil ameliorants. Imported feed was also an important contributor to total S imports, notably in specific by-products and forage and grain.

**Potassium** In general, the largest source of K imported on to dairy farms was inorganic fertiliser, while by-products and forages such as hay and silage were also important sources. Grain and other fertiliser types also have the potential to contribute significantly to K imports on to dairy farm.

**Calcium and Magnesium** Calcium and magnesium imports on to dairy farms may be skewed due to some very large inputs resulting from applications of lime, or irrigation water with high cation concentrations. Calcium and Mg was also imported onto farms in important amounts in inorganic fertiliser and grain, and to a lesser extent in forages.

## 7.2 Nutrient emissions from cows

The efficiency with which dairy cows utilise feed nutrients to make milk (i.e. the ratio of milk nutrients exported to total feed nutrients consumed) significantly influences the amounts of nutrients excreted. The nutrients excreted from cows on a herd basis were determined from the difference between estimates of the total nutrient intake in the diet and nutrient exported in milk production.

On average across all the herds studied, almost half a kilogram of nitrogen was excreted daily by each cow, about 7 times the amount of phosphorus excreted. Therefore, for an average herd size of 250 cows over a 300 day lactation, 32.4 and 4.6 T of N and P are excreted by cows in particular places around a dairy farm.

Supplementation of pasture-based diets by farmers aiming to increase milk production results in greater dietary crude protein and phosphorus, potassium and sulphur intake. However, as nutrient intakes increased, so did excretion of these nutrients, associated with reduced efficiency of nutrient use.

High nutrient concentrations in excreta result in reduced nutrient recycling, due to high loading rates and uneven distribution of nutrients, and greater losses through volatilization, leaching and runoff processes.

Feed nutrient use efficiencies less than 20% are considered to be low, while 30 to 35% is considered above average (Chase 2003). On average across all the participating dairy farms, close to 80% of intake N and 75% of P were excreted. Despite this, on some farms as much as 35% of feed intake N was used to produce milk, although these high nutrient use efficiencies were not consistent across sampling times. While the ranges (minimum to maximum) in utilisation of nutrients, particularly N and P by Australian dairy herds are comparable to that cited internationally, there is scope to increase nutrient use efficiency and maximise milk production, per unit of nutrient input.

## 7.3 Nutrient loading rates from excreta in specific locations

The calculated nutrient excretion rates for the lactating herds on each farm at each survey visit were combined with farm-specific spatial and temporal cow movement data to determine nutrient deposition loads (kg/day) to locations around dairy farms. The largest amounts of nutrients were recycled by cows to grazed paddocks, but these were not uniformly returned, with accumulations occurring in specific parts of the landscape.

Overall, cows spent the greatest proportion of their time in pasture paddocks. As the reliance on imported feed increases, the proportion of time spent in other places such as feed pads increases. Paddocks closest to the dairy had a much greater proportion of time than those further away and subsequently had much greater nutrient loading rates. For example, the amount of nitrogen deposited in night paddocks was 1.5 times greater than that deposited in day paddocks.

Overall, nutrients excreted in holding yards were significantly greater than that deposited within the dairy shed, feed pads and laneways. Consequently, nutrients deposited in yards need to be managed carefully as, based on project data, some farmers only wash their yards once daily despite collecting excreta deposited in the dairy shed at each milking. Nitrogen excreted in yards is therefore particularly susceptible to volatilisation losses, based on the nutrient loads returned by dairy cows in this location.

Although slightly smaller mean nutrient loads were deposited in feed pads and holding areas, the mean depositions were 10.3 and 26.5 kg/day respectively. The deposition rates in these generally small areas constitute very high loading rates on a ‘per ha’ basis. Excreta in these areas are also infrequently collected and nutrients losses through gaseous emissions, runoff and leaching can be large.

Improved nutrient capture and recycling will depend on an understanding of the spatial and temporal patterns of animal movement and the implications for nutrient return and accumulation.

## 7.4 Nutrient balances and use-efficiencies on Australian dairy farms

The whole-farm nutrient balances determined for each participating dairy farm provide specific information relating to the various nutrient imports and exports, the potential surplus or deficit of N, P, K, S, Ca and Mg, and a measure of the relative conversion of imported nutrients into exported products. As such, they integrate farm-scale information into productivity and environmental performance indicators (Oenema et al., 2003).

It is important to note that the study was designed to describe nutrient flows and transformations across the breadth of the Australian dairy industry and enable the comparison of contrasting Australian dairy systems. We therefore used a stratified-random process to select the participating dairy farms, and not a random sampling approach. Nevertheless, the large number of farms included in this study, the recognition of key farm characteristics, and the diversity of systems selected have provided an industry-wide assessment of current nutrient status and environmental challenges at a range of scales. This is demonstrated by the similar range of dairy farm characteristics and the description of the median farm in this study (Table 4) and those describing the Australian dairy industry as a whole (Dairy 2010 Situation & Outlook - Annual Report 2010, http://www.dairyaustralia.com.au/Situation-and-Outlook).

Overall, nutrient imports onto dairy farms in Australia are generally much higher than those exported, in products. As presented in section 6.3.4, whole-farm **nitrogen** surplus determined in this study ranged from 47 to 600 kg N/ha and nitrogen use efficiency ranged from 14 to 50%. While these ranges in N surplus/ha and N use efficiency appear broad, they reflect the varying levels of intensity and inputs present in both Australian and international dairy production systems.

For example, a recent study in Western European titled 'Green dairy' involving 130 commercial dairy systems also found a similar range in N surplus/ha, with average regional N surpluses between 93 and 502 kg N/ha and nitrogen use efficiencies ranging between 19 and 40% (Raison et al., 2006). Similar ranges in N surpluses and N use efficiencies have been measured in more regionally focused studies in the USA (Rotz et al., 2006; Histov et al., 2006), The Netherlands (van deer Meer, 2001 ), New Zealand (Ledgards et al., 2004) and in Western Australia (Ovens et al., 2008).

Whole-farm **phosphorus** surplus across the 41 independent dairy farms ranged from -7 to + 133 kg P/ha. Five of the 41 farms were in net P deficit, while 8 farms had high P surpluses of > 50 kg P/ha. Interestingly, 3 of the high P surplus farms were 'organic', with high P surpluses reflecting a relatively low P export and high P import, primarily in the form of rock phosphate. The overall phosphorus use efficiencies across the 41 dairy farms ranged from 6 to 158%. When the 5 high P surplus farms were excluded, the range of P surpluses determined in this study were again similar to the 4 - 36 kg P/ha range determined within the Green Dairy project (Raison et al., 2006).

It is clear from these balances that P inputs are very different between farms. Negative P balance farms have greater exports of P than imports (in any form), and are subsequently 'mining' P from the system, presumably the soil. This approach appears to be warranted in some systems where existing soil test levels were above recommended thresholds of adequacy (Gourley et al., 2006), and existing nutrient reserves can be utilised without affecting potential production. However, other farms were also continuing to mine P, but the low soil P reserves were likely to be limiting potential pasture production and therefore milk production.

In contrast, most farms had a net P surplus. Continuing high P surpluses will result in increasing soil P levels, which is well supported by the high and sometimes extremely high soil P test values measured on these farms (section 6.3.6). However, due to varying P fixation processes in soils, and the heterogeneous distribution of manure deposition, relatively small P surpluses may maintain existing P supply and will not necessarily mean an increase in soil test levels.

Whole-farm **potassium** balances indicated that all the dairy farms in the study were in potassium surplus, ranging between 13 and 452 kg K/ha. The median K surplus for the 41 farms was 74 kg K/ha. The potassium use efficiency across the 41 dairy farms ranged from 9 to 48%, with a median value of 20%. The wide range in K surplus/ha and K use efficiency principally reflect the varying levels (and relatively high concentration of K) in fertiliser and feed inputs in these Australian production systems. Whole-farm **sulphur** balances were closely associated with P balances, with S surpluses ranging from -1 to 184 kg

S/ha, and a median value of 27 kg S/ha. The overall sulphur use efficiency across the 41 dairy farms ranged from 6 to 110%, with a median value of 21%.

While there are clearly substantial differences in K and S inputs and balances between dairy farms, the majority of dairy farms, had substantial K and S surpluses, well in excess of any agronomic recommendations (Gourley et al., 2006).

The current and ongoing reliance on imported feed on many dairy farms has resulted in substantial K (and to a lesser extent S) imports. In contrast, the relatively low concentration of K and S in milk means that high rates of K and S are being deposited by animals in excreta. This potentially large source of import is rarely considered in light of fertiliser requirements and K surpluses in particular are often exacerbated by further fertiliser applications.

The imbalance in K and S determined in this study is supported by the high and sometimes extremely high soil K and S test values measured on these farms (section 6.3.6). While both K and S can accumulate in soil and act as a future pool in a readily available form, excess K and S can reduce animal production and cause severe metabolic disorders. Many paddocks, particularly those closest to the dairy shed, receiving shed effluent and used for feeding animals, have very elevated nutrient levels overall. Of particular concern is the very high soil K levels which when combined with luxury uptake in pastures, can cause significant and sometimes fatal metabolic disorders such as grass tetany and milk fever.

The whole-farm **calcium** and **magnesium** balances indicated a huge array of surplus values, ranging from 1 to 1327 kg Ca/ha, and 6 to 222 kg Mg/ha. The median surplus values were 84 kg Ca/ha and 32 kg Mg/ha. Where farms had highly elevated Ca and Mg surpluses, specific sources of farm imports were largely responsible.

Some of these imports occurred due to informed management decisions, such as the application of lime, while other imports were as a consequence of other activities, such as the use of bore water for irrigation with high Ca and Mg concentrations, and were therefore outside the control of the farmer. Consequently, the Ca and Mg efficiency estimates, which for example ranged from 1- 97% for Ca, provide limited insight into improving management practices. It is therefore important to identify the sources and reasons behind such significant imports of C and Mg, other than the more normal amounts imported in feed and fertiliser.

As is normal practice, the whole-farm nutrient balances determined in this project have not directly quantified (environmental) losses through runoff, leaching, denitrification or volatilisation pathways. However, the transient nature of **nitrogen** on dairy farms, and the generally 'steady-state' of soil nitrogen pools in established pasture soils, means that farm-scale surplus N is well recognised as an estimate of total N lost to the broader environment (Oenema et. al, 2003).

In contrast, whole-farm surpluses, deficits and efficiencies of P, K, S, Ca and Mg use should also be considered in light of the overall productivity of the dairy farm, existing nutrient stores available (i.e. soil fertility levels) and their potential rate of decline. For example, surplus **phosphorus** may accumulate in surface and sub-surface soil through soil P fixation processes, and unless the associated soil P storage capacity and potentially losses are known, P surplus estimates alone should not be considered as an immediate and directly quantifiable loss.

The **potassium** and **sulphur** losses are not usually associated with environmental impacts and therefore K and S balances and efficiencies of agronomic use in determining nutrient removal and accumulation and assisting in determining potential fertiliser requirements. **Calcium** and **magnesium** losses and imbalances are however sometimes used as environmental indicators for soil acidification (Ca) processes and indicators for potential animal metabolic disorders. The Ca and Mg balances determined are therefore useful in demonstrating associated Ca and Mg flows and removal within different dairy farming systems, and may assist in informing management decisions about soil amelioration with lime or gypsum.

**Confidence in nutrient balance outputs**

The integrated uncertainties around the various 'grouped sources' of nutrient imports or exports were in general relatively small and usually less than 10% of the estimated mass of nutrient imported or exported. An exception to this was N imported with N fixation, where estimates are widely recognised as having a high degree of uncertainty and reflected in its high predicted variance. However, as these collective variance estimates were further integrated and the relative weighting from individual contributions included, the total uncertainty of the balance estimates were less that 11% overall and most commonly less than 5%.

Despite some individual sources of nutrients having high levels of uncertainty (i.e. some feed nutrient concentrations, N fixation), multiple measurements and the relatively small contribution of some individual sources, reduced this impact as a proportion of the overall variance (for more details see 'central limit theorem').

Correspondingly, the determined uncertainties around quantification of individual sources of imported and exported nutrients and the overall uncertainty for the whole-farm nutrient balances provided a high degree of confidence in the estimates of surplus/deficit and nutrient use efficiencies, and subsequently the corresponding and implications. This analysis of uncertainties also helped identify where further information is required and therefore further research needs.

**Benefits of directly measuring nutrient concentrations rather than using published values**

There are often good reasons to use published data on nutrient concentrations (book values) or established algorithms to estimate nutrient concentrations and nutrient loads. This is appropriate where there is little variation in concentrations in nutrient sources and a high level of confidence in the provided information. For example it is generally accepted that book values provide a reliable assessment of nutrient concentrations for inorganic fertiliser, livestock and milk (Oenema et al., 2003; Mulier et al., 2003). Additionally, the use of book values may be appropriate when data is difficult to directly measure, as in the case of atmospheric deposition for example, or where the contribution is likely to be small relative to other components, such as from bedding or irrigation water. In contrast, nutrient concentrations in grains and forages can vary substantially and may have a large impact on the resultant farm-gate nutrient budget outcomes.

In this current study we substituted book values for measured values in the nutrient balance model on 5 contrasting dairy farms to assess the impact on the resulting N balance and efficiency estimates. This was done for major inputs/outputs and a selection of minor ones, according to availability of data (section 6.3.4 and Table 25).

Overall there were only relatively small changes to the estimates of nutrient imports and balances, and negligible changes in nutrient exports. Not surprisingly, the magnitude of these differences was largely driven by the relative contribution from imported (and exported) feed onto each farm, as other key nutrient sources such as fertiliser imports were already determined from published nutrient concentrations. However, this comparison also indicated that particular farms and feed types can have substantially different nutrient balance estimates, when using book values or farm-based data. For example farm 32 had an N surplus of 351 kg N/ha or 315 kg N/ha (a 10.3% difference) depending whether farm-based or book values were used in the calculations.

The detailed analysis of nutrient concentrations amongst different dairy feed types indicated that when estimating some nutrient contents in some feed types, greater accuracy would sometimes result from segmenting data based on regions, irrigation status, organic status, sampling time and even individual farms (section 6.3.3). In contrast milk nutrient concentrations, no doubt due to the biologically regulated nature of milk production, were highly consistent and supported the use of a standardised set of national nutrient concentrations.

While using the most accurate data is desirable, the scarcity of nutrient concentration data for different regions or sampling times, and the difficulties in practically implementing non-uniform nutrient concentrations for particular feed types or other nutrient sources, may prevent this degree of detail.

Additionally, when interpreted in the broader context of an overall whole-farm balance, these nutrient concentration differences may become less important. The purpose of estimating nutrient balances will, to a large extent, dictate the accuracy of the balance outputs required and therefore influence the balance between practicality and precision.

An important exception to the use of industry-wide book values however should be made when estimating nutrient concentrations on organic dairy farms. A key point of difference between organic and conventional systems is the specifically selected and used feeds and fertilisers that are required to meet organic standards. The measured differences in specific nutrient concentrations in this project, when comparing the organic and conventional dairy farms, notable P, N and S, would support the use of specific nutrient concentrations for a range of feed and fertiliser types in organic system. This could easily be achieved by using a separate set of organic nutrient standards.

**Standardisation in the design of Australian dairy nutrient balance calculations**

There are a surprising number of modifications, assumptions, exclusions and inclusions associated with individual approaches used nationally and internationally, to undertake dairy nutrient balances. While dairy farm nutrient balances almost always include inputs such as feed, milk and fertiliser, other nutrient sources such as bedding, N fixation, atmospheric deposition, and irrigation may not be included. Exclusions are often justified when they are not relevant to the farm operations, (i.e., no irrigation, no legumes grown), or a relatively small overall contribution (i.e., bedding, atmospheric deposition), while others may rely on the assumption of steady state conditions (i.e., animal numbers and mass) (Gourley et al., 2007). These assumptions are sometimes incorrect, leading to inaccurate estimates of nutrient surplus and efficiency, and also do not allow for a more standardised and flexible farm-gate assessment. While most nutrient budgeting includes N fixation from legumes, it is interesting to note that the widely cited MINAS system from the Netherlands was primarily focussed on manure, feed and fertiliser, and did not account for N fixation (van der Meer 2001). Another widely used nutrient balancing program 'N-CyCle' does not account for bedding, irrigation, or atmospheric deposition (Wattiaux 2001).

The accounting for nutrients project has identified that there is a wide variety of sources that contribute to nutrient inputs and ultimately to nutrient use efficiencies and surpluses (or deficits). These different nutrient sources demonstrate the diversity of farming systems and the range of necessary nutrient inputs required to accurately determine nutrient balances in Australian dairy production systems.

While imports of all major nutrients in sources such as grain, forages and fertilisers are well recognised and clearly important, some significant loads of nutrients can also be imported in sources such as irrigation water, feed by-products, soil ameliorants, and animals. The importance of N fixation was also demonstrated in this study, particularly where farms are not using high amounts of N fertilisers.

Even though specific dairy farm systems such as organic farms did not use inorganic manufactured N or P fertilisers, the importance of alternative fertiliser forms demonstrated the need to include all fertiliser and soil ameliorant types within the whole-farm balances. It is interesting to note that the organic farms imported relatively high rates of P, K and S on a per-ha basis, when compared to the conventional farms.

An important part of standardising nutrient balances in Australia will be to ensure that the corresponding land-base used to estimate per-ha nutrient loads and surpluses/deficits is well defined and uniformly applied. We found that defining the land area that contributed to the production system was not a straight forward task. Many of the dairy production systems involved in this study had separate land units which contributed to milk production or feeding dry stock, and providing forage or crops. We recommend that the 'contact-land' area be used as the basis of standardising the land area and nutrient balance information. We defined the contact-land base as the 'Total land area contacted by lactating and dry cows within the home farm, including grazed paddocks, laneways, feeding areas, calving areas, and sick paddocks. This was selected as all dairy farms process this land base, it is the principal management area contributing to milk production, it is the major land area where nutrient cycling and deposition is occurring, and it is the likely area contributing to the greatest levels of nutrient accumulation and losses.

## 7.5 Relationships between key farm management factors and nutrient use efficiency, balances and surpluses

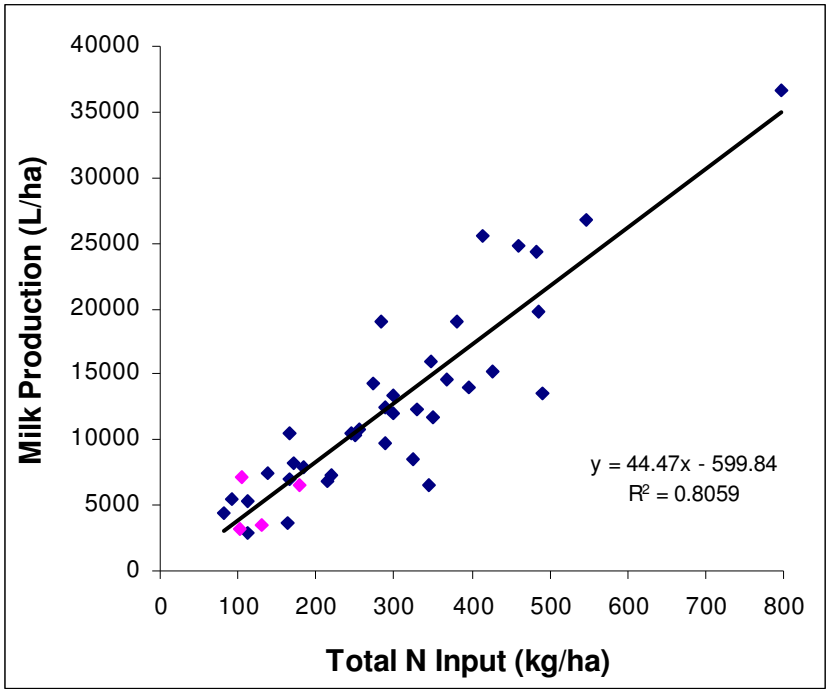
The diversity of these dairy farms has facilitated the assessment of relevant relationships between nutrient loading rates, nutrient use efficiencies and surpluses (or deficits) and important management factors such as milk production, stocking rate and reliance on imported feed. At a farm level, this improved understanding will enable more targeted mitigation strategies, improving both profitability and environmental outcomes. At an industry level and government, these relationships will assist in identifying current and future challenges facing the dairy industry, specifically relating to nutrient losses from dairy farms, and help shape strategic policy directions.

A greater emphasis has been directed towards nitrogen inputs, balances and efficiencies than the other nutrients studied. This is because of the greater magnitude of N flows and surpluses at the farm scale, the transient nature of N in dairy farms, the recognition of measured surplus as a quantifiable loss to the broader environment, and the strong relationships between various farm management factors and productivity. This does not however reduce the importance of other nutrient balance assessments, in terms of an environmental performance indicator (particularly in the case of P) and its usefulness in improving management decisions such as fertiliser and effluent applications.

**Nitrogen** There was a very strong relationship between the total milk production and the total amount of nitrogen imported onto the farm (Figure 34). While recognising the potential associative relationship between other important drivers of milk production such as grain importation, this relationship demonstrates the interdependency of milk productivity and N inputs.

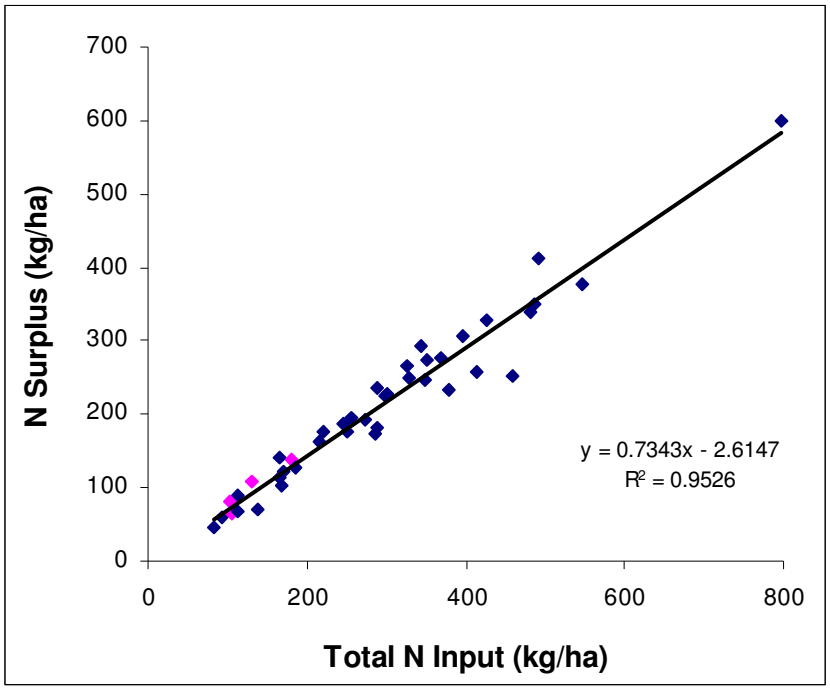
The average national milk productivity of around 12,000 L/ha (Dairy Australia, 2010) was mirrored by the productivity of the 41 dairy farms in this study (median 10866 L/ha and 12388 L/ha). Twelve dairy farms had milk productivity of < 7200 L/ha, and while all 4 organic farms were within this low production category, their production was comparable to other low stocked and low input conventionally managed dairy production systems.

**Figure 34. The relationship between total milk production and total nitrogen imports for the 41 participating dairy farms. The 4 organic farms are highlighted.**



The whole-farm nitrogen surplus determined in this study ranged from 47 to 600 kg N/ha and was strongly related to the level of N inputs (Figure 35). The majority of this imported N was in imported feed and fertiliser. The slope of this strongly linear relationship provides an industry estimate of the proportion of N not exported in product (73.4%). Therefore a corresponding overall industry estimate of N use efficiency equals the equivalent proportion of exported (26.6%). This estimate corresponds well this the average N use efficiency of 28.1% determined for the 41 dairy farms in this study.

**Figure 35. The relationship between total nitrogen imports and nitrogen surplus for the 41 participating dairy farms. The 4 organic farms are highlighted.**




The determined nitrogen surplus was also strongly related to the level of milk production for each farm (Figure 36). As milk production increases per ha, so does the N imported and corresponding N surplus. The slope of this linear relationship (0.0145) provides a national industry estimate of the N surplus (kg) per litre of milk produced. This equates to a net N surplus of 14.5 g of nitrogen per litre of milk produced. A very similar linear relationship was described in the EU Green Dairy project (Raison et. al., 2006) with milk production ranging from 3000 to 50,000 L/ha, N surplus ranging from 70 to 745 kg N/ha and a corresponding slope of 0.0122 (12.2 g N/L).

The 41 dairy farms investigated had contrasting feed production systems with the calculated reliance on imported feed ranging from 3 to 66% of the total dietary metabolisable energy. Interestingly there was no significant relationship between the N surplus/ha and the reliance on imported feed (Figure 37). This may be explained by defining the systems and the potential N use efficiencies, loss pathways and potential for N recycling. For example, it would be expected that a major source of N input on farms with low reliance on imported feed would be from N fertiliser used to drive greater home-grown pasture and crop production. Conversely, farms with high feed N inputs would correspond with more complex feeding systems and animals spending a larger proportion of time on feed pads. Each system would initially specific N loss pathways, i.e. leaching and volatilisation losses from fertiliser N; N volatilization losses in urinary deposition on feed pads and effluent storage. However as the cow consumes this feed, be it imported or home-grown, the issues associated with N intake, the use efficiency in conversion of feed to milk, the places of manure deposition, and the opportunity for recycling of excreted N, are common, and influenced by 'between farm management variation' rather than a single factor such as reliance on imported feed.

**Figure 36. The relationship between total milk production and total nitrogen surplus for the 41 participating dairy farms. The 4 organic farms are highlighted.**

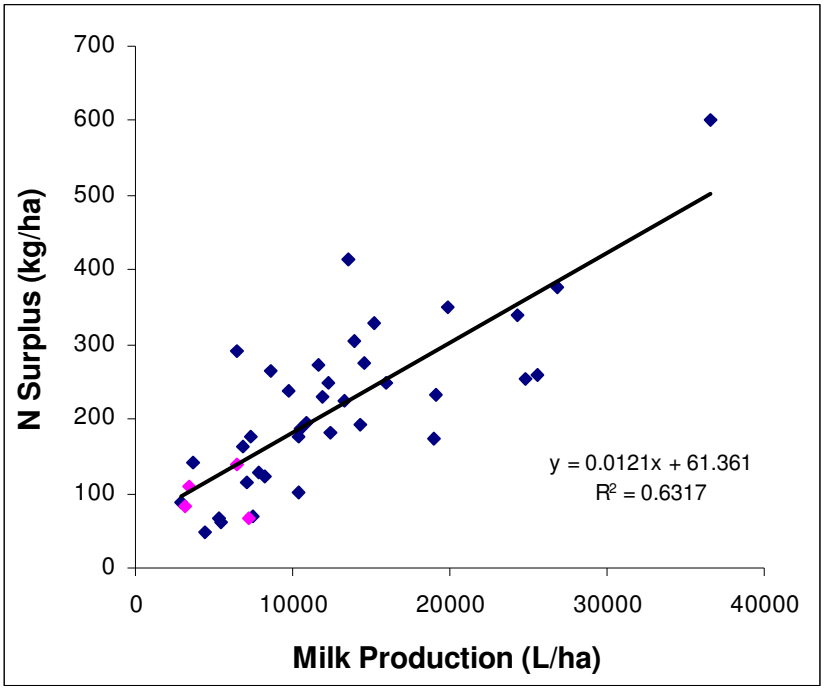
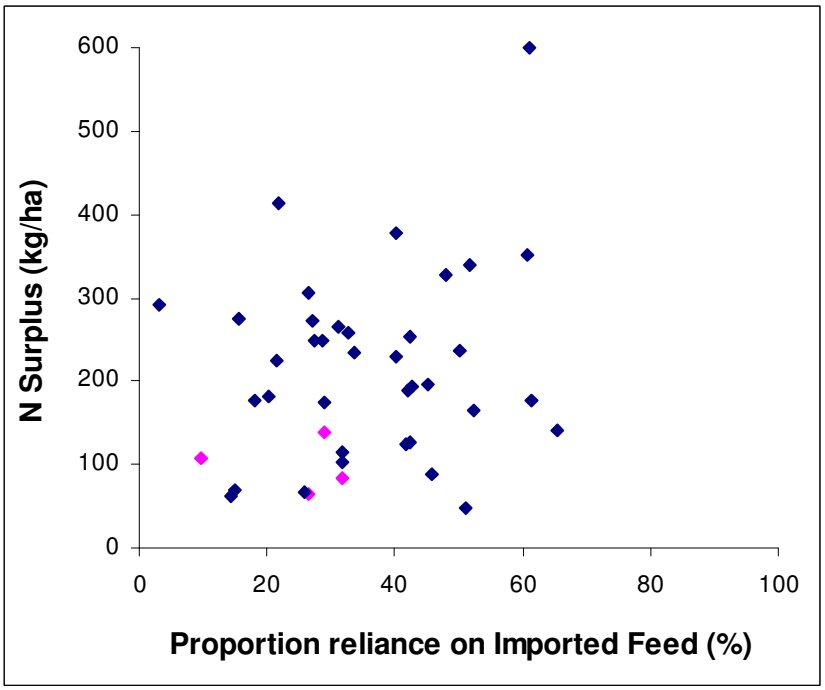
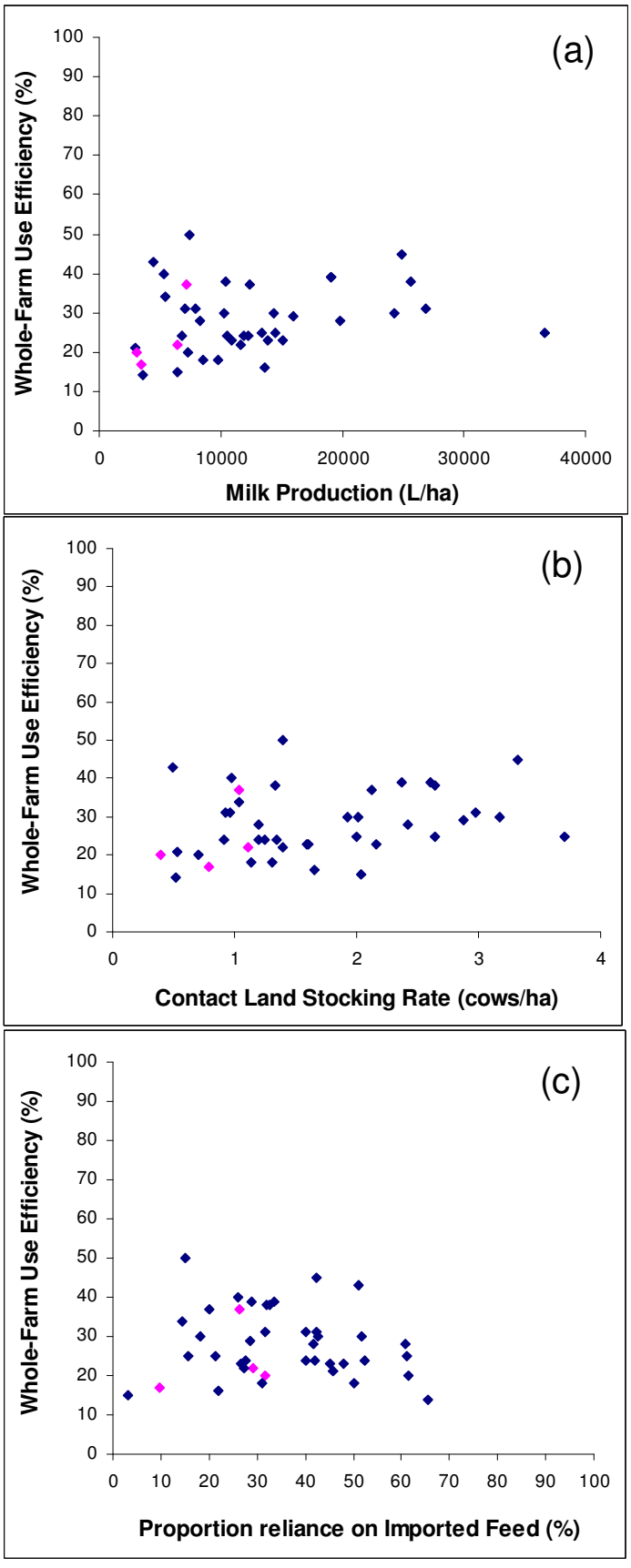



**Figure 37. The relationship between total nitrogen surplus and reliance on imported feed for the 41 participating dairy farms. The 4 organic farms are highlighted.**




Overall, the nitrogen use efficiency determined on the 41 dairy farms ranged from 14 to 50%. Basic characteristics of the dairy farms such as milk production per ha (Figure 38a), stocking rate (Figure 38b) and reliance on imported feed (Figure 38c) were not good predictors of overall N use efficiency. This suggests that within-farm management factors such as the effectiveness of recycling animal excreted nutrients, timing and rate of N fertiliser applications, as well as soil and climatic factors, are more likely to be key drivers of N use efficiency, than prescriptive farm characteristics.

**Figure 38. The relationship between Whole-farm N efficiency and total milk production (a), stocking rate (b), and reliance on imported feed (c) for the 41 participating dairy farms. The 4 organic farms are highlighted.**



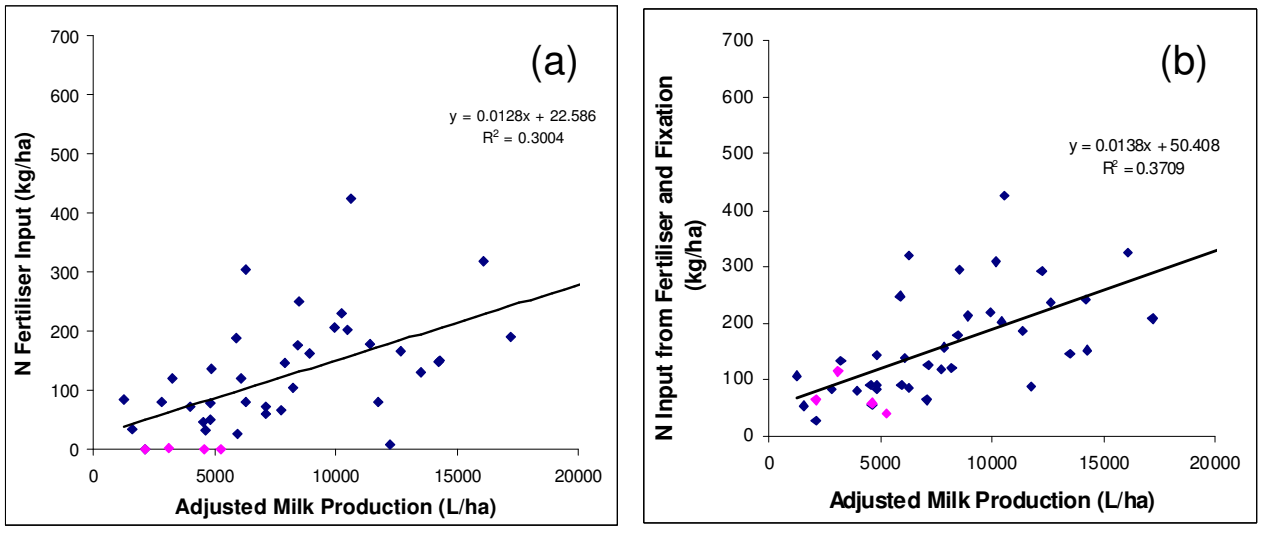
Whole-farm nitrogen use efficiency on dairy farms appears to be limited by the biological potential of cows to transform feed N into milk and for pastures and crops to utilise applied fertiliser and recycled nitrogen in manure (Powell et al., 2010).

The variability in the N efficiencies determined within this study and internationally, as well as the potential N use efficiencies that have been determined under experimental conditions, suggest that there are substantial opportunities to improve N use, resulting in enhanced productivity and reduced N losses to the broader environment.

For example, the effectiveness of N fertiliser applications was investigated from the relationship between total fertiliser N applied and 'adjusted' total milk production. The adjustment was a discounting of the total milk production by subtracting predicted milk production from purchased imported feed, as the purpose of applying N fertiliser to grow feed for milk production. While there was a significant relationship between N fertiliser imported and adjusted milk production (Figure 39a), there was a high degree of variation around the predicted regression line. For example, the average annual N fertiliser application of 150 kg N/ha, resulted in a milk production range of around 5000 to 15000 L/ha. While the importance of factors outside of management control, such as climate and soil characteristics, will be influencing the efficiency of N use from applied fertiliser, there is likely to be substantial improvements in management of applied N, such as timing and application rates, as well as the efficiency of utilisation of grown pasture and crops.

Five dairy farms involved in the study, including the 4 organic farms (Figure 39a) and one conventional farm did not apply any N fertiliser during the monitored production period. These farms appear to fall outside the predicted relationship and suggest alternative sources of N are influencing milk production other than fertiliser N. A review of the individual whole-farm balances indicated that N fixation was a significant source of imported N on all of these farms. The subsequent inclusion of estimated N fixation with total N fertiliser (Figure 39b) demonstrated the improved relationship and the potentially important role that N fixation can play on particular dairy farms as a cost effective N source.

**Figure 39. The relationship between adjusted milk production from home-grown feed, total fertiliser N imported (a) and total fertiliser N imported plus N fixation (b) for the 41 participating dairy farms. The 4 organic farms are highlighted.**

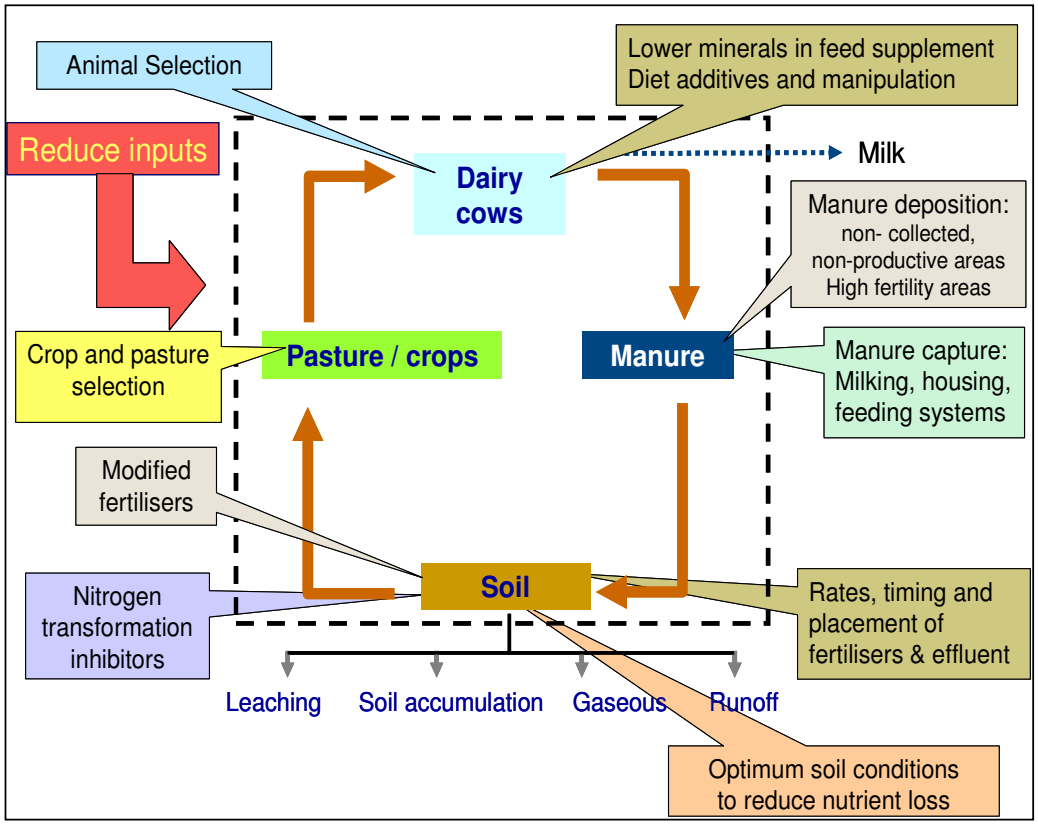



On a whole-farm basis, substantial increases in nitrogen use efficiency and reductions in N surplus and environmental N loss have been demonstrated by adopting current 'best-practice' recommendations (Appendix 8). Figure 40 indicates the major components of N cycling within dairy farms and potential intervention strategies which may impact on N use efficiency. For example, over a six-year period, dairy farms in The Netherlands were able to increase whole-farm N use efficiency (from 15% to 30%) through improved N fertiliser applications and effluent management (Groot et al., 2006). Other successful

strategies to improve whole-farm N use efficiency have included feeding more dietary fibre and less crude protein (Kohn et al., 1997), and N losses through ammonia volatilisation (Oenema et al., 2009).

However, simple and effective assessment methods are needed to understand the potential efficiency of N use in our dairy systems and to set realistic goals for improved N efficiencies and N surpluses. A key outcome from this current project has been the blend of scientific and farmer knowledge through collaborative on-farm research. For example the 'snap-shot' assessments of N use during the farm monitoring program have provided important industry-based information which can be used to determine benchmark N use efficiency values. However further work is needed to better understanding of how and why on-farm management decisions impact actual N use efficiencies and to quantify the actual productivity gains from capturing more N in productivity drivers. This information can then be used to develop and apply recommendations that have a high probability of being implemented on dairy farms.

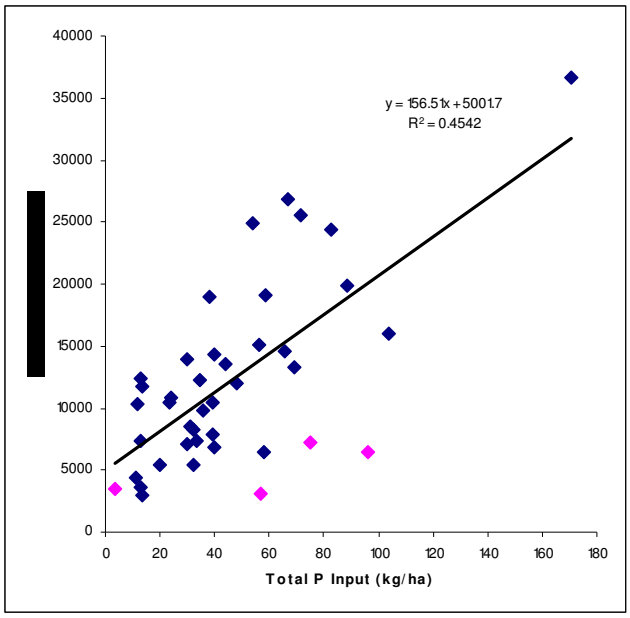
**Figure 40. Components of nitrogen cycling within dairy farms and key points of intervention to increase the efficiency of N use**




**Phosphorus** While whole-farm phosphorus surplus can be impacted significantly by P imported in feeds, notably in by-products, the most common source of imported P was inorganic fertiliser, mainly as highly soluble NPKS blends but also in alternative fertiliser products, such as rock phosphate and poultry manure. In general the farms which did not apply P fertiliser in any form had very low P surpluses (< 10 kg P/ha) or deficits.

Unlike the strong relationship between nitrogen import and milk production, there was a poor correlation between total milk production and total amount of phosphorus imported onto the farm (Figure 41). The wide variation determined between imported P and milk production is well demonstrated at an imported P level of around 50 kg P/ha, where milk production varying between 3000 and 27000 L/ha. Conversely, farms with low milk production per ha, including the 4 organic dairy farms, had varying P imported, ranging between 3 and 93 kg P/ha.

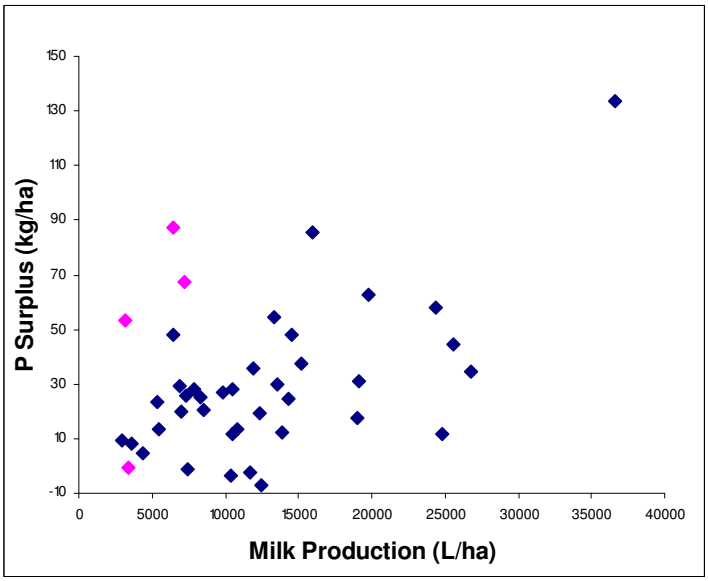
**Figure 41. The relationship between milk production and total P imported for the 41 participating dairy farms. The 4 organic farms are highlighted.**




The determined phosphorus surplus was also poorly related to the level of milk production for each farm (Figure 42). While the dairy farmers with the highest milk production had the highest P surplus per ha, the remaining farms had highly variable P surpluses. Of note are 3 of the low production organic farms, which had P surplus values above 50 kg P/ha. In contrast, farms with milk production around the national average (12,000 - 15,000 L/ha) had P surpluses ranging from -8 to +90 kg P/ha).

Similarly to N fertiliser, the effectiveness of P fertiliser applications was investigated from the relationship between total fertiliser P applied and the 'adjusted' total milk production (Figure 43). In contrast to N fertiliser however, there was a poor relationship between imported P fertiliser and milk production attributed to home-grown pasture and crops. Of note is the high range in adjusted milk production when P fertiliser inputs were small (< 10 kg P/ha).

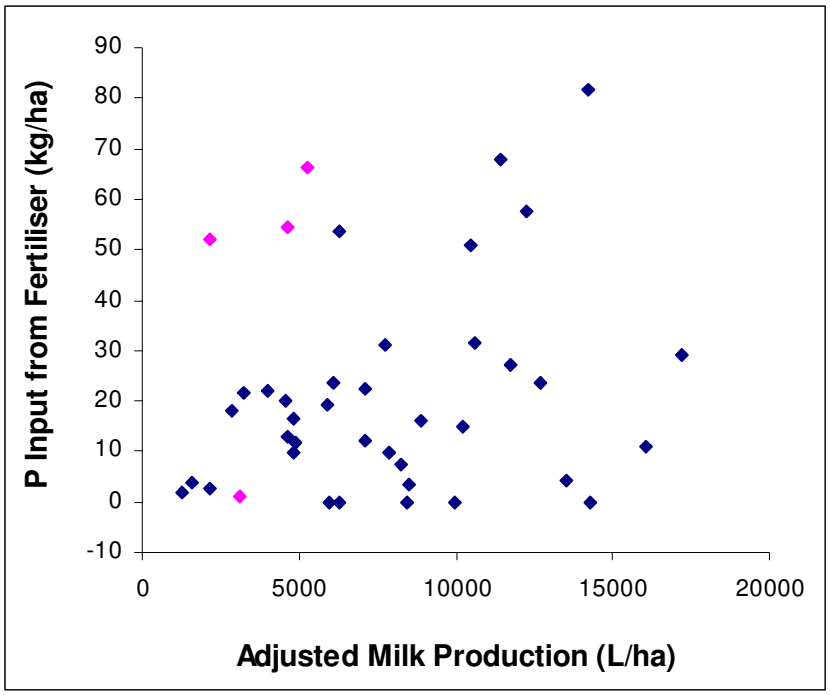
**Figure 42. The relationship between milk production and total P surplus for the 41 participating dairy farms. The 4 organic farms are highlighted.**



The lack of a defined relationship between P imports and productivity is supported by the generally high levels of soil P measured across a broad range of dairy farms. The mean soil level for Colwell P and Olsen P, determined across the conventional dairy farms was 127 and 36 mg/kg respectively, well above the agronomic soil test targets of 45 – 90 mg/kg for Colwell P and 20 mg/kg for Olsen P, recommended for pasture production (Gourley et. al., 2006). Under these high soil P conditions, additional pasture and crop production from the application of P fertiliser would not be expected and therefore neither would an associated increase in milk production from home grown feed. Moreover, the milk production from farms with low or no P imports but with adequate levels of soil P, suggest that these soil P reserves can be utilised without a resulting decline in milk production.

Despite the generally low soil P levels on the organic dairy farms, and the relatively high P surpluses caused by the use of rock phosphate on some of these farms, there did not appear to be a resultant increase in home-grown feed and subsequently in milk production. These results support the low levels of P availability from rock phosphate and the limited pasture (and crop) yield responses that may occur unless soil and climatic conditions favour its use (Bolland et. al, 1988).

**Figure 43. The relationship between adjusted milk production from home-grown feed and total fertiliser P imported for the 41 participating dairy farms. The 4 organic farms are highlighted.**



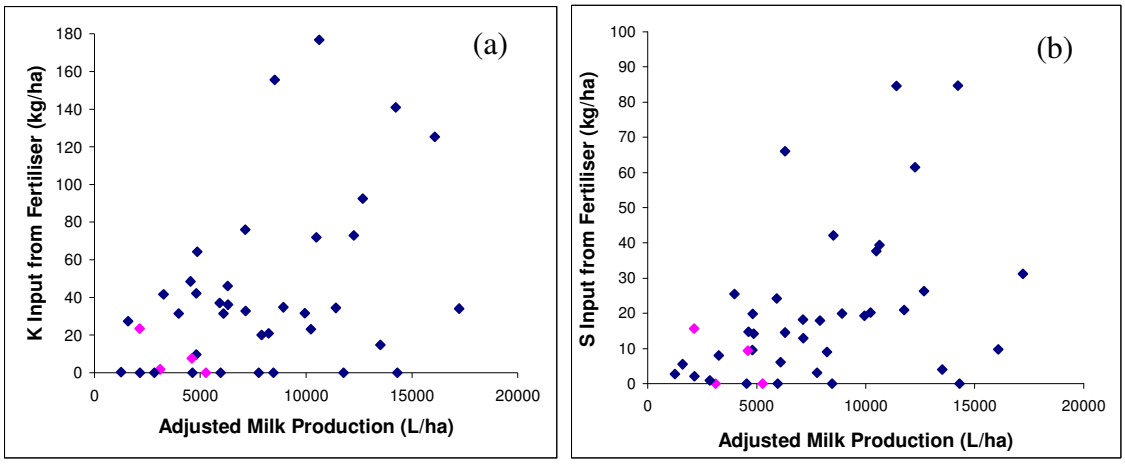

**Potassium and sulphur** While sulphur imports were generally influenced by fertiliser imports, the largest source of imported potassium was generally from hay and silage. Like phosphorus, farms that did not apply inorganic fertiliser generally had lower S surpluses (< 10 kg S/ha) or deficits. In contrast, there was significantly more potassium imported overall with all farms, including low input systems, having K surpluses.

Both K and S imports generally increased with increasing milk production, reflecting greater imports of both feed and fertiliser as dairy systems become more intensive. However like phosphorus, these relationships were poorly defined.

The effectiveness of K and S fertiliser applications was also investigated from the relationship between total fertiliser K or S applied and the 'adjusted' total milk production (Figure 44a and b). Again, there were no apparent relationships between imported K or S fertiliser and adjusted milk production (milk production attributed to home-grown pasture and crops). The lack of defined relationships between K and S imports in fertiliser and adjusted milk production is explained by the generally high levels of soil K and S measured on most of these dairy farms.

The mean soil levels for K (Colwell K) and S (KCl-40), determined across the conventional dairy farms were 296 and 23 mg/kg respectively, well above the agronomic soil test targets of 120 – 180 mg/kg for K and 10 mg/kg for S recommended for pasture production (Gourley et. al., 2006). Based on the generally high soil K and S levels on these dairy farms (with the exception of some of the low input systems), we would not expect increasing pasture and crop productivity and therefore increasing milk production from fertiliser K and S applications.

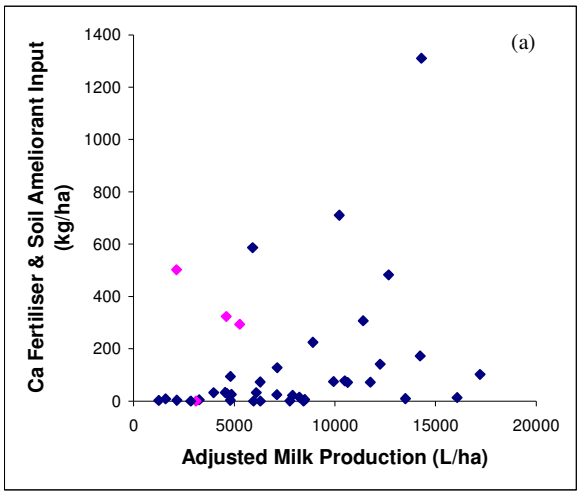
**Figure 44. The relationship between adjusted milk production from home-grown feed, total potassium (a) and sulphur (b) fertiliser imported, for the 41 participating dairy farms. The 4 organic farms are highlighted.**

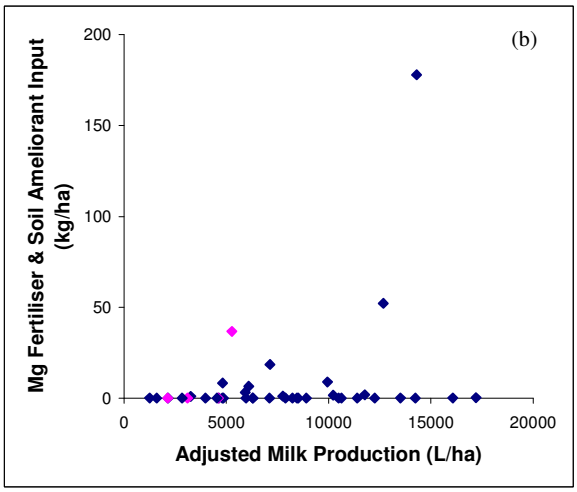



**Calcium and magnesium** While there may be large amounts of calcium and magnesium imported onto dairy farms, they are rarely applied in fertilisers or soil ameliorants with the aim to overcome pasture or crop Ca and Mg deficiencies, but rather to address indirect soil chemical or physical constraints such as soil acidity or structural decline. The very high loads of Ca and Mg imported onto some dairy farms in this study reflected the Ca and Mg content of lime, or were as a consequence of high concentrations in bore water.

It is not surprising therefore to find that there were no apparent relationships between total Ca or Mg imported onto farms in fertiliser and soil ameliorants and 'adjusted' total milk production (Figure 45). However, the magnitude of the fluxes of Ca and Mg in feed, standard fertilisers and milk under 'normal' dairy farm conditions (when large importations of Ca and Mg are excluded), serve to quantify imports, exports and balances and provide a benchmark against which remedial applications of Ca and Mg can be assessed.

**Figure 45. The relationship between adjusted milk production from home-grown feed, total Ca (a) and Mg (b) imported in fertiliser and soil ameliorants, for the 41 participating dairy farms. The 4 organic farms are highlighted.**


(a)
Ca Fertiliser & Soil Ameliorant Input (kg/ha)
1400
1200
1000
800
600
400
200
0
0
5000
10000
15000
20000
Adjusted Milk Production (L/ha)


(b)
Mg Fertiliser & Soil Ameliorant Input (kg/ha)
200
150
100
50
0
0
5000
10000
15000
20000
Adjusted Milk Production (L/ha)

## 7.6 Benefits and challenges of national on-farm research

The Accounting for Nutrients on Australian Dairy Farms project was a large, complex and ambitious national project. Commitment was sort from a large number of participating organisations, including industry stakeholders and state government agencies in the form of collaboration agreements. The commitment varied between cash funding support, in-kind contributions of staff time and more passive commitments such as attending meetings (appendix 1).

A key part of the program focused on collecting information from 44 individual participating farms and involved around 22 staff. The players associated with the on-farm research program were principally sourced from State supported agencies. Ownership of the overall project and the level of commitment varied amongst participating state agencies and their staff. As the project management team was almost entirely based within DPI Victoria, there was a feeling of the project being a Victorian project. Staff at times had competing roles, with other tasks sometimes taking priority.

The project management also had no direct line management of collaborating staff and the selection of replacement staff. Staff allocated to the project were sometimes less than ideally skilled to undertake the tasks at hand, and the lack of continuity lead to further training needs and potential inconsistencies in sampling and processing. Despite this, the national on-farm team members' staff were highly committed. An important part of this was the individual loyalty and goodwill provided by participating individuals. Relying on the commitments of participating agencies carried some risks. The 3 training and delivery 2-day workshops, where on-farm team members were able to meet, learn new standardised techniques and importantly share experiences and data, were an important part of gaining commitment this and therefore a successful completion for data collection and sampling.

The commitment of the farmers was very strong throughout the 2 years that the project teams interacted with them. This may well have been because they wanted to be involved in something they saw as important and also because the originally nominated farmers understood the commitments associated with the project. A number of participating farmers already had strong relationships with some of the project on-farm team players, and proven track records as collaborators. In most cases, the record keeping undertaken by the farmers was of high quality. The regular contact with the farmers at least every 3 months, no doubt assisted in establishing good lines of communication and expectations from both sides. These attributes were vital in ensuring that the expectations from both sides were clearly understood. The final visit in December 2009 with the participating farmers focused on the provision of completed data and interpretation. This visit marked the completion of the farmers' contribution to the project and they received some very useful products in return. This information has also been used in farmer field days on the participating farms and other extension related activities.

The level of complexity and time requirements for data collection was underestimated by the project management team in the development stage of the project. The project team had planned to collect two years data from each farm; however, it became apparent during the mid-term review that this task was overly ambitious and demanding, particularly for the on-farm teams and the farmers. The decision to consolidate the monitoring period to 1 full year was based on a detailed analysis of the data and its ability to meet the projects needs. An important benefit to the consolidation of the monitoring period was that the attrition rate of participating farmers and on-farm team members was very low, with only 3 farms unable to complete the data collection process out of the initial 44 farms. Feedback from the farmers and the on-farm team members indicated that many farmers would have withdrawn from the project, and there would have been much greater problems with incomplete and inaccurate records, if farm monitoring program, had continued into another year.

There is little doubt that the national approach adopted in this project has had major benefits in terms of being able to collect diverse and reflective national dairy industry information. Many projects tend to study a limited range of systems, which in turn has the potential to limit the conclusions that can be drawn. Conversely, conclusions are drawn that are not applicable to a large portion of the national industry. Along with the ability to collect national data, this project had the ability to bring key scientists in the field together. This had massive investment benefits in terms of in-kind contributions as well as more subtle and longer term benefits. It allowed expertise to be harnessed in a truly collaborative fashion that only generally happens with a very limited number of project collaborators.

The National Project was challenging from a management perspective and required strong and collaborative leadership. The role of an independent project manager was highly beneficial to instil a national and collective team spirit. There was also a need for strong, committed and highly skilled core team members to serve as the engine room of the project and the reference point for all other participants. These roles were well served by key players in DPI Victoria in well defined roles, but other involved other highly committed senior players in other states. While there are additional costs in operating a national approach, such as the need participants to travel around Australia, and also additional workshops, etc, these costs are more than offset by the benefits that are accrued.

Many of the identified risks to the Accounting for Nutrients project were associated with the on-farm monitoring program, most notably data collection and collation, as well as sample handling and provision to a central location for processing and analysis. While these risks were well managed overall, it is worthwhile sharing some of these potential risks and planned responses (Table 32) with project stakeholders in an attempt to assist future national on-farm research programs.

**Table 32. Potential risks and proposed responses.**

| Potential Risk | Response |
|---|---|
| Change in landholders' priorities. Participating farmers choose to opt out of the study. | The participating dairy farmers were selected based on a number of criteria including their commitment to regular data collection and monitoring. Developing and communicating procedures for participating farmers and staff regarding responsibilities and appropriate actions on landholders' properties. Regular communication with farmers is maintained to manage concerns and to be able to put alternative data acquisition strategies in place. |
| Participating farmers do not collect the requested farm data. | Regular communication with landholders. Targeted communication approaches to manage these concerns and put alternative data acquisition strategies in place. |
| Team members don't meet timelines for visits, sample handling or data provision. | Regular communication with on-farm team members through emails and importantly direct phone conversations at least bi-monthly. Targeted approaches to on-farm players who are not meeting commitments. Additional farm training visits from science leadership team to work. Directly contacting managers in other regions/states to put pressure on team members or request alternative players available. |
| Team members do not provide the quality of information required. | Direct communication with on-farm team members to identify improvements required. Targeted approaches to on-farm players who are not meeting commitments. Additional farm training visits from science leadership team to on-farm teams. Directly contacting managers in other regions/states to put pressure on team members or request alternative players. |
| Loss of key project staff | Processes developed to respond to temporary or permanent loss of project staff, including backfilling position from within the project team, appointment of replacement staff. |
| Unfavourable climatic conditions and emergency response. | There are no controls for climatic / environmental conditions such as flood, fire, and drought. Potential impacts may mean that project staff are seconded to emergency response and can not undertake project work as planned. |

# 8. Achievements

## 8.1 Innovative approaches

- **The Accounting for nutrients project developed a large and diverse network.** More than 50 scientists, technicians, industry and policy players from across Australia, all with an interest in improving nutrient management on dairy farms, were brought together within the Accounting for Nutrients project. This network has provided large benefits at a range of levels (i.e. skills, knowledge, standardisation of information and approaches, sources of information, future implications) that couldn't have been realised without this national approach.

- **There was a high level of engagement with stakeholders.** Early and continued engagement with key players from dairy and fertiliser companies, government policy, extension and research groups, and project funders, representing more than 15 stakeholder organisations, has provided important feedback and on-ground relevance for project activities and outcomes. Evaluation of these stakeholders has demonstrated strong and ongoing support for the A4N project and future activities.

- **The Accounting for Nutrients project has demonstrated the benefits of a successful on-farm research program**. The project focused on data collected from a broad range of dairy farms operating across Australia with the aim of collecting information that was highly relevant to current and future dairy systems and provided the ability to draw out significant industry implications. The quality and quantity of information was excellent. The collaborating farmers became active and enthusiastic partners in the research project; providing an essential part of data collection and offering valuable feedback and advice to the project teams. The project structure and methodology will provide a useful template for future on-farm research programs.

- **The project design allowed for detailed investigation of nutrient flows at multiple levels of complexity.** Preliminary planning and review prior to the on-farm research allowing for well defined investigations of nutrient fluxes and transformations at the component scale, farm scale and industry scale.

- I**nnovative and thorough data gathering methods were developed.** The time and effort assigned to developing and reviewing detailed protocols for all facets of data collection was essential. The development of standardised data collection methods and the corresponding relational data base allowed for a thorough process of data quality assurance, collation and interpretation. A number of these protocols, whilst designed specifically for the project, should serve as important templates for future on-farm research work. Additionally, the strengths and weaknesses of the data collection methods have significant implications for industry based programs which rely on gathering, processing and interpreting information from dairy farms.

## 8.2 New knowledge

- **New and comprehensive knowledge of nutrient contents and sources of variation in Australian dairy feed, milk, and manure**. Project activities included the collection and analysis of over 2000 dairy feed, milk and manure samples from diverse commercial dairy farms. The subsequent collation and statistical interpretation has resulted in the creation of important new knowledge about the nutrient content of dairy feeds (including measures of feed quality), and milk and manure. This information will serve as a valuable resource for a range of current and future projects.

- **Key nutrient flows and transformations on Australian dairy farms have been identified and accurately quantified.** The detailed recording of sources of nutrient imports and exports enabled an accurate assessment of all nutrient flows and the identification of major sources responsible for nutrient imbalances. The data has shown that for any individual dairy farm, the contributions of individual components to nutrient imports were variable and determined by stocking rate and the source of imported feed.

- **The efficiency of feed nutrient use has been quantified**. The efficiency with which dairy cows utilise feed nutrients to make milk was measured on a diverse range of farms using different feed types, and during different seasons. Overall, feed nutrient use efficiencies were less than 20% and considered to be low, when compared to international studies. There is scope to increase nutrient use efficiency and maximise milk production, per unit of nutrient input.

- **Nutrients spatially and temporally deposited in excreta have been quantified**. The nutrient excretion loads from cows at each survey visit were combined with farm-specific spatial and temporal cow movement data to determine nutrient deposition loads to specific locations around dairy farms. This improved understanding of the spatial and temporal patterns of animal movement and the resultant nutrient deposition, accumulation and losses have important implications for targeted improvements in nutrient capture and recycling.

- **The most accurate and broad-scale assessment of soil fertility levels on dairy farms has been undertaken.** The comprehensive analysis of soil nutrient levels across a broad range of dairy farms with over 2000 paddocks sampled has provided an important insight into current soil nutrient status and fertiliser requirements on dairy farms. The statistical analysis of relationships between soil fertility and various management factors allows for a comprehensive exploration of the factors influencing nutrient accumulation.

- **The rigorous and comprehensive approach adopted in the A4N project has provided a solid 'best science' platform for determining nutrient flows and balances**. This comprehensive work on 44 operationally and climatically diverse dairy farms provides unique and defendable data on nutrient status, flows, balances and efficiencies a range of scales applicable to the Australian dairy industry. This new Australian-based data, in association with other national and international information, significantly improves our ability to define 'achievable' nutrient management targets on dairy farms.

## 8.3 New products

- **The A4N project has generated Australia's most comprehensive data base of dairy feed, milk, and manure nutrient concentrations**. The collation of over 2000 dairy feed, milk and manure samples from diverse commercial dairy farms has resulted in the creation of Australia's largest available data set of dairy feeds milk and manure. This database will serve as a valuable resource for a range of current and future dairy projects, and will also provide an important data source for determining nutrient concentrations of various dairy imports and exports in nutrient balance calculations.

- **The dairy farm data base.** The custom built A4N database, designed to manage the collated dairy farm data, includes extensive information on operational characteristics, productivity and related nutrient management metrics. This comprehensive commercial dairy farm information and the ease of accessibility of the database provide a valuable resource to other dairy projects; especially modelling based projects.

- **The nutrient accounting empirical model.** The development of the nutrient balance empirical model allowed the research team to calculate nutrient balances and nutrient use efficiencies for the participating dairy farms. This model incorporates best science and includes over 30 data input sheets that are designed to accept data generated in custom queries in our database. While specifically designed as a research tool, the model has the potential to form the basis of a nutrient budgeting tool for industry.

- **On-farm rapid assessment techniques.** Rapid assessment techniques, involving structured questionnaires and data pro-forma, were a key tool in gathering information on dairy farms. These tools and techniques can serve as a valuable resource for future on-farm research.

- **Soil nutrient mapping standards.** The development of nutrient distribution maps within the A4N project has lead to a wide-scale adoption of this approach in delivering soil test information. A number of important projects nation-wide have implemented soil nutrient mapping. As a result, the national standards for soil test interpretation have been more effectively implemented at a national scale.

- **National dairy nutrient management planning guidelines.** The direct involvement of the A4N project team has assisted in the development of draft national 'Nutrient management planning' guidelines. The guidelines describe the components of a structured, rational and comprehensive nutrient management plan for dairy farms that includes economic and environmental objectives.

- **The Nutrient accounting technical manual.** A key deliverable for the A4N project has been the creation of a manual that outlines and justifies the requirements for determining a nutrient balance on dairy farms, how to accurately calculate balances and efficiencies based on the best available data, and how to interpret this information.

- **Scientific publications.** Seven scientific publications have already been written, and peer reviewed for publication in scientific journals and science forums. The publishing of information from the A4N project is an important final step in consolidating the scientific basis of the projects information and industry implications. It is anticipated that another 4 scientific papers will be produced over the forth-coming 12 months.

# 9. Dairy industry implications and opportunities

The intensification of the Australian dairy industry has exacerbated nutrient surplus at the farm scale. This excessive and inefficient use of nutrients, specifically N and P, can significantly degrade air and water quality. Moreover, the dairy industry continues to intensify in the cooler and wetter southern coastal areas which also support important waterways and a rapidly growing urban population. Increasing societal expectations for clean water and air, along with competition for land and water resources is likely to increase the tension between these competing interests.

**Current soil fertility levels on dairy farms**

1) Soil nutrient levels are generally high on Australian dairy farms. The dairy industry is recognised both nationally and internationally as having high levels of nutrient inputs and as a consequence having the potential to accumulate and lose a significant amount of nutrients to the broader environment. If current nutrient management practices persist and dairy farms continue to intensify, soil nutrient levels are likely to further increase with even greater nutrient losses to the broader environment.
2) More than three quarters of the paddocks sampled had fertility levels above agronomic requirements. At these levels most pastures and crops are very unlikely to produce additional dry matter from fertiliser inputs. An exception to this are specific dairy farm systems, such as organic or biodynamic systems, which are likely to have substantially lower soil phosphorus and sulphur fertility, with pastures and crops potentially responsive to soluble fertiliser inputs.
3) Key indicators of elevated soil fertility are overall milk production, nutrient surplus and stocking rate. These indicators reflect the overall intensity of the dairy operation and the likely higher amounts of nutrients imported in feed and fertiliser.
4) Within farm nutrient heterogeneity is substantial irrespective of dairy farm system. Higher soil nutrient levels of P, K and S are driven by paddock stocking density, proximity to the dairy, frequency of effluent applications, and feeding strategies.
5) High nutrient loading from the deposition of animal excreta is clearly a key driver of elevated soil nutrient levels. Paddocks with high densities of animals per ha can have very high nutrient accumulation while those infrequently visited and with low stock densities will generally have lower soil nutrient levels. These high nutrient areas can also have degraded soil structure and low plant cover, increasing the risk of nutrient losses. An additional negative consequence from high potassium loads, resulting from manure, effluent applications and fertiliser, is high plant K uptake and potential metabolic disorders in livestock.
6) There are substantial opportunities on dairy farms to reduce or exclude fertiliser inputs. The relatively small costs associated with undertaking a strategic and on-going soil sampling program are likely to be returned many times through the potential savings in unnecessary fertiliser expenditure. In many cases, high levels of soil test P, K and S may supply necessary plant nutrients for a number of years before maintenance nutrient applications may be required.
7) Where fertiliser applications are warranted for increasing pasture and crop productivity, a more strategic approach should be undertaken. The clear link between spatial distribution of nutrient levels and animal management provides informed guidance on where nutrients are likely to be required.

**Nutrient surpluses and efficiencies**

8) Whole-farm nitrogen surplus is strongly linked to milk production. Whole-farm nitrogen surpluses ranged from 47 to 600 kg N/ha and was strongly related to the level of milk production and the amount of N imported. The overall industry estimate of whole-farm N use efficiency (the proportion of imported N exported in product) was 26%. Based on these findings, an average of

14.5 g of N is lost to the broader environment for each litre of Australian milk produced. This appears to apply irrespective for the intensity of the dairy systems.

9) Nitrogen losses from Australian dairy farms are comparable to similar systems internationally. While potential N losses appear high, the Australian results are comparable to those measures in other industrialised dairy industries such as New Zealand, the EU and USA. However, two thirds of the farms assessed had annual N surpluses above that acceptable under current European Union compliance standards and this proportion is a likely reflection of the Australian dairy industry as a whole. While no such standards currently exist in Australia, international markets may assign similar standards in the future.

10) Nutrient use efficiency in dairy production is limited by the biological potential of cows to transform feed nutrients into milk and of crops and pastures to convert applied nutrients into forage and other agronomic products. However, the variability in actual nutrient use efficiencies achieved by dairy producers and the disparity between actual and potential efficiencies determined under experimental conditions indicates that substantial improvements in nutrient use efficiency can be made on many commercial dairy farms.

**Linking nutrient inputs, efficiency and productivity**

11) The link between nitrogen input and milk production is strong. While it is clear that large amounts of nitrogen are imported onto dairy farms, the system does not adequately store available N reserves. It is therefore not surprising that there is a strong correlation between nitrogen inputs and milk production. The correlation may in part be indirectly caused by the link between the energy and nitrogen content of feed, as well as the direct impact of nitrogen in increasing pasture, crop and animal production. The use of nitrogen fertiliser was also strongly related to milk production from home-grown pasture and crops, but on specific farms, the role of N fertiliser was substituted by biological N fixation by legumes.

12) The link between phosphorus, potassium and sulphur inputs and milk production is weak. Farms that produced greater amount of milk per ha also had increasing levels of imported P, K and S. However, this appeared to result from greater importation in feed as well as fertiliser. Unlike nitrogen, there was no definable relationship between P, K and S fertiliser inputs and milk production from home-grown pasture and crops. This is not surprising given the high soil P, K and S levels and the relative availability and supply capacity that already exist on these dairy farms.

13) Substantial productivity gains can result from increasing nutrient use efficiency. The range of nutrient use efficiencies on an individual farms indicate that significant increases in efficiency are possible. While general farm characteristics, such as stocking rate, reliance on imported feed, etc, did not directly influence nutrient use efficiency, 'good management' rather than fixed farm characteristics are likely to result in greater nutrient use efficiency outcomes.

14) While decreasing the surplus and increasing the efficiency of P, K and S is likely to result in a cost savings in fertiliser costs, increasing N use efficiency has the potential to significantly increase pasture and crop productivity and milk production, and decrease the amount of purchased feed.

15) There are some substantial opportunities to improve nutrient cycling; enhancing overall productivity and the proportion of nutrients ending up in product. This has been demonstrated by the quantification of various nutrient pools, fluxes and transformations that occur within Australian dairy farms. Opportunities to improve nutrient cycling and nutrient use efficiency identified within this study include a reduction in dietary nutrient concentrations, increasing the capture, storage and redistribution of excreted nutrients, and more strategic fertiliser and effluent applications.

# 10. Recommendations

Improved nutrient management of Australian dairy farms requires solutions to an array of system problems, such as excessive use of fertiliser, excess feeding of nutrients, poor manure collection, storage, and land-spreading techniques, heterogeneous nutrient distribution through animal management, grazing and forage harvesting practises, and in some cases, farmers' unwillingness and/or inability to adopt improved practices.

Over the past 20 years, a range of environmental policies have been developed and implemented in Europe and the US, and more recently in New Zealand, with the aim of reducing nutrient losses from dairy farms to the environment. Central to many of these policy approaches has been the development and on-farm implementation of nutrient accounting tools, used in association with traditional soil testing and interpretation. These tools provide a more holistic approach to dairy farm nutrient management, encompassing the needs of farmers to meet production goals, assisting to identify opportunities for improvements in nutrient use efficiencies, and reducing the risk of off-farm nutrient impact.

From a research perspective, there is a need to identify the on-farm management decisions which result in greater nutrient use efficiencies, assess the labour and financial costs associated with improved practices, and to quantify the actual productivity gains from capturing more nutrients in productivity drivers. This information will be critical in developing and applying recommendations that have a high probability of being implemented on dairy farms.

**More specifically we recommend that:**

- All dairy industry stakeholders, including farmers, government and commercial players are aware of excess nutrients on dairy farms, the major causes, and opportunities for mitigation and enhanced production, particularly as the dairy industry continues to strive for increased productivity.

- The uneven distribution of nutrients within dairy farm is highlighted to dairy farmers and service providers. Of particular note are nutrient hotspots which may contribute to significant nutrient exports to the environment. There is an ongoing need to promote strategic soil testing and more spatially and temporally specific fertiliser and effluent applications.

- Specific farm systems with likely high nutrient surpluses and losses in environmentally sensitive areas are identified and provided with targeted assistance to increase nutrient use efficiency and decrease nutrient losses.

- Standardised approaches are used to determine nutrient budgets across the Australian dairy industry. Major industry stakeholders have explicitly requested that a comprehensive decision support tool be developed which utilises the recently determined approaches and data delivered by the Accounting for Nutrients project. Prior to this undertaking, improved approaches can be achieved through the endorsement and promotion of the Nutrient Accounting Technical Manual produced by this project. In the future, there would be significant benefits in the availability of a comprehensive nutrient accounting web-based decision support tool.

**Future research needs**

- Specific management strategies that have the potential to improve nutrient use efficiency on dairy farms should be investigated on commercial dairy farms to determine the economic, environmental and social costs and benefits.

    - Productivity and efficiency gains from improved nutrient management should be identified using existing data from Australian dairy farms and international expertise. Using dairy industry models such as Udder and DairyMod in the first instance would offer a cost effective research strategy prior to any field studies.

    - A review be undertaken of innovative N fertiliser technologies being developed and trialled internationally, and where appropriate evaluate their applicability in Australian dairy systems.

- A range of on-farm interventions need to be assessed and achievable targets demonstrated for improved nutrient use efficiencies and nutrient surpluses on a range of commercial dairy farms with different production systems, soils and climatic conditions.

- The comprehensive information collected from the participating dairy farms be further utilised to evaluate the corresponding energy, carbon, greenhouse and water footprints of these systems, to ensure that industry recommendations on nutrient management don't conflict with goals relating to other environmental objectives.

- The potential productivity, profitability and N efficiency of pasture legumes as a nitrogen source in contrast to nitrogen fertiliser be investigated.

- New and existing research projects should take the opportunities to link production and environmental outcomes and the implications at the farm level through the use of the nutrient accounting empirical model.

- Trials should be established that demonstrate that there are financial benefits and no reduction in productivity when fertiliser inputs are reduced on soils with high soil fertility levels.

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# APPENDICES

**Appendix. 1. Project plan/timeline**

**Appendix. 2. Relationships between stakeholder groups**

**Appendix 3. Accounting for Nutrients Project Products**

**Appendix 4. Summary of data types and linkages required for determining Whole-farm and Farm – System nutrient balances and efficiencies.**

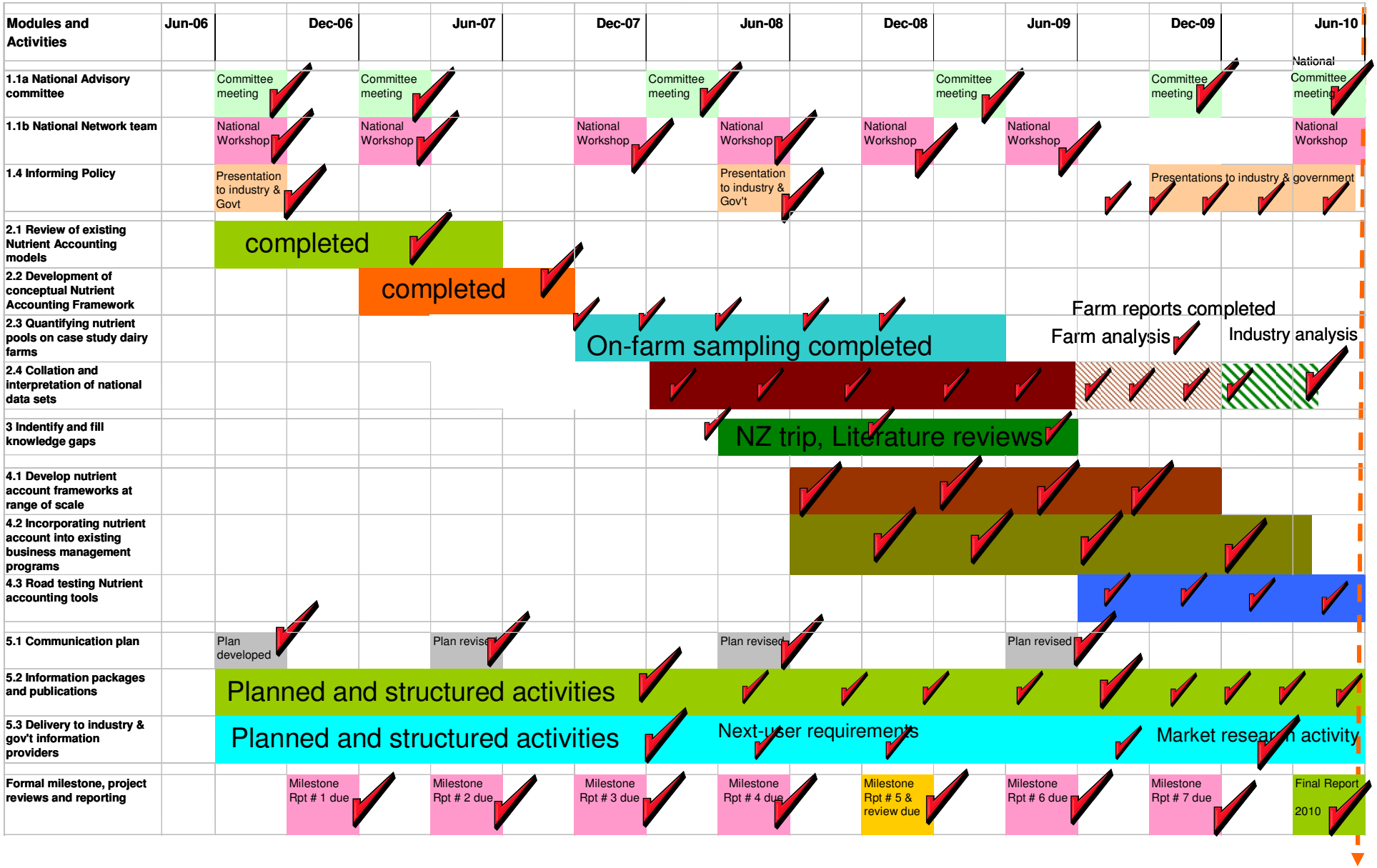
**Appendix 5. Feed types identified and sampled from the 44 monitored dairy farms and corresponding mineral characteristics**

**Appendix 6. Whole-farm nutrient surpluses and efficiencies for N, P, K, S, Ca and Mg**

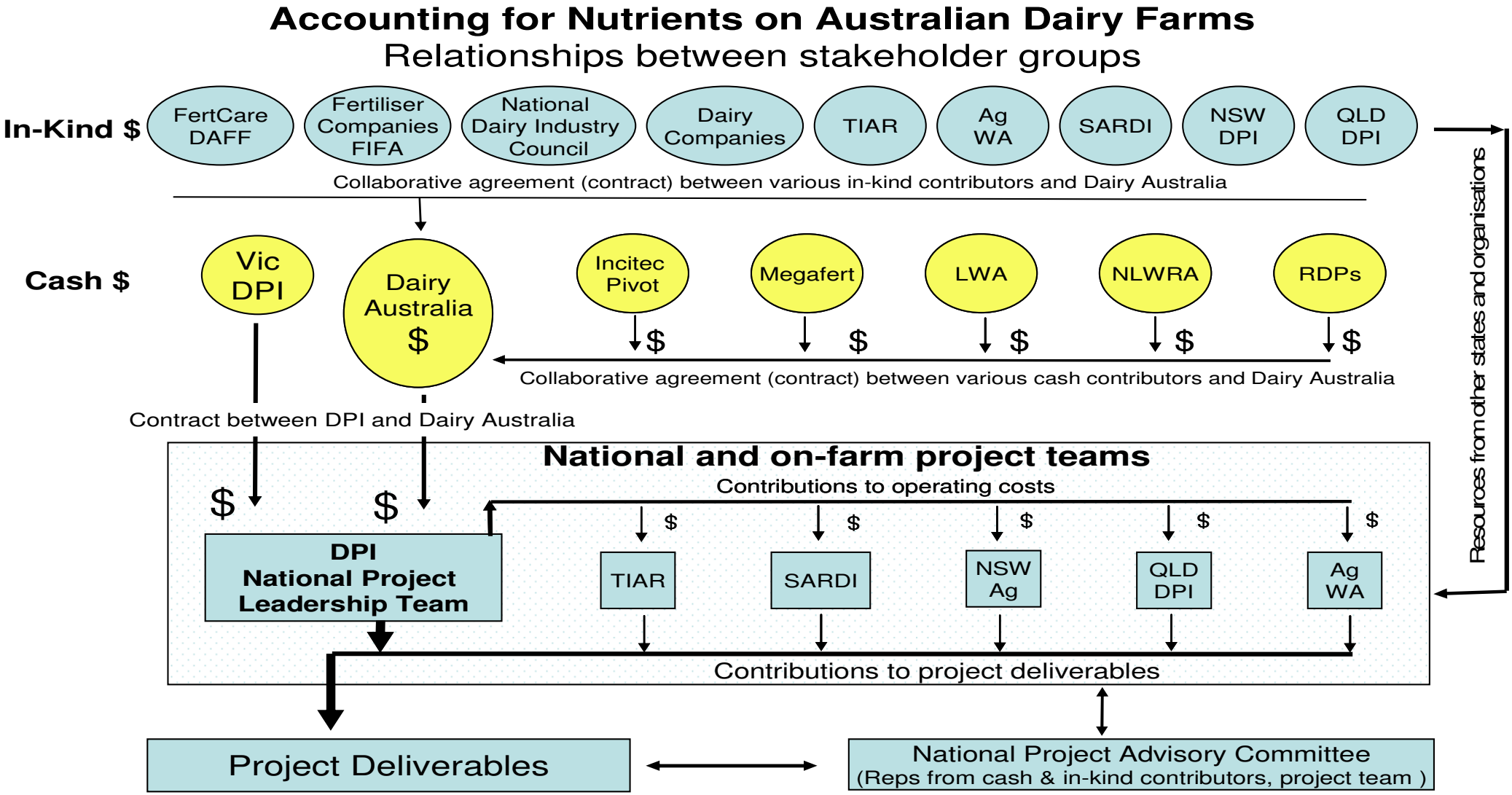
**Appendix 7. Example of information provided to participating dairy farms**

**Appendix 8. Options for improving nutrient use efficiency and reducing nutrient losses to the environment**

# Appendix 1. Proposed project plan, achievements to date, and likely timelines for completion of project components.



## Appendix. 2. Relationships between stakeholder groups



# Appendix 3. Accounting for Nutrients Project Products

## Introduction

Accounting for Nutrients on Australian Dairy Farms is a large national research project with the aim of generating new knowledge and specialist research outputs that will contribute to improved nutrient management on dairy farms. The project outputs are described in detail in the document 'Accounting for Nutrients on Australian Dairy Farms – Project Outputs' which can be found at http://www.accounting4nutrients.com.au/projectreports.aspx.

The objective of this brief document is to provide a short and concise description of the components of the Nutrient Accounting Package, their uses and resulting outcomes or outputs. Key components of the Accounting for Nutrients (A4N) project are shown in Figure 1 below. Note that the current phase of A4N will not deliver a decision support tool for routine calculation of nutrient budgets by next users such as farmers, consultants and other technical advisers.

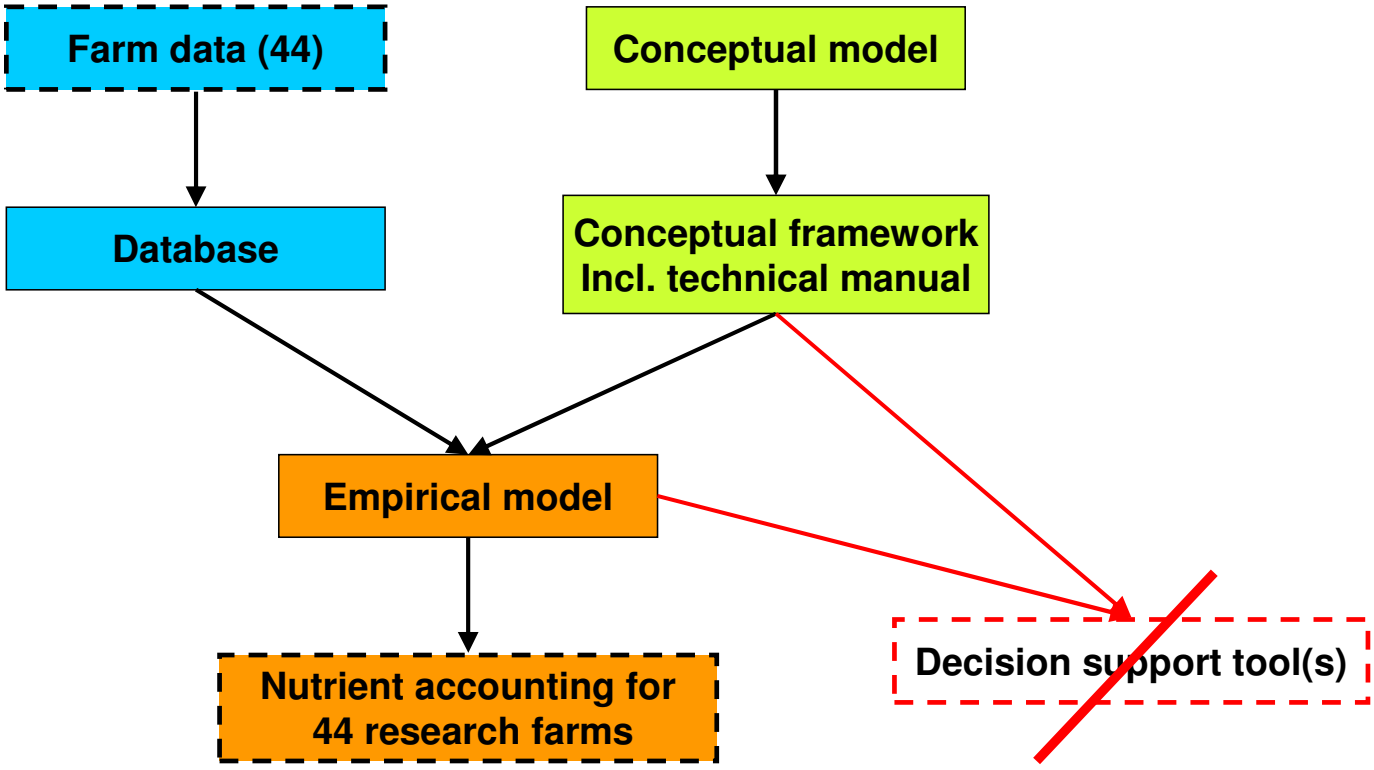



**Figure 1. Outline of key components of a nutrient accounting framework.**

## KEY COMPONENTS DESCRIBED.

**FARM DATA** relevant to nutrient accounting is being collected from 44 dairy farms across Australia for a period of two years. This data which includes farm characteristic data and all data such as inputs and outputs, grazing patterns and nutrient contents of inputs and outputs will be stored in the database described below and will be used in the calculation of nutrients budgets for each of the farms.

The **DATABASE** is the repository for the data from the 44 farms monitored as part of Accounting for Nutrients. It is being used currently by the A4N project team to organise and store data.

The database and nutrient accounting data such as nutrient contents of feeds will be available to all stakeholders at the end of the project although the identity of farms will be protected. Stakeholders may use the data for a range of purposes as they see fit such as developing a better understanding regional differences in nutrient dynamics etc.

The **CONCEPTUAL MODEL** is a simple schematic description of the key nutrient flow pathways into, out of and within a dairy farm. Its purpose was to help clarify key pathways and data collection requirements in the early phase of A4N and it has since been used to inform the more detailed Conceptual Framework described below.

The **CONCEPTUAL FRAMEWORK** is a detailed schematic and mathematical description of all of the possible pathways and pools of nutrients on dairy farms. It includes a mathematical description of all of the calculations and methods of estimating nutrient contents etc. All of the justifications for these and assumptions will be listed, e.g. references to origin, to ensure complete transparency of the nutrient accounting approach being developed. The 'Conceptual Framework' will be documented in a Technical manual and will be freely available to all stakeholders who may subsequently use it to inform development of their own nutrient budgeting systems. It will be the key reference document in the development of an industry Decision Support System.

The **EMPIRICAL MODEL** will be an research nutrient accounting model designed specifically for the use of the A4N team to calculate nutrient budgets for the 44 dairy farms being studied. It will be built using the information contained in the 'Conceptual Framework' document. It is nominally a research tool and will not have a user-friendly front end that would allow users to routinely use it for commercial farm management analysis as might be expected from a practical DSS

A **DECISION SUPPORT TOOL** is not one of the outputs of the A4N project. A decision support tool would allow users to routinely calculate nutrient accounts for farms. It would have a user-friendly interface with pull down menus etc. in a 'windows' type format. It would provide management guidance such as required fertiliser rates.

# Appendix 4. Feed types identified and sampled from the 44 dairy farms and corresponding nutrient characteristics.

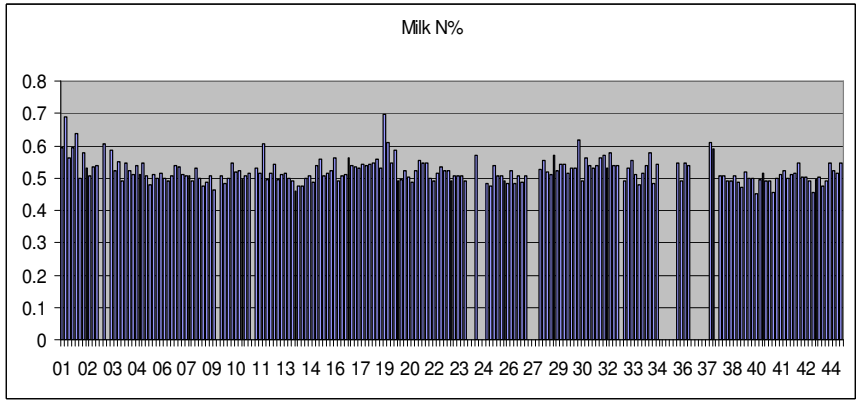
| | CP% | | | | | | N% | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | n | Min | Max | Mean | Std.Dev | CV | n | Min | Max | Mean | Std.Dev | CV |
| Almond shells | 2 | 3.400 | 7.500 | 5.450 | 2.899 | 53.195 | 2 | 0.544 | 1.200 | 0.872 | 0.464 | 53.195 |
| Annual pasture (RG) | 5 | 16.500 | 26.400 | 23.360 | 4.002 | 17.130 | 5 | 2.640 | 4.224 | 3.738 | 0.640 | 17.130 |
| Barley grain | 30 | 8.000 | 18.200 | 12.013 | 2.595 | 21.599 | 30 | 1.280 | 2.912 | 1.922 | 0.415 | 21.599 |
| Black Cake | 4 | 29.700 | 37.300 | 33.675 | 3.225 | 9.578 | 4 | 4.752 | 5.968 | 5.388 | 0.516 | 9.578 |
| Brassica crop | 11 | 11.100 | 26.900 | 19.000 | 4.976 | 26.190 | 11 | 1.776 | 4.304 | 3.040 | 0.796 | 26.190 |
| Bread | 4 | 17.100 | 18.600 | 17.750 | 0.723 | 4.076 | 4 | 2.736 | 2.976 | 2.840 | 0.116 | 4.076 |
| Brewers grain | 13 | 19.800 | 30.400 | 24.723 | 2.537 | 10.261 | 13 | 3.168 | 4.864 | 3.956 | 0.406 | 10.261 |
| Buffer | 0 | | | | | | 0 | | | | | |
| Canola hay | 3 | 13.900 | 20.100 | 16.633 | 3.164 | 19.024 | 3 | 2.224 | 3.216 | 2.661 | 0.506 | 19.024 |
| Canola meal | 26 | 32.400 | 45.500 | 39.715 | 2.690 | 6.774 | 26 | 5.184 | 7.280 | 6.354 | 0.430 | 6.774 |
| Canola silage | 1 | 17.200 | 17.200 | 17.200 | | | 1 | 2.752 | 2.752 | 2.752 | | |
| Cereal hay | 48 | 1.900 | 15.400 | 10.052 | 3.142 | 31.254 | 48 | 0.304 | 2.464 | 1.608 | 0.503 | 31.254 |
| Cereal pasture | 0 | | | | | | 0 | | | | | |
| Cereal silage | 19 | 7.100 | 18.100 | 11.384 | 3.111 | 27.332 | 19 | 1.136 | 2.896 | 1.821 | 0.498 | 27.332 |
| Cereal straw | 34 | 1.300 | 7.700 | 4.026 | 1.549 | 38.472 | 34 | 0.208 | 1.232 | 0.644 | 0.248 | 38.472 |
| Citrus pulp | 6 | 6.100 | 9.200 | 8.167 | 1.125 | 13.781 | 6 | 0.976 | 1.472 | 1.307 | 0.180 | 13.781 |
| Clover hay | 2 | 18.000 | 18.700 | 18.350 | 0.495 | 2.697 | 2 | 2.880 | 2.992 | 2.936 | 0.079 | 2.697 |
| Corn grain | 10 | 8.900 | 10.600 | 9.670 | 0.577 | 5.972 | 10 | 1.424 | 1.696 | 1.547 | 0.092 | 5.972 |
| Cotton seed meal | 9 | 48.200 | 51.400 | 50.056 | 1.067 | 2.131 | 9 | 7.712 | 8.224 | 8.009 | 0.171 | 2.131 |
| Forage blend | 10 | 8.300 | 16.100 | 12.060 | 3.315 | 27.485 | 10 | 1.328 | 2.576 | 1.930 | 0.530 | 27.485 |
| Fruit | 2 | 17.100 | 18.600 | 17.850 | 1.061 | 5.942 | 2 | 2.736 | 2.976 | 2.856 | 0.170 | 5.942 |
| Grain + Min | 82 | 6.300 | 32.200 | 16.228 | 4.828 | 29.750 | 81 | 1.008 | 5.152 | 2.587 | 0.773 | 29.866 |
| Grain husks | 1 | 19.400 | 19.400 | 19.400 | | | 1 | 3.104 | 3.104 | 3.104 | | |
| Kikuyu pasture | 7 | 10.000 | 29.900 | 20.371 | 7.676 | 37.679 | 7 | 1.600 | 4.784 | 3.259 | 1.228 | 37.679 |
| Kikuyu silage | 1 | 10.600 | 10.600 | 10.600 | | | 1 | 1.696 | 1.696 | 1.696 | | |
| Legume pasture | 1 | 28.700 | 28.700 | 28.700 | | | 1 | 4.592 | 4.592 | 4.592 | | |
| Linseed meal | 1 | 0.930 | 0.930 | 0.930 | | | 1 | 0.149 | 0.149 | 0.149 | | |
| Lucerne hay | 47 | 14.600 | 30.200 | 21.457 | 3.252 | 15.156 | 47 | 2.336 | 4.832 | 3.433 | 0.520 | 15.156 |
| Lucerne pasture | 6 | 27.000 | 34.000 | 29.400 | 2.573 | 8.751 | 6 | 4.320 | 5.440 | 4.704 | 0.412 | 8.751 |
| Lucerne silage | 16 | 8.800 | 26.600 | 19.775 | 5.080 | 25.687 | 16 | 1.408 | 4.256 | 3.164 | 0.813 | 25.687 |
| Lucerne&sorghum past | 2 | 23.700 | 28.800 | 26.250 | 3.606 | 13.738 | 2 | 3.792 | 4.608 | 4.200 | 0.577 | 13.738 |
| Lupins grain | 14 | 19.700 | 32.800 | 25.979 | 4.199 | 16.162 | 14 | 3.152 | 5.248 | 4.157 | 0.672 | 16.162 |
| Maize silage | 17 | 5.100 | 10.500 | 7.447 | 1.130 | 15.178 | 17 | 0.816 | 1.680 | 1.192 | 0.181 | 15.178 |
| Millett crop | 4 | 15.100 | 21.400 | 18.650 | 2.651 | 14.217 | 4 | 2.416 | 3.424 | 2.984 | 0.424 | 14.217 |
| Mineral additive | 22 | 0.000 | 36.400 | 18.915 | 14.785 | 78.163 | 22 | 0.000 | 5.824 | 3.026 | 2.366 | 78.163 |
| Mineral mined | 0 | | | | | | 0 | | | | | |
| Mix Beef | 4 | 16.700 | 18.800 | 17.875 | 0.869 | 4.864 | 4 | 2.672 | 3.008 | 2.860 | 0.139 | 4.864 |
| Mix Calf | 17 | 14.600 | 23.700 | 19.018 | 2.910 | 15.301 | 17 | 2.336 | 3.792 | 3.043 | 0.466 | 15.301 |
| Mixed grain | 36 | 8.700 | 35.600 | 15.775 | 5.597 | 35.479 | 36 | 1.392 | 5.696 | 2.524 | 0.895 | 35.479 |
| Molasses | 2 | 2.200 | 5.600 | 3.900 | 2.404 | 61.645 | 2 | 0.352 | 0.896 | 0.624 | 0.385 | 61.645 |
| Oat Hay | 44 | 5.500 | 15.500 | 8.968 | 2.386 | 26.606 | 44 | 0.880 | 2.480 | 1.435 | 0.382 | 26.606 |
| Oats & peas silage | 1 | 14.100 | 14.100 | 14.100 | | | 1 | 2.256 | 2.256 | 2.256 | | |
| Oats grain | 1 | 10.700 | 10.700 | 10.700 | | | 1 | 1.712 | 1.712 | 1.712 | | |
| Oats pasture | 1 | 33.100 | 33.100 | 33.100 | | | 1 | 5.296 | 5.296 | 5.296 | | |
| Palm kernels | 6 | 15.200 | 16.900 | 16.033 | 0.662 | 4.131 | 6 | 2.432 | 2.704 | 2.565 | 0.106 | 4.131 |
| Palm oil | 0 | | | | | | 0 | | | | | |
| Paspalum silage | 3 | 9.600 | 14.100 | 11.933 | 2.255 | 18.894 | 3 | 1.536 | 2.256 | 1.909 | 0.361 | 18.894 |
| Pasture hay | 201 | 4.600 | 20.300 | 10.703 | 3.146 | 29.389 | 201 | 0.736 | 3.248 | 1.713 | 0.503 | 29.389 |
| Pasture silage | 240 | 5.900 | 27.200 | 16.475 | 4.176 | 25.344 | 240 | 0.944 | 4.352 | 2.636 | 0.668 | 25.344 |
| Pea pollard | 1 | 13.200 | 13.200 | 13.200 | | | 1 | 2.112 | 2.112 | 2.112 | | |
| Pellets Calf | 26 | 11.200 | 23.100 | 18.315 | 3.061 | 16.715 | 26 | 1.792 | 3.696 | 2.930 | 0.490 | 16.715 |
| Pellets Dairy | 90 | 7.700 | 24.000 | 14.756 | 2.877 | 19.499 | 90 | 1.232 | 3.840 | 2.361 | 0.460 | 19.499 |
| Pellets Springer | 29 | 11.200 | 25.100 | 16.283 | 3.338 | 20.500 | 29 | 1.792 | 4.016 | 2.605 | 0.534 | 20.500 |
| Pellets Weaner | 12 | 14.900 | 20.500 | 16.758 | 1.670 | 9.964 | 12 | 2.384 | 3.280 | 2.681 | 0.267 | 9.964 |
| Prairie grass silage | 1 | 11.900 | 11.900 | 11.900 | | | 1 | 1.904 | 1.904 | 1.904 | | |
| Ryegrass pasture | 222 | 6.100 | 35.300 | 23.004 | 5.811 | 25.262 | 222 | 0.960 | 5.648 | 3.665 | 0.946 | 25.819 |
| Safflower grain | 1 | 16.800 | 16.800 | 16.800 | | | 1 | 2.688 | 2.688 | 2.688 | | |
| Salt | 0 | | | | | | 0 | | | | | |
| Seteria silage | 3 | 8.600 | 15.300 | 11.900 | 3.351 | 28.161 | 3 | 1.376 | 2.448 | 1.904 | 0.536 | 28.161 |
| Sorghum crop | 5 | 11.800 | 19.000 | 14.480 | 3.203 | 22.118 | 5 | 1.888 | 3.040 | 2.317 | 0.512 | 22.118 |
| Sorghum grain | 3 | 10.000 | 12.100 | 11.267 | 1.115 | 9.897 | 3 | 1.600 | 1.936 | 1.803 | 0.178 | 9.897 |
| Sorghum hay | 4 | 10.400 | 14.400 | 12.375 | 1.694 | 13.688 | 4 | 1.664 | 2.304 | 1.980 | 0.271 | 13.688 |
| Sorghum/millet hay | 2 | 7.500 | 11.600 | 9.550 | 2.899 | 30.357 | 2 | 1.200 | 1.856 | 1.528 | 0.464 | 30.357 |
| Sorghum/millet sil | 3 | 7.500 | 11.000 | 8.867 | 1.872 | 21.110 | 3 | 1.200 | 1.760 | 1.419 | 0.299 | 21.110 |
| Soybean meal | 7 | 40.400 | 53.400 | 48.129 | 6.103 | 12.681 | 7 | 6.464 | 8.544 | 7.701 | 0.977 | 12.681 |
| Sugar | 0 | | | | | | 0 | | | | | |
| TMR forage only | 11 | 8.900 | 22.000 | 15.027 | 4.003 | 26.635 | 11 | 1.424 | 3.520 | 2.404 | 0.640 | 26.635 |
| TMR with minerals | 6 | 7.200 | 25.100 | 14.950 | 5.989 | 40.060 | 6 | 1.152 | 4.016 | 2.392 | 0.958 | 40.060 |
| Triticale grain | 20 | 11.700 | 16.400 | 13.710 | 1.221 | 8.908 | 20 | 1.872 | 2.624 | 2.194 | 0.195 | 8.908 |
| Turnip crop | 8 | 9.000 | 19.300 | 13.100 | 4.242 | 32.384 | 8 | 1.440 | 3.088 | 2.096 | 0.679 | 32.384 |
| Vegetable oil | 0 | | | | | | 0 | | | | | |
| Vetch hay | 5 | 16.000 | 25.500 | 21.340 | 3.723 | 17.444 | 5 | 2.560 | 4.080 | 3.414 | 0.596 | 17.444 |
| Wheat grain | 24 | 9.199999809 | 19.2 | 13.7875 | 2.43780251 | 17.681251 | 24 | 1.472 | 3.072 | 2.206 | 0.390048 | 17.68125 |
| Yeast | 0 | | | | | | 0 | | | | | |
| total sample number | 1499 | | | | | | | | | | | |

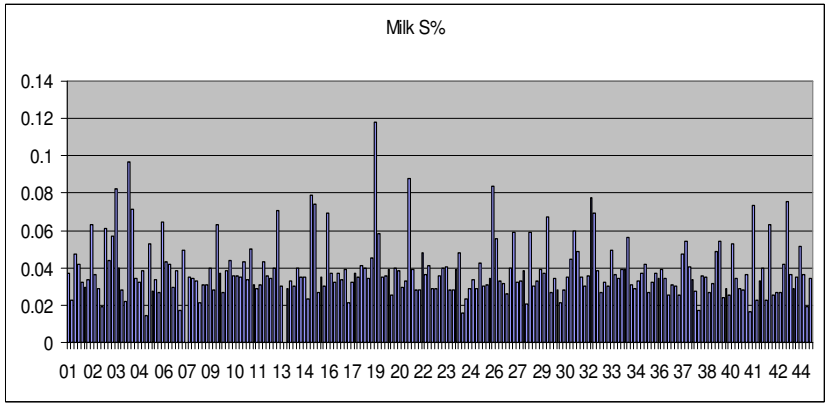
| | P% | | | | | | K% | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | n | Min | Max | Mean | Std.Dev | CV | n | Min | Max | Mean | Std.Dev | CV |
| Almond shells | 3 | 0.119 | 0.275 | 0.204 | 0.079 | 38.694 | 3 | 2.520 | 3.280 | 2.957 | 0.392 | 13.274 |
| Annual pasture (RG) | 4 | 0.466 | 0.753 | 0.554 | 0.134 | 24.186 | 4 | 1.150 | 3.430 | 2.330 | 1.218 | 52.264 |
| Barley grain | 30 | 0.241 | 0.573 | 0.364 | 0.082 | 22.623 | 28 | 0.429 | 0.676 | 0.518 | 0.064 | 12.292 |
| Black Cake | 5 | 0.166 | 0.252 | 0.207 | 0.038 | 18.548 | 5 | 0.025 | 0.125 | 0.059 | 0.041 | 70.380 |
| Brassica crop | 8 | 0.249 | 0.517 | 0.346 | 0.088 | 25.354 | 11 | 1.700 | 4.420 | 3.013 | 0.703 | 23.320 |
| Bread | 4 | 0.255 | 0.358 | 0.293 | 0.045 | 15.259 | 4 | 0.269 | 0.344 | 0.306 | 0.031 | 10.030 |
| Brewers grain | 13 | 0.455 | 0.930 | 0.679 | 0.162 | 23.881 | 14 | 0.046 | 1.730 | 0.511 | 0.538 | 105.242 |
| Buffer | 2 | 0.135 | 0.150 | 0.143 | 0.011 | 7.443 | 2 | 0.005 | 0.047 | 0.026 | 0.029 | 112.154 |
| Canola hay | 3 | 0.262 | 0.404 | 0.314 | 0.079 | 25.027 | 3 | 1.970 | 2.650 | 2.407 | 0.379 | 15.748 |
| Canola meal | 27 | 0.937 | 1.390 | 1.158 | 0.109 | 9.416 | 26 | 1.170 | 1.550 | 1.346 | 0.093 | 6.930 |
| Canola silage | 1 | 0.297 | 0.297 | 0.297 | | | 1 | 2.880 | 2.880 | 2.880 | | |
| Cereal hay | 41 | 0.046 | 0.387 | 0.206 | 0.064 | 31.258 | 46 | 0.630 | 2.960 | 1.705 | 0.565 | 33.143 |
| Cereal pasture | 1 | 0.523 | 0.523 | 0.523 | | | 1 | 4.470 | 4.470 | 4.470 | | |
| Cereal silage | 22 | 0.153 | 0.473 | 0.326 | 0.089 | 27.200 | 22 | 0.877 | 3.250 | 1.978 | 0.761 | 38.486 |
| Cereal straw | 33 | 0.014 | 0.240 | 0.095 | 0.060 | 62.654 | 28 | 0.540 | 1.620 | 1.169 | 0.280 | 23.905 |
| Citrus pulp | 5 | 0.173 | 0.471 | 0.280 | 0.119 | 42.576 | 6 | 0.901 | 1.310 | 1.132 | 0.148 | 13.118 |
| Clover hay | 2 | 0.276 | 0.498 | 0.387 | 0.157 | 40.563 | 2 | 2.510 | 2.730 | 2.620 | 0.156 | 5.938 |
| Corn grain | 11 | 0.258 | 0.362 | 0.309 | 0.035 | 11.260 | 11 | 0.284 | 0.432 | 0.359 | 0.053 | 14.756 |
| Cotton seed meal | 10 | 1.240 | 1.560 | 1.409 | 0.097 | 6.909 | 10 | 1.500 | 1.740 | 1.610 | 0.071 | 4.431 |
| Forage blend | 10 | 0.163 | 0.525 | 0.353 | 0.113 | 31.941 | 9 | 1.290 | 2.460 | 1.754 | 0.461 | 26.278 |
| Fruit | 2 | 0.477 | 0.496 | 0.487 | 0.013 | 2.762 | 2 | 3.660 | 5.010 | 4.335 | 0.955 | 22.021 |
| Grain + Min | 78 | 0.250 | 1.110 | 0.547 | 0.197 | 36.020 | 81 | 0.258 | 0.905 | 0.572 | 0.122 | 21.330 |
| Grain husks | 1 | 1.240 | 1.240 | 1.240 | | | 1 | 1.380 | 1.380 | 1.380 | | |
| Kikuyu pasture | 7 | 0.277 | 0.548 | 0.418 | 0.104 | 24.855 | 7 | 1.650 | 4.190 | 2.990 | 0.960 | 32.101 |
| Kikuyu silage | 1 | 0.386 | 0.386 | 0.386 | | | 1 | 1.850 | 1.850 | 1.850 | | |
| Legume pasture | 1 | 0.674 | 0.674 | 0.674 | | | 1 | 2.470 | 2.470 | 2.470 | | |
| Linseed meal | 1 | 0.538 | 0.538 | 0.538 | | | 1 | 0.847 | 0.847 | 0.847 | | |
| Lucerne hay | 41 | 0.148 | 0.659 | 0.360 | 0.117 | 32.487 | 44 | 0.763 | 3.200 | 2.041 | 0.557 | 27.315 |
| Lucerne pasture | 6 | 0.365 | 0.454 | 0.413 | 0.030 | 7.152 | 5 | 1.040 | 2.270 | 1.724 | 0.466 | 27.029 |
| Lucerne silage | 15 | 0.239 | 0.585 | 0.423 | 0.125 | 29.563 | 14 | 1.270 | 2.880 | 2.152 | 0.560 | 26.019 |
| Lucerne&sorghum past | 2 | 0.239 | 0.511 | 0.375 | 0.192 | 51.289 | 2 | 1.360 | 3.320 | 2.340 | 1.386 | 59.228 |
| Lupins grain | 13 | 0.291 | 0.440 | 0.365 | 0.046 | 12.692 | 14 | 0.662 | 0.961 | 0.810 | 0.097 | 11.981 |
| Maize silage | 18 | 0.195 | 0.360 | 0.264 | 0.043 | 16.344 | 18 | 0.702 | 1.520 | 1.031 | 0.227 | 22.024 |
| Millett crop | 2 | 0.275 | 0.312 | 0.294 | 0.026 | 8.914 | 4 | 1.380 | 6.520 | 3.575 | 2.178 | 60.927 |
| Mineral additive | 39 | 0.045 | 4.650 | 1.710 | 1.201 | 70.250 | 40 | 0.038 | 1.300 | 0.562 | 0.365 | 64.895 |
| Mineral mined | 0 | | | | | | 0 | | | | | |
| Mix Beef | 4 | 0.572 | 0.976 | 0.721 | 0.180 | 24.892 | 4 | 0.546 | 0.723 | 0.640 | 0.080 | 12.564 |
| Mix Calf | 17 | 0.296 | 0.860 | 0.594 | 0.166 | 27.918 | 17 | 0.509 | 1.100 | 0.785 | 0.210 | 26.804 |
| Mixed grain | 31 | 0.222 | 0.785 | 0.390 | 0.118 | 30.338 | 33 | 0.388 | 0.815 | 0.551 | 0.110 | 19.952 |
| Molasses | 14 | 0.087 | 0.362 | 0.217 | 0.105 | 48.315 | 14 | 2.340 | 5.140 | 4.171 | 0.948 | 22.721 |
| Oat Hay | 43 | 0.090 | 0.422 | 0.218 | 0.081 | 37.210 | 46 | 0.823 | 3.150 | 1.771 | 0.637 | 35.989 |
| Oats & peas silage | 1 | 0.389 | 0.389 | 0.389 | | | 1 | 2.430 | 2.430 | 2.430 | | |
| Oats grain | 1 | 0.345 | 0.345 | 0.345 | | | 1 | 0.542 | 0.542 | 0.542 | | |
| Oats pasture | 1 | 0.946 | 0.946 | 0.946 | | | 1 | 5.940 | 5.940 | 5.940 | | |
| Palm kernels | 6 | 0.553 | 0.742 | 0.651 | 0.076 | 11.694 | 6 | 0.606 | 0.739 | 0.692 | 0.050 | 7.284 |
| Palm oil | 0 | | | | | | 0 | | | | | |
| Paspalum silage | 3 | 0.295 | 0.346 | 0.324 | 0.026 | 8.124 | 3 | 1.480 | 2.710 | 2.057 | 0.619 | 30.077 |
| Pasture hay | 204 | 0.097 | 0.504 | 0.258 | 0.083 | 31.967 | 206 | 0.620 | 3.850 | 1.764 | 0.600 | 34.008 |
| Pasture silage | 233 | 0.130 | 0.759 | 0.385 | 0.106 | 27.472 | 233 | 0.990 | 4.850 | 2.709 | 0.828 | 30.579 |
| Pea pollard | 1 | 0.250 | 0.250 | 0.250 | | | 1 | 0.956 | 0.956 | 0.956 | | |
| Pellets Calf | 28 | 0.193 | 0.975 | 0.636 | 0.202 | 31.766 | 27 | 0.597 | 1.030 | 0.808 | 0.102 | 12.596 |
| Pellets Dairy | 92 | 0.141 | 1.220 | 0.583 | 0.188 | 32.305 | 90 | 0.371 | 0.949 | 0.632 | 0.109 | 17.274 |
| Pellets Springer | 29 | 0.091 | 0.821 | 0.583 | 0.157 | 26.989 | 29 | 0.337 | 1.050 | 0.654 | 0.216 | 33.051 |
| Pellets Weaner | 12 | 0.580 | 1.040 | 0.780 | 0.122 | 15.679 | 12 | 0.542 | 0.893 | 0.752 | 0.112 | 14.917 |
| Prairie grass silage | 1 | 0.186 | 0.186 | 0.186 | | | 1 | 1.300 | 1.300 | 1.300 | | |
| Ryegrass pasture | 221 | 0.114 | 0.783 | 0.449 | 0.132 | 29.311 | 223 | 0.479 | 4.890 | 2.773 | 0.821 | 29.597 |
| Safflower grain | 1 | 0.474 | 0.474 | 0.474 | | | 1 | 0.692 | 0.692 | 0.692 | | |
| Salt | 1 | 0.037 | 0.037 | 0.037 | | | 1 | 0.022 | 0.022 | 0.022 | | |
| Seteria silage | 3 | 0.209 | 0.340 | 0.267 | 0.067 | 25.010 | 3 | 1.080 | 3.090 | 2.163 | 1.014 | 46.878 |
| Sorghum crop | 5 | 0.251 | 0.546 | 0.396 | 0.114 | 28.890 | 5 | 1.510 | 3.480 | 2.558 | 0.879 | 34.380 |
| Sorghum grain | 2 | 0.313 | 0.336 | 0.325 | 0.016 | 5.012 | 2 | 0.290 | 0.373 | 0.332 | 0.059 | 17.704 |
| Sorghum hay | 4 | 0.309 | 0.388 | 0.347 | 0.034 | 9.738 | 4 | 1.420 | 2.500 | 1.903 | 0.485 | 25.514 |
| Sorghum/millet hay | 2 | 0.365 | 0.405 | 0.385 | 0.028 | 7.347 | 2 | 1.720 | 2.280 | 2.000 | 0.396 | 19.799 |
| Sorghum/millet sil | 3 | 0.288 | 0.452 | 0.364 | 0.083 | 22.668 | 3 | 1.230 | 3.360 | 2.270 | 1.066 | 46.955 |
| Soybean meal | 8 | 0.564 | 0.921 | 0.777 | 0.116 | 14.969 | 8 | 1.960 | 2.730 | 2.369 | 0.325 | 13.720 |
| Sugar | 3 | 0.068 | 0.320 | 0.165 | 0.136 | 82.284 | 3 | 0.030 | 0.514 | 0.194 | 0.277 | 143.293 |
| TMR forage only | 10 | 0.278 | 0.548 | 0.402 | 0.082 | 20.477 | 10 | 1.030 | 2.310 | 1.580 | 0.402 | 25.455 |
| TMR with minerals | 6 | 0.221 | 0.436 | 0.345 | 0.088 | 25.538 | 7 | 1.160 | 3.060 | 1.739 | 0.694 | 39.900 |
| Triticale grain | 21 | 0.223 | 0.439 | 0.311 | 0.064 | 20.706 | 19 | 0.388 | 0.549 | 0.475 | 0.038 | 8.048 |
| Turnip crop | 7 | 0.204 | 0.371 | 0.265 | 0.072 | 27.316 | 7 | 2.000 | 4.010 | 2.910 | 0.791 | 27.166 |
| Vegetable oil | 1 | 0.008 | 0.008 | 0.008 | | | 1 | 0.007 | 0.007 | 0.007 | | |
| Vetch hay | 4 | 0.265 | 0.290 | 0.277 | 0.011 | 3.922 | 5 | 2.240 | 2.570 | 2.408 | 0.142 | 5.883 |
| Wheat grain | 25 | 0.215 | 0.418 | 0.3364 | 0.051143 | 15.202965 | 25 | 0.336 | 0.53 | 0.43204 | 0.053685 | 12.425848 |
| Yeast | 1 | 0.886 | 0.886 | 0.886 | | | 1 | 1.16 | 1.16 | 1.16 | | |

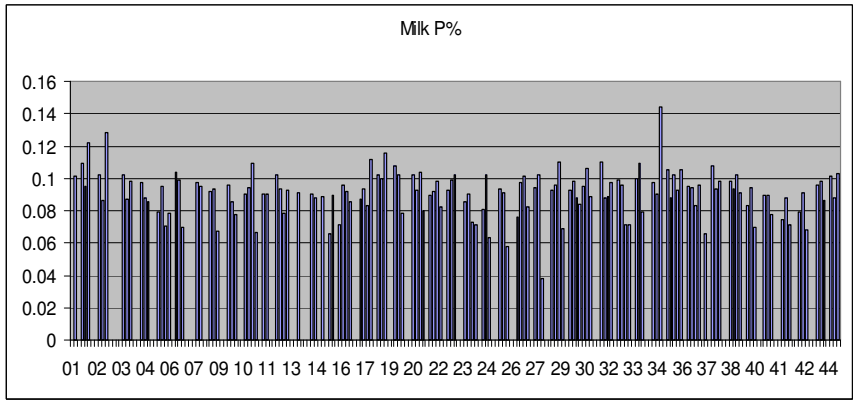
| | S% | | | | | | Ca% | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | n | Min | Max | Mean | Std.Dev | CV | n | Min | Max | Mean | Std.Dev | CV |
| **Almond shells** | 3 | 0.025 | 0.038 | 0.031 | 0.006 | 20.923 | 2 | 0.245 | 0.329 | 0.287 | 0.059 | 20.696 |
| **Annual pasture (RG)** | 5 | 0.187 | 0.377 | 0.279 | 0.067 | 24.118 | 5 | 0.560 | 0.914 | 0.785 | 0.143 | 18.227 |
| **Barley grain** | 27 | 0.101 | 0.176 | 0.135 | 0.021 | 15.730 | 24 | 0.028 | 0.077 | 0.043 | 0.010 | 22.598 |
| **Black Cake** | 5 | 0.688 | 0.781 | 0.728 | 0.037 | 5.062 | 5 | 0.009 | 0.051 | 0.018 | 0.018 | 103.439 |
| **Brassica crop** | 11 | 0.323 | 1.050 | 0.633 | 0.251 | 39.648 | 8 | 0.925 | 1.850 | 1.473 | 0.299 | 20.294 |
| **Bread** | 4 | 0.173 | 0.219 | 0.199 | 0.021 | 10.509 | 4 | 0.214 | 0.251 | 0.235 | 0.016 | 6.676 |
| **Brewers grain** | 14 | 0.178 | 0.354 | 0.276 | 0.055 | 20.024 | 14 | 0.101 | 0.255 | 0.163 | 0.045 | 27.317 |
| **Buffer** | 1 | 0.000 | 0.000 | 0.000 | | | 2 | 0.010 | 0.398 | 0.204 | 0.274 | 134.219 |
| **Canola hay** | 3 | 0.457 | 0.638 | 0.530 | 0.095 | 18.008 | 3 | 0.752 | 0.876 | 0.812 | 0.062 | 7.647 |
| **Canola meal** | 26 | 0.530 | 0.842 | 0.701 | 0.076 | 10.843 | 25 | 0.512 | 0.743 | 0.623 | 0.068 | 10.846 |
| **Canola silage** | 1 | 0.505 | 0.505 | 0.505 | | | 1 | 0.968 | 0.968 | 0.968 | | |
| **Cereal hay** | 48 | 0.058 | 0.251 | 0.161 | 0.041 | 25.337 | 43 | 0.091 | 0.477 | 0.228 | 0.088 | 38.600 |
| **Cereal pasture** | 1 | 0.432 | 0.432 | 0.432 | | | 1 | 0.276 | 0.276 | 0.276 | | |
| **Cereal silage** | 22 | 0.104 | 0.239 | 0.182 | 0.041 | 22.393 | 20 | 0.173 | 0.428 | 0.274 | 0.063 | 22.899 |
| **Cereal straw** | 32 | 0.027 | 0.148 | 0.091 | 0.031 | 34.405 | 32 | 0.124 | 0.324 | 0.209 | 0.058 | 27.748 |
| **Citrus pulp** | 6 | 0.061 | 0.136 | 0.107 | 0.031 | 29.080 | 6 | 0.622 | 1.170 | 0.914 | 0.193 | 21.071 |
| **Clover hay** | 2 | 0.106 | 0.251 | 0.179 | 0.103 | 57.440 | 2 | 1.280 | 1.420 | 1.350 | 0.099 | 7.333 |
| **Corn grain** | 11 | 0.089 | 0.157 | 0.117 | 0.020 | 16.801 | 11 | 0.003 | 0.010 | 0.006 | 0.002 | 39.555 |
| **Cotton seed meal** | 10 | 0.464 | 0.619 | 0.526 | 0.050 | 9.426 | 9 | 0.175 | 0.266 | 0.236 | 0.027 | 11.498 |
| **Forage blend** | 10 | 0.164 | 0.284 | 0.200 | 0.038 | 19.090 | 10 | 0.289 | 1.010 | 0.564 | 0.254 | 45.042 |
| **Fruit** | 2 | 0.209 | 0.224 | 0.217 | 0.011 | 4.899 | 2 | 0.120 | 0.175 | 0.148 | 0.039 | 26.367 |
| **Grain + Min** | 80 | 0.108 | 0.440 | 0.218 | 0.066 | 30.383 | 78 | 0.009 | 2.190 | 0.719 | 0.530 | 73.734 |
| **Grain husks** | 1 | 0.213 | 0.213 | 0.213 | | | 1 | 0.101 | 0.101 | 0.101 | | |
| **Kikuyu pasture** | 7 | 0.154 | 0.386 | 0.269 | 0.077 | 28.662 | 7 | 0.321 | 0.785 | 0.532 | 0.168 | 31.617 |
| **Kikuyu silage** | 1 | 0.127 | 0.127 | 0.127 | | | 1 | 0.193 | 0.193 | 0.193 | | |
| **Legume pasture** | 1 | 0.206 | 0.206 | 0.206 | | | 1 | 2.220 | 2.220 | 2.220 | | |
| **Linseed meal** | 1 | 0.286 | 0.286 | 0.286 | | | 1 | 0.200 | 0.200 | 0.200 | | |
| **Lucerne hay** | 47 | 0.141 | 0.415 | 0.279 | 0.063 | 22.381 | 46 | 0.216 | 2.160 | 1.216 | 0.344 | 28.280 |
| **Lucerne pasture** | 7 | 0.198 | 0.503 | 0.380 | 0.097 | 25.600 | 4 | 1.350 | 1.460 | 1.405 | 0.047 | 3.313 |
| **Lucerne silage** | 15 | 0.146 | 0.390 | 0.278 | 0.079 | 28.504 | 14 | 0.622 | 1.560 | 0.984 | 0.277 | 28.188 |
| **Lucerne&sorghum past** | 2 | 0.269 | 0.451 | 0.360 | 0.129 | 35.748 | 2 | 0.476 | 1.260 | 0.868 | 0.554 | 63.868 |
| **Lupins grain** | 13 | 0.154 | 0.229 | 0.199 | 0.025 | 12.512 | 12 | 0.158 | 0.329 | 0.249 | 0.046 | 18.366 |
| **Maize silage** | 18 | 0.055 | 0.175 | 0.106 | 0.028 | 26.143 | 17 | 0.111 | 0.342 | 0.197 | 0.075 | 38.094 |
| **Millett crop** | 4 | 0.298 | 0.718 | 0.439 | 0.195 | 44.474 | 4 | 0.360 | 0.630 | 0.530 | 0.125 | 23.596 |
| **Mineral additive** | 38 | 0.000 | 3.020 | 1.029 | 0.897 | 87.125 | 41 | 0.226 | 17.600 | 7.924 | 5.452 | 68.804 |
| **Mineral mined** | 0 | | | | | | 0 | | | | | |
| **Mix Beef** | 4 | 0.247 | 0.304 | 0.274 | 0.025 | 9.253 | 4 | 0.362 | 1.550 | 1.048 | 0.528 | 50.421 |
| **Mix Calf** | 14 | 0.161 | 0.315 | 0.231 | 0.048 | 20.808 | 17 | 0.234 | 1.800 | 0.947 | 0.434 | 45.871 |
| **Mixed grain** | 32 | 0.096 | 0.249 | 0.167 | 0.039 | 23.275 | 29 | 0.034 | 0.721 | 0.181 | 0.188 | 103.938 |
| **Molasses** | 12 | 0.390 | 0.610 | 0.491 | 0.060 | 12.148 | 13 | 0.769 | 1.150 | 0.993 | 0.120 | 12.127 |
| **Oat Hay** | 44 | 0.088 | 0.284 | 0.147 | 0.045 | 30.760 | 41 | 0.120 | 0.426 | 0.236 | 0.082 | 34.798 |
| **Oats & peas silage** | 1 | 0.195 | 0.195 | 0.195 | | | 1 | 1.880 | 1.880 | 1.880 | | |
| **Oats grain** | 1 | 0.146 | 0.146 | 0.146 | | | 1 | 0.039 | 0.039 | 0.039 | | |
| **Oats pasture** | 1 | 0.353 | 0.353 | 0.353 | | | 1 | 0.534 | 0.534 | 0.534 | | |
| **Palm kernels** | 6 | 0.182 | 0.236 | 0.212 | 0.020 | 9.329 | 6 | 0.321 | 0.466 | 0.410 | 0.052 | 12.770 |
| **Palm oil** | 0 | | | | | | 0 | | | | | |
| **Paspalum silage** | 3 | 0.165 | 0.221 | 0.184 | 0.032 | 17.417 | 3 | 0.331 | 0.977 | 0.710 | 0.337 | 47.500 |
| **Pasture hay** | 204 | 0.095 | 0.396 | 0.220 | 0.064 | 28.997 | 197 | 0.139 | 0.972 | 0.488 | 0.186 | 38.007 |
| **Pasture silage** | 239 | 0.098 | 0.426 | 0.279 | 0.066 | 23.576 | 235 | 0.159 | 1.220 | 0.593 | 0.221 | 37.303 |
| **Pea pollard** | 1 | 0.100 | 0.100 | 0.100 | | | 1 | 0.339 | 0.339 | 0.339 | | |
| **Pellets Calf** | 28 | 0.140 | 0.366 | 0.261 | 0.060 | 23.066 | 28 | 0.404 | 2.640 | 1.464 | 0.620 | 42.392 |
| **Pellets Dairy** | 91 | 0.107 | 0.347 | 0.205 | 0.045 | 21.966 | 91 | 0.133 | 2.800 | 1.438 | 0.690 | 48.018 |
| **Pellets Springer** | 26 | 0.458 | 1.610 | 1.083 | 0.305 | 28.115 | 29 | 0.361 | 2.250 | 1.168 | 0.518 | 44.343 |
| **Pellets Weaner** | 11 | 0.136 | 0.410 | 0.262 | 0.076 | 28.891 | 10 | 1.060 | 2.280 | 1.710 | 0.311 | 18.211 |
| **Prairie grass silage** | 1 | 0.143 | 0.143 | 0.143 | | | 1 | 0.685 | 0.685 | 0.685 | | |
| **Ryegrass pasture** | 226 | 0.124 | 0.597 | 0.343 | 0.089 | 26.051 | 217 | 0.106 | 1.080 | 0.550 | 0.191 | 34.654 |
| **Safflower grain** | 1 | 0.149 | 0.149 | 0.149 | | | 1 | 0.238 | 0.238 | 0.238 | | |
| **Salt** | 1 | 1.890 | 1.890 | 1.890 | | | 1 | 0.222 | 0.222 | 0.222 | | |
| **Seteria silage** | 2 | 0.150 | 0.155 | 0.153 | 0.004 | 2.318 | 2 | 0.154 | 0.173 | 0.164 | 0.013 | 8.217 |
| **Sorghum crop** | 5 | 0.090 | 0.212 | 0.149 | 0.049 | 33.010 | 5 | 0.303 | 0.565 | 0.446 | 0.129 | 28.894 |
| **Sorghum grain** | 3 | 0.090 | 0.157 | 0.121 | 0.034 | 27.916 | 2 | 0.014 | 0.151 | 0.082 | 0.097 | 117.737 |
| **Sorghum hay** | 4 | 0.108 | 0.157 | 0.134 | 0.026 | 19.508 | 4 | 0.513 | 0.655 | 0.568 | 0.064 | 11.299 |
| **Sorghum/millet hay** | 2 | 0.052 | 0.379 | 0.215 | 0.232 | 107.528 | 2 | 0.287 | 0.466 | 0.377 | 0.127 | 33.618 |
| **Sorghum/millet sil** | 3 | 0.091 | 0.396 | 0.202 | 0.169 | 83.491 | 3 | 0.257 | 0.497 | 0.417 | 0.138 | 33.186 |
| **Soybean meal** | 8 | 0.343 | 0.457 | 0.392 | 0.035 | 8.986 | 8 | 0.183 | 0.356 | 0.279 | 0.072 | 25.958 |
| **Sugar** | 3 | 0.020 | 0.155 | 0.066 | 0.077 | 118.025 | 3 | 0.007 | 0.066 | 0.038 | 0.030 | 77.617 |
| **TMR forage only** | 11 | 0.123 | 0.303 | 0.224 | 0.051 | 22.620 | 11 | 0.330 | 1.230 | 0.674 | 0.323 | 48.014 |
| **TMR with minerals** | 6 | 0.127 | 0.255 | 0.190 | 0.048 | 25.258 | 7 | 0.289 | 1.230 | 0.757 | 0.372 | 49.216 |
| **Triticale grain** | 20 | 0.102 | 0.199 | 0.158 | 0.023 | 14.877 | 20 | 0.026 | 0.077 | 0.042 | 0.014 | 32.195 |
| **Turnip crop** | 8 | 0.417 | 0.692 | 0.549 | 0.085 | 15.397 | 6 | 0.706 | 1.190 | 0.950 | 0.183 | 19.242 |
| **Vegetable oil** | 1 | 0.000 | 0.000 | 0.000 | | | 1 | 0.000 | 0.000 | 0.000 | | |
| **Vetch hay** | 5 | 0.184 | 0.310 | 0.233 | 0.055 | 23.795 | 5 | 0.568 | 1.290 | 0.907 | 0.318 | 35.055 |
| **Wheat grain** | 25 | 0.114 | 0.217 | 0.15804 | 0.025458 | 16.108749 | 23 | 0.01 | 0.0794 | 0.045309 | 0.01586 | 35.004657 |
| **Yeast** | 1 | 0.298 | 0.298 | 0.298 | | | 1 | 0.136 | 0.136 | 0.136 | | |

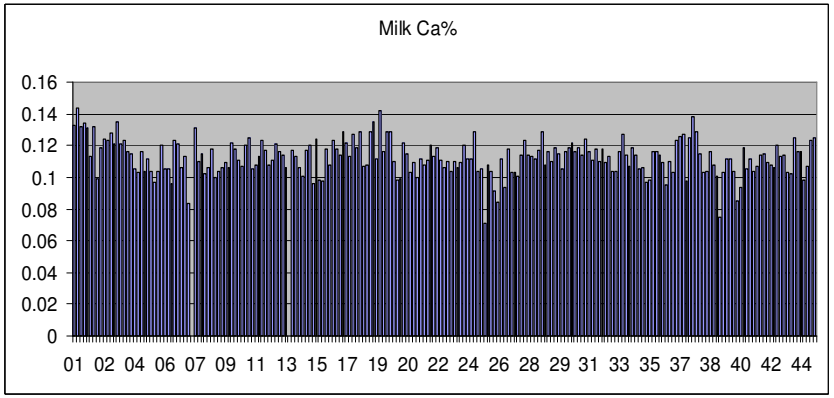
| | Mg% | | | | | | Metabolisable energy (MJ) | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | n | Min | Max | Mean | Std.Dev | CV | n | Min | Max | Mean | Std.Dev | CV |
| Almond shells | 3 | 0.090 | 0.129 | 0.112 | 0.020 | 18.175 | 0 | | | | | |
| Annual pasture (RG) | 5 | 0.184 | 0.320 | 0.243 | 0.050 | 20.535 | 3 | 9.199 | 11.021 | 10.337 | 0.992 | 9.595 |
| Barley grain | 28 | 0.094 | 0.200 | 0.131 | 0.025 | 19.134 | 24 | 11.714 | 12.833 | 12.338 | 0.215 | 1.742 |
| Black Cake | 5 | 0.009 | 0.038 | 0.019 | 0.012 | 64.033 | 2 | 13.141 | 13.461 | 13.301 | 0.226 | 1.701 |
| Brassica crop | 10 | 0.298 | 0.815 | 0.501 | 0.180 | 35.900 | 11 | 8.279 | 12.683 | 10.604 | 1.093 | 10.305 |
| Bread | 4 | 0.058 | 0.081 | 0.070 | 0.009 | 13.139 | 1 | 13.192 | 13.192 | 13.192 | | |
| Brewers grain | 14 | 0.109 | 0.359 | 0.227 | 0.092 | 40.594 | 12 | 10.504 | 12.987 | 11.286 | 0.680 | 6.026 |
| Buffer | 2 | 0.001 | 42.200 | 21.100 | 29.839 | 141.416 | 0 | | | | | |
| Canola hay | 3 | 0.318 | 0.426 | 0.383 | 0.057 | 14.989 | 2 | 8.247 | 9.870 | 9.059 | 1.148 | 12.674 |
| Canola meal | 27 | 0.446 | 0.670 | 0.554 | 0.064 | 11.506 | 0 | | | | | |
| Canola silage | 1 | 0.362 | 0.362 | 0.362 | | | 1 | 9.453 | 9.453 | 9.453 | | |
| Cereal hay | 47 | 0.075 | 0.204 | 0.139 | 0.028 | 20.395 | 45 | 5.292 | 9.716 | 8.455 | 1.121 | 13.262 |
| Cereal pasture | 1 | 0.195 | 0.195 | 0.195 | | | 0 | | | | | |
| Cereal silage | 20 | 0.115 | 0.295 | 0.179 | 0.055 | 30.479 | 19 | 8.015 | 10.160 | 9.078 | 0.582 | 6.412 |
| Cereal straw | 33 | 0.066 | 0.187 | 0.109 | 0.028 | 25.787 | 33 | 4.264 | 8.639 | 5.979 | 0.936 | 15.663 |
| Citrus pulp | 6 | 0.101 | 0.165 | 0.142 | 0.024 | 16.821 | 3 | 9.753 | 10.681 | 10.109 | 0.500 | 4.943 |
| Clover hay | 2 | 0.235 | 0.243 | 0.239 | 0.006 | 2.367 | 2 | 8.923 | 9.313 | 9.118 | 0.276 | 3.025 |
| Corn grain | 10 | 0.082 | 0.141 | 0.118 | 0.016 | 13.424 | 9 | 12.726 | 12.998 | 12.874 | 0.113 | 0.877 |
| Cotton seed meal | 9 | 0.713 | 0.788 | 0.746 | 0.029 | 3.852 | 0 | | | | | |
| Forage blend | 7 | 0.165 | 0.226 | 0.199 | 0.023 | 11.305 | 10 | 8.152 | 9.841 | 8.823 | 0.606 | 6.870 |
| Fruit | 2 | 0.204 | 0.234 | 0.219 | 0.021 | 9.686 | 1 | 11.598 | 11.598 | 11.598 | | |
| Grain + Min | 78 | 0.032 | 0.905 | 0.338 | 0.191 | 56.320 | 40 | 11.261 | 13.119 | 12.551 | 0.405 | 3.229 |
| Grain husks | 1 | 0.475 | 0.475 | 0.475 | | | 0 | | | | | |
| Kikuyu pasture | 6 | 0.277 | 0.422 | 0.331 | 0.058 | 17.572 | 7 | 8.296 | 11.191 | 9.820 | 1.052 | 10.710 |
| Kikuyu silage | 1 | 0.155 | 0.155 | 0.155 | | | 1 | 10.163 | 10.163 | 10.163 | | |
| Legume pasture | 1 | 0.294 | 0.294 | 0.294 | | | 1 | 11.882 | 11.882 | 11.882 | | |
| Linseed meal | 1 | 0.410 | 0.410 | 0.410 | | | 1 | 10.754 | 10.754 | 10.754 | | |
| Lucerne hay | 47 | 0.143 | 0.709 | 0.388 | 0.149 | 38.381 | 45 | 7.820 | 11.235 | 9.641 | 0.675 | 7.005 |
| Lucerne pasture | 6 | 0.261 | 0.523 | 0.395 | 0.091 | 23.170 | 5 | 10.465 | 11.504 | 11.136 | 0.406 | 3.648 |
| Lucerne silage | 14 | 0.251 | 0.810 | 0.476 | 0.206 | 43.249 | 14 | 7.763 | 10.375 | 9.334 | 0.925 | 9.908 |
| Lucerne&sorghum past | 2 | 0.314 | 0.617 | 0.466 | 0.214 | 46.026 | 2 | 10.715 | 11.048 | 10.881 | 0.236 | 2.164 |
| Lupins grain | 14 | 0.152 | 0.198 | 0.169 | 0.011 | 6.707 | 0 | | | | | |
| Maize silage | 19 | 0.126 | 0.326 | 0.201 | 0.053 | 26.150 | 10 | 9.293 | 10.252 | 9.748 | 0.308 | 3.160 |
| Millett crop | 4 | 0.246 | 0.983 | 0.603 | 0.307 | 50.911 | 4 | 9.148 | 10.212 | 9.761 | 0.457 | 4.685 |
| Mineral additive | 42 | 0.214 | 8.810 | 2.965 | 2.545 | 85.851 | 10 | 2.115 | 14.120 | 9.775 | 4.516 | 46.204 |
| Mineral mined | 0 | | | | | | 0 | | | | | |
| Mix Beef | 4 | 0.422 | 0.964 | 0.637 | 0.256 | 40.170 | 1 | 12.594 | 12.594 | 12.594 | | |
| Mix Calf | 16 | 0.158 | 0.724 | 0.337 | 0.184 | 54.647 | 3 | 11.893 | 13.045 | 12.630 | 0.640 | 5.066 |
| Mixed grain | 31 | 0.114 | 0.556 | 0.223 | 0.118 | 52.999 | 23 | 11.391 | 13.241 | 12.617 | 0.394 | 3.121 |
| Molasses | 12 | 0.352 | 0.398 | 0.375 | 0.015 | 4.084 | 2 | 12.675 | 12.877 | 12.776 | 0.142 | 1.115 |
| Oat Hay | 44 | 0.085 | 0.261 | 0.154 | 0.040 | 26.214 | 40 | 7.074 | 9.776 | 8.453 | 0.637 | 7.537 |
| Oats & peas silage | 1 | 0.181 | 0.181 | 0.181 | | | 1 | 9.159 | 9.159 | 9.159 | | |
| Oats grain | 1 | 0.134 | 0.134 | 0.134 | | | 1 | 12.641 | 12.641 | 12.641 | | |
| Oats pasture | 1 | 0.305 | 0.305 | 0.305 | | | 1 | 11.528 | 11.528 | 11.528 | | |
| Palm kernels | 6 | 0.295 | 0.356 | 0.316 | 0.024 | 7.657 | 3 | 7.171 | 8.940 | 7.915 | 0.917 | 11.587 |
| Palm oil | 0 | | | | | | 0 | | | | | |
| Paspalum silage | 3 | 0.215 | 0.286 | 0.251 | 0.036 | 14.148 | 3 | 7.282 | 8.850 | 8.207 | 0.821 | 10.009 |
| Pasture hay | 200 | 0.100 | 0.338 | 0.204 | 0.049 | 23.830 | 196 | 6.124 | 9.478 | 8.103 | 0.666 | 8.224 |
| Pasture silage | 236 | 0.132 | 0.395 | 0.243 | 0.045 | 18.353 | 230 | 6.748 | 11.129 | 9.220 | 0.721 | 7.821 |
| Pea pollard | 1 | 0.255 | 0.255 | 0.255 | | | 1 | 9.869 | 9.869 | 9.869 | | |
| Pellets Calf | 24 | 0.247 | 0.484 | 0.307 | 0.056 | 18.173 | 0 | | | | | |
| Pellets Dairy | 91 | 0.150 | 1.310 | 0.477 | 0.245 | 51.316 | 11 | 11.838 | 12.944 | 12.480 | 0.336 | 2.694 |
| Pellets Springer | 24 | 0.296 | 1.080 | 0.540 | 0.185 | 34.195 | 6 | 12.040 | 13.086 | 12.616 | 0.381 | 3.021 |
| Pellets Weaner | 11 | 0.200 | 0.458 | 0.301 | 0.089 | 29.471 | 0 | | | | | |
| Prairie grass silage | 1 | 0.274 | 0.274 | 0.274 | | | 1 | 8.514 | 8.514 | 8.514 | | |
| Ryegrass pasture | 219 | 0.107 | 0.546 | 0.294 | 0.080 | 27.060 | 199 | 6.666 | 12.449 | 10.494 | 1.041 | 9.916 |
| Safflower grain | 1 | 0.236 | 0.236 | 0.236 | | | 1 | 8.995 | 8.995 | 8.995 | | |
| Salt | 1 | 1.520 | 1.520 | 1.520 | | | 0 | | | | | |
| Seteria silage | 3 | 0.130 | 0.190 | 0.154 | 0.032 | 20.452 | 3 | 7.553 | 8.518 | 8.094 | 0.493 | 6.093 |
| Sorghum crop | 5 | 0.266 | 0.899 | 0.616 | 0.260 | 42.118 | 5 | 7.420 | 10.246 | 8.966 | 1.065 | 11.878 |
| Sorghum grain | 3 | 0.150 | 0.173 | 0.164 | 0.013 | 7.609 | 1 | 12.634 | 12.634 | 12.634 | | |
| Sorghum hay | 4 | 0.718 | 0.868 | 0.765 | 0.071 | 9.273 | 4 | 7.184 | 8.177 | 7.744 | 0.426 | 5.500 |
| Sorghum/millet hay | 2 | 0.292 | 0.494 | 0.393 | 0.143 | 36.345 | 2 | 7.224 | 8.458 | 7.841 | 0.873 | 11.130 |
| Sorghum/millet sil | 3 | 0.334 | 0.424 | 0.376 | 0.045 | 12.077 | 2 | 7.534 | 9.088 | 8.311 | 1.099 | 13.224 |
| Soybean meal | 7 | 0.246 | 0.352 | 0.309 | 0.039 | 12.610 | 1 | 14.648 | 14.648 | 14.648 | | |
| Sugar | 3 | 0.004 | 0.126 | 0.044 | 0.071 | 158.628 | 0 | | | | | |
| TMR forage only | 11 | 0.244 | 0.407 | 0.287 | 0.045 | 15.649 | 8 | 8.235 | 10.222 | 9.556 | 0.629 | 6.579 |
| TMR with minerals | 7 | 0.227 | 0.504 | 0.344 | 0.100 | 29.100 | 3 | 9.229 | 10.519 | 9.750 | 0.679 | 6.969 |
| Triticale grain | 21 | 0.104 | 0.159 | 0.128 | 0.013 | 10.213 | 17 | 12.542 | 12.841 | 12.674 | 0.095 | 0.749 |
| Turnip crop | 8 | 0.257 | 0.351 | 0.300 | 0.033 | 10.943 | 8 | 9.767 | 11.305 | 10.810 | 0.551 | 5.101 |
| Vegetable oil | 1 | 0.001 | 0.001 | 0.001 | | | 0 | | | | | |
| Vetch hay | 5 | 0.233 | 0.320 | 0.270 | 0.037 | 13.836 | 5 | 9.328 | 10.766 | 10.027 | 0.629 | 6.274 |
| Wheat grain | 23 | 0.1 | 0.153 | 0.125478 | 0.012158 | 9.6889789 | 24 | 12.38626 | 13.33229 | 12.86099 | 0.184629 | 1.435575 |
| Yeast | 1 | 0.332 | 0.332 | 0.332 | | | 0 | | | | | |

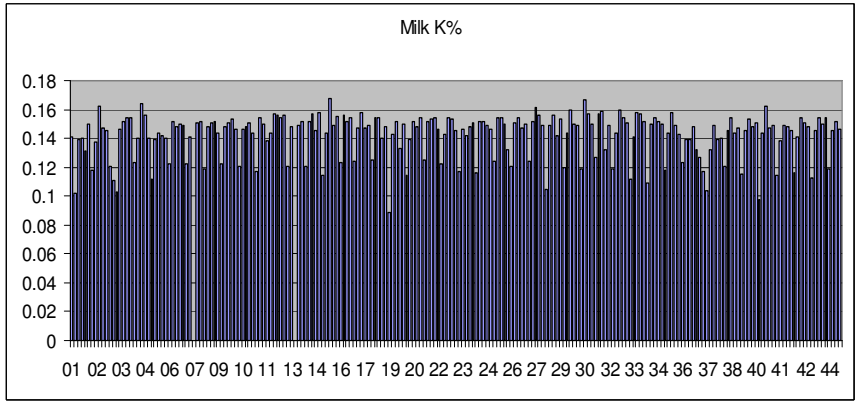
# Appendix 5. Milk nutrient concentrations from quarterly milk samples collected from participating dairy farms.

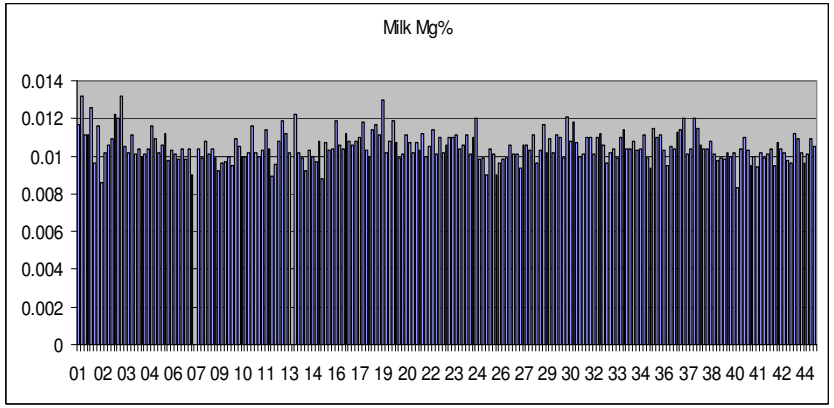


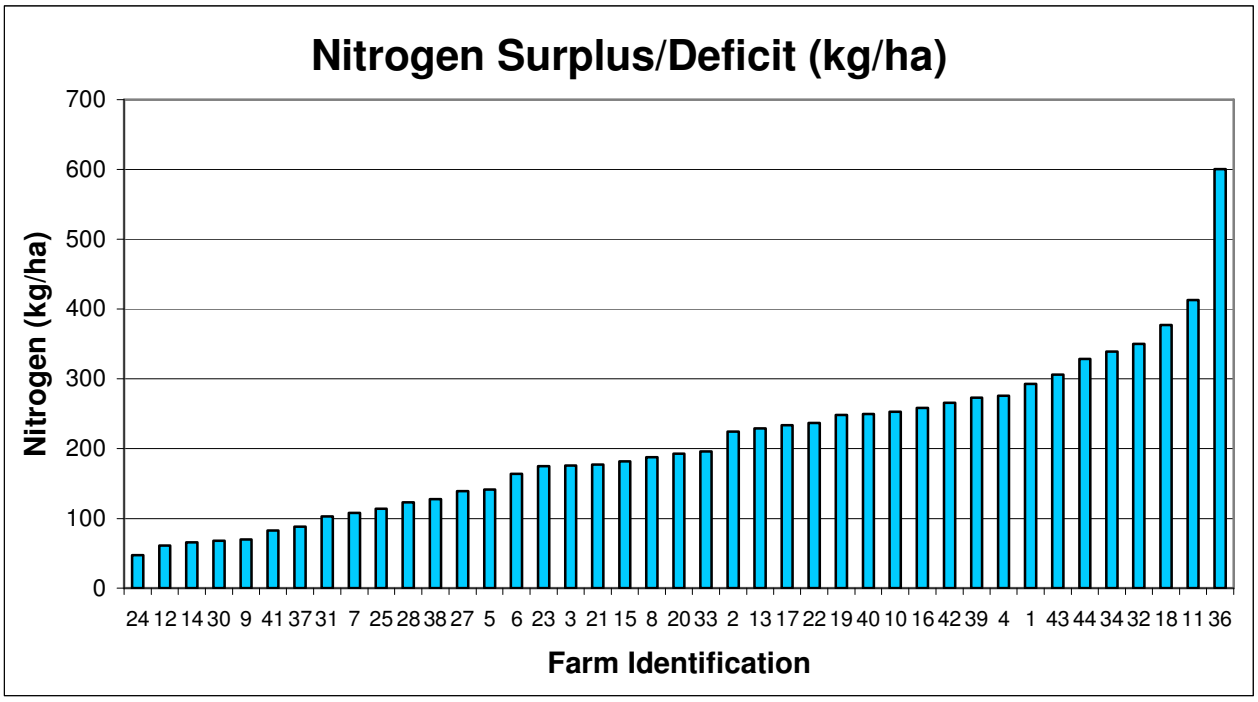



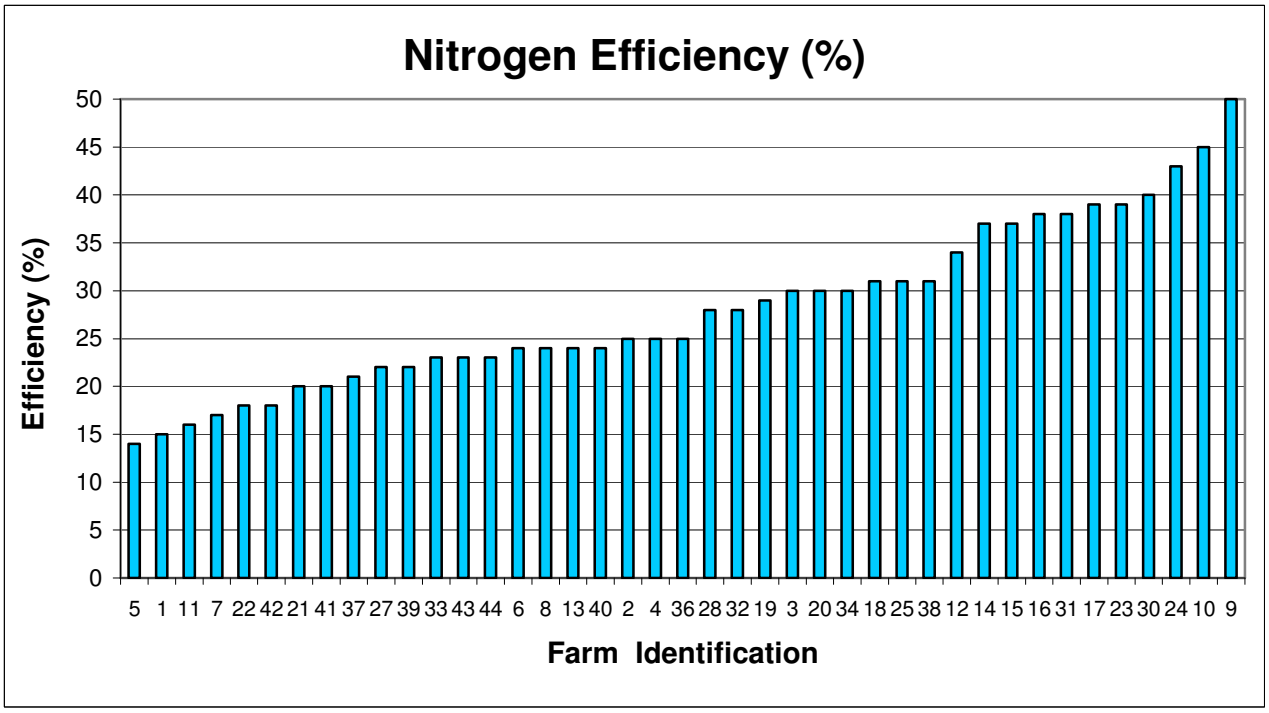



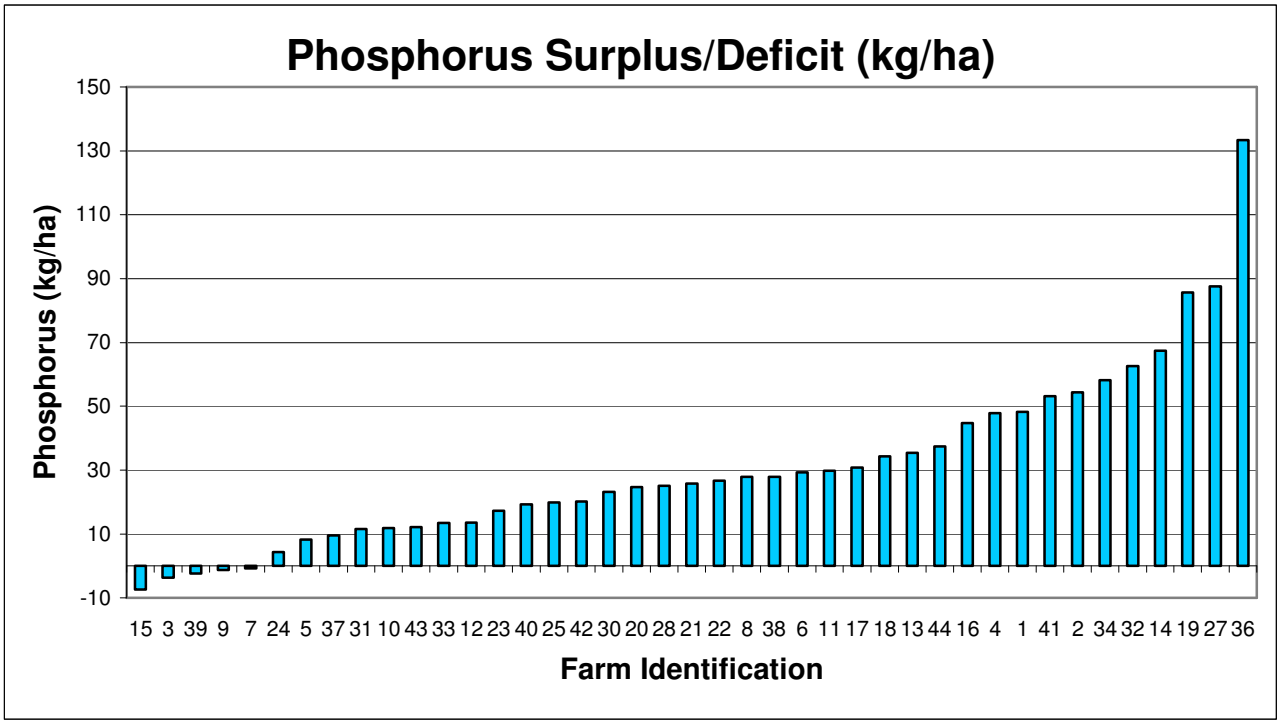



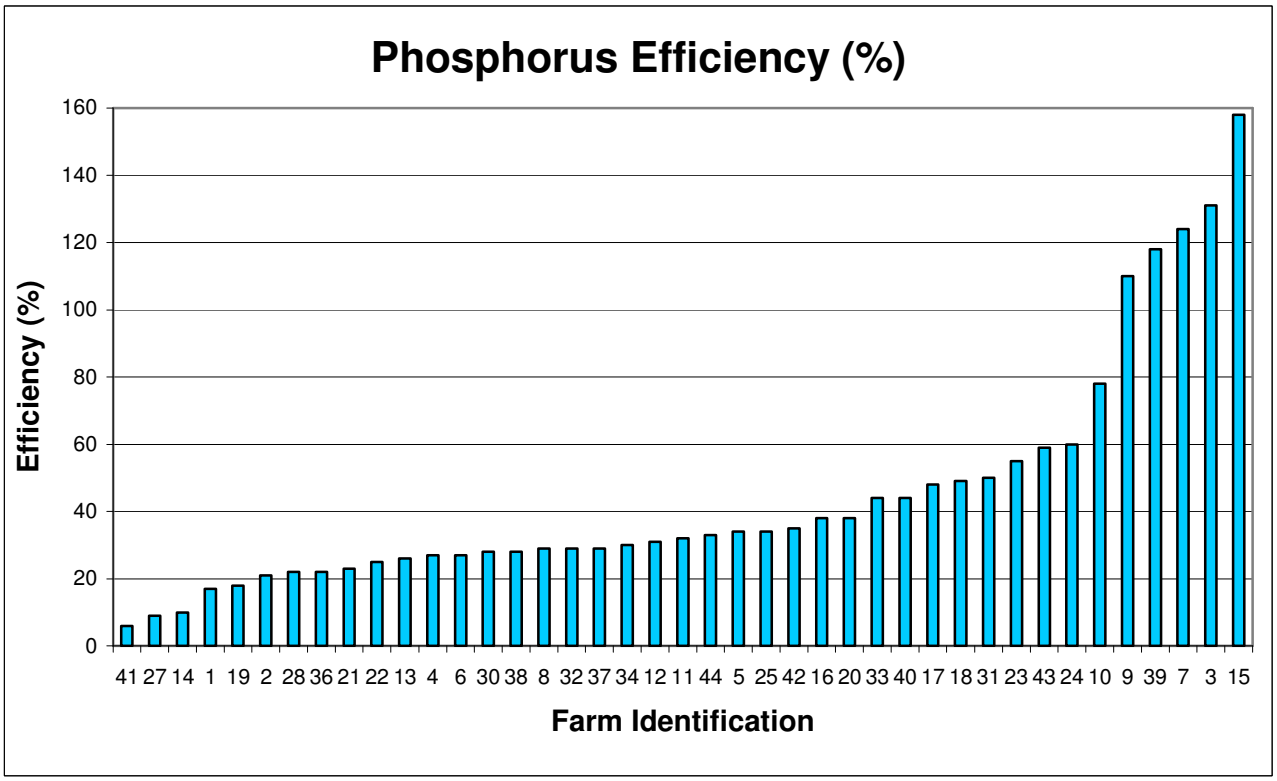



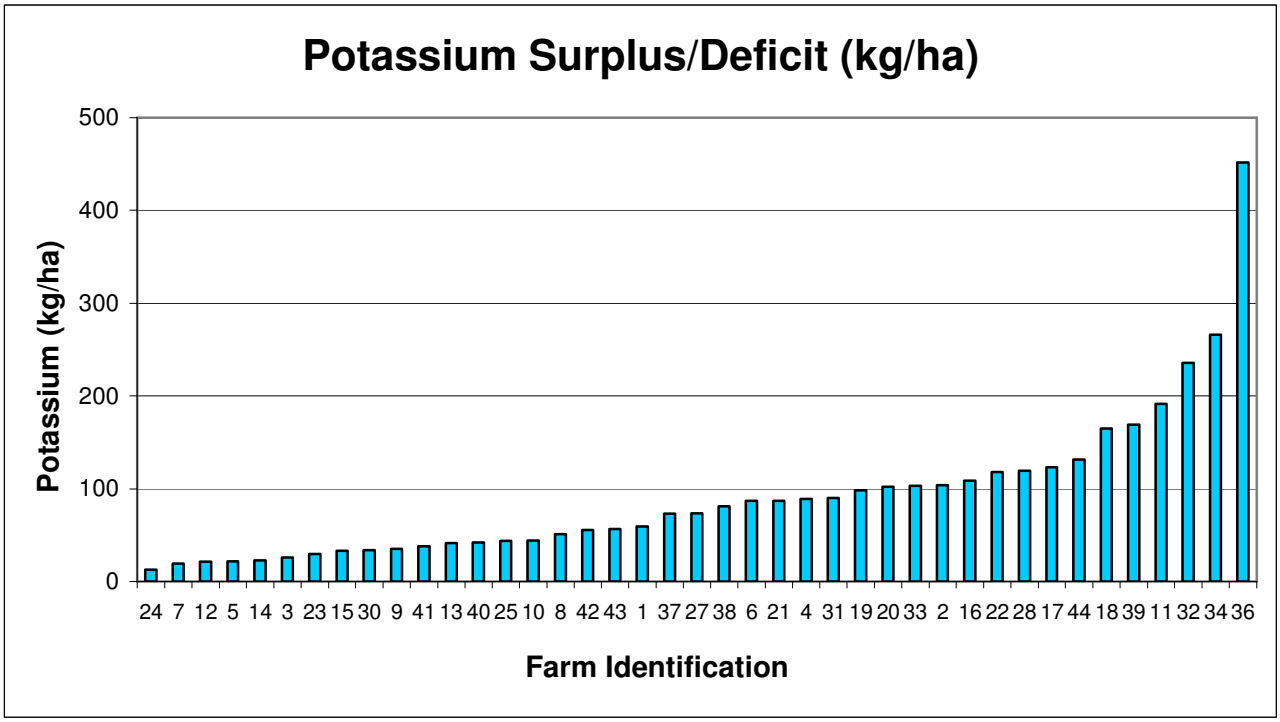
# Appendix 6. Nutrient surpluses and efficiencies determined for 41 dairy farms involved in the study.

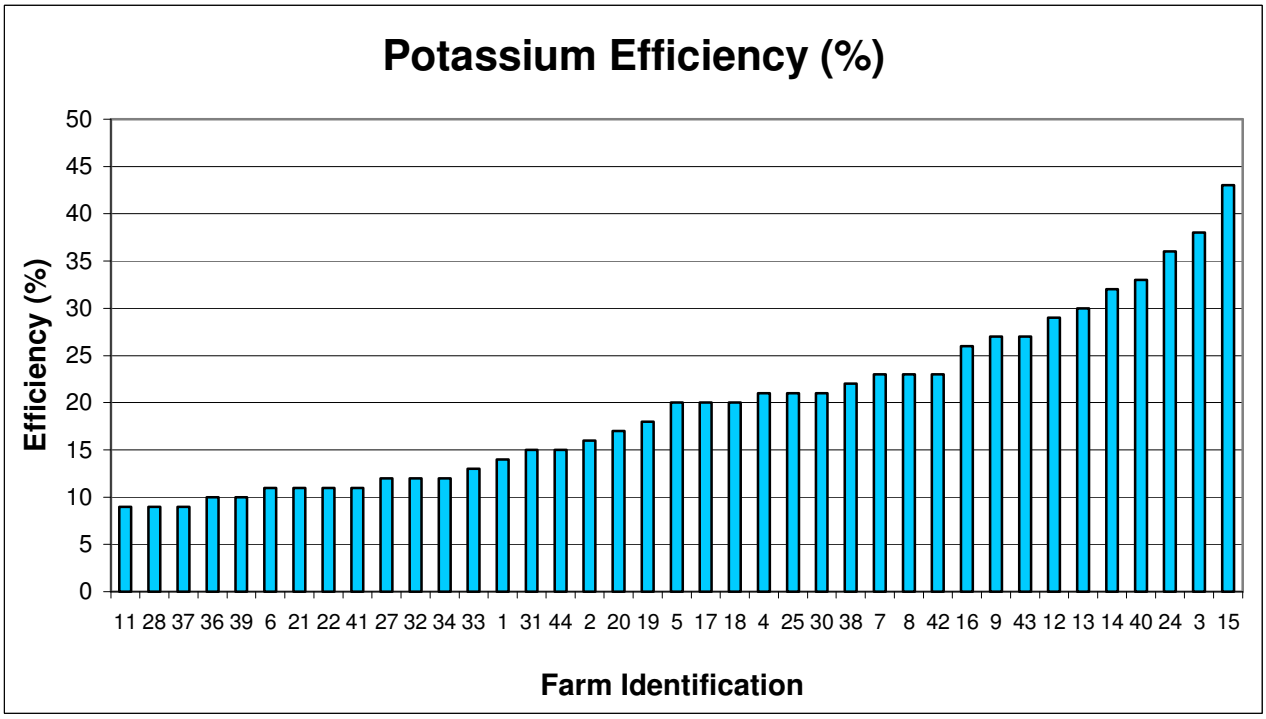


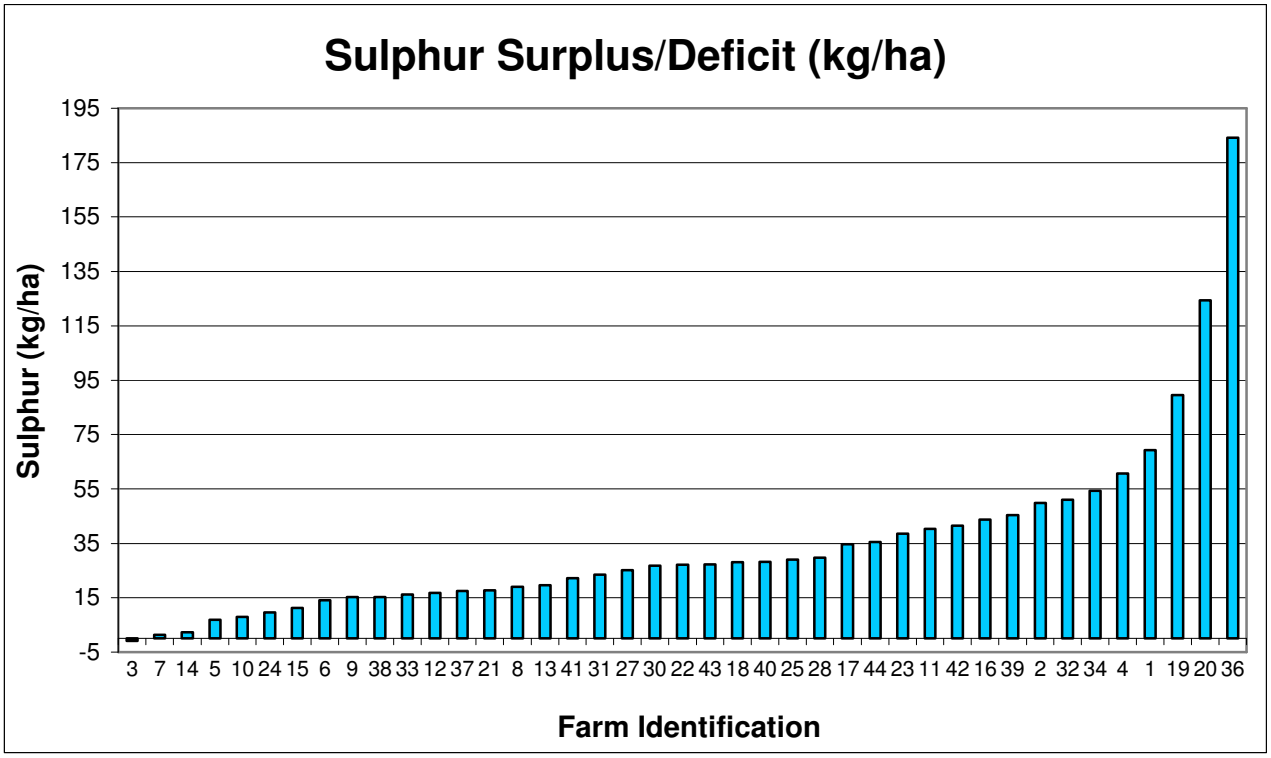



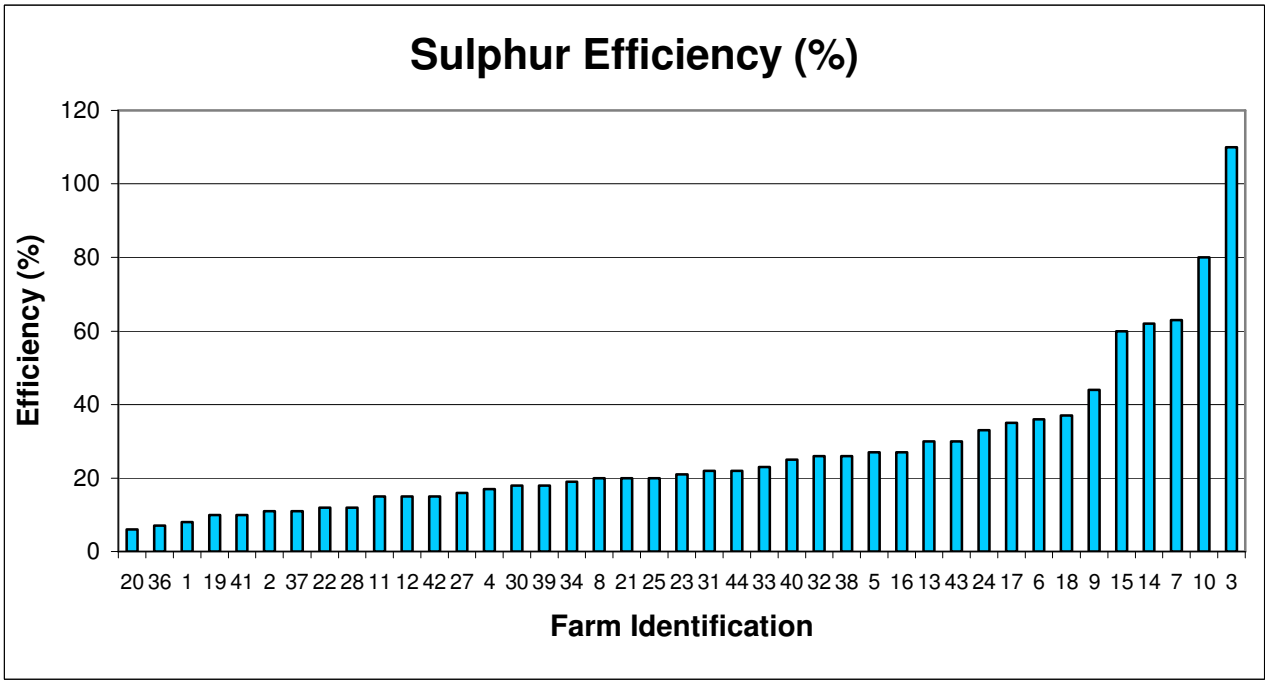

Phosphorus Surplus/Deficit (kg/ha)
Phosphorus (kg/ha)
150
130
110
90
70
50
30
10
-10
15 3 39 9 7 24 5 37 31 10 43 33 12 23 40 25 42 30 20 28 21 22 8 38 6 11 17 18 13 44 16 4 1 41 2 34 32 14 19 27 36
Farm Identification

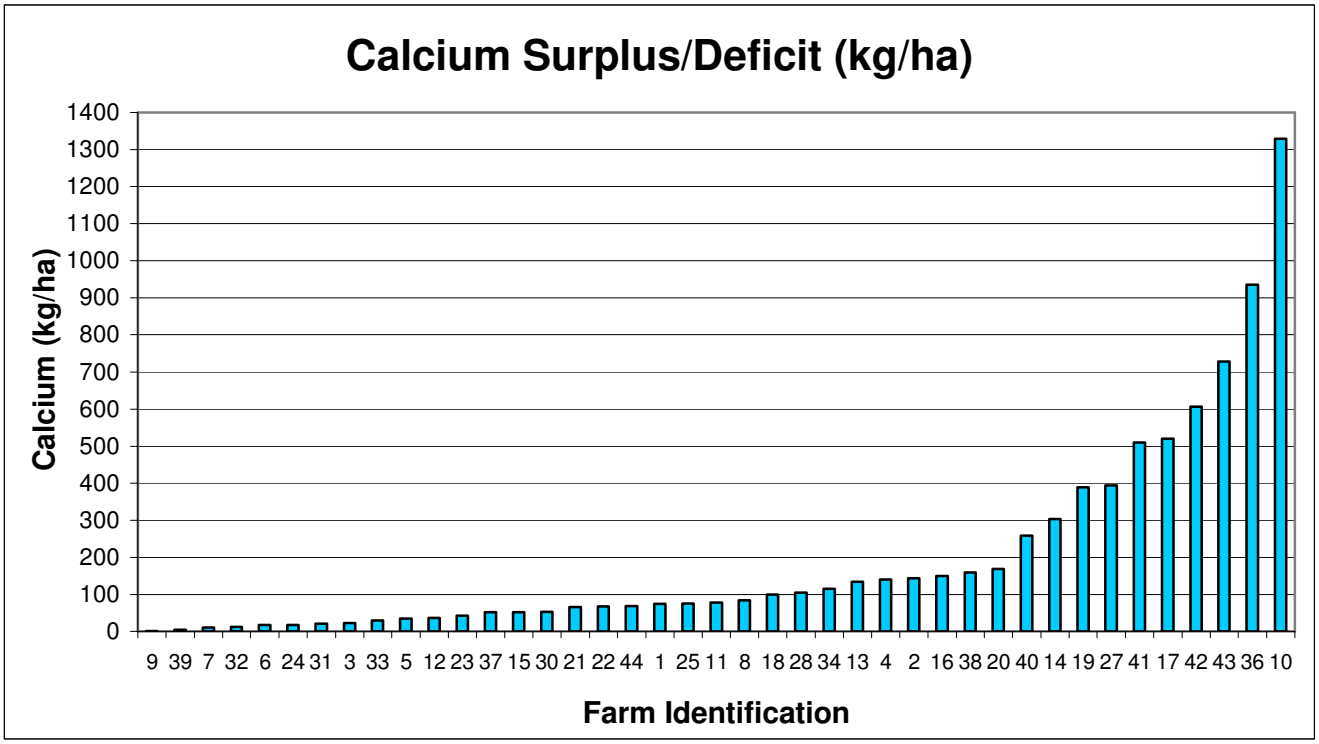

Phosphorus Efficiency (%)
Efficiency (%)
160
140
120
100
80
60
40
20
0
41 27 14 1 19 2 28 36 21 22 13 4 6 30 38 8 32 37 34 12 11 44 5 25 42 16 20 33 40 17 18 31 23 43 24 10 9 39 7 3 15
Farm Identification

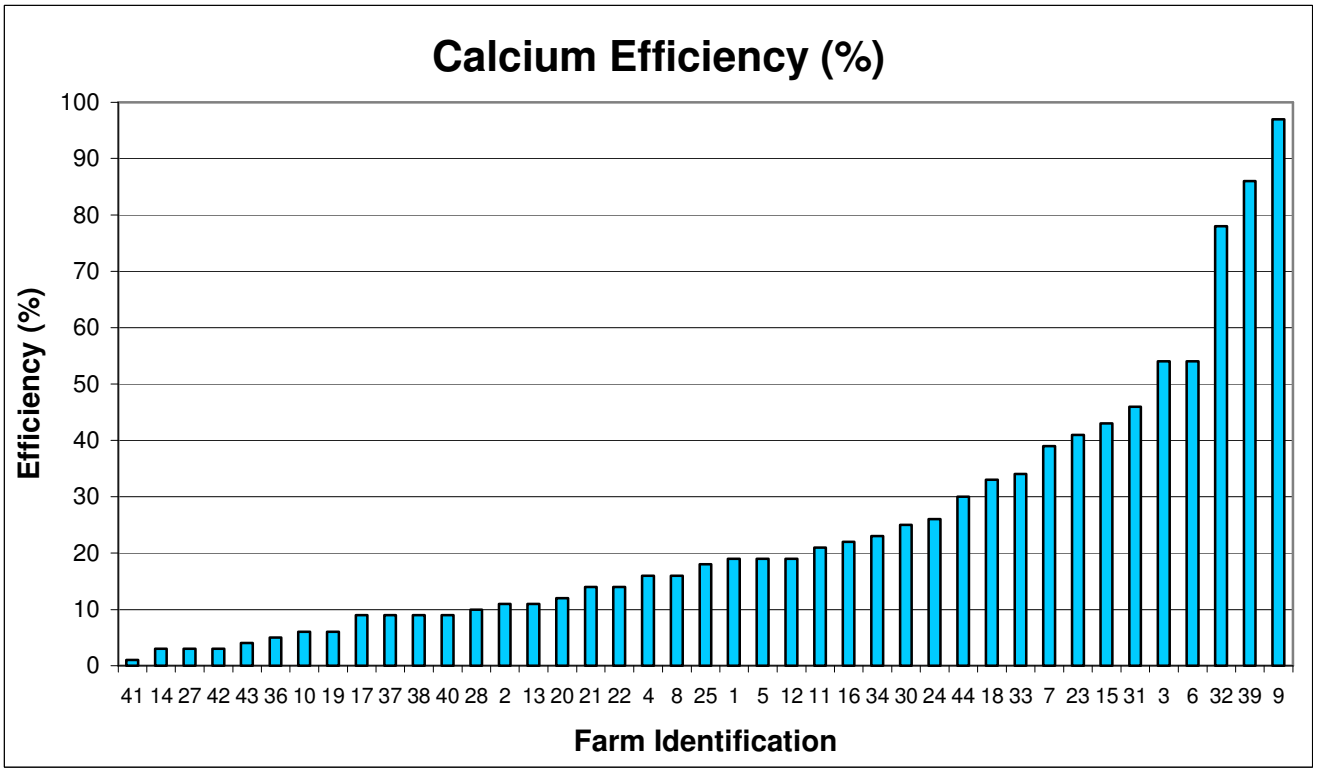

Potassium Surplus/Deficit (kg/ha)
Potassium (kg/ha)
500
400
300
200
100
0
24 7 12 5 14 3 23 15 30 9 41 13 40 25 10 8 42 43 1 37 27 38 6 21 4 31 19 20 33 2 16 22 28 17 44 18 39 11 32 34 36
Farm Identification

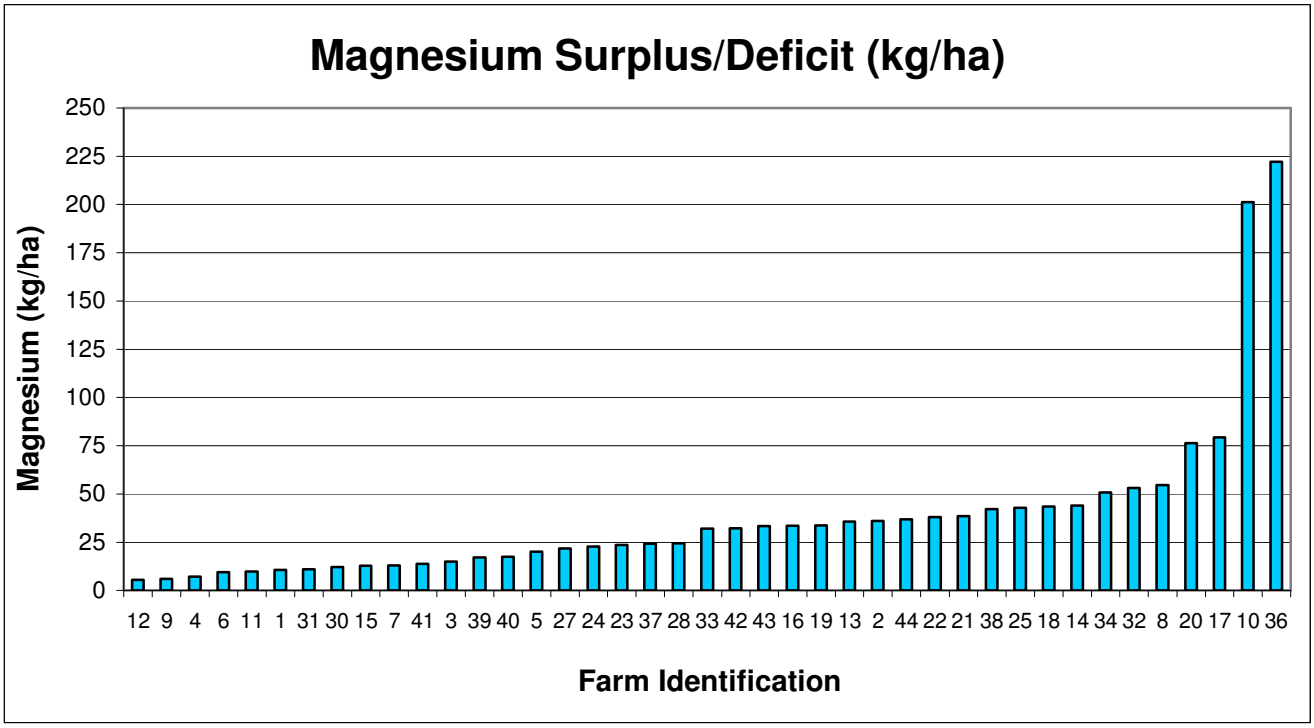

Potassium Efficiency (%)
Efficiency (%)
50
45
40
35
30
25
20
15
10
5
0
11 28 37 36 39 6 21 22 41 27 32 34 33 1 31 44 2 20 19 5 17 18 4 25 30 38 7 8 42 16 9 43 12 13 14 40 24 3 15
Farm Identification

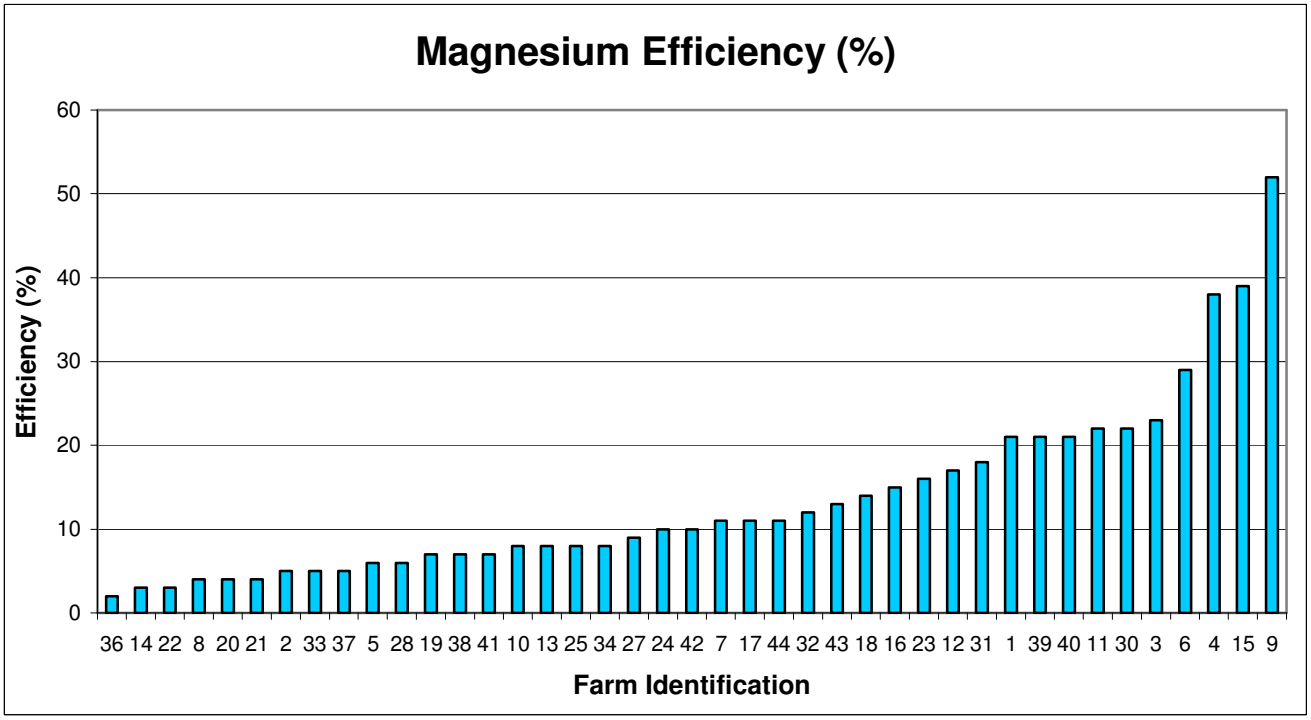

Sulphur Surplus/Deficit (kg/ha)
Sulphur (kg/ha)
195
175
155
135
115
95
75
55
35
15
-5
3 7 14 5 10 24 15 6 9 38 33 12 37 21 8 13 41 31 27 30 22 43 18 40 25 28 17 44 23 11 42 16 39 2 32 34 4 1 19 20 36
Farm Identification


Sulphur Efficiency (%)
Efficiency (%)
120
100
80
60
40
20
0
20 36 1 19 41 2 37 22 28 11 12 42 27 4 30 39 34 8 21 25 23 31 44 33 40 32 38 5 16 13 43 24 17 6 18 9 15 14 7 10 3
Farm Identification


Calcium Surplus/Deficit (kg/ha)
Calcium (kg/ha)
1400
1300
1200
1100
1000
900
800
700
600
500
400
300
200
100
0
9 39 7 32 6 24 31 3 33 5 12 23 37 15 30 21 22 44 1 25 11 8 18 28 34 13 4 2 16 38 20 40 14 19 27 41 17 42 43 36 10
Farm Identification

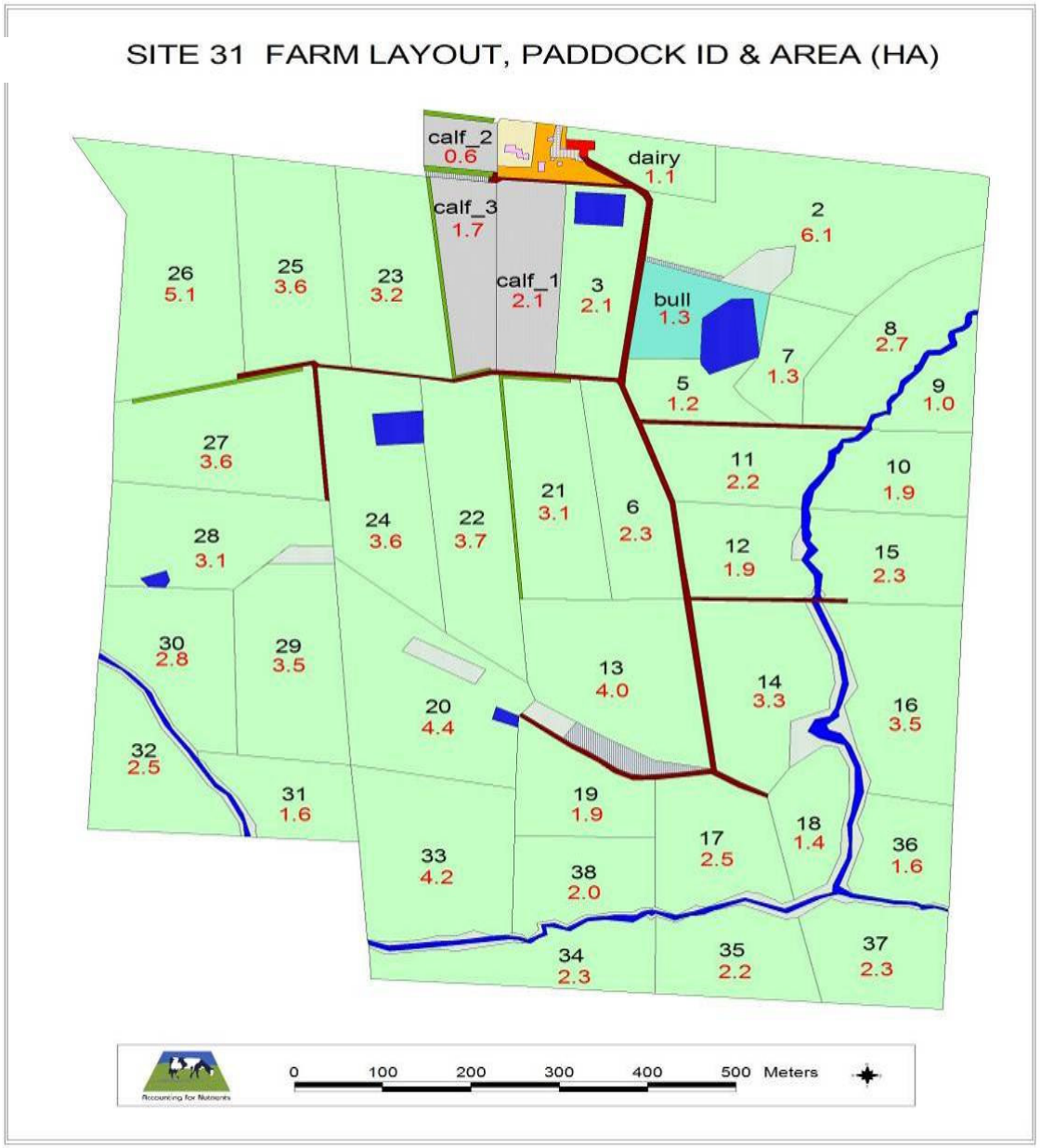

Calcium Efficiency (%)
Efficiency (%)
100
90
80
70
60
50
40
30
20
10
0
41 14 27 42 43 36 10 19 17 37 38 40 28 2 13 20 21 22 4 8 25 1 5 12 11 16 34 30 24 44 18 33 7 23 15 31 3 6 32 39 9
Farm Identification


Magnesium Surplus/Deficit (kg/ha)
Magnesium (kg/ha)
250
225
200
175
150
125
100
75
50
25
0
12 9 4 6 11 1 31 30 15 7 41 3 39 40 5 27 24 23 37 28 33 42 43 16 19 13 2 44 22 21 38 25 18 14 34 32 8 20 17 10 36
Farm Identification


Magnesium Efficiency (%)
Efficiency (%)
60
50
40
30
20
10
0
36 14 22 8 20 21 2 33 37 5 28 19 38 41 10 13 25 34 27 24 42 7 17 44 32 43 18 16 23 12 31 1 39 40 11 30 3 6 4 15 9
Farm Identification

# Appendix 7. Example of information provided to participating dairy farms



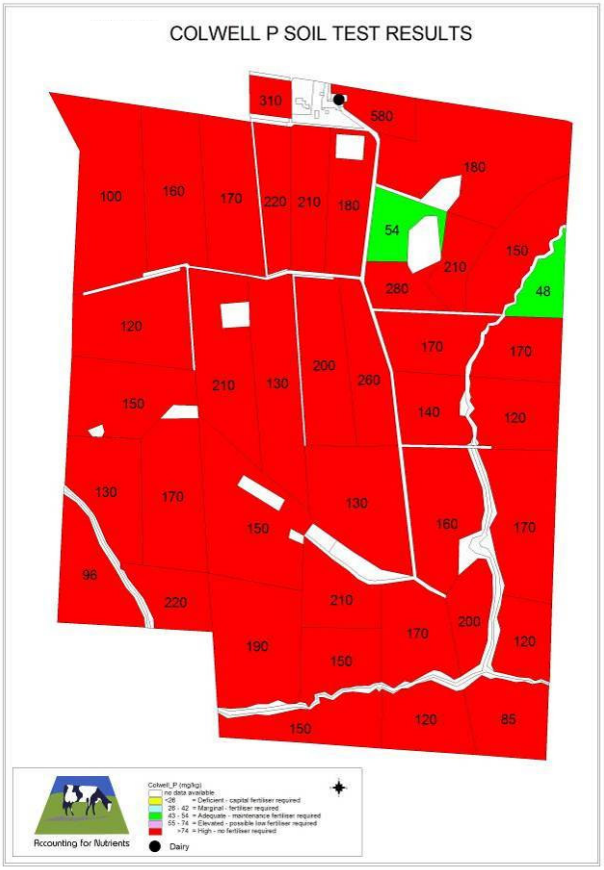

SITE 31 FARM LAYOUT, PADDOCK ID & AREA (HA)
calf_2
0.6
dairy
1.1
calf_3
1.7
calf_1
2.1
3
2.1
bull
1.3
2
6.1
26
5.1
25
3.6
23
3.2
8
2.7
7
1.3
5
1.2
9
1.0
27
3.6
11
2.2
10
1.9
21
3.1
6
2.3
24
3.6
22
3.7
28
3.1
12
1.9
15
2.3
30
2.8
29
3.5
13
4.0
14
3.3
16
3.5
20
4.4
32
2.5
31
1.6
19
1.9
18
1.4
17
2.5
36
1.6
33
4.2
38
2.0
34
2.3
35
2.2
37
2.3
0
100
200
300
400
500 Meters

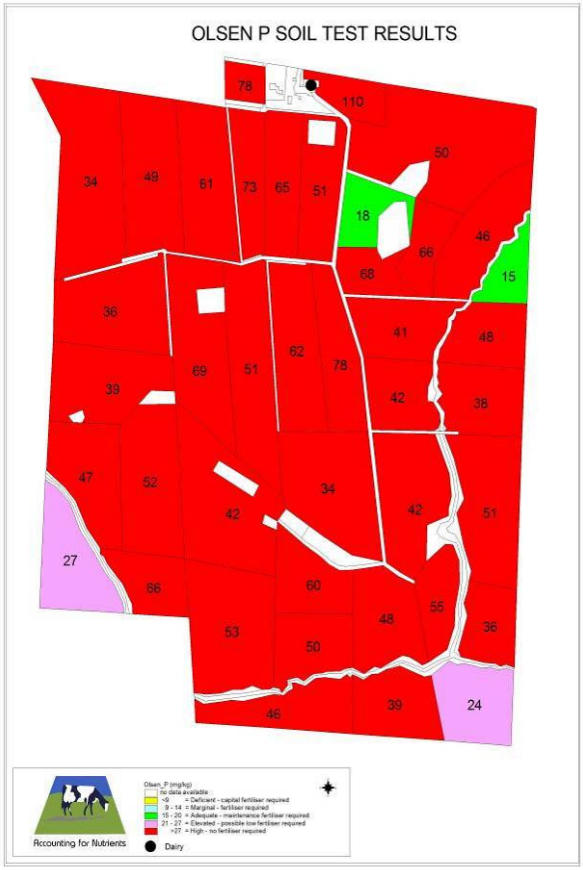
## Whole farm nutrient balance (total kg nutrient) - farm #

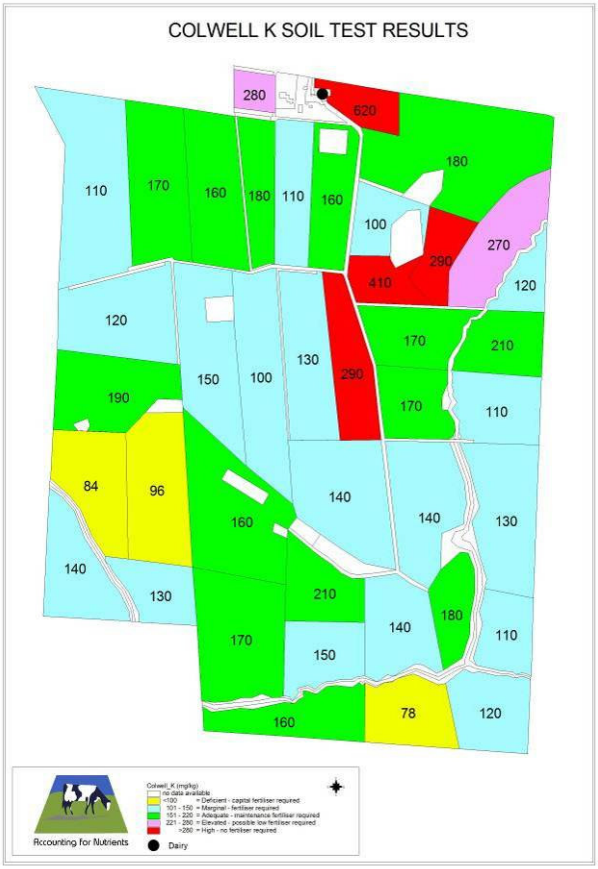
| | Nitrogen (kg) | Phosphorus (kg) | Potassium (kg) | Sulphur (kg) | Calcium (kg) | Magnesium (kg) |
|---|---|---|---|---|---|---|
| **Imports** | | | | | | |
| Silage | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Hay | 1138 (98) | 183 (18) | 1077 (67) | 114 (11) | 341 (72) | 110 (16) |
| Concentrates/grains | 8868 (862) | 902 (81) | 1501 (143) | 675 (71) | 190 (24) | 412 (39) |
| Feed minerals | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Byproducts | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| NPKS blends | 0 (0) | 1216 (34) | 7600 (215) | 1824 (52) | 2432 (69) | 0 (0) |
| Urea/Liquid N | 6028 (98) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Other fertilisers | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Soil ameliorants | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Animals | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Irrigation | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| N fixation | 138 (54) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Atmospheric deposition | 100 (11) | 10 (1) | 50 (6) | 50 (6) | 50 (6) | 50 (6) |
| Other | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| **Total Imported** | **16273** (875) | **2310** (90) | **10228** (267) | **2663** (88) | **3013** (102) | **572** (43) |
| **Exports** | | | | | | |
| Milk | 5717 (116) | 998 (67) | 1514 (73) | 456 (80) | 1230 (55) | 110 (4) |
| Animals | 695 (16) | 179 (4) | 50 (1) | 199 (3) | 497 (7) | 124 (0) |
| Forages | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Manure | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Other | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| **Total exported** | **6413** (117) | **1177** (67) | **1564** (73) | **654** (80) | **1727** (55) | **234** (4) |
| **BALANCE** | **9860** (883) | **1134** (112) | **8664** (276) | **2009** (119) | **1287** (116) | **337** (43) |
| **Efficiency (%)** | **39** | **51** | **15** | **25** | **57** | **41** |

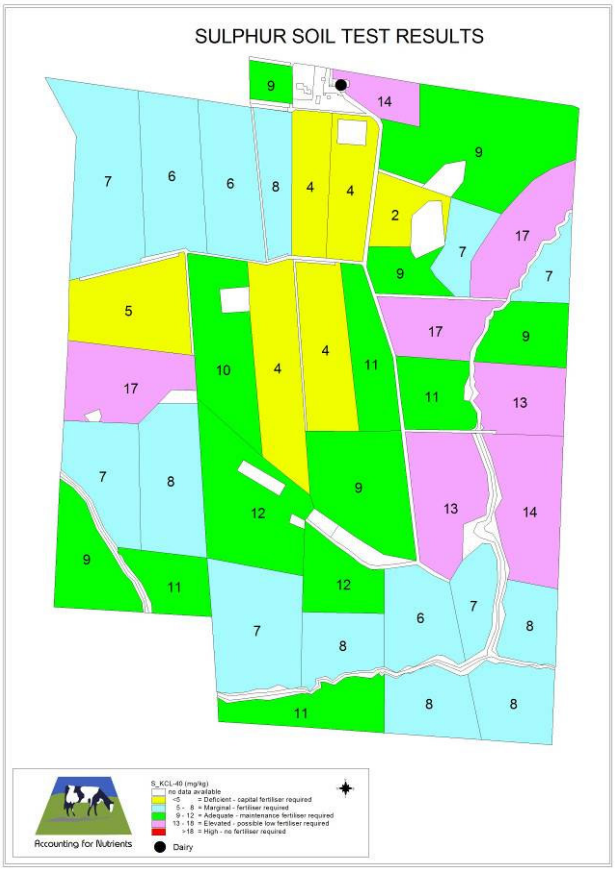
## Whole farm nutrient balance (kg/ha nutrient) - farm #

| | N (kg/ha) | P (kg/ha) | K (kg/ha) | S (kg/ha) | Ca(kg/ha) | Mg(kg/ha) |
|---|---|---|---|---|---|---|
| **Imports** | | | | | | |
| Silage | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Hay | 11 (1) | 2 (0) | 11 (1) | 1 (0) | 3 (1) | 1 (0) |
| Concentrates/grains | 89 (9) | 9 (1) | 15 (1) | 7 (1) | 2 (0) | 4 (0) |
| Feed minerals | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Byproducts | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| NPKS blends | 0 (0) | 12 (0) | 76 (2) | 18 (1) | 24 (1) | 0 (0) |
| Urea/Liquid N | 60 (1) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Other fertilisers | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Soil ameliorants | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Animals | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Irrigation | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| N fixation | 1 (1) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Atmospheric deposition | 1 (0) | 0 (0) | 1 (0) | 1 (0) | 1 (0) | 1 (0) |
| Other | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| **Total Imported** | **163** (9) | **23** (1) | **102** (3) | **27** (1) | **30** (1) | **6** (0) |
| **Exports** | | | | | | |
| Milk | 57 (1) | 10 (1) | 15 (1) | 5 (1) | 12 (1) | 1 (0) |
| Animals | 7 (0) | 2 (0) | 0 (0) | 2 (0) | 5 (0) | 1 (0) |
| Forages | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Manure | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| Other | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) | 0 (0) |
| **Total Exported** | **64** (1) | **12** (1) | **16** (1) | **7** (1) | **17** (1) | **2** (0) |
| **BALANCE** | **99** (9) | **11** (1) | **87** (3) | **20** (1) | **13** (1) | **3** (0) |
| **Efficiency (%)** | **39** | **51** | **15** | **25** | **57** | **41** |


COLWELL P SOIL TEST RESULTS
310
580
180
100
160
170
220
210
180
54
150
210
280
48
120
170
170
200
210
130
260
150
140
120
130
170
150
130
160
170
96
220
210
200
190
170
120
150
150
120
85
Accounting for Nutrients
Dairy

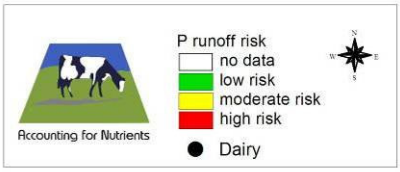

OLSEN P SOIL TEST RESULTS
78
110
50
34
49
61
73
65
51
18
46
66
68
15
36
41
48
62
78
69
51
39
42
38
47
52
34
42
51
42
27
66
60
55
48
53
50
36
46
39
24
Accounting for Nutrients
Dairy


COLWELL K SOIL TEST RESULTS
280
620
180
110
170
160
180
110
160
100
270
290
410
120
120
170
210
150
100
130
290
190
170
110
84
96
140
140
160
130
210
140
180
130
140
170
150
110
160
78
120
Accounting for Nutrients
Dairy

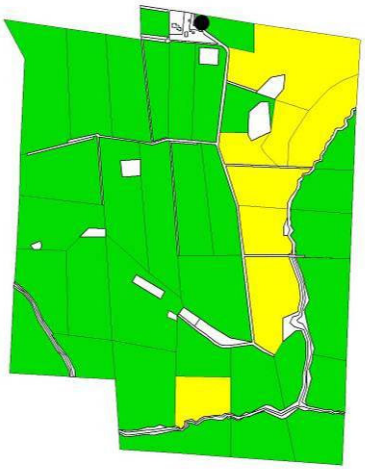

SULPHUR SOIL TEST RESULTS
9
14
9
7
6
6
8
4
4
2
17
7
9
7
5
17
9
4
4
11
10
17
11
13
7
8
9
13
14
12
9
11
12
6
7
8
7
8
11
8
8
Accounting for Nutrients
Dairy

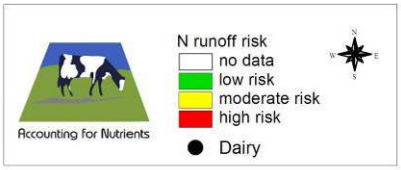
Phosphorus surface runoff risk






Nitrogen surface runoff risk





# Appendix 8.

## Options for improving nutrient use efficiency and reducing nutrient losses to the environment

Review this list of management options when making nutrient decisions on your farm.

**MANAGEMENT OPTIONS**

| |
|---|
| **N fertiliser management**<br>❖ Time N-fertiliser application for periods when pasture and crop N demand is greatest<br>❖ Apply optimum N applications (30-50 kg N/ha) per application<br>❖ Adjust N-fertiliser rates & timings seasonally to respond to actual or expected production demand (seasonal variations)<br>❖ Avoid N-applications when soils are saturated (i.e. when leaching or runoff risk is high & low plant demand).<br>❖ Avoid N-applications when soils are dry and plants are under moisture stress (plant N-uptake low). |
| **P, K, S fertiliser management**<br>❖ Consider nutrient levels already in soil when deciding on fertiliser types and rates<br>❖ Consider nutrient loss pathways when deciding where to apply fertiliser or effluent<br>❖ Use adequate buffer distance from waterways (+20 m) when applying fertiliser |
| **Effluent ponds and manure management**<br>❖ Ensure effluent storage ponds are of a sufficient size and do not over flow ( particularly in winter)<br>❖ Ensure effluent storage ponds are sealed<br>❖ Recycle effluent to land rather than pond treatment & disposal to waterways<br>❖ Ensure runoff from tracks/lanes is not channelled into streams near crossings<br>❖ Use adequate buffer distance from waterways (+20 m) when applying effluent<br>❖ Use a sealed wintering/standing pad with effluent collection and storage system and ensure runoff from yards, feed pads, etc. does not go directly into waterways |
| **Supplementary feeding areas and night paddocks**<br>❖ Store leaky supplementary feeds (e.g. silage) on a sealed base with an effluent collection/storage/disposal system<br>❖ Avoid the continued use of the same areas as night-paddocks and feed-out paddocks |